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Report No: 36421

PROJECT APPRAISAL DOCUMENT

ON A

PROPOSED LOAN

IN THE AMOUNT OF US$74 MILLION

TO THE

REPUBLIC OF PARAGUAY

FOR A

ROAD MAINTENANCE PROJECT

July 24, 2006

Finance, Private Sector Development and Infrastructure Department
Argentina, Chile, Paraguay and Uruguay Country Management Unit
Latin America and the Caribbean Region

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CURRENCY EQUIVALENTS

(Exchange Rate Effective June 30, 2006)

Currency Unit = Paraguayan Guaranies
US$1 = PYG 5,570

FISCAL YEAR
January 1 – December 31

ABBREVIATIONS AND ACRONYMS

CAS	Country Assistance Strategy
CIF	Cost, Insurance and Freight
DAI	MOPC's Internal Audit Directorate *(Dirección de Auditoría Interna)*
DAJ	VMOPC's Legal Directorate *(Dirección de Asuntos Jurídicos)*
DAT	VMOPC's Technical Assistance Directorate *(Dirección de Asesoría Técnica)*
DC	Road Maintenance Department *(Departamento de Conservación)*
DCP	VMAF's Aid Management Directorate *(Dirección de Crédito Público)*
DCST	Traffic Security and Control Directorate *(Dirección de Control y Seguridad de Tránsito)*
DCV	Directorate of Rural Roads *(Dirección de Caminos Vecinales)*
DF	VMAF's Finance Directorate *(Dirección de Finanzas)*
DINATRAN	National Directorate of Transport *(Dirección Nacional de Transporte)*
DPE	VMAF's Economic Planning Directorate *(Dirección de Planificación Económica)*
DV	Directorate of Highways *(Dirección de Vialidad)*
EP	Pavement Evaluation Department *(Departamento de Evaluación de Pavimento)*
EU/UA	MOPC's Environmental Unit *(Unidad Ambiental)*
FOB	Free On Board
GMANS	Performance-based Road Maintenance Contracts *(Contrato de Gestión y Mantenimiento por Nivel de Servicio)*
GOP	Government of Paraguay
GTZ	German Technical Cooperation Agency *(Gessellschaft fur Technische Zusammenarbeit)*

HDM	Highways Design & Maintenance Standards Model
IBRD	International Bank for Reconstruction and Development
ICB	International Competitive Bidding
IDB	Inter-American Development Bank
IERR	Internal Economic Rate of Return
ILO	International Labor Organization
IRI	International Roughness Index
JBIC	Japan Bank for International Cooperation
LAC	Latin America and the Caribbean
MOF	Ministry of Finance *(Ministerio de Hacienda)*
MOPC	Ministry of Public Works and Communications *(Ministerio de Obras Públicas y Comunicaciones)*
PC	Traffic Police *(Policía Caminera)*
PSP	Private Sector Participation
SIAMV	Integrated Road Maintenance Management System
SIVIPAR	Paraguay's Road Infrastructure System *(Sistema de Infraestructura Vial del Paraguay)*
UBI	Patrimony Unit *(Unidad de Bienes Inmobiliarios)*
UOC	MOPC's Procurement Unit *(Unidad Operativa de Contrataciones)*
UP	Toll Unit *(Unidad de Peajes y Pesajes)*
UPGP	Project Preparation Unit *(Unidad de Preparación y Gestión del Proyecto)*
VMOPC	MOPC's Vice-Ministry of Public Works and Communications *(Vice-Ministerio de Obras Públicas y Comunicaciones)*
VMAF	MOPC's Vice-Ministry of Administration and Finance *(Vice-Ministerio de Administración y Finanzas)*

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Country Director:	Axel van Trotsenburg
Sector Director	Makhtar Diop
Sector Manager:	José Luis Irigoyen
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PARAGUAY
Road Maintenance

CONTENTS

Page

A. STRATEGIC CONTEXT AND RATIONALE 1
1. Country and sector issues 1
2. Rationale for Bank involvement 4
3. Higher level objectives to which the project contributes 5

B. PROJECT DESCRIPTION 5
1. Lending instrument 5
2. Project development objective and key indicators 5
3. Project components 7
4 Lessons learned and reflected in the project design 9
5. Alternatives considered and reasons for rejection 10

C. IMPLEMENTATION 12
1. Partnership arrangements 12
2. Institutional and implementation arrangements 13
3. Monitoring and evaluation of outcomes/results 13
4. Sustainability 14
5. Critical risks and possible controversial aspects 15
6. Loan/credit conditions and covenants 17

D. APPRAISAL SUMMARY 18
1. Economic and financial analyses 18
2. Technical 19
3. Fiscal 20
4. Fiduciary 20
5. Social 20
6. Environment 22
7. Governance 23
8. Safeguard policies 24
9. Policy Exceptions and Readiness 24

Annex 1: Country and Sector Background 25

Annex 2: Major Related Projects Financed by the Bank and/or other Agencies 35

Annex 3: Results Framework and Monitoring 36

Annex 4: Detailed Project Description 43

Annex 5: Project Costs 53

Annex 6: Implementation Arrangements 54

Annex 7: Financial Management and Disbursement Arrangements 59

Annex 8: Procurement Arrangements 67

Annex 9: Economic and Financial Analysis 74

Annex 10: Safeguard Policy Issues 92

Annex 11: Infrastructure and Public Spending 114

Annex 12: Social Evaluation of the Component of Improvement and Maintenance of the Unpaved Road Network 119

Annex 13: Improved Governance Framework and Project Action Plan 135

Annex 14: Project Preparation and Supervision 160

Annex 15: Documents in the Project File 161

Annex 16: Statement of Loans and Credits 164

Annex 17: Country at a Glance 165

Annex 18: Letter of Sector Policy 167

Annex 19: Maps IBRD 34798, IBRD 34799, IBRD 3480, IBRD 3481 171

PARAGUAY
ROAD MAINTENANCE
PROJECT APPRAISAL DOCUMENT
LATIN AMERICA AND CARIBBEAN
LCSFT

Date: July 24, 2006	Team Leader: Andres G. Pizarro
Country Director: Axel van Trotsenburg	Sectors: Roads and highways (70%); General transportation sector (30%)
Sector Manager/Director: Jose Luis Irigoyen	Themes: Trade facilitation and market access (P); Infrastructure services for private sector development (S); Rural services and infrastructure (S)
Project ID: P082026	Environmental screening category: Partial Assessment
Lending Instrument: Specific Investment Loan	

Project Financing Data

[X] Loan [ ] Credit [ ] Grant [ ] Guarantee [ ] Other:
For Loans/Credits/Others:
Total Bank financing (US$m.): 74.00
Proposed terms: Fixed Spread Loan with principal repayments linked to commitments, payable in 23 years, including a 5-year grace period with a level repayment schedule with repayments on each April 15 and October 15 respectively.

Financing Plan (US$m)

Source	Local	Foreign	Total
BORROWER	31.41	0.00	31.41
INTERNATIONAL BANK FOR RECONSTRUCTION AND DEVELOPMENT	74.00	0.00	74.00
Total:	105.41	0.00	105.41

Borrower:
Ministerio de Hacienda
Chile 128
Asunción, Paraguay
Tel: 595-21-448-283 Fax: 595-21-412-6004
www.hacienda.gov.py
Responsible Agency:
Ministry of Public Works and Communications
Oliva y Alberdi
Asunción, Paraguay
Tel: (595) (21) 414 9000
ministro@mopc.gov.py
www.mopc.gov.py

Estimated disbursements (Bank FY/US$m)

FY	2007	2008	2009	2010	2011	2012	0	0	0
Annual	8.47	19.87	19.12	12.41	9.42	4.71	0.00	0.00	0.00
Cumulative	8.47	28.34	47.46	59.87	69.29	74.00	74.00	74.00	74.00

Project implementation period: Start September 1, 2006 End: August 31, 2011
Expected effectiveness date: January 1, 2007
Expected closing date: June 30, 2012

Does the project depart from the CAS in content or other significant respects? ***Ref. PAD A.3***	[ ]Yes [X] No
Does the project require any exceptions from Bank policies? ***Ref. PAD D.7***	[ ]Yes [X] No
Have these been approved by Bank management?	[ ]Yes [ ] No

Is approval for any policy exception sought from the Board?	[ ]Yes [X] No
Does the project include any critical risks rated "substantial" or "high"? ***Ref. PAD C.5***	[X]Yes [ ] No
Does the project meet the Regional criteria for readiness for implementation? ***Ref. PAD D.7***	[X]Yes [ ] No

Project development objective: ***Ref. PAD B.2, Technical Annex 3***
The project's development objective is to establish a sustainable road management strategy that ensures the upgrading and maintenance of the road network through a rational and balanced use of scarce resources.

Project description: ***Ref. PAD B.3.a, Technical Annex 4***
Component 1: Strengthening Strategic Planning and Road Management (Estimated cost US$6.5 million (not including contingencies): This component would finance the strengthening of strategic planning, road conservation, and budget management and will be subdivided into 4 subcomponents.
- Sub-component 1.1: Strengthening of Strategic Planning;
- Sub-component 1.2: Strengthening Budget Preparation and Execution;
- Sub-component 1.3: Strengthening Road Management and Maintenance;
- Sub-component 1.4: Monitoring and Evaluation of the Project;

Component 2: Improvement and Maintenance of the Paved Road Network (Estimated cost US$43.3 million (not including contingencies): This component has two subcomponents which complement each other (i) performance based contracts; and (ii) toll system optimization.
- Sub-component 2.1: Performance-based Maintenance Contracting ($39.67million): This sub-component would finance four contracts that include initial rehabilitation works to meet the minimum level of service, road safety works, vertical and horizontal signaling, and maintenance works of the national paved road network, through performance-based maintenance contracts (GMANS).
- Sub-component 2.2: Toll System Optimization ($3.6 million): This sub-component would finance (i) A Toll System Optimization Master Plan, and (ii) Investments to implement the Master Plan.

Component 3: Improvement and Maintenance of Unpaved Road Network (Estimated cost US$42.9 million (excluding contingencies): This component will finance the rehabilitation and conservation of unpaved national roads that connect to the national paved network, and secondary roads connecting rural communities to these; and propose a comprehensive solution for improved access to isolated rural communities. It is subdivided into two sub-components:
- Sub-component 3.1: Road improvement and maintenance works.
- Sub-component 3.2: Capacity Building and Development Program for MOPC District Offices and Regional and Local Governments.

Which safeguard policies are triggered, if any? ***Ref. PAD D.6, Technical Annex 10***
(i) Environmental Assessment; (ii) Involuntary Resettlement; (iii) Indigenous Peoples

Significant, non-standard conditions, **if any**, for: ***Ref. PAD C.7***

Board presentation: NONE

Loan/credit effectiveness: NONE

Covenants applicable to project implementation:

- Operational Manual: the Borrower will implement the Project according to the Operational Manual agreed at negotiations and acceptable to the Bank.
- Road and Traffic Survey: the Borrower to carry out road and traffic surveys on an annual basis and furnish to the Bank the corresponding data no later than June 30 of the following year (starting 2008).
- Safeguards and Social Participation Framework: (i) the Borrower to designate a sociologist or anthropologist in charge of supervising the Indigenous People's Plan implementation, and coordinate the Social Participation Framework.
- Improved Governance Framework and Project Action Plan: (i) the Borrower to take all necessary measures to comply with the Mitigation Actions contained in the Table A13-2 of the Improved Governance Action Plan (Annex 13); (ii) the Borrower, through the MOPC's Internal Audit Unit, to provide the Bank with semi-annual reports on the Alert Indicators established in the Project Action Plan; and (iii) the Borrower,

through the MOPC and the Ministry of Finance, to hold annual reviews with the Bank to update the provisions of the Improved Governance Framework and Project Action Plan.

- Procurement: (i) the Borrower to implement a project management information system satisfactory to the Bank; (ii) special procurement provisions to be reflected in Loan Agreement; and (iii) the Borrower to update the Procurement Plan in accordance to Bank Guidelines, and furnish said updated plan to the Bank for approval, no later than 12 months after the date of the preceding Procurement Plan.
- Financial Management: (i) the Borrower to create within the first six months of Project implementation, a separate budgetary line to allow report based disbursements against quarterly budget allocation of funds.
- Financial Report: The Borrower to prepare and furnish to the Bank an interim unaudited financial report, satisfactory to the Bank, no later than 45 days after each subsequent calendar semester.
- Monitoring: the Borrower to prepare project progress reports on a semi-annual basis and submit them to the Bank for review. These reports should refer to progress made under the different components of the project and measure performance against the results indicators established in the results framework.
- Evaluation: the Borrower to prepare under terms of reference acceptable to the Bank an assessment of the impacts of the Project (enhanced road condition, reduced transport costs, enhanced production opportunities, job generation, increased road safety). For this purpose baseline indicators will be required no later than six months after effectiveness. The Borrower shall present the results at the Mid-Term Review meeting and at Project end.

A. STRATEGIC CONTEXT AND RATIONALE

1. Country and sector issues

General political and economic context

1. The current Administration, that took office in August 2003, decided to focus its development agenda on fighting corruption and getting Paraguay out of the economic stagnation it had been suffering in the past 5 years and that had reduced its per capita income by 15%. The specific objectives of the Administration's policy are the following: (i) restore confidence in public institutions through a constant fight against corruption and the modernization of public administration; (ii) sustain economic growth by shifting from an economy based on informal commerce associated with the import and re-export of consumer products and public employment, to one centered on agro-industry and diversified exports; (iii) increase human capital through increased investment in education and health, and through policies that improve equity and access to these services.

2. The economic performance of Paraguay has improved in the last three years as evidenced by the return to positive rates of economic growth, which reached 3.8, 4.1 and 2.9 percent in 2003, 2004 and 2005[1], respectively; the decline of inflation; and the overall improvement in fiscal accounts. This recovery was prompted by a combination of positive developments, which included a pronounced expansion in agricultural output driven by the exceptionally high commodity prices, the institution of more rational spending policies, and enhancements in tax administration.

3. Despite these important advances, economic growth is still lower than required for the country to make up for the time it lost during the second half of the nineties, when poverty rates and income inequality increased considerably. In a constrained fiscal framework, Paraguay will have to reconcile the need for increased investment with prudent macroeconomic management, and enhance the impact of limited resources by improving the efficiency of public expenditure. Infrastructure development is a top priority for the Government of Paraguay (GOP) due to its positive impact on economic growth and social inclusion and the considerable gap in quality and coverage that exists between Paraguay and other countries in the Latin America and the Caribbean (LAC) region. The challenge is to improve investment climate, target investments to help the poor, improve the efficiency of public expenditure, and continue fighting corruption.[2] The road transport sector is fundamental for achieving all of these development objectives through its linkages to competitiveness, social inclusion, public management and private sector participation (PSP), and transparency.

4. The improvement of Paraguay's efficiency in public expenditure will depend highly on its ability to fight corruption and enhance its governance environment and practices. In this respect, the current government's development strategy led by President Duarte Frutos has as one of its main objectives the recovery of confidence in state institutions through among other measures a sustained fight against corruption and a greater participation of civil society in the formulation of public policy and the control of public expenditure. Following the GOP's initiatives of improving transparency and governance overall, this project has been designed from the outset on building an integrated approach that seeks to enhance both the efficacy and impact of the project and the GOP's governance practices and processes. To this end, the project includes an Improved Governance Action Plan that has been designed in close collaboration with the GOP and the World Bank's Department of Institutional Integrity, following global practices in this field. The objective of this plan is to strengthen governance, participation and

[1] International Monetary Fund.
[2] Paraguay: Development Policy Review, June 30, 2004.

transparency around the project and as a result prevent corruptive practices, so that the full impact potential of the project is attained.

The road sector and Paraguay's development agenda

5. ***Road infrastructure development is critical for the competitiveness of Paraguayan producers.*** Given its landlocked status, unless air or river transport is involved, the external trade of Paraguay must include a land segment, which tends to increase the cost of transport and insurance and pushes up the price of imports. About 85 percent of the total volume of domestic and international goods (soybean and cotton in particular) is transported by roads, which provide critical links to ports in neighboring countries and export markets in the Mercosur region.[3] The competitiveness of the agro-industry continues to be constrained by transport costs, which remain among the highest in the LAC region. Although many other factors besides road transport are involved in the determination of the cost of insurance and freight, an indication of the importance of transport for Paraguay's development can be found in the average CIF/FOB ratio. This ratio has been found to be close to 9 percent for Paraguayan imports from all destinations, one of the highest in the region. The same applies to homogeneous goods originating in the region, for which the cost of insurance and transport in Paraguay has also been found to be one of the highest in LAC. [4]

6. ***Road infrastructure investments in rural Paraguay can have a positive impact on poverty alleviation by reducing the cost of accessing basic services and creating business opportunities for the poor.*** Poverty is more widespread in rural than in urban areas: 75 percent of the extreme poor and 56 percent of the poor lived in rural areas in 2001. The improvement of secondary and rural roads can promote social inclusion by reducing the cost of accessing educational and health services in rural communities. It can also contribute to link small farmers, who are mostly poor, to markets in the same way that major commercial agricultural sites are now connected to export gateways, and thus help them increase their productivity and go beyond subsistence production.

7. ***Road infrastructure investments account for a sizeable portion of public investment in Paraguay.*** The budget of the DV, the specialized sector agency under the MOPC, accounted for 10 percent of the overall budget of the central administration in 2004. The sector also employs a large number of public servants (2 percent of the total), making it central to all efforts to improve the efficiency of public expenditure. Efficiency could be enhanced through improvements in planning and programming, increasing allocations for maintenance, improving priority setting framework for rehabilitation and new projects, and effective monitoring of expenditures.

Main Sector Issues and strategic choices

8. Given the economic and sector context, the strategic importance of the road transport sector for Paraguay's sustainable development, and the road sector strategy, the following are the issues on which the GOP is seeking the support of the Bank.

a. ***Addressing improvement needs of the unpaved national primary and secondary roads.*** No hierarchical legal definition exists of the road network other than urban and inter-urban roads; the MOPC being legally responsible for the entire inter-urban road network. Paraguay has 60,000 km of inter-urban roads of which around 4,000 km are paved (asphalt mainly), 4,500 km are all weather (stone paved or gravel), and the remaining 51,500 km are earth roads with no treatment. Around

[3] Paraguay: Country Assistance Strategy, November 26, 2003.
[4] ECLAC (2001)

15,000 km of roads could be considered the main road network, and the remaining 45,000 km rural road network. The road network endowment seems comparable to equivalent countries and adequately covers the national territory, but the quantity of paved roads and its distribution is uneven. The great majority of paved roads are located in the eastern part of the country while in the western region (Paraguayan Chaco) there are: (i) the road to the Argentinean frontier via Puerto Falcon; and (ii) the TransChaco highway to the Bolivian border, paved to *Estancia La Patria* and under construction from *Estancia La Patria* to *Infanta Rivarola* (border with Bolivia). In the western region, inhabited by 3% of Paraguay's population, there is also a program of improvement and maintenance of roads that benefits 2,000 km of paved and unpaved roads. Nevertheless overall, the amount of paved roads in Paraguay compares badly in the region: Paraguay has 0.77 km/1000 inhabitants of paved roads against 3 km/1000 inhabitants for Argentina, and 4 km/1000 inhabitants for Uruguay. This results explains the historic MOPC policy of paving and network development to reduce the gap of unmet needs, and currently reflected in the construction of the TransChaco highway and new projects to pave highways that may be considered missing links in the main network such as: the route 11 linking Santa Rosa-Puerto Antequera, and the route linking Caazapá-Cnel. Bogado. However, the efforts deployed to pave the network have not improved its overall quality as the condition of the main road network has deteriorated in recent years reaching 14% good condition, 56% fair condition and 30% bad condition, illustrating the inadequacy of preventive maintenance mechanisms and management of road assets. Additionally, too little has been invested on improving unpaved roads, the ratio of all-weather roads to paved roads in Paraguay is 1 paved km to 0.75 km all-weather road, while what is used as a rule of thumb for this ratio is generally 1 paved to 3 all-weather indicating gradual investments according to needs. As a result, many communities, including one Department capital (San Pedro), do not have all weather access, constituting one of the greatest hindrances to development of the country[5]. Finally, it seems that, in light of inadequate road maintenance mechanisms, the authorities are pushed primarily to pave roads rather than to seek low-cost all-weather solutions that with the design and management of adequate maintenance programs would suffice in some situations, partly explaining the unbalanced ratio of paved to all-weather roads.

b. ***Maintenance expenditures and mechanisms are insufficient to preserve the condition of the road network.*** Experience shows that the optimal way of preserving road assets involves allocating sufficient resources to road maintenance as a priority and the surplus to road network extension. The practice today in Paraguay has been the opposite. The annual budget for the Road Directorate over the last four years has averaged around US$150 million, of which US$8 million have been destined to maintenance. However, the amount effectively invested in maintenance is less than half that amount. When only the minimum investments are considered, the maintenance gap is estimated conservatively at US$34 million per year. If this situation does not change, Paraguay will face an increasing loss in the value of road assets and will incur in the greater costs associated with the reconstruction of deteriorated infrastructure. GOP needs to adopt a policy of optimizing resources by giving priority to road maintenance. As a result, the decentralized MOPC Districts are faced with a significant amount of demands related to the unstabilized unpaved road network, that span from main roads to small rural roads. The relatively low-level of these roads makes them very vulnerable to weather related emergencies and as a result, the Districts end up dealing mainly with emergencies on the unpaved network rather that carrying out a routine maintenance program on the whole of the paved road network.

c. ***It is unlikely that the Road Fund Law (SIVIPAR) will become implemental in its present form, and thus provide new resources to the sector, as it has not achieved a consensus within the Government, because of its deficiencies.*** The SIVIPAR Law was adopted in 2003, but it has not been regulated or implemented. It provides for a wide range of uses, with the exception of maintenance

[5] Several rural water sub-projects financed by the Bank have had difficulties due to lack of accessibility to communities.

expenditure, and gives the MOPC significant flexibility in the choice of allocation. The sources of the funds for SIVIPAR are the existing ones (tax on petroleum derivatives, car registration tax etc.), and no additional resources have been included. The appropriate regulation of this law or the redrafting of a new law, reserving a significant amount of its proceeds to maintenance expenditures, is paramount to the sustainability of the road sector. The project team has contributed to the definition of the regulation by proving expertise and examples of Road Funds and has contributed to the debate on the subject. After many internal debates within MOPC, the Minister has opted for the redrafting of a new and improved law rather than attempting to regulate this existing and relatively poor SIVIPAR text. Additional work is needed on new regulations to be carried out during the execution of the Project to ensure sufficient funds for road maintenance expenditures.

d. ***Despite abundant donor assistance, planning in the transport sector remains weak.*** There is a lack of correlation between investment, maintenance and rehabilitation priorities (determined by some consistently applied criteria, whether economic or not) and budgets. A mixture of tradition, institutional set-up, and a lack of capacity, resources and incentives have rendered the planning process irrelevant. First of all, the elimination of the Transport Infrastructure Planning Office (OPIT), previously under the dismantled Vice-Ministry of Transport and its establishment as an independent, but under funded agency, severely affected the capacity of the MOPC to undertake investment planning. Also, the budgeting process is not conducive to inculcate planning processes in the DV's work procedures. The nature of the intervention carried out by both the Ministry of Finance (MH) and the Congress in budget approval, as well as the limited capabilities at the MOPC, limit the relevance of extended use of planning tools. Finally, the DV lacks an updated multi-annual plan for the development of the road sector.

e. ***The deficient system for results-based management and monitoring of projects in the road sector limits transparency and accountability in the use of resources.*** The lack of adequate information systems and of incentives for adequate monitoring results in poor management of programs and projects. Given the limitations of resources and the increased attention paid by GOP to transparency and efficiency, the MOPC, which is the main recipient of treasury transfers, is under increasing pressure by the Ministry of Finance to improve its monitoring of the results and impact of projects.

f. ***Lack of access in rural areas is contributing to social exclusion and most recently social unrest.*** Efforts to expand accessibility into rural areas have been uncoordinated and incoherent. Some donor programs are addressing both the municipal road network (JBIC, IDB) and the MOPC rural unpaved network. The type of roads and the level of service to which they are being rehabilitated do not follow a national strategy that is in line with future maintenance possibilities. The results have been costly in some cases due to excessive standards and difficult to replicate in others due to lack of institutional capacity in rural areas.

2. Rationale for Bank involvement

9. In this context, the Bank decided to renew its dialogue and assistance to the Government of Paraguay (GOP) through its Country Assistance Strategy (CAS) discussed on December 16, 2003. The CAS objectives are closely linked to the Administrations' strategy and seek to restore confidence in the economy and prepare a sound basis for broad-based economic growth and poverty reduction. To this end the CAS supports (i) the fiscal and financial reforms that the Administration has sent to Congress; (ii) improved governance and transparency in public administration; (iii) the promotion of sustained growth, particularly in the rural areas where poverty is concentrated; and (iv) social inclusion, to improve the

coverage and efficiency of basic social services to help Paraguay meet the Millennium Development Goals (MDGs).

10. The approach to road infrastructure sought through this Project will contribute to objectives (iii) and (iv) of the CAS, since improved road infrastructure ensures the territorial integration necessary for social inclusion and access to basic services in rural areas, and sustained economic growth is reliant on the stable condition and quality of the main road infrastructure network to facilitate productive and commercial transactions. Finally, the Project will also contribute to CAS objective (ii) as increased transparency and efficiency in the way in which rehabilitation, maintenance, management and financing of road infrastructure will be pursued.

3. Higher level objectives to which the project contributes

11. The project's higher level objective is to support economic growth and poverty alleviation. The improvement of secondary and rural roads reduces the cost of accessing basic services in rural communities and links small farmers to markets. Additionally, by ensuring improved and stable condition of the main paved road network the competitiveness of Paraguayan producers, who are highly reliant on land transport, will be increased and will contribute to promote export-led economic growth.

B. PROJECT DESCRIPTION

1. Lending instrument

12. The GOP has requested a Specific Investment Loan (SIL) for US$74 million to finance the enhancement, management and maintenance of its road infrastructure. The loan will be disbursed over five years (2007-2011) years.

2. Project development objective and key indicators

13. The project's development objective is to establish a sustainable road management strategy that ensures the upgrading and maintenance of the road network through a rational and balanced use of scarce resources. The guiding principle of this approach will be improved road conservation within a comprehensive vision that ensures the sustainability of road investments and the preservation of existing road assets. The project's specific objectives will be: (i) developing the institutional capacities of the MOPC for strategic planning and management of the road network; (ii) arresting the deterioration of Paraguay's priority road network, composed of international and regional road corridors, and maintaining its level of service as perceived by road users; (iii) improving access for the poorest and most excluded rural communities to the primary paved road network in the Departments of San Pedro, Caaguazú and Caazapá.

14. The achievement of the project's development objectives will be assessed through the following key indicators: (i) percentage of MOPC's portfolio executed following results-based management practices; (ii) creation of MOPC's Road Strategic Planning Unit and implementation of the Road Strategy; (iii) levels of users' satisfaction; (iv) average costs if vehicle operation; (v) average transportation times; and (vi) number of public transportation vehicles that use the unpaved roads of the targeted departments.

15. The project will help to establish the strategic and programmatic principles that will allow MOPC to coordinate all efforts and financing in a more rational manner, taking into account fiscal realities. The

project will be national in character providing a national logical framework, but will be implemented for some of its main paved road network of the Eastern region of Paraguay, and its rural access component in Departments that rank high in terms of poverty and have low levels of accessibility. The approach, however, may be replicated throughout the country. The project will be used to catalyze a working relationship between the Ministries of Finance and Public Works on prudent fiscal management.

16. The approach will seek to promote, as a guiding theme in all components: ***(i) results-focus***, through the identification of targets for each intervention, to be monitored and whose attainment will be the indication of success; ***(ii) accountability***, through planning and programming all interventions based on objective and quantifiable criteria that can be presented for public scrutiny, and ***(iii) participation and transparency*** through mechanisms designed to facilitate stakeholders' having the opportunity to influence and share control over the decision making about the initiatives that affect them and with the flow of timely and reliable information on the progress of activities and results obtained made accessible to all relevant stakeholders.

17. The project approach has, during its preparation, guided the MOPC towards the preparation of a **Road Sector Strategy** that is consistent with this project and, in fact, generalizes it. The Road Sector Strategy sets road maintenance as a priority and sets budgetary targets for each road network type, thus organizing the decision making process with objective criteria at a macro level and linking it with the budget (see Annex 18).

18. To implement the Road Sector Strategy and render it credible a **Road Social Contract** has been established. It is the culmination of the participatory planning process that has marked the preparation of the project. The Road Social Contract will be signed by the MOPC with representatives of the road users, the contractors, local governments and congressmen. It will symbolize the MOPC adherence to the Road Strategy, its commitment to it and specific actions towards the different stakeholders in exchange of contributions by the stakeholder towards the strategy. In this way there is adherence and commitment to the strategy by all concerned.

Box 1: Road Social Contract

The Road Social Contract will be signed during the year 2006, between the MOPC and the main stakeholders that play an important role in the use and maintenance of the road network. The objective of the contract is to define the intentions of the parties which will then be reflected in bilateral agreements that will include said intentions with regards to very concrete and specific issues between the MOPC and the other party and that will materialize through a specific action mutually agreed upon. The bilateral accords do not explicitly necessarily refer to the Road Strategy but are in-line and contribute implicitly to it.

The consensus for the elaboration of the Road Social Contract will develop over a number of meetings and workshops, on a multilateral and bilateral basis with the stakeholders to arrive at an agreement on the wording and content of the mutual intentions of the parties.

3. Project components

19. The project cost is estimated at about US$105 million (including contingencies); with a Bank loan amount of US$74 million. The GOP contribution includes counterpart funds for investments and for taxes and duties (including value added tax of 10%).

20. ***Component 1: Strengthening Strategic Planning and Road Management (Estimated cost US$6.5 million (not including contingencies), of which US$4.55 million would be financed by the Bank Loan).*** This component would finance the strengthening of strategic planning, road conservation, and budget management within a framework of fiscal constraints, in particular, it will support a series of consulting services, training and goods that may be grouped into four subcomponents:

21. ***Sub-component 1.1: Strengthening of Strategic Planning;*** This subcomponent will finance consulting services, goods and training essentially to (i) improve the legal framework of the road sector and road classification; (ii) set-up a strategic planning unit and equip it with the necessary instruments, knowledge base and training for it to begin performing its tasks; (iii) introduce and implement a traffic counting system; (iv) strengthen the procurement unit with a software system to improve its contract and procurement monitoring ; (v) provide strengthening training for the MOPC.

22. ***Sub-component 1.2: Strengthening Budget Preparation and Execution:*** This subcomponent will finance consulting services to (i) develop a management of works and budget execution monitoring system; (ii) develop and implement an automated works certification and payment system.

23. ***Sub-component 1.3: Strengthening Road Management and Maintenance:*** This sub-component will finance consulting services, training and goods to (i) strengthen management of road improvement and maintenance contracts; (ii) update the pavement management system and the road inventory; (iii) develop and implement a communication strategy on road maintenance.

24. ***Sub-component 1.4: Monitoring and Evaluation of the Project:*** This sub-component will finance works and consulting services to set-up baselines and mechanisms to monitor outcome and impact indicators. These activities include (i) measuring vehicle operating costs; (ii) socioeconomic impact surveys, baseline and monitoring; (iii) baseline and surveys of user satisfaction; (iv) strengthening and training UGPG in project coordination and Bank procedures etc.; (v) implementation of Indigenous People's Plan; (vi) implementation of Environmental Management Plan; and (vii) implementation of the Improved Governance Framework and Project Action Plan.

25. The detailed activities are in the annexes.

26. ***Component 2: Improvement and Maintenance of the Paved Road Network (Estimated cost US$43.3 million (not including contingencies), of which US$30.34 million would be financed by the Bank Loan):*** This component has two subcomponents which complement each other (i) performance based contracts; (ii) toll system optimization. The toll system optimization is estimated to increase revenue collection from US$7 million a year to around US$12 million, however the amount may be significantly higher as the amount of existing fraud is difficult to estimate. The improvement of revenue collection would improve resources available for the maintenance of the paved road network. The optimization of the toll system and toll increasing strategies are tied to road section improvements and user satisfaction, as such both programs will be tied and coordinated in implementation as to obtain the desired effects.

27. ***Sub-component 2.1: Performance-based Maintenance Contracting ($39.27 million):*** This sub-component would finance: (i) a training program on performance-based contract preparation and

monitoring for MOPC staff; and (ii) contracts that include initial rehabilitation works to meet the minimum level of service, road safety works, vertical and horizontal signaling, and maintenance works of the national paved road network, through performance-based maintenance contracts (GMANS). The prospective road networks have been identified and contracts have been structured into seven GMANS. The project will finance four of the identified GMANS with a total of 968 km: GMANS 1 (381 km), covering routes PY02, PY03 and PY08; GMANS 2 (337km), covering Southern Access to Asuncion, route PY 01, PY 04; GMANS Urban (108 km), covering part of route PY02 and part of Southern Access to Asuncion; and GMANS 5 (142 km), covering the San Ignacio – Pilar section of PY04. The seven GMANS that have been prepared cover the entire main paved road network of the Eastern region of Paraguay. The purpose of this approach was to remain holistic and programmatic despite the fact that the loan would not finance all the needs. The MOPC will seek financing from other sources to cover the remaining contracts, but following the procedures and mechanisms developed by this project.

28. ***Sub-component 2.2: Toll System Optimization ($3.6 million):*** This sub-component would finance (i) a Toll System Optimization Master Plan, and (ii) investments to implement the Master Plan. The Toll System Master Plan would ultimately define which tolls would be relocated, what sequencing would be used to start charging in both directions and what investments in infrastructure and systems are necessary. The loan would finance part of the Master Plan, concentrating on the tolls that are on the routes that will benefit from a performance-based contract.

29. ***Component 3: Improvement and Maintenance of Unpaved Road Network (Estimated cost US$42.90 million (excluding contingencies), of which US$30.03 million would be financed by the Bank Loan):*** This component will finance the rehabilitation and conservation of unpaved national roads that connect to the national paved network, and secondary roads connecting rural communities to these; and propose a comprehensive solution for improved access to isolated rural communities.

30. ***Sub-component 3.1: Road improvement and maintenance works***. The general approach for the preparation of this component has been to adopt solutions whose conservation is sustainable in the long run financially; this has been done by matching the proposed works to historical expenditures and looking at available resources of local governments. Also the whole network has been taken into account so as to give signs of transparency and possibility of participation of regional and local stakeholders. The road network included was considered an entity by Department (the regional administrative division). So initially a ranking of all the Departments in the country was carried out on the basis of objective criteria (rural population, poverty, km of unpaved network), the top three ranked Departments were selected: Caaguazú, San Pedro and Caazapá. The candidate road network in each Department was then chosen through a clear and participatory process to determine the priority segments in each Department. The strategy has sought a progressive approach for improving the roads level of service, in line with fiscal realities, and avoiding creating expectations that cannot be met. This framework was chosen so that it could be applied permanently in the future and MOPC could use it to guide future interventions.

31. A variety of interventions will be financed. As for the main unpaved road network in each Department, interventions may entail the contracting of road rehabilitation and maintenance; or contracting arrangements for specific works (bridges, drainage, etc.) leaving the rest to be carried out by force account. For community access roads, low cost solutions will be sought through participatory processes. These would be implemented by the communities themselves, the *Intendencias*, or the *Gobernaciones* through agreements (Convenios) with the MOPC.

32. ***Sub-component 3.2: Capacity Building and Development Program for MOPC District Offices and Regional and Local Governments.*** This subcomponent will finance works, goods and consultant services to (i) improve the infrastructure, systems and equipment in MOPC District offices to improve road management and contacting of maintenance; (ii) capacity building for MOPC District staff on

managing maintenance contracts; (iii) capacity building program beginning on planning and management on roads maintenance for Regional and Local Governments.

33. The structure of the works components has been designed in such a way as to be adopted by the MOPC as its framework to deal with that type of road network. In essence, the paved road component, aims through the preparation of several contracts, even though the loan will not finance all, to set a precedent and a methodology to deal with the road maintenance of the main road network. In this way, all the roads will be maintained either with other donors or with local funds using this approach. It will be extremely difficult to step-out of this framework once it is operational, thus ordering the MOPC strategy and rationalizing expenditures for the paved road network. The assumption of producing and re-ordering of the sector has already produced a result, as IDB has asked to finance the remaining performance-based contracts and has imposed a minimum of 20% expenditures to go to road maintenance in its future loan.

34. The structure of the unpaved road network component, which has been developed through a participatory planning methodology, is also inscribed in a general framework that can be applied by the MOPC in the future. The approach ordered all Departments on objective criteria to provide a priority list, on this basis; the first three Departments were chosen to be financed. Within these Departments a process of participatory planning with the local communities and governments was undertaken to arrive at a Road Master Plan for each Department. The approach again sets a methodological precedent which on the objective criteria defines the road sections which are a priority in each Department. It then becomes extremely difficult to justify intervening in a road section that is not a part of the Department's Road Master Plan; in this way arbitrary decisions are curtailed. The methodology has been highly appreciated by local governments and the community, and other Departments are already demanding similar treatment, which in itself is an indication of success and the birth of another framework with which to deal with another road network type.

4 Lessons learned and reflected in the project design

35. The proposed project is different to previous Bank projects due to its focus on maintenance. However, its design builds on lessons learned from the implementation of the Eighth Highway Project (Loan 3685-PA), several operations of other donors, and Bank experience in Paraguay and in the region. The resulting lessons assimilated for project design are summarized below:

36. ***It is difficult to establish and retain the appropriate planning and budgetary priority for road maintenance.*** The sustainability of the Eighth Highway project was rated unlikely, despite the fact that overall the project was rated satisfactory and during the project execution the GOP had on average complied with the Loan Agreement condition of allocating US$14.2 million annually to finance maintenance expenditure. The reasons for this are that after the end of the project the budgetary allocation for road maintenance had dropped to US$8 million a year, and no mechanism for ensuring that this level of expenditure was maintained remained. Furthermore, the project invested in several significant highways, contributing at the same time to concentrating resources in construction and rehabilitation rather than trying to assure a reasonable equilibrium between infrastructure expansion and maintenance in the light of limited resources. This project will attempt to reverse this logic, through a more balanced program for road maintenance and rehabilitation.

37. ***Weak implementation and management capacity, together with insufficient attention given to strengthening institutional arrangements hampered the execution of the Eighth Highway project.*** In the context of relatively weak public sector institutional capacity and the execution of a complex project, the shortcomings of the MOPC in the management of the project were evident. The project was executed by the Directorate of Highways (DV) using the MOPC internal procedures. The DV's own workload together with the additional procedures for communicating with the Bank strained DV's capacities. As a

consequence, this project will work with a Project Preparation and Coordination Unit (UPGP) that will be the Bank's interlocutor, to reduce the problems encountered in the previous project. This unit will be made up of Ministry staff, remunerated at the regular level, and is intended only to coordinate tasks and maintain communication with the Bank, leaving project implementation and other project management functions (procurement, financial management, etc) to be mainstreamed within existing MOPC units. The selected approach was the result of a series of internal discussions within MOPC and two inter-ministerial workshops that were also supported by the Ministry of Finance, specifically the Central Public Investment Unit (UCIP), who is trying to improve project management in line-ministries. It will not be set-up, therefore, along the lines of traditional Project Implementation Units (PIU).

38. ***Link Project Objectives to Budget Allocations to Enhance Counterpart Fund Availability.*** The shortage of counterpart funds that plagued the Eighth Highway project was aggravated by a deteriorating macro-economic situation during its execution, but as observed in the Paraguay Public Expenditure Review[6] these shortages stem mainly from management inefficiencies, a convoluted budget process, and inexistent pre-investment planning. The project will aim to mitigate the effects of shortages in counterpart funds by strengthening planning and management in the MOPC and by preparing a five-year investment plan to guide the use of the MOPC budget.

39. ***Lack of a clear sector policy and a dependency on external financing for investments has resulted in the multiplicity of interventions by financial institutions whose objectives, requirements and procedures are filling the sector policy void without adequate coherence.*** The recent Bank Public Expenditure Review showed that all sector investments (construction and rehabilitation) are mainly financed by external funds. Weak institutional capacity together with an unclear sector policy have resulted in externally financed projects that, in order to fit-in without overlapping with each other are concentrating on different road sub-networks; IDB is working in Chaco and in rural roads; CAF in several national road improvements; JBIC in several national road improvements; and FONPLATA in national road improvements. No project is dealing with across-the-board sector policy or priorization of resource allocation, nor with a national road maintenance policy, and several sub-networks in this distribution are being left out such as the national and departmental unpaved road networks. This project aims to fill in this void by addressing across-the-board policies that would be the basis for the MOPC strategy for future years and that would help to coordinate donor assistance to obtain more effective results. As a result of the projects preparation activities, and in particular the recent forum of road maintenance[7], a seminal road policy has been prepared by the Vice Minister of Public Works and is summarized in Annex 1. The project seeks to build upon this already important progress and institutionalize and disseminate a coherent and fiscally-responsible road sector policy and strategy.

5. Alternatives considered and reasons for rejection

40. The approach followed is based on a strong concentration on maintenance and results based management. The idea is to reverse a tradition of arbitrary and discretional decision making that has been suboptimal in resource allocation at best or contributed to corruption at worst and heavily concentrate on the conservation of existing assets before their expansion. Any other approach would have contributed to the increase of road assets without a sustainable basis.

41. ***Specific Investment loan (SIL) instead of an Adjustable Program Loan (APL):*** The use of an Adjustable Program Loan (APL) that would disburse in two phases was discussed. The idea would have been to complete a number of reforms and capacity improvements before major investments were made. This would have contributed to establishing additional management and performance related capacity and

[6] Paraguay Public Expenditure Review, 2005, on-going.
[7] Dialogue for Road Asset Management and Development in Paraguay, June 2005, MOPC and World Bank.

tools before dealing with new innovative contracts. However, the particular nature of performance-based contracts which are based on a detailed inventory of existing conditions makes any delay in contracting maintenance difficult as the base-line inventory changes with the delays. In addition to this, all four GMANS contracts that will be executed under this project are planned to be bid during the first year of implementation of the project. Furthermore, since the other main investment component which relates to unpaved roads was prepared and developed within a highly participative framework, delays in implementation were seen to be counterproductive with regards to the expectations created within the beneficiaries. Finally, another important consideration with respect to this issue was that the second APL would need a new approval by the Paraguayan Legislative Branch, which would imply an additional delay for the project in the middle of its implementation.

42. ***Road maintenance instead of capital investments in new construction:*** Although it is clear that Paraguay needs to finish completing a few links of its priority network to be more in-line with its neighbors in terms of the coverage of its network; the main deficiencies lay squarely in road maintenance. The existence of a plethora of donors all contributing to the expansion of the road network, and none contributing significantly to the road maintenance agenda provided an opportunity where this project could contribute to address and advance an agenda. Finally, the structure of MOPC budget, where historically all capital investments are financed by external sources reduces significantly the margin of maneuver the MOPC has in rebalancing its budget from investments in new roads to road maintenance contracts if it is not initially supported by a donor project.

43. ***Maintenance by contract instead of force account.*** Traditionally road maintenance has been carried out by force account by the MOPC. The analysis of the sector shows that the efficiency of this approach has gradually been reduced as the size of the road network has increased and the MOPC has not been able to propose higher remuneration or renew staff. It has become evident that in order to manage ever increasing networks the contracting out of activities, originally carried out by MOPC has become necessary. Although the project does not intend to do away with force account maintenance, it seeks to introduce maintenance by contract as an alternative.

44. ***Performance based maintenance contracts instead of traditional road maintenance contracts.*** Performance-based maintenance contracts have the characteristic of being easily monitored as the outputs are known in advance and are linked to levels of service that can be apprehended, understood and assessed by users. Consequently, they are the most transparent type of contract. In addition, the level of service, which is the final output can easily be adjusted to available resources, thus the level of service obtained is directly linked to what the country can afford. In this way, these contracts are a better guarantee of improved resource allocation. Given the governance environment and the clear need to link resource allocations to outputs, it is believed that this type of contract can more efficiently contribute to the objectives the project pursues.

45. ***Bottom-up preparation and project design instead of a top-down approach.*** The project has been designed in each of its components through a highly participatory approach. This is particularly the case in the unpaved road component where, the beneficiaries largely defined the scope of the component. The approach is a response to the need to increase transparency and accountability to the GOP actions corresponding to the strategy the current administration is said to be pursuing and consistent with CAS objectives.

C. IMPLEMENTATION

1. Partnership arrangements

46. Historically there have been a significant number of donors (ie: IDB, JBIC, FONPLATA, EU) working in the road sector in Paraguay without an explicit direction set by the MOPC. Instead there has been a tendency for donors to develop projects that have been isolated from each other to facilitate implementation from the donor's point of view. This project fills a particular niche that no other project or donor was dealing with: road maintenance. The approach adopted has been to set-up a strategy that could be applied by the MOPC for its main networks: (i) paved; (ii) improved (all-weather); (iii) unpaved (earth); and would address the main activities; (i) new construction; (ii) improvements; (iii) periodic maintenance; and (iv) routine maintenance. The project is helping to develop the Road Strategy and Road Social Contract, and is preparing subprojects that may be financed by other sources, equally the methodology used for the preparation of the Departmental eligible Unpaved Road Networks is comprehensive and aims at being used for all the remaining Departments that were not covered by this project.

47. As a result of this approach, the Task Team and MOPC counterparts believe that a framework is being created with which the MOPC can align donors on a coherent approach, rather than donors defining what they will support. Already during preparation, the approach is proving fruitful; IDB has joined this initiative and decided to adjust a loan in preparation of US$150 million, to dedicate 20% of it to finance the remaining GMANS prepared by the MOPC under this project. IDB has participated in several of the workshops carried out to define the Departmental Road Master Plans and has adhered to the methodology. MOPC will likely impose this methodology, to whoever finances, for the development of future Departmental Road Master Plans.

48. The MOPC will also be receiving *Mercosur Structural Funds* (FOSEM), and the MOPC is defining the possible projects to be financed by FOSEM in coherence with this project. MOPC is considering financing additional Road Master Plans in the Departments with this financing.

49. Regarding the Unpaved Road Network component several partnership possibilities have arisen and are being explored. The component has Caazapá as one of its Departments, where GTZ has assisted the local government in elaborating a Departmental Development Plan for 2003-2013[8]. The Plan however did not develop the road infrastructure aspects; as a result it has been convened that the Plan will incorporate the Road Master Plan developed by MOPC for Caazapá under this project. GTZ in turn would assist MOPC in the implementation of the project component in Caazapá with its existing facilities, knowledge and contacts. An agreement is being prepared in this respect.

50. The Unpaved Road Network component has also potentially a possibility of developing labor-intensive activities through routine maintenance and in particular with microenterprises. As a result, the Task Team and the MOPC counterparts have been approached by the International Labor Organization (ILO) and the World Confederation of Labor (WCL), to analyze possibilities of participation in enhancing these aspects of the project. ILO has particularly worked in this manner with or following Bank projects of similar nature with significant results (Peru, for example).

[8] Plan de Desarrollo Departamental de Caazapá 2003-2013, Gobernación de Caazapá, GTZ, Ministry of Agriculture

2. Institutional and implementation arrangements

51. The Borrower created a Project Preparation and Coordination Unit (UPGP) in the Ministry for project preparation on September 9, 2004 by Resolution No. 523. Said resolution was later amended by Resolution No. 593 dated August 24, 2005 to extend the UPGP's functions and responsibilities to include project implementation and evaluation. The Unit has been staffed with five Ministry staff members who have shown capacity and interest in the project. During project identification, two inter-ministerial workshops were carried out to define within the Ministry the shape and attributions of the UPGP. Furthermore, it was decided that the UPGP would be the main promoter of project activities and sole interlocutor with the Bank, but would use the MOPC divisions to prepare the project, thus the linkages and the responsible staff in all the MOPC dependencies that might deal with the project have been worked-out through this participatory process. The Ministry of Finance, which is promoting the improvement of pre-investment and project management in the line ministries, participated in the workshops and is an integral actor of the final definition of the shape and functions that the UPGP has taken.

52. The MOPC will be ultimately responsible for the implementation of the project, through both the Viceministry of Public Works, for the majority of the project, but also through the Viceministry of Administration and Finance. In the Viceministry of Public Works, the Directorate of Highways (Dirección de Vialidad), the Directorate of Rural Roads (Caminos Vecinales), and the Directorate of Technical Assistance (Dirección de Asesoría Técnica) will be the main instances responsible for project execution. In the Viceministry of Administration and Finance, the Directorate of Finance (Dirección de Finanzas) will be in charge of implementing the subcomponent of Toll Modernization.

3. Monitoring and evaluation of outcomes/results

53. The monitoring and evaluation system is designed to assess whether or not the Project is being implemented in line with the proposed objectives and to measure fulfillment of agreed targets. Project progress reports will be prepared by the UPGP initially on a semi-annual basis and submitted to the Bank for review and comments. These reports should indicate the progress made under the different components of the Project and measure performance against the results indicators established in the results framework (Annex 3). In particular, progress reports will include information regarding: (i) disbursement performance over the period covered by the report and updated disbursement calendar; (ii) updated procurement plan for activities under each of the project's components; (iii) the execution of the improvement and maintenance programs with detailed information of allocated budget, scope of the activities and results achieved; (iv) progress achieved in the implementation of the environmental and social measures that have been taken as part of the implementation of the project; and (v) potential developments that could affect project implementation, which should consist of a review of the main risks and the impact of mitigation measures envisioned at appraisal (See section on risks).

54. For investment components, implementation progress will be measured against performance indicators measured routinely by the Directorate of Highways (DV). The performance of institutional strengthening activities and results will be evaluated with respect to road management indicators adopted. During Project preparation, a participatory workshop on the logical results framework of the Project was held in order to define the indicators corresponding to the Project's development objective and components. Through this participatory process, feasible indicators were selected, in accordance with both needs and practicality. As shown in Annex 3, the indicators that have been identified correspond to the Project's Development Objective and the outcomes of each Project component. Some of these indicators will be measured directly by MOPC, and some will require consultant services. The Borrower will prepare under terms of reference acceptable to the Bank an assessment of the impacts of the Project (enhanced road condition, reduced transport costs, enhanced production opportunities, job generation, and

increased road safety). For this purpose baseline indicators will be required no later than six months after effectiveness.

55. With respect to the Higher Level objectives of the Project, which is to support economic growth and poverty alleviation, a specific methodology of social economic evaluation has been designed to monitor and measure the impact that the Project may have on economic growth and poverty alleviation, and is described in the first section of Annex 3. The proposed evaluation is based on the standard methodology for estimating the causal effects of public policies, social programs and other forms of public interventions (i.e. the model of potential outcomes). This evaluation will focus, in particular, on the three Departments targeted by component 3 (Improvement and Maintenance of Unpaved Road Network), namely Caaguazú, San Pedro and Caazapá, due to the expected major socio-economic impacts of this component.

4. Sustainability

56. The basis of this project is to establish a sustainable road strategy which is based on prioritizing road maintenance and respecting budgetary constraints to improve resource allocation. As a result, sustainability of the project will depend on the maintained commitment by the MOPC in the pursuance of this approach. To reach a point where the strategy fixes frameworks that are the basis of the decision making process, that begin to impose themselves over political prerogatives and discretionality in project selection. In this respect, the development of the Road Strategy with appropriate budgetary targets for different road network types and for new construction, improvements and maintenance is a first step towards setting guidelines to ensure that decisions will be more consistent with the policies and strategies designed in the framework of this project. Additionally, the signature of the Road Social Contract has meant disclosing MOPC intentions, policy and projects with stakeholders, and requiring of stakeholders participation and involvement in the strategy. As a result, the MOPC will be held accountable for the strategy and the stakeholders are encouraged to participate, and provide support, thus improving the chances of success.

57. The works project components have also been designed in such a way as to become frameworks to deal with the particular issue they are addressing over and above this particular project, and may be adopted by the MOPC as the way of implementing the overall Road Strategy. In particular, in the paved road component seven performance-based maintenance contracts were prepared covering 2,300 km of the 4,000 km paved road network. The project will finance part of the contracts that will maintain part of this road network; the MOPC will have to find other sources of financing to fund the remainder. The IDB has expressed its intention to finance some of these contracts. As a consequence of this approach, the MOPC has established a way of maintaining its paved road network and should manage to align all potential donors on this approach.

58. In the unpaved road network component, the project has designed a methodology and an overall approach that considered the totality of Departments and set a priority list on the basis of objective criteria. The project then developed Road Master Plans for the first three Departments on a participatory planning basis, which has ensured transparency, accountability, and broad participation, promoting ownership of the project by the different stakeholders. This approach has incited much interest from the remaining Departments and other donors. There have already been preliminary talks with IDB as it has shown interest in this approach and using the UPGP for the preparation of a similar component. In other words, a methodology is in place that will be scaled-up to other Departments; this ensures that the rationale of this project may be extended providing sustainability to the strategy.

59. The successful adoption of strategic planning tools, built capacity and the adoption of road management practices that link road management activities to budget preparation will be a key to the

sustainability of the project. It is paramount that the strengthening component is successful, so that MOPC gradually orients its management towards results and performance both in execution and management; and develops strategic and rational road programs. The MOPC has already provided positive signs in this respect, as it is willing to create a Strategic Planning Unit in the very short term.

60. The gradual balancing of the road sector budget in favor of road maintenance to fill the historic financing gap that has separated investments in new construction from road maintenance will be the ultimate indication of the success of this project, and in particular the convergence toward the targets set in the new MOPC road sector strategy. This project itself will provide more funds for road maintenance and its approach has incited IDB to dedicate 20% of its future loan to road maintenance (potentially $30 million over four years). Furthermore, the toll system subcomponent seeks to improve revenue collection from an existing source that could potentially be dedicated to the maintenance of the paved road network.

5. Critical risks and possible controversial aspects

61. The project faces several risks due to the poor governance and weak institutions in Paraguay[9], but given the current motivation of the GOP to make decisive improvements in governance and resource allocation and the central role the MOPC may have in this endeavor, there are significant benefits from engagement in Paraguay with this project. For that reason, an Improved Governance Action Plan has been included as part of the project design to reinforce and improve governance measures during project implementation.

62. Some of the main risks and mitigation measures are given in the following table; the specific process risks as identified in the Improved Governance Action Plan are detailed in Annex 13:

Risk	Risk Minimization Measure	Level of Risk
From Outputs to Objective		
Varying support at the Ministerial and Vice Ministerial level resulting in lack of commitment and ownership of a sustainable road strategy.	The preparation activities have involved a significant amount of participatory activities, including workshops, resulting in large dissemination of the project objectives and scope, support for the project and some amount of political capital gain by MOPC. In turn, this forces MOPC to meet expectations that the project has created. The participatory approach will continue throughout the project. The MOPC has as a result of project preparation activities developed a Road Sector Strategy based on results and targets that prioritize road maintenance. This Strategy involves the adoption of road management practices that link road management activities to budget preparation aimed at ensuring the sustainability of the project. The project will continue to solicit the participation of the Ministry of Finance as it is already doing in the design of the UPGP, so that	M

[9] According to the Bank's Paraguay Country Financial Accountability Assessment (CFAA), March 2004, and the Country Assistance Strategy (CAS), November 2003, which disclose the problems of governance and institutional weaknesses in Paraguay.

	the project could improve visibility and support at that level.	
Weak institutional and managerial capacities for project implementation in particular due to cumbersome procedures for the flow of funds, limited disbursement capacity, and limited knowledge of Bank procedures.	The project has considered specific actions during implementation to enhance the Ministry's capacity as part of project component 1. The UPGP that has been created is made up of staff members who come from the ranks of the Ministry and are already committed to the objectives sought by the project. A process of internal dissemination of the project and the functions, name and final composition of the UPGP has been discussed in two workshops. As a result other Ministry staff which will work on the project from their divisions have been clearly implicated in the project. A capacity building program will be aimed at UPGP, the procurement unit and the Public Debt unit on Bank procedures. Specific measures regarding automation of works certification and payments have also been included in component 1. Institutional strengthening of MOPC has been included in component 1 to deal with specifically small procedural improvements.	H
Lack of expertise and institutional capacity within the MOPC to manage, supervise and monitor performance-based contracts.	There is a strong emphasis in capacity building on this issue both at preparation stage and during implementation on the basis of seminars and training. In addition, a permanent consultant with expertise in this field was hired to support the UPGP during project preparation. Components 1 and 2 include capacity building and training in performance-based contract management.	H
Limited leverage of the project in the context of a large amount of donors may limit the efficacy of the dialogue.	The project will construct the necessary planning tools and capacities so that technically the right choices can be made, and the MOPC can as a result organize and guide donors in its best interest. Furthermore, the team will continue the discussions and round table with all the willing donors and try to foster a climate of continued communication between donors and MOPC to reduce overlapping risks and enhance the projects leverage. In this respect, progress has already been made with renewed interest by IDB in this project design and possible adoption of part of its framework. Additionally, GTZ and ILO have requested participation in the project as possible partners supporting the unpaved road network component and its possible positive effects on rural employment.	H
From Components to Outputs		
Difficulties in the procurement process of performance-based	Project preparation and bidding documents are being prepared by consulting firms with significant experience in this type of contract.	M

contracts due to lack of experience in Paraguay	An advisor is assisting UPGP in the supervision of these consultant services. It has been decided to conduct at least three workshops with interested firms in order to clearly explain the concept of performance-based contracts, and walk them through the indicators chosen, how they are measured and the structure of the bidding documents. It was decided to avoid prequalification of these contracts to avoid a shortlist which may be more vulnerable to collusion. Furthermore, international firms will be actively sought to participate in the bidding.	
Lack of counter-part funds due to budget constraints, cumbersome budgeting process, and reliance on undetermined funding sources.	An attempt to improve the MOPC planning capacity with budget constraint as one of its main objectives by introducing long-term budgeting. Component 1 included explicitly strengthening of the budget preparation and monitoring process. The preparation of a Road Sector Strategy prioritizing road maintenance and accrued support for this approach will mitigate this issue. Increasing awareness in the Ministry of Finance on the importance of road maintenance and the need to protect road maintenance expenditures will be further pursued.	S
Deficient control environment increases financial management risks.	Training of financial management MOPC staff on Bank procedures. Implementation of an automated certification and payment system. Implementation of a management of works and budget execution monitoring system.	H
Overall Risk Rating		H

Note: The Bank's risks ratings are: High (H), Substantial (S), Moderate (M) and Low (L).

6. Loan/credit conditions and covenants

63. ***Effectiveness conditions.*** There are no additional effectiveness conditions.

64. ***Other Covenants.***

- Operational Manual: the Borrower will implement the Project according to the Operational Manual agreed at negotiations and acceptable to the Bank.

- Road and Traffic Survey: the Borrower to carry out road and traffic surveys on an annual basis and furnish to the Bank the corresponding data no later than June 30 of the following year (starting 2008).

- Safeguards and Social Participation Framework: (i) the Borrower to designate a sociologist or anthropologist in charge of supervising the Indigenous People's Plan implementation, and coordinate the Social Participation Framework.

- Improved Governance Framework and Project Action Plan: (i) the Borrower to take all necessary measures to comply with the Mitigation Actions contained in the Table A13-2 of the Improved Governance Action Plan (Annex 13); (ii) the Borrower, through the MOPC's Internal Audit Unit, to provide the Bank with semi-annual reports on the Alert Indicators established in the Project Action Plan; and (iii) the Borrower, through the MOPC and the Ministry of Finance, to hold annual reviews with the Bank to update the provisions of the Improved Governance Framework and Project Action Plan.

- Procurement: (i) the Borrower to implement a project management information system satisfactory to the Bank; (ii) special procurement provisions to be reflected in Loan Agreement; (iii) the Borrower to update the Procurement Plan in accordance to Bank Guidelines, and furnish said updated plan to the Bank for approval, no later than 12 months after the date of the preceding Procurement Plan.

- Financial Management: (i) the Borrower to create within the first six months of Project implementation, a separate budgetary line to allow report based disbursements against quarterly budget allocation of funds.

- Financial Report: the Borrower to prepare and furnish to the Bank an interim unaudited financial report, satisfactory to the Bank, no later than 45 days after each subsequent calendar semester.

- Monitoring: the Borrower to prepare project progress reports on a semi-annual basis and submit them to the Bank for review. These reports should refer to progress made under the different components of the project and measure performance against the results indicators established in the results framework

- Evaluation: the Borrower to prepare under terms of reference acceptable to the Bank an assessment of the impacts of the Project (enhanced road condition, reduced transport costs, enhanced production opportunities, job generation, and increased road safety). For this purpose baseline indicators will be required no later than six months after effectiveness. The Borrower shall present the results at the Mid-Term Review meeting and at Project end.

D. APPRAISAL SUMMARY

1. Economic and financial analyses

Cost benefit NPV=US$ **47.9** (million); ERR = **33** % (see Annex 9).

65. The MOPC performed the economic analysis of the civil works project components: (i) performance-based maintenance contracting program (GMANS contracts) and (ii) improvement and maintenance of unpaved road network and bridges program. The economic evaluation covered altogether the rehabilitation of bridges and road works on 1,164 km that consist of 826 km paved roads, comprising the three GMANS contracts financed under the project, and 338 km unpaved roads, comprising the first

year improvement program of unpaved roads located at the Caaguazú, San Pedro, and Caazapá Departments, representing a total investment of US$28.3 million.

66. The MOPC collected detailed road characteristics and traffic data and performed a Project level economic evaluation with the objective of: (a) identifying sections to receive only routine maintenance over the 2007-2010 period and sections in which some kind of periodic maintenance, rehabilitation, or improvement is economically justified; (b) computing the net benefits of possible Project alternatives; (c) ensuring that the investments yield a rate of return higher than 12% for roads that serve an economic function, with traffic higher than about 50 vehicles per day, and a rate of return higher than 10% for very low volume roads, with traffic less than about 50 vehicles per day, that serve a social function or are needed for regional integration; and (d) estimating the network condition over the 2007-2010 period considering the Project investments. For paved roads, net benefits were evaluated using the Highway Development and Management Model (HDM-4) Version 1.3, which simulates life cycle conditions and costs and provides economic decision criteria for multiple road design and maintenance alternatives. For unpaved roads, net benefits were evaluated using Roads Economic Decision Model (RED) Version 3.2, which adopts the same economic analysis principles and road user costs as HDM-4, being tailored for the economic evaluation of unpaved roads.

67. The overall Net Present Value (NPV), at 12% discount rate, is US$47.9 million and the Economic Rate of Return (ERR) is 33%. Compared with the without project scenario, the MOPC will increase road works expenditures by US$22.7 million, in present value economic terms over the evaluation period, while road users will save US$70.5 million over the same period, which means that every dollar spent by the MOPC results in around 3.1 dollars saved by road users. Under a worst-case scenario of benefits dropping to 80% of the current level and 20% increase in costs, the Project continues to yield satisfactory benefits, represented by an economic rate of return of 24%. The analysis of switching values of critical items indicates that to yield a net present value equal to zero, for the performance-based contracts, investment costs would need to be multiplied by 6.60 or benefits multiplied by 0.19. For the improvement and maintenance of unpaved road network and bridges, investment costs need to be multiplied by 2.22 or benefits multiplied by 0.45.

2. Technical

68. The design of rehabilitation and maintenance works was based on sound technical criteria and is in line with international best practices. Intervention strategies for the paved road network were selected considering the main factors that affect pavement performance: surface condition, structural strength, traffic characteristics and climatic conditions. The selected road sections are not expected to show signs of distress and high roughness values in the short to medium term, which would warrant excessive repairs or strengthening within the next 5 to 7 years.

69. The GMANS approach was preferred over other methods, such as road concessions, because it optimizes the choice of technology, shortens response times, reduces design and supervision demands, and can be implemented even in low traffic sub-networks. The payment schedule, based on performance standards, ensures that the network is maintained after rehabilitation for the full duration of the contract. Furthermore, the choice of standards can be easily adapted to the available resources making the project more adaptable.

70. The road segments included in the Departmental Unpaved Road Networks of component 3 were selected according to their compliance with a set of eligibility criteria, and a series of workshops with the local communities and stakeholders. The selected networks sections were all evaluated using economic analysis to aid the selection of the final intervention, the approach is consistent with international practice and the rates of return for the proposed networks are adequate.

3. Fiscal

71. Previous analysis[10] by the Bank has shown that Paraguay's external public debt-to-GDP ratio is relatively low for regional standards, and almost all of it is owed to multilateral and bilateral institutions, resulting in low interest rates and long maturities. The public external debt-to-GDP ratio declined from 50.3 percent of GDP in 2002 to an estimated 34.4 percent in 2005, and is projected to fall to 28.9 percent in 2006[11]. Paraguay is also fortunate to enjoy a relatively benign medium-term public debt sustainability outlook. Nevertheless, debt sustainability analysis and fiscal projections point to the limited fiscal space available to Paraguay and indicate that sustained primary surpluses are needed to stabilize public debt at manageable levels. These surpluses are achievable if current policies stressing fiscal prudence and economic growth are maintained.

4. Fiduciary

72. ***Financial Management.*** The assessment conclusion is that the proposed financial management arrangements, considering the proposed action plan, would meet minimum Bank requirements and the MOPC has sufficient capacity and resources to implement them.

73. ***Procurement.*** During Project preparation, the Task Team noted delays in the procurement of Bank funded consulting services. It appeared that the arrangements implemented by the MOPC in application of Paraguay's Law No. 2051/2003 on public procurement as applied in the MOPC were particularly cumbersome. Furthermore, the MOPC was using these administrative steps also for procurement of Bank funded activities. In order to address the concerns raised by the Task Team with regards to these deficiencies, the MOPC issued Resolution No. 146 dated March 28, 2006, pursuant to which the MOPC will implement reduced administrative steps outlined by the Resolution. An in-depth assessment of the MOPC's PPU capacity was carried out by the team and a detailed action plan was prepared to address all risks identified and to provide some opportunities for improvement in procurement matters, as discussed in Annex 8. The overall procurement risk has been rated as "high". The following are the findings of the assessment related to the staff in: (i) the Operational Procurement Unit (UOC) of the MOPC, (ii) the four sub-units (SUOCs), that will be working with the project (Directorate of Highways (DV), Directorate of Rural Roads (DCR), Directorate of Technical Assistance (DTA) and the Directorate of Finance (DF), and (iii) the Project Preparation Unit (PPU). All of them have limited experience with Bank's procurement guidelines, procedures, documents and in contract information to ensure implementation of the project according to Bank requirements. The key features of the procurement risk mitigation strategy for this project include: (i) actions designed to strengthen the MOPC's procurement capacity, procedures and controls, and (ii) enhanced technical and fiduciary supervision during implementation. Full implementation of the Action Plan will provide the MOPC adequate arrangements to meet the Bank's procurement requirements for the proposed project.

5. Social

74. The project will have a positive impact on the living standards of the country's population through its direct effect on competitiveness, employment, and economic growth. The negative impact of the proposed project on the local population and cultural property would be minimal, as most of the proposed roadwork will consist of the rehabilitation and maintenance of already existing road networks. The Project's expected social results are circumscribed to the indirect social impacts to be generated by the road accesses in the beneficiary communities. Nevertheless, a Social Evaluation of the component 3

[10] The World Bank. (2006). Paraguay Public Expenditure Review. Report No. 32797-PY.
[11] International Monetary Fund.

of the Project of improvement and maintenance of the unpaved road network focused on three beneficiary Departments: Caaguazú, Caazapá, and San Pedro was carried out during project preparation due to its possible major social impacts, and is included in Annex 13. The objective of this evaluation was to identify opportunities, barriers and risks that could complement previous social analysis carried out during project preparation, and to serve as a basis for reinforcing the social impacts of this component's intervention strategy, and complement other social analysis of the component 3 and the Project related to the indigenous people plan, the environmental assessment, and the resettlement plan.

75. ***Social Evaluation of the Component of Improvement and Maintenance of the Unpaved Road Network.*** The main findings of the social assessment indicate that the key social issues related to the project's component 3 objectives are: (i) two groups of beneficiaries with different economical and social assets in a context of high levels of rural poverty. A small number of people with the highest assets coexisting with a large number of poor people, most of them small farmers, including vulnerable groups such as women, children and youth, unemployed, and indigenous communities; (ii) a challenging mechanism of institutional arrangements (national and sub-national governments) with weak experience and managerial capacities in participatory processes within a context of low levels of transparency and accountability; (iii) some stakeholders' lack of culture of road conservation activities (mainly private contractors); and (iv) low but increasing levels of social participation. These findings will be addressed through measures foreseen in the Social Participation Framework.

76. ***Social Participation Framework for the Project.*** A Social Participation Framework (EPS) has been set up from the outset of the Project. The EPS has been used during project preparation, and is expected to continue during the all its cycle. The EPS contemplates three components: (1) participation; (2) communication; and (3) transparency and accountability. Each of these three components has their specific objectives as described in Annex 12. The overall main objectives of the EPS can be summarized as follows: (i) to guarantee a broad participation of the different stakeholders aiming at among other objectives, establishing their roles and responsibilities in realist and fair terms; (ii) to create awareness of the importance of expenditures and mechanisms for road maintenance to preserve the road network's condition; (iii) to disseminate the project's objectives and achievements, and its contribution to the strengthening of the MOPC; and (iv) to increase project's transparency and accountability throughout its cycle. This framework resulted from a series of consultation with key stakeholders aimed at ascertaining the local existing experience in similar processes. As part of this framework, the project will implement the following measures: (i) consolidating the Participation Mechanisms of the EPS with differentiated intervention strategies according to the needs and capacities of all relevant stakeholders, and mainly the more vulnerable beneficiary groups, developing a wealth of feasible mechanisms ranging from one-way communication, such as information meetings, to collaborative mechanisms (shared control over decision making) and empowerment towards sustainability of the Project if possible (transfer of control over decisions and resources) (ii) strengthening of the Communication Activities of the EPS with public awareness campaigns, and training of different stakeholders according to their needs and cultural characteristics; (iii) developing of Transparency and Accountability Measures of the EPS in a comprehensive Improved Governance Framework and Project Action Plan to ensure fluent flow of reliable information to relevant stakeholders, continued feedback, social oversight; and clear signs of accomplishment of obligation of power-holders to take responsibility for their actions; and (iv) an Institutional Strengthening Component aimed at facilitating the institutional arrangement for implementation of these measures. All of these measures have been developed in the Operational Manual of the Project. Many activities of this framework have been developed during the Project preparation, among them the selection of the *Basic Rural Unpaved Road Network* in the 3 Departments to be supported by the Project in line with fiscal realities by regional and local governments and civil society organization. Moreover, participation is envisaged to continue for the expansion of the network.

77. ***Indigenous Peoples Plan.*** The Indigenous Peoples Plan (IPP), which is included in Annex 10.B, was designed in accordance with OP/BP 4.10 (Indigenous Peoples) with the objective of effectively promoting the participation of indigenous peoples in the programs of execution of maintenance works in the rural road network, within a framework of equality and respect. The IPP identifies the way in which the Project can respond to some of the concerns raised by the indigenous peoples located in the intervention area. The IPP was designed in collaboration with representatives of the indigenous communities, indigenous associations, and non-governmental organizations working with indigenous peoples, in the Departments where the Project will be implemented, namely the Departments of Caazapá, Caaguazú and San Pedro. As a result of the participation of the indigenous communities in the project, the IPP aims at achieving a sustainable improvement of the quality of life of the targeted indigenous communities, through the strengthening of the organizational and managerial capacity of these communities, and by training their members and generating employment.

78. ***Resettlement Framework.*** The MOPC prepared a Resettlement Action Plan (RAP) for the highway segments for which maintenance contracts will be implemented during Year I of the project and a Resettlement Framework (RF) for subsequent highway segments, which was prepared in accordance with OP/BP 4.12 (Involuntary Resettlement) with assistance from the Bank team and consultants. The RF carefully balances the need to provide adequate highway maintenance and safety with the principle of minimizing the amount of involuntary resettlement and disruption to peoples' lives. The resettlement affects only illegal occupants of the right of way that may affect maintenance works. The idea is to avoid stoppages of maintenance works, or situations where the contractors may justly require compensation due to obstacles to their activities. The Ministry has backed this process with the parallel issuance and dissemination of a Ministerial Resolution reminding all that it is illegal to occupy the right of way. The RF and RAP concern only the paved highways because there are no resettlement issues on the unpaved highways that would be maintained during the project. The RF and RAP are included in Annex 10.C.

6. Environment

79. ***Environmental classification.*** The project has been classified as "Category B", following the Bank's Operational Guidelines [OP 4.01], as no significant negative impacts that could jeopardize the natural environment of its area of influence are foreseen. This classification takes into account the fact that the proposed project will support the rehabilitation and maintenance of existing roads, either paved (national road network) or unpaved (rural road network), and consequently will not involve new heavy construction activities. The main issues that have to be considered to ensure an adequate environmental management of the subprojects include: stabilizing slopes, recovering areas for the exploitation and deposit of materials, controlling existing erosion processes, and incorporating road safety measures. Prevention, mitigation and/or compensation measures for these issues are straightforward.

80. ***Environmental Assessment.*** The Bank's environmental assessment focused on the main aspects agreed upon with the Quality Assurance Team (QAT) of the Bank: (i) analysis of the mechanisms established to address the issues that trigger Bank safeguards; (ii) evaluation of the environmental management capacity of the executing agency; (iii) environmental evaluation of the first package of projects in each component; and (iv) evaluation of overall compliance with the national environmental legislation. The environmental evaluation for the road infrastructure components of this project was conducted by the Environmental Unit of the MOPC and reviewed by the Bank as part of the project's due diligence. Individual subprojects were evaluated using the screening methodology established in the QAT's *Guide for Rapid Environmental and Social Evaluation of Road Projects,* with the purpose of ensuring that environmental and social issues were appropriately considered in their design. This environmental assessment determined that, since the proposed Project involves rehabilitation and maintenance works in existing roads, no major environmental impacts that could endanger the natural surroundings of the area of influence, or the local population are to be expected. Due to the size and

quantities involved in the works to be executed, no significant negative impacts are expected that might not be easily reverted or mitigated. Additionally, due to the inventory of environmental liabilities, it is expected that the project will help eliminate existing environmental liabilities not attributable to the project, such as problems of erosion of accesses, sedimentation, canalization of entries and exits of draining works, among others. The operational procedures that the UA require for the development of construction activities, such as inspection and supervision of works are considered adequate for the modalities proposed in the routine improvement and maintenance of unpaved roads and the performance-based maintenance of paved roads. However, these procedures can be improved through the implementation of measures that can contribute to the institutional strengthening of the UA, as described in Annex 10.A. In sum, none of the sub-projects have critical environmental and social risks. Therefore, the project is environmentally viable and fulfils the Bank's Environmental Safeguards Policy.

7. Governance

81. The efficiency of public expenditure depends highly on the governance environment and practices, therefore any effort to improve resource allocation must be accompanied by an enabling governance framework[12] that facilitates accrued efficiency, and inhibits corrupt practices[13]. In recent years, the Ministry of Public Works of Paraguay has been assigned an average of around 10% of the total annual public expenditure. Within the Ministry more than 90% is dedicated to the road sector. As a result, improved efficiency in the use of these resources is essential for improving the overall public expenditure of the Paraguayan Government.

82. All preliminary assessments showed that if the project was to increase its chance to attain the development objectives it set itself, it would have to be structured in such a way as to strengthen the governance environment of the sector and the project. As a result, the Paraguay Road Maintenance project has been based from the outset, on an integrated approach that seeks to enhance the efficacy and impact of the project on the one hand, and on the other to introduce practices and processes to the MOPC that would help it to gradually reform to improved governance practices, which aim to increase the capacity of the government to effectively manage its resources and implement sound policies in the road sector, ensuring increased efficiency and success of the project.

83. In addition to the structure of the project, which is based on (i) increased accountability by setting budgetary linked policy commitments; (ii) improving efficiency by working on the basis of performances and results; (iii) ensuring transparency of the decision making process, objectives and commitments; and which will in itself improve governance, the Project is accompanied by an Improved Governance Action Plan.

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[12] The World Bank Institute (WBI) defines governance as the traditions and institutions by which authority in a country is exercised for the common good. This includes (i) the process by which those in authority are selected, monitored and replaced, (ii) the capacity of the government to effectively manage its resources and implement sound policies, and (iii) the respect of citizens and the state for the institutions that govern economic and social interactions among them.

[13] Corruption on the other hand, is generally defined as "the abuse of public office for private gain."[13] In an environment of weak governance, it is believed that corruption may thrive; however, weak governance itself is not necessarily synonymous with corruption, but rather with different manners of inefficiencies. For instance, due to budgetary structure, bureaucracy, scarce resources or inadequate management capacity, agencies may distort priorities, or otherwise not adequately prepare programs – all of which lead to inefficiency, but not necessarily as a result of or to corrupt practices. Thus it is important to distinguish between corruption – intent and action for private gain with public goods – and a range of governance failures that likely provide opportunities if not inducements for rent seeking behavior.

84. The objective of this Plan is to strengthen governance around the project and as a result limit corruptive practices, so that the full impact potential of the project is attained. It is based on prevention measures, deterrence measures, and detection measures. It essentially regroups all the project cycle and areas that may be victim to governance weaknesses, integrating mitigation measures from policy formulation and planning, budgeting, procurement, execution and supervision, financial management, revenue collection, political interference, capacity limitations, and project impact enhancement. The detection measures rely on "alert" indicators that should preempt additional care and observation from project supervision. This additional supervision would seek to establish the existence of red flags on the basis of a monitoring framework. The detailed Plan is presented in Annex 13.

8. Safeguard policies

Safeguard Policies Triggered by the Project	Yes	No
Environmental Assessment (OP/BP 4.01)	[X]	[ ]
Natural Habitats (OP/BP 4.04)	[ ]	[X]
Pest Management (OP 4.09)	[ ]	[ ]
Cultural Property (OPN 11.03, being revised as OP 4.11)	[ ]	[X]
Involuntary Resettlement (OP/BP 4.12)	[X]	[ ]
Indigenous Peoples (OP/BP 4.10)	[X]	[ ]
Forests (OP/BP 4.36)	[ ]	[X]
Safety of Dams (OP/BP 4.37)	[ ]	[X]
Projects in Disputed Areas (OP/BP 7.60)*	[ ]	[X]
Projects on International Waterways (OP/BP 7.50)	[ ]	[X]

9. Policy Exceptions and Readiness

85. The Project does not warrant any exceptions to Bank policies and is deemed to be ready for implementation.

* *By supporting the proposed project, the Bank does not intend to prejudice the final determination of the parties' claims on the disputed areas*

Annex 1: Country and Sector Background

PARAGUAY: Road Maintenance

A. Country economic context

1. Paraguay, a landlocked country located in central South America with Bolivia to the north, Brazil to the east, and Argentina to the south and west, has a relatively new democratic government. Paraguay experienced a 35 year military dictatorship that ended in 1989. Its highly centralized government was fundamentally changed by the 1992 constitution which provides a division of powers and grants a democratic system of government. The second president elected under this new state in 1998 resigned after diverse internal conflicts; in 1999 the President of the Senate assumed the presidency leading a government that was accused of corruption. Finally, in 2003 a new president was elected and currently leads the Paraguayan government. The new government aims to fight corruption and improve economic management.

2. After a period of crisis and stagnation up until 2003, the Paraguayan economy is beginning to recover. While in 2000 real GDP declined by 3.3% and the economy experienced zero growth in 2002, in 2004 real GDP presented the highest level, 4.1%, in the last fourteen years. It is estimated that real GDP growth was 2.8% in 2005.

3. Paraguayan economy is driven by agricultural, industrial production and commercial activities; the transportation sector represents only 4% of the total real GDP (Table A1-1). It is important to mention that real GDP per capita reached an estimated US$1,416 level in 2005 (constant 1994 US$; population is around 5.7 million individuals), but it is still below its 1995 level of US$1,531.

Table A1-1
Structure of total real GDP divided by economic activity presenting yearly growth.

Sector	2000	2001	2002	2003	2004
Agriculture	14.35%	16.70%	17.52%	18.85%	18.65%
Industrial production	15.63%	15.20%	15.00%	14.55%	14.44%
Commerce	18.70%	18.32%	18.00%	18.34%	18.56%
Total GDP*	**7,251,642**	**7,401,336**	**7,397,741**	**7,681,785**	**7,999,448**
GDP growth	**-3.3%**	**2.1%**	**0.0%**	**3.8%**	**4.1%**
*Thousands of 1994 constant US$					

Source: Author's calculation using data published by Central Bank of Paraguay in its Economic Report, November 2005.

4. Inflation and interest rates have decreased significantly under the present administration. The inflation rate in 2002 was 14.6% in contrast with the 2.8% and 9.9% annual inflation rate experienced in 2004 and 2005, respectively. Interest rates diminished in 2005 to half their value in 2003, from 50% to 25% (Table A1-2). This denotes a significant impact of Central Bank actions towards the stabilization of Paraguayan prices.

Table A1-2
Inflation and interest rates in Paraguay from 2003-2005.

Variable / Year	2003	2004	2005*
General consumer's inflation rate (e-o-p)	9.3%	2.8%	9.9%
General producer's inflation rate	26.4%	5.8%	8.5%
Interest rate (national currency)	50.0%	33.5%	25.3%
Interest rate (foreign currency)	10.4%	8.1%	7.7%
* Does not include December			

Source: Author using data published by Central Bank of Paraguay in its Economic Report, November 2005.

B. International Economic Context

5. The international trade of goods and services continues to be one of the most important factors in the Paraguayan economy. Exports represent about one fifth of the economy, while imports almost two fifths of its GDP (Table A1-3). Paraguay maintains international trade agreements with several countries, with Mercosur being the treaty with its main trading partners: Argentina, Brazil and Uruguay. Table A1-4 presents the structure of Paraguay's international trade.

Table A1-3
Balance of Trade as a share of Paraguayan GDP.

Year	Balance of Trade as a % of GDP		
	Exports	Imports	Balance
2002	19%	30%	-11%
2003	22%	33%	-11%
2004*	23%	38%	-15%
2005*	20%	37%	-17%

Source: Author's calculation using data published by Central Bank of Paraguay in its Economic Report, November 2005.

Table A1-4
Structure of Paraguayan international trade by country

	Structure of Exports by Country							
Year	Argentina	Brazil	Uruguay	Total Mercosur	Rest of the world	Total exports		
2002	4%	37%	17%	58%	42%	950.6		
2003	5%	34%	20%	59%	41%	1241.504		
2004*	6%	19%	28%	53%	47%	1625.683	Balance of Trade	Year
2005*	6%	19%	29%	54%	46%	1553.001	-559.649	2002
	Structure of Imports by Country						-623.812	2003
Year	Argentina	Brazil	Uruguay	Total Mercosur	Rest of the world	Total imports	-1026.043	2004*
2002	20%	32%	4%	56%	44%	1510.249	-1297.076	2005*
2003	22%	33%	3%	58%	42%	1865.316		
2004*	23%	31%	2%	57%	43%	2651.726		
2005*	21%	28%	2%	50%	50%	2850.077		
* Preliminary data (year 2005 does not include December)								

Source: Author's calculation using data published by Central Bank of Paraguay in its Economic Report, November 2005.

6. In particular, Paraguay's exports are heavily concentrated on a few agricultural commodities, including soybeans, cotton, grains, cattle, timber and sugar, as well as on electricity generation. It must be

noted that the low unit price of Paraguay's exports magnifies the incidence of transport costs in the economic equation. Paraguay has vast hydroelectric resources, including the world's largest hydroelectric generation facility built and operated jointly with Brazil (Itaipú Dam). The structure of Paraguay's international trade, both by product and country, clearly shows that Paraguay's external market is vulnerable to the hazards of weather and to the fortunes of the Brazilian and Argentine economies (Table A1-4).

7. Paraguay presently runs a deficit in the trade of goods (Table A1-5), but maintains a current account surplus due to a large surplus in services, reflecting large exports of electricity. International reserves during 2003 increased sharply, going from US$641.3 million in December 2002 up to US$983.4 million in December 2003, and continued this trend in 2004 reaching US$1,168 million (Table A1-5).

Table A1-5
Paraguayan Balance of Payments composition

Balance of Payments in Paraguay (millions of US$)			
Account	**2002**	**2003**	**2004**
Current Account	92.6	130.7	13.9
Goods	-279.90	-276.1	-400.6
Services	213.8	249	251.4
Rent	42.8	-6.7	-31.1
Transfers	115.9	164.5	194.2
Financial and Capital Account	92.10	246.30	45.4
Capital Account	4	15	16
Financial Account	88.1	231.3	29.4
Errors and Omissions	-268.5	-75.8	120.2
Variation in reserves of assets incr(-) decr(+)	83.8	-301.3	-179.5
Total International Reserves	641.3	983.4	1168

Source: Author's calculations using data published by Central Bank of Paraguay in its Economic Report, November 2005.

8. Previous analysis14 by the Bank has shown that Paraguay's external public debt-to-GDP ratio is relatively low for regional standards, and almost all of it is owed to multilateral and bilateral institutions, resulting in low interest rates and long maturities. The public external debt-to-GDP ratio declined from 50.3 percent of GDP in 2002 to an estimated 34.4 percent in 2005, and is projected to fall to 28.9 percent in 200615. Paraguay is also fortunate to enjoy a relatively benign medium-term public debt sustainability outlook. Nevertheless, debt sustainability analysis and fiscal projections point to the limited fiscal space available to Paraguay and indicate that sustained primary surpluses are needed to stabilize public debt at manageable levels. These surpluses are achievable if current policies stressing fiscal prudence and economic growth are maintained.

[14] The World Bank. (2006). Paraguay Public Expenditure Review. Report No. 32797-PY.
[15] International Monetary Fund.

Table A1-6
Public external debt of Paraguay

Public External Debt (thousands of US$)		
Year	**Balance**	**% of GDP**
2002	2,283,051	50.3%
2003	2,477,573	44%
2004	2,390,687	36%
2005*	2,265,235	34.4%
*Does not include December		

C. Road Sector Background

9. Given the nature of the productive activity in Paraguay, with agriculture, industry and commerce generating more than 50% of total Gross Domestic Product (GDP), the role of the transport sector in the country's economy is of paramount relevance enabling regional integration and facilitating access to international markets.

10. Given its landlocked status, unless air or river transport are involved, the external trade of Paraguay must include a land segment, which tends to increase the cost of transport and insurance and pushes up the price of imports. About 90 percent of the total volume of domestic and international goods (soybean and cotton in particular) is transported by roads, which provide critical links to ports in neighboring countries and export markets in the Mercosur region.[16]

Table A1-7
Total Exports (Tons)

Transport	Exports	%
Road	2,049,296	46%
Fluvial	2,201,261	50%
Rail	176,473	4%
Total	**4,427,030**	**100%**

11. The competitiveness of the agro-industry continues to be constrained by transport costs, which remain among the highest in the LAC region. They represent an annual cost of US$90 million and around 40US$/tn. Moreover, Paraguayan producers need to travel an average of 1,325 km to the nearest international port from Asuncion (Table A1-8). Although many other factors besides road transport are involved in the determination of the cost of insurance and freight, an indication of the importance of transport for Paraguay's development can be found in the average CIF/FOB ratio. This ratio has been found to be close to 9 percent for Paraguayan imports from all destinations, one of the highest in the region. The same applies to homogeneous goods originating in the same region, for which the cost of insurance and transport in Paraguay has also been found to be one of the highest in LAC. [17]

[16] Paraguay: Country Assistance Strategy, November 26, 2003.
[17] ECLAC (2001)

Table A1-8
Main Corridors

Main Corridors	Distance (kms)
Asunción - Ciudad del Este - Paranagua	1,050
Asunción - Buenos Aires, through Puerto Falcón	1,191
Asunción - Nueva Palmira	1,351
Asunción - Antofagasta	1,861
Asunción - Porto Alegre (RGS), through Encarnación	1,175
Average Distance to export ports	**1,325**

12. The road transportation defines two types of roads in Paraguay: urban and inter-urban roads; the MOPC is legally responsible for the entire inter-urban road network. Paraguay has 60,000 km of inter-urban roads of which around 4,000 km are paved (asphalt mainly), 1,500 km are all weather (stone paved or gravel), and the remaining 54,500 km are earth roads with no treatment. Around 15,000 km of roads could be considered the main road network managed by the Directorate of Highways, and the remaining 45,000 km rural road network managed by the Rural Roads Directorate (*Dirección de Caminos Vecinales*).

Table A1-9
Paraguay's Road Network

Roads	Paved	Stone Paved	Gravel	Earth	Total
Primary	3.525,5	8,6	512,0	5.500,0	9.546,1
Secondary	542,6	180,8	926,8	3.900,0	5.550,2
Terciary			561	44.439,0	45.000,0
Total	**4068,1**	**189,4**	**1165,8**	**55.838,8**	**60.096,3**

13. The road network endowment seems comparable to equivalent countries and adequately covers the national territory, but the quantity of paved roads and its distribution is uneven. The great majority of paved roads, which have increased in average 100 km per year in the last 20 years, are in the eastern part of the country, while the Chaco area (the western part) can only boast the road to the Argentinean frontier via Puerto Falcón and the TransChaco highway under construction; this limited endowment, however, is the true reflection of the small population that resides in the region.

14. Nevertheless, overall the amount of paved roads in Paraguay compares badly with the rest of the region: Paraguay has 0.77 km/1000 inhabitants of paved roads, against 3 km/1000 inhabitants for Argentina and 4 km/1000 inhabitants for Uruguay. This result explains the historic MOPC policy of paving and network development to reduce the gap of unmet need, and is currently reflected in the construction of the TransChaco highway and new projects to pave highways that may be considered missing links in the main network such as the route 11 linking Santa Rosa-Puerto Antequera, and the route linking Caazapá-Cnel. Bogado.

15. However, the efforts deployed to pave the network have not improved its overall quality as the condition of the main road network has deteriorated in recent years, reaching 14% in good condition, 56% in fair condition and 30% in bad condition, illustrating the inadequacy of preventive maintenance

mechanisms and management of road assets. Additionally, too little has been invested on improving unpaved roads: the ratio of all-weather roads to paved roads in Paraguay is 1 paved km to 0.75 km all-weather road, while what is used as a rule of thumb for this ratio is generally 1 paved to 3 all-weather indicating gradual investments according to needs. It seems that, in light of inadequate road maintenance mechanisms, the authorities are pushed primarily to pave roads rather than to seek low-cost, all-weather solutions that with the design and management of adequate maintenance programs would suffice in some situations, partly explaining the unbalanced ratio of paved to all-weather roads.

16. As a result, many communities, including one Department capital (San Pedro), do not have all weather access, constituting one of the greatest hindrances to development of the country[18].

17. Poverty is more widespread in rural than in urban areas: 75 percent of the extreme poor and 56 percent of the poor lived in rural areas in 2001. The improvement of secondary and rural roads can promote social inclusion by reducing the cost of accessing educational and health services in rural communities. It can also contribute to link small farmers, who are mostly poor, to markets in the same way that major commercial agricultural sites are now connected to export gateways, and thus help them increase their productivity and go beyond subsistence production.

18. The investment performed in road maintenance is under the minimum standards. The annual budget for the Directorate of Highways over the last four years has averaged around US$150 million, of which US$8 million has been destined to maintenance. However, the amount effectively invested in maintenance is less than half that amount. When only the minimum investments are considered, the maintenance gap is estimated conservatively at US$34 million per year. If this situation does not change, Paraguay will face an increasing loss in the value of road assets and will incur in greater costs associated with the reconstruction of deteriorated infrastructure. The GOP needs to adopt a policy of optimizing resources by giving priority to road maintenance. As a result, the decentralized MOPC Districts are faced with a significant amount of demands related to the untreated unpaved road network, that span from main roads to small rural roads. The relatively low-level of these roads makes them very vulnerable to weather related emergencies and as a result, the Districts end up dealing mainly with emergencies on the unpaved network rather that carrying out a routine maintenance program on the whole of the paved road network.

19. The budget of the Directorate of Highways (DV), the specialized sector agency under the MOPC, accounted for 10 percent of the overall budget of the central administration in 2004. The sector also employs a large number of public servants (2 percent of the total), making it central to all efforts to improve the efficiency of public expenditure. Efficiency could be enhanced through improvements in planning and programming, increasing allocations for maintenance, improving priority setting framework for rehabilitation and new projects, and effective monitoring of expenditures.

20. The SIVIPAR Law was adopted in 2003, but it has not been regulated or implemented. It provides for a wide range of uses, with the exception of maintenance expenditure, and gives the MOPC significant flexibility in the choice of allocation. The sources of the funds for SIVIPAR are the existing ones (tax on petroleum derivatives, car registration tax, etc.), and no additional resources have been included. The appropriate regulation of this law, so that it reserves a significant amount of its proceeds to maintenance expenditures is paramount to the sustainability of the road sector, or in the event of the impossibility of application of this regulation a new law should be considered to reserve adequate and increased funding for road maintenance. Recently, the MOPC has decided to scrap this law and redraft a better one.

[18] Several rural water sub-projects financed by the Bank have had difficulties due to lack of accessibility to communities.

Main Sector Issues and strategic choices

21. Given the economic and sector context, the strategic importance of the road transport sector for Paraguay's sustainable development, and the road sector strategy, the following are the issues on which the GOP is seeking the support of the Bank.

a. ***Addressing improvement needs of the unpaved national primary and secondary roads.*** The efforts deployed by the GOP to pave the network have not improved its overall quality as the condition of the main road network has deteriorated in recent years reaching 14% good condition, 56% fair condition and 30% bad condition, illustrating the inadequacy of preventive maintenance mechanisms and management of road assets. Additionally, too little has been invested on improving unpaved roads, the ratio of all-weather roads to paved roads in Paraguay is 1 paved km to 0.75 km all-weather road, while what is used as a rule of thumb for this ratio is generally 1 paved to 3 all-weather indicating gradual investments according to needs. In light of inadequate road maintenance mechanisms, the authorities are pushed primarily to pave roads rather than to seek low-cost all-weather solutions that with the design and management of adequate maintenance programs would suffice in some situations, partly explaining the unbalanced ratio of paved to all-weather roads.

b. ***Maintenance expenditures and mechanisms are insufficient to preserve the condition of the road network.*** Experience shows that the optimal way of preserving road assets involves allocating sufficient resources to road maintenance as a priority and the surplus to road network extension. The practice today in Paraguay has been the opposite. When only the minimum investments are considered, the maintenance gap is estimated conservatively at US$34 million per year. If this situation does not change, Paraguay will face an increasing loss in the value of road assets and will incur in the greater costs associated with the reconstruction of deteriorated infrastructure. The GOP needs to adopt a policy of optimizing resources by giving priority to road maintenance.

c. ***It is unlikely that the Road Fund Law (SIVIPAR) will see the day in its present form, and thus provide new resources to the sector, as it has not achieved a consensus within the Government, because of its deficiencies.*** The SIVIPAR Law was adopted in 2003, but it has not been regulated or implemented. It provides for a wide range of uses, with the exception of maintenance expenditure, and gives the MOPC significant flexibility in the choice of allocation. The sources of the funds for SIVIPAR are the existing ones (tax on petroleum derivatives, car registration tax etc.), and no additional resources have been included. The appropriate regulation of this law or the redrafting of a new law, reserving a significant amount of its proceeds to maintenance expenditures, is paramount to the sustainability of the road sector. The project team has contributed to the definition of the regulation by proving expertise and examples of Road Funds and has contributed to the debate on the subject. However, a lot more will have to be done during the preparation of the project to ensure that the SIVIPAR law or a new Road Fund Law and regulation assure sufficient funds for road maintenance expenditures. After many internal debates within MOPC, the Minister has opted for the redrafting of a new and improved law rather than attempting to regulate this existing and relatively poor SIVIPAR text.

d. ***Despite abundant donor assistance, planning in the transport sector remains weak.*** There is a lack of correlation between investment, maintenance and rehabilitation priorities (determined by some consistently applied criteria, whether economic or not) and budgets. A mixture of tradition, institutional set-up, and a lack of capacity, resources and incentives have rendered the planning process irrelevant. First of all, the elimination of the Transport Infrastructure Planning Office (OPIT), previously under the dismantled Vice-Ministry of Transport and its establishment as an independent, but under funded agency, severely affected the capacity of the MOPC to undertake investment planning. Also, the budgeting process is not conducive to inculcate planning processes in the DV's work procedures. The nature of the intervention carried out by both the Ministry of Finance (MH), and the Congress in budget approval, as

well as the limited capabilities at the MOPC, limit the relevance of extended use of planning tools. Finally, the DV lacks an updated multi-annual plan for the development of the road sector.

e. ***The deficient system for results-based management and monitoring of projects in the road sector limits transparency and accountability in the use of resources.*** The lack of adequate information systems and of incentives for adequate monitoring results in poor management of programs and projects. Given the limitations of resources and the increased attention paid by GOP on transparency and efficiency, the MOPC, which is the main recipient of treasury transfers, is under increasing pressure by the Ministry of Finance to improve its monitoring of the results and impact of projects.

f. ***Lack of access in rural areas is contributing to social exclusion and most recently social unrest.*** Efforts to expand accessibility into rural areas have been uncoordinated and incoherent. Some donor programs are addressing both the municipal road network (JBC, IDB) and the MOPC rural unpaved network. The type of roads and the level of service to which they are being rehabilitated do not follow a national strategy that is in line with future maintenance possibilities. The results have been costly in some cases due to excessive standards and difficult to replicate in others due to lack of institutional capacity in rural areas.

D. Road Strategy

22. The MOPC is increasingly conscious of the need to develop a more integrated strategy of road network development as the competing needs are increasing, and the criteria for choice of limited resource allocation has become increasingly complex. As a result the MOPC has presented the skeleton of a strategy for the development and maintenance of the road network which has been defined around 5 major objectives:

1. Maintain the existent network
2. Complete the remaining missing links of priority main road network
3. Develop the all-weather road network
4. Undertake minimal improvements on targeted earth roads
5. Improve the services to the population

23. The main novelty of this strategy is the management of assets approach whose most important aspect is to strengthen and preserve the existing network.

24. The strategy proposed advances targets for the development of each road sub-networks; (i) to reach a target objective of 3 km of all-weather roads for each km of paved roads; and (ii) to double the length of the paved road network by 2025. The all-weather roads will see the highest increase (16 times between 2005 and 2025) while the length of roads in "earth" condition will decrease by almost half, resulting from the improvement of the other roads condition. The road network development by class of roads is presented below:

Table A1-10: Paraguay's Road Network Strategy

Network\ Year	2005	2015	2025
Paved Roads (km)	4,000	5,000	8,000
Improved Roads(km)	1,500	12,000	24,000
Earth Roads (km)	54,500	47,000	39,500

Source: Taller Diálogo sobre la Gestión Vial para el Desarrollo del Paraguay, MOPC 2005

25. The proposed strategy was refined to consider the different activities to be carried out in the object networks: (i) paved; (ii) improved (all-weather); and (iii) earth. In this refinement of the strategy, it was considered that not all the earth road network is a candidate for improvement and maintenance, given the budgetary realities and the importance of that network. The activities, thus considered are: (i) new construction or improvement; (ii) periodic maintenance; and (iii) routine maintenance.

26. Given the output targets set, the MOPC calculated an average annual budget and it came up with US$179.2 million. This amount corresponds to the higher end average budget allocation the MOPC has been assigned in previous years. A comparison exercise was carried out, between the potential budgetary needs if the strategy had been applied in 2006 and the actual budget allocations of that year. The result indicates that the approved budget (US$167 million) was only US$12 million under what would have been required were the proposed Road Strategy to be applied. However, the comparison also showed that the 2006 budget, through commitment to existing programs, over-budgets in construction of new paved roads, rehabilitation (or periodic maintenance) of paved roads, improvement of earth roads, and under-budgets in everything else. Everything else includes: routine maintenance of paved roads, of improved (all weather roads), and of earth roads, but also periodic maintenance of improved roads (all weather roads).

27. As a result, it can be said that the targets proposed by the road strategy are in line on average with the historical budgetary envelope for the road sector. To be successful, the MOPC will have to reassign the budget and improve its execution.

Table A1-11: Paraguay's Road Network Strategy Output Targets

Network\ Year	Existing Length (km)	Activity	Annual Target Length (km)	Unit cost (US$)	Total cost (US$ million)	2006 Budget
Paved Roads (km)	4,000	New construction	100	400,000	40.0	85.5
		Periodic Maintenance	260	100,000	26.0	32.1
		Routine Maintenance	3,734	3,000	11.2	2.1
Improved Roads (all-weather) (km)	4,500	Periodic Maintenance	2,000	10,000	20.0	0.0
		Routine Maintenance	6,000	2,000	12.0	2.3
Earth Roads (km)	21,500	Improvement to all-weather	750	40,000	30.0	36.1
		Minimum works	20,000	1,000	20.0	0.0
		Routine Maintenance	20,000	1,000	20.0	9.0
Total					179.2	167.3

Source: Estrategia Vial del Paraguay, MOPC, 2006.

28. With this tool the MOPC will redirect commitments and align expenditures with the proposed strategy targets. It is understood that the alignment will be gradual as commitments will have to be complied with. This project, will contribute to the proposed budgetary alignment by providing funds for routine maintenance.

29. The implementation of this new strategy, and in particular, the alignment of funds that have historically gone to the main road network, benefiting the same type of construction company and resulting in a particular level of service, and involving a particular type of action by MOPC, will not be easy to carry out. For this reason, and following the project's overall approach, the MOPC will seek to involve relevant stakeholders to buy into this approach and help MOPC to implement the strategy. Thus a

Road Social Contract will be prepared, discussed, negotiated and signed between stakeholders and the MOPC, throughout 2006 and prior to project execution.

30. The **Road Social Contract** seeks with clear objectives, good will and mutual agreement to ensure efficiency, transparency and participation. The stakeholders that have been identified to participate in the contract are: individual beneficiaries as represented by consumer associations or the like, contractors as represented by construction associations (CAVIALPA, etc), professional road users as represented by transport companies (freight and passengers), local government as represented by Governors and local Mayors, national representation as represented by congressmen. A series of consultations, discussions are being conducted by MOPC with the identified stakeholders and will culminate in the signature of the Road Social Contract indicating the responsibilities and obligations of everybody involved.

31. It is anticipated that the road users will be involved in the decentralization and identification of needs and will contribute to take care of the network, as well as of informing the authorities of deficiencies. The politicians will participate in enacting road laws, more particularly ensuring sufficient budget, financing, road funds, Road Fund Law and others. Contractors will participate by adapting their offer to new types of contracts – including performance-based contracts. The Ministry of Public Works will improve management and the provision of information to users through the execution according to the budget, access to the accounts, and transparency. Finally, the government and municipalities will co-participate in agreements, contributions, promotion, identification of needs and control.

Annex 2: Major Related Projects Financed by the Bank and/or other Agencies

PARAGUAY: Road Maintenance

The World Bank Group		
Sector Issue	**Project**	**Latest Supervision (PSR)**
Public Sector Projects		(IP) (DO)
Strengthening of MOPC and improvement of roads (120km)19	Eighth Highway Project (3685-PA)	M S
Other Development Agencies		
Inter-American Development Bank (IDB)		
Sector Issue	**Project**	**Approval Year**
Public Sector Projects		
Improving Route No. 3, Emboscada – San Estanislao (130 km); implement a road maintenance and an environment protection program.	Road Corridors Program (933/OC-PR)	1997
Improving main inter-urban roads (730 km); phase 1 1170 km; and phase 2 Management maintenance plan (6000 km); Institutional Strengthening.	National Program of Rural Roads – Second Part (Phases 1 and 2) (1230/OC-PR)	2003
Rehabilitation of Route No. 9, Transchaco (258.56 km) and improvement of corridors (787.56 km)	West Integration Corridor Program (1278/OC-PR)	2001
Andean Development Corporation (CAF)		
Sector Issue	**Project**	**Approval Year**
Public Sector Projects		
West region road maintenance (2000 km)	West Integration Corridor Program (Loan No. 311)	2002
Cuenca de Plata Development Fund		
Sector Issue	**Project**	**Approval Year**
Public Sector Projects		
Environment management plan and support to districts	West Integration Corridor Program (PAR-17)	2002
Japan Bank Cooperation International (JBIC)		
Sector Issue	**Project**	**Approval Year**
Public Sector Projects		
Improvement of Paraguari – Villarrica and La Colmena Tebicuary (118.76km); recarpeting of paved rodas Yby Yau – Pedro Juan Caballero Bella Vista (148km); San Juan Bautista Encarnación and First part of la Transchaco; enlargement of bridges in Paraguari – San Juan Bautista, Carapeguá-Acahay-La Colmena (332.5)	Improvement Roads Program II (PG-P13)	2004
National Social and Economic Development Bank (BNDES)		
Sector Issue	**Project**	**Approval Year**
Public Sector Projects		
Construction, improvement and rehabilitation of Route No. 10, Tacuara – Katuete (Crossing Carumbey; 182.5 km)	Improvement of Route No. 10, Tacuara - Katuete	2001

[19] This was the last project carried out by the World Bank in the road sector in Paraguay. This project was closed in 2001and its Implementation Completion Report was prepared in 2003.

Annex 3: Results Framework and Monitoring

PARAGUAY: Road Maintenance

Project Impact Results Framework – Part I

Project Higher Level Objectives	Outcome Indicators	Use of Outcome Information
Support economic growth and poverty alleviation	Higher levels of income More diversified sources of income Higher levels of expenditure Increased employment opportunities and labor force participation Improved access to schools and higher schooling attendance and enrollment Improved access to health facilities, greater use of health services and lower prevalence of certain diseases Higher social and political participation Improved access to market and lower prices	Income levels and composition by sources from household survey data (HSD) Expenditure levels from HSD Occupational status, type of employment and labor supply (labor force participation and hours of work) from HSD Schooling status (enrollment, attendance) reasons for non-attendance and access to school (transport time, monetary costs, etc.) from HSD Access to health facilities (transport time, monetary costs, etc.) and prevalence of certain diseases (e.g. gastrointestinal, respiratory, etc.) from HSD Self-reported participation in political activities and social groups from HSD Access to the market and prices from HSD

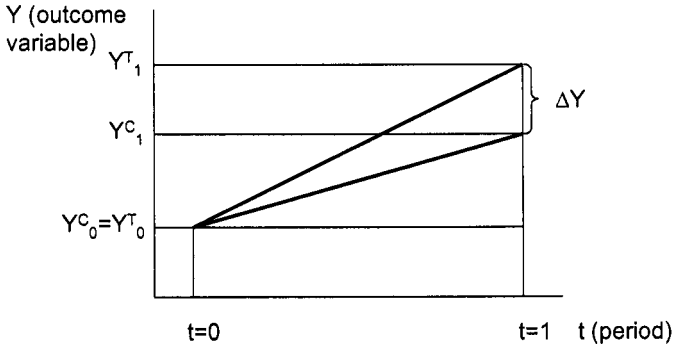
1. In accordance with the World Bank's renewed emphasis on evaluating the projects finances, a socioeconomic impact evaluation of the Paraguay Road Maintenance Project will be performed. The proposed impact evaluation will contribute to the overall evaluation of the project by estimating its causal impact on several socioeconomic indicators. The evaluation's objectives are: (i) to assess the extent to which the project attains its higher-level objective of supporting economic growth and poverty alleviation; and (ii) to increase the Bank's current knowledge about the impact of rural roads projects to poverty alleviation.

2. It is important to underscore that the proposed impact evaluation will be concerned with measuring the indirect effects of the project on outcome indicators (i.e. on several socioeconomic variables of interest). Thus, the project is not expected to have a direct effect on these variables, as is the case of the output indicators discussed in the Results Framework and Monitoring below.

3. The proposed impact evaluation is based on the model of potential outcomes, which is the standard methodology for estimating the causal effects of public policies, social programs and other forms of public interventions. This methodology has been widely used to evaluate projects financed by the Bank, including those discussed in Baker (2002), as well as road transport projects financed by the Bank similar to the Paraguay project, such as those discussed in Grootaert (2002).

4. Given that the selection of the roads to be improved and maintained has already been taken, a non experimental or quasi experimental design or approach is proposed. The evaluation will use the difference in differences (or double differences) estimator, along with propensity score matching for finding an appropriate control group, in order to estimate the causal impact of the project.

5. The estimate of the causal impact (ΔY) project will be obtained as follows. First, the difference for the treatment group in the average level of the socioeconomic variable of interest (Y^T) after the project has been implemented (Y^T_1) and the average level before the beginning of the project (Y^T_0) is obtained. Second, the difference for the control group in the average level after the project has been implemented (Y^C_1) and the average level before the beginning of the project (Y^C_0) is subtracted from the first difference (see figure below). This second difference serves as a control for other simultaneous factors that may affect the socioeconomic variable of interest.



6. The control group will consist of communities which are not affected by the project but whose observable characteristics are comparable to those of the communities which will be affected by the project (treatment group). This part of the evaluation would first select control communities using propensity score matching (PSM) based on road characteristics as well as community-level characteristics from the latest census data. The 2002 Census will be extensively relied on for the proposed surveys in terms of: (i) identifying the treatment communities using the area of influence of the project once the unpaved roads which will be maintained and repaired are selected; (ii) serve as the basis on which the treatment and control communities will be matched; and (iii) serve as the sampling frame for drawing the random samples of households within the treatment and control communities.

7. After consultation with the Paraguayan Statistical Agency (Dirección General de Estadística, Encuestas y Censos (DGEEC), Secretaría Técnica de Planificación), it was determined that currently no survey or census in Paraguay is appropriate for performing the proposed impact evaluation of the project. Thus, a specific household survey will need to be collected. In terms of the type of data to be collected, the survey would collect panel data over a period of seven years: a baseline survey in 2006, an initial follow-up survey in 2008, and a final follow up survey in 2013. Based on the costs provided by Baker (2000), it is estimated that the total cost of the evaluation, of which the data collection would represent more than half of the cost, would be somewhere between US$200,000 and US$450,000.

Results Framework – Part II

PDO	Outcome Indicators	Use of Outcome Information
Improvement of Road Service: Basic transport infrastructure is maintained in a condition that facilitates the movement of freight and passengers at a cost-efficient level of service by arresting any further deterioration due to fiscal restrictions, and ensuring the preservation of existing infrastructure assets in the long term, through the rationalization of public expenditures of the road transport sector.	**Cost-efficiency** Reduction in average costs of vehicle operation. **Time-efficiency** Reduction in average transportation times, and time of trips. **Number of Public Transportation Vehicles** Increase in the number of public transportation vehicles that use the unpaved roads of the targeted departments. **Arresting Deterioration** Increase in the levels of user satisfaction.	Assess the impact of improved road conditions on the competitiveness of Paraguayan producers, economic growth, and reduction of poverty.
Intermediate Results One per Component	**Results Indicators for Each Component**	**Use of Results Monitoring**
Component 1: Strengthening MOPC's Strategic Planning and Road Management.	**Component 1:** Number of laws, decrees, and regulations related to the project are enacted. Creation of Road Strategic Planning Unit. Road strategy and five-year plan implemented. Programming and follow-up of the result-based management and budget implemented. Road monitoring system implemented. Strategy of internal and external communication on road maintenance implemented.	**Component 1:** MOPC to follow up on the pace of implementation of the DV's program.
Component 2: Primary paved road network rehabilitated and maintained at an adequate level of service through performance-based contracts.	**Component 2 :** Km of paved network maintained under level of service. Percentage of compliance with the standards of the kilometers contracted (service standards are specified in the performance-based contracts, and include the conditions of 5 items: road surface, hard shoulder, road safety, structures, and right-of-way).	**Component 2:** Measure the performance of the program in order to identify potential problems and make the necessary adjustments; and monitor its impact on road condition.

Component 3: Unpaved national routes and secondary roads connecting rural communities too national network rehabilitated and maintained.	Km of the unpaved road network improved. Km of the unpaved road network maintained at appropriate standards. Decrease in critical points of difficult transit.	Evaluate the performance of the program and make adjustments aimed at improving its impact.

Arrangements for results monitoring

		Target Values					Data Collection and Reporting		
Project Outcome Indicators	**Baseline**	**YR1**	**YR2**	**YR3**	**YR4**	**YR5**	**Frequency and Reports**	**Data Collection Instruments**	**Responsibility for Data Collection**
PDO: Improvement of Road Service:									
Average costs of vehicle operation*	G/km/ Vehicle	G/km/ vehicle	G/km/ vehicle	G/km/ vehicle	G/km/ vehicle	G/km/ vehicle	Annual	Transport company survey	MOPC/UPGP obtains data from DV and prepares report
Average transportation times*	Km/h	Km/h	Km/h	Km/h	Km/h	Km/h		Transport company survey	
Number of public transportation vehicles that use the unpaved roads of the targeted departments:								MOPC and DINATRAN data	
• Caazapá	16 veh/km			18 veh/km		20 veh/km			
• Caaguazú	23 veh/km			25 veh/km		28 veh/km			
• San Pedro	40 veh/km			44 veh/km		49 veh/km			
Levels of users' satisfaction*	Bad Fair Good V. Good	Bad Fair Good V. Good	Bad Fair Good V. Good	Bad Fair Good V. Good	Bad Fair Good V. Good	Bad Fair Good V. Good		Users' Surveys	
Intermediate Outcome Indicators									
Component 1:									
Number of laws, decrees, and regulations related to the project enacted.	0%	75%	100%				Annual	Legislative databases; Progress Implementation Report	MOPC/UPGP obtains date from DAJ and DCST and prepares report
Creation of Road Strategic Planning Unit	No unit	UPE created	Operational UPE				Annual	Ministerial Resolution	VMOPC/DAT

			equipped and trained					creating the Unit; Progress Implementation Reports	
Road strategy and five-year plan implemented	No strategy			Road Strategy elaborated and Published	Monitoring of Physical and Financial Execution of the Road Strategy	Monitoring of Physical and Financial Execution of the Road Strategy	Annual	5-year Plan published on MOPC website; Progress Implementation Reports	VMOPC/DAT
Programming and follow-up of the result-based management and budget implemented	Non-existent	Budget Preparation (BP) based on performance	BP reports published	BP reports published	BP reports published	BP reports published	Semi-Annual	Budget Execution Reports published on MOPC website	VMAF/DPE
Road monitoring system implemented	No System	System established	Reports on % of compli-ance with Strategy	Reports on % of compli-ance with Strategy	Reports on % of compli-ance with Strategy	Reports on % of compli-ance with Strategy	Annual	Progress Implementation Report published on MOPC website	VMOPC/DV
Strategy of internal and external communication on road maintenance implemented	No strategy	Strategy Designed	Reports on % of monitored roads published	Reports on % of monitored roads published	Reports on % of monitored roads published	Reports on % of monitored roads published	Annual	Strategic Plan published and disseminated	VMOPC/DV
Component 2:									
Km of paved network maintained under level of service.	0km	523 km	860 km	968 km	1218 km	2016 km	Semi-Annual	Signed contracts in execution; Visual road inspections and activity reports	MOPC/UPGP obtains data from DV and prepares report
Percentage of compliance with the standards of the kilometers contracted.	0%	% in conformity with standards	% in conformity with standards	% in conformity with standards	% in conformity with standards	% in conformity with standards	Semi-Annual	Contracts supervision and visual road inspections reports	MOPC/UPGP obtains data from DV and prepares report

Component 3:									
Km of the unpaved road network improved.	0km	377.73 km	597.21 km	876.37 km	876.37 km	876.37 km	Semi-Annual	Signed contracts; visual road inspections and activity reports	MOPC/UPGP obtains data from DV and DCV and prepares report
Km of the unpaved road network maintained at appropriate standards.	0 km	0 km	260.67 km	672.27 km	893.14 km	893.14 km	Semi-Annual	Visual road inspections and activity reports	MOPC/UPGP obtains data from DV and prepares report
Number of critical points of difficult transit.	18	18	10	8	0	0	Semi-Annual	Visual road inspections and activity reports	MOPC/UPGP obtains data from DV and prepares report

* Baseline data will be provided by the Transport Company Survey

Annex 4: Detailed Project Description

PARAGUAY: Road Maintenance

A. Project Design

1. During the preparation of the Project, and in order to define with more clarity and precision the orienting and instrumental mechanisms that intervene in the planning of this Project, the Task Team and the MOPC carried out a workshop on the Project's Logical Framework (*Taller de Marco Lógico*). The workshop was focused on the explanation of the logical framework as a methodology and tool for project planning and design. Subsequently, the workshop also addressed the Project's more specific problems and characteristics, within a participatory mechanism, including the following issues: (i) analysis of stakeholders; (ii) problem tree or diagram (Figure A4-1); (iii) solution tree or diagram (Figure A4-2); (iv) analysis of alternatives; (v) logical framework matrix, which consisted of the definition of the outcomes, outputs, purposes, components, activities, indicators, means of verification, and assumptions (this matrix is reflected in this document in the tables contained in Annex 3). Finally, based on the activities defined in the logical framework, the workshop participants elaborated the Project's schedule, detailing the timing and chronological sequences of the Project's activities. The approach was participatory and collaborative with all relevant MOPC staff, and helped ultimately design the final details of the project.

Figure A4-1
Projects' Problem Tree

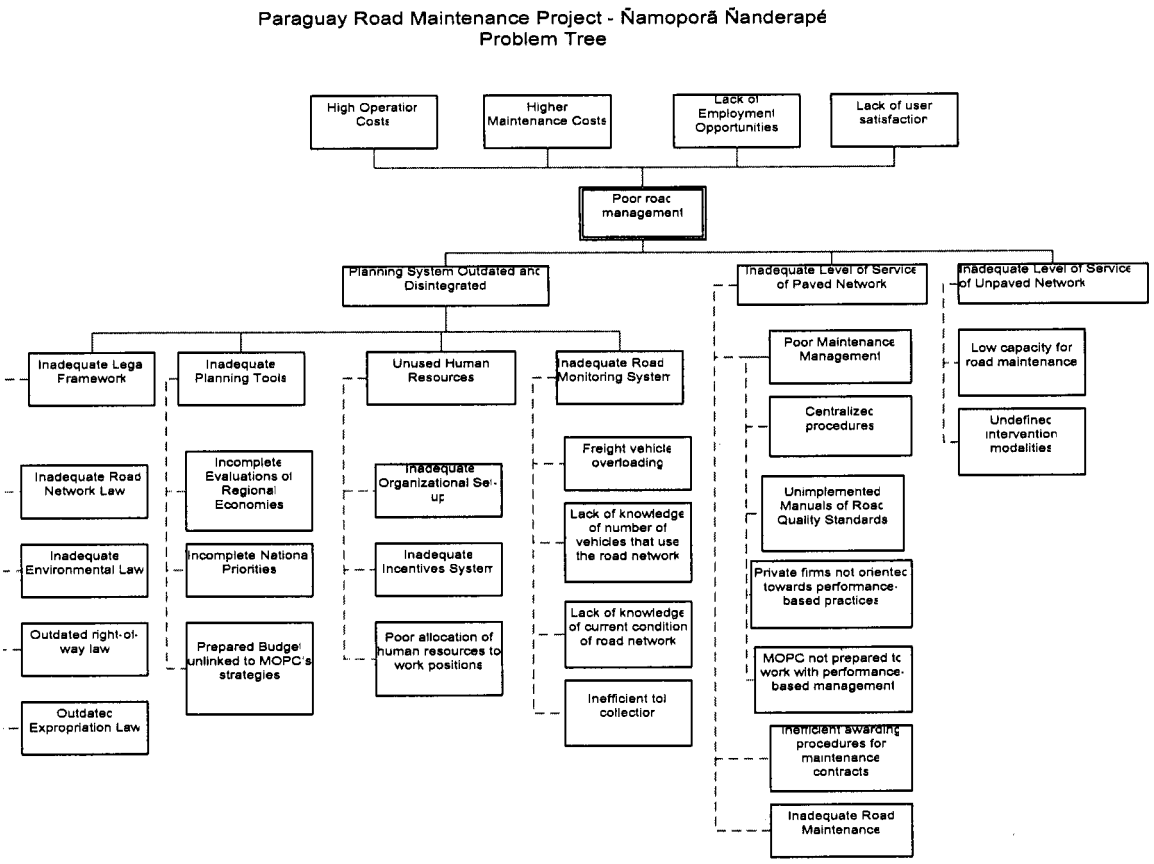
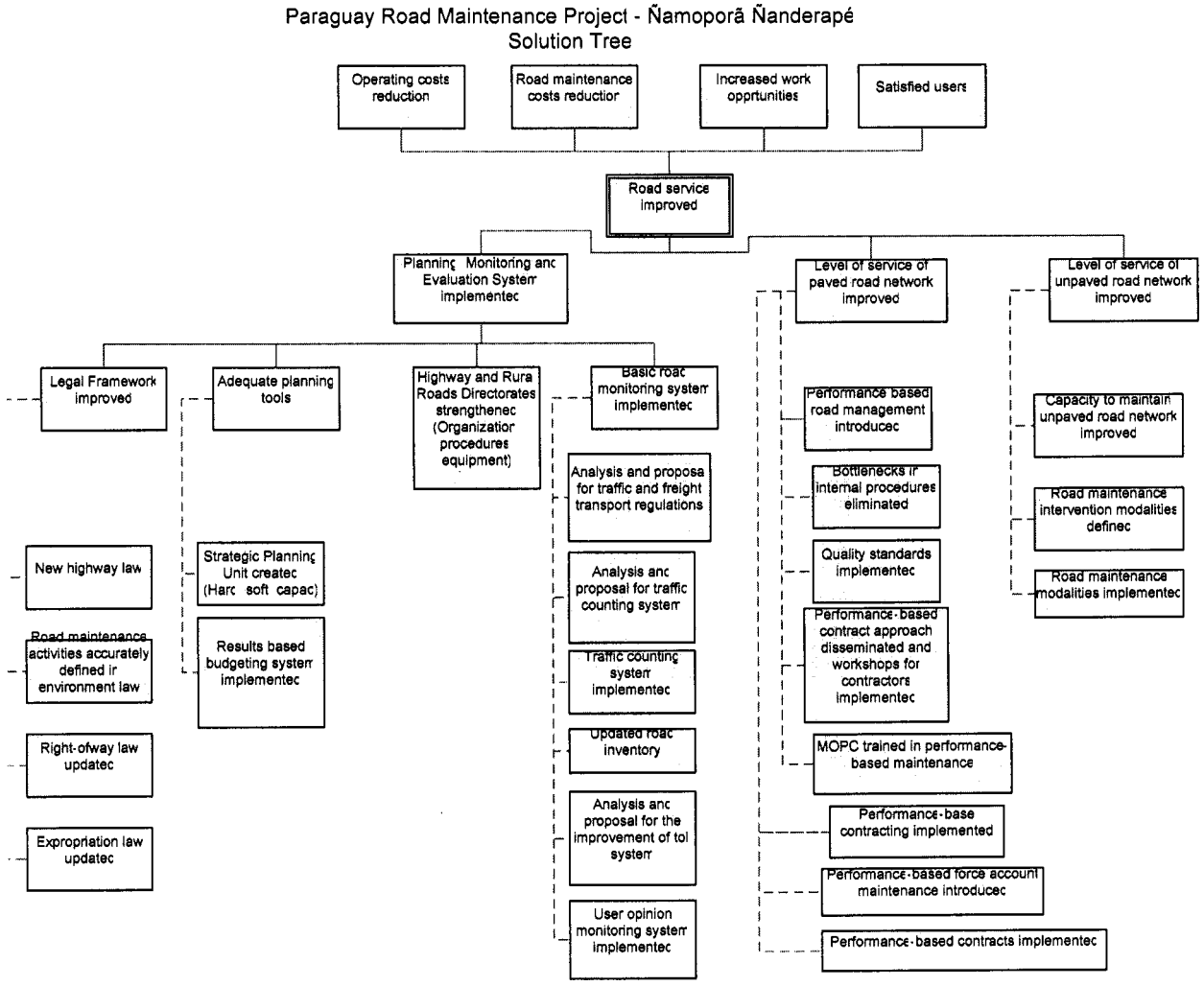


Figure A4-2
Project's Solution Tree



B. Detailed Project Costs

2. The project comprises three components, namely: (i) Strengthening Strategic Planning and Road Management; (ii) Improvement and conservation of the fundamental road network; (iii) Improvement and conservation of rural access. The overall project's development objective is to establish a sustainable road management strategy that ensures the upgrading and maintenance of the road network through a rational and balanced use of scarce resources. The guiding principle of this approach will be improved road conservation within a comprehensive vision that ensures the sustainability of road investments and the preservation of existing road assets.

3. The overall cost of the project is estimated at US$92.753 million (not including contingencies) of which US$86.246 million correspond to the maintenance and rehabilitation components; of this US$43.346 million will be destined to the maintenance and rehabilitation of paved roads, and US$42.900 to the maintenance and rehabilitation of unpaved roads; and the remaining US$6.507 million correspond

to the institutional strengthening components. The Bank would finance US$64.927 million and the GOP the remaining US$27.826 million (excluding contingencies).

Table A4-1
Detailed Project Costs

PROJECT COMPONENTS	Total	Bank	GOP
1. Strengthening of Road Management and Strategic Planning	**6.507**	**4.555**	**1.952**
1.1 Strengthening of Road Management and Planning	3.471	2.429	1.041
1.2 Strengthening of Budget Processing and Preparation	0.077	0.054	0.023
1.3 Road Maintenance strengthening	1.760	1.232	0.528
1.4 Project Monitoring and Evaluation	1.199	0.839	0.360
2. Improvement and Maintenance of the Paved Road Network	**43.346**	**30.343**	**13.004**
2.1 a) Performance-based maintenance contracts (including supervision)	39.276	27.494	11.783
2.1 b) Capacity building in performance-based contracts	0.440	0.308	0.132
2.2 Toll Master Plan and Toll optimization	3.630	2.541	1.089
3. Improvement and Maintenance of the Unpaved Road Network	**42.900**	**30.030**	**12.870**
3.1 a) Upgrading and maintenance works (including supervision)	41.213	28.849	12.364
3.1 b) Pre-investment and preparation studies for future year programs	0.792	0.554	0.238
3.2 a) Capacity building of MOPC Districts and Local Governments	0.165	0.116	0.050
3.2 b) MOPC District office improvement and equipment	0.233	0.163	0.070
3.3. Implementation of Indigenous People Plan and Environmental Management Plan	0.497	0.348	0.149
TOTAL BASELINE COST	**92.753**	**64.927**	**27.826**
Physical Contingencies(1)	8.787	6.258	2.529
Price Contingencies(2)	3.692	2.630	1.062
TOTAL COST	**105.232**	**73.815**	**31.417**
Front end fee	0.185	0.185	0.000
TOTAL FINANCING REQUIRED	**105.417**	**74.000**	**31.417**

1) Approximately 10% of the infrastructure components

2) Approximately 1.5% per year of the total Baseline cost and physical contingencies

* Total costs of each sub-component in this table include a 10% VAT, which is included exclusively in the amount financed by the GOP.

4. On September 9, 2005 the MOPC established a Unit for Project Coordination and Preparation by ministerial resolution No. 523. This unit would be the main promoter of project activities and sole interlocutor with the Bank, but would use the MOPC divisions to prepare the project. Thus, the linkages

and the responsible staff in all the MOPC dependencies that might deal with the project have been worked-out through this participatory process.

C. Detailed Project Components

Project component 1: Strengthening Strategic Planning and Road Management (US$6.5 million, excluding contingencies):

5. This component would finance the strengthening of management and road conservation within a framework of fiscal constraints, in particular, it will support:

(1) The development, promotion and dissemination of a National Transport Plan for investments in ten years and the implementation and strengthening of the MOPC project cycle (including the strengthening of the DCP, the UOC and the DAI).

(2) The strengthening of capacities, methodologies, and tools for economic evaluation and the prioritization and selection of projects using advanced modeling and software programs such as HDM, EBM, and others.

(3) The development of data bases and information systems for planning road conservation activities, monitoring road network condition, evaluating conservation based on performance, and monitoring costs of road construction, rehabilitation and maintenance.

(4) The evaluation of the financing sources of the road sector and analysis of alternative sources, mechanisms for optimizing existing ones, and future projections of resources and needs.

Project component 2: Improvement and conservation of the fundamental road network (US$43.3 million, excluding contingencies):

Sub-component 2.1: Performance-based maintenance contracts and MOPC's capacity building in performance-based contracts management.

6. This component would finance the rehabilitation, periodic maintenance and signaling of segments of the national fundamental road network, through performance-based maintenance contracts (GMANS). This type of contract requires the contractor to rehabilitate and then maintain a network of roads over a period of five years for a lump sum amount. They further specify required road service outputs, and use incentive-based payments to ensure the quality of the works[20] in order to improve the service provided to paved road network users to the standards established by the MOPC.

7. The general objectives of this component are to prepare the GMANS contracts (including the execution of specific technical studies for each segment and the preparation of the bidding documents) for a total of 968 km of Paraguay's paved network, and to strengthen MOPC's capacity in the management of performance-based maintenance contracts.

[20] Throughout the contract period, rehabilitation works must meet or exceed the minimum thickness of overlay and must comply with the maximum level of roughness, rut depth, cracking, or raveling. Regular monthly visual inspections of maintenance activities focus on a few essential items (about 10) in ensuring compliance with the specifications: no potholes or unsealed cracks, no excessive rutting, good condition maintained on shoulders, culverts and drains, guardrails, vertical and horizontal signs, as well as on the road side environment.

8. Some factors considered when elaborating the list of prospective paved roads to be covered by GMANS contracts include the identification of maintenance and improvement projects programmed for the next two years and information collected on the overall condition of the road network.

9. From these considerations it was possible to suggest seven contracts. These contracts were similarly balanced in terms of the maintenance works and budget disbursements needed; also they were at a distance that could facilitate the road works. These seven contracts would be defined in the following way:

Table A4-2:
Performance-Based Maintenance Contracts (including supervision 7% and excluding contingencies)

GMANS Contract	Segment	Length (Km)	Estimated Cost (US$ million)
I	PY 02: Caacupé – Inicio concesión Tapé Porá PY 03: Limpio – Empalme PY 08 PY 08: Empalme PY 03 – Coronel Oviedo	381	10.3
II	South Access: Desvío a Guarambaré - Itá PY 01: Itá – Encarnación	337	7.8
III	PY 06: Encarnación – Empalme PY 07	250	7.2
IV	PY 08: Empalme PY 03 – Yby Yau PY 05: Pedro Juan Caballero - Concepción	393	10.6
V	PY4: San Ignacio- Pilar	142	10.7
VI	PY 07: Hernandarias – Empalme PY 03 PY 03: Empalme PY 08 – Saltos del Guaira	400	11.6
URBAN	PY 02: San Lorenzo – Caacupé Acceso Sur: 4 Mojones – Desvío a Guarambaré	108	4.6
TOTAL		**2,011**	**62.8**

10. The prospective road networks have been identified and GMANS contracts also. This Project will finance four GMANS (Urban, I, II and V), for a total length of 968 km, with a total investment of US$39.3 million (including supervision 7% and excluding contingencies) for 5 years, where US$2.57 million would be allocated to supervision (7%), US$19.31 million for rehabilitation, US$0.97 million to point improvements and safety, and $16.45 million to routine maintenance.

Sub-component 2.2: Toll Master Plan and Toll Optimization.

11. This component would also include a sub-component of Toll Master Plan and Toll Optimization with a total cost of US$3.63 million, of which US$2.54 million will be financed by the Bank and US$1.09 million will be finance by the GOP. This sub-component was designed as part of a strategy to ensure the sustainability of the Project.

12. The toll system in Paraguay has 15 toll booths that only collect in one direction, and the majority of which has obsolete equipment which does not allow for exhaustive controls, leading to an inefficient

collection system. It is expected that the modernization and enhancement of the toll system will generate the necessary resources for the regular maintenance of the national paved road network.

13. This sub-component will include: (i) a consultancy study for the diagnosis of the system and for the elaboration of a Toll Master Plan; (ii) adequate facilities and control equipment for each toll post to improve toll collection; (iii) the implementation of a two-way toll fee once the expected level of service is achieved through the GMANS contracts.

Project component 3: Improvement and Maintenance of the Unpaved Road Network (US$42.9 million, excluding contingencies):

Sub-component 3.1: Upgrading and maintenance works and pre-investment and preparation studies.

14. This component would finance the rehabilitation and conservation of unpaved national routes that connect to the national paved network, and secondary roads connecting rural communities to these; and propose a comprehensive solution for improved access to isolated and marginalized rural communities. The approach will include different levels of intervention depending on the level of importance of the segments included in the program.

15. The selection criterion of the Paraguayan Departments to be covered under this component was based on an index that considers three relevant socioeconomic variables. These variables are the density of unpaved roads (defined as the number of km of unpaved roads per square km), the density of the rural population (defined as the rural population per square km), and a poverty index. Each one of the first two factors received a weight of 25% in the index generated, while the poverty index represents the most important variable for this analysis with a weight of 50%. The following table presents the 17 Departments of Paraguay ordered by the index proposed.

Table A4-3
Departments' Importance Index

Índice de Importancia de los Departamentos

				Peso 0.25		Peso 0.25		Peso 0.50		
				Red No Pavimentada por km2		Población Rural por km2		Índice Pobreza		Índice de
Departamento	Región del pais	Proy. DCV BID 1230	Proy. DCV JBIC PG.P14	(km/1000km2)	Factor	(#)	Factor	(%)	Factor	Importancia
Caaguazu	Oriental	Contrato	Mejoramiento	61	1.00	27	0.35	60.6	0.97	0.82
San Pedro	Oriental	Contrato	Mejoramiento	56	0.92	13	0.17	62.2	1.00	0.77
Caazapa	Oriental		Mejoramiento	45	0.74	12	0.16	48.1	0.77	0.61
Central	Oriental			39	0.65	77	1.00	20.7	0.33	0.58
Cordillera	Oriental			45	0.74	31	0.41	35.2	0.57	0.57
Concepción	Oriental			36	0.60	6	0.08	47.8	0.77	0.55
Canindeyu	Oriental	Convenio		50	0.83	7	0.09	38.5	0.62	0.54
Itapua	Oriental	Convenio		43	0.70	19	0.25	36.8	0.59	0.53
Paraguari	Oriental			42	0.69	20	0.26	34.5	0.55	0.52
Guaira	Oriental	Contrato		20	0.32	30	0.39	40.7	0.65	0.51
Ñeembucu	Oriental			47	0.77	3	0.04	36.5	0.59	0.50
Amambay	Oriental			32	0.53	3	0.04	39.4	0.63	0.46
Misiones	Oriental			26	0.43	5	0.07	38.3	0.62	0.43
Alto Paraguay	Occidental			16	0.26	0	0.00	44.6	0.72	0.42
Boquerón	Occidental			26	0.42	0	0.00	32.9	0.53	0.37
Pdte.Hayes	Occidental			14	0.23	1	0.01	36.2	0.58	0.35
Alto Paraná	Oriental	Convenio		19	0.31	13	0.17	26.3	0.42	0.33

16. According to this index, Caaguazú, San Pedro and Caazapá should be the Departments covered by this project component. These three Departments already benefit from the programs of the Rural Roads Directorate (*Dirección de Caminos Vecinales*) financed by the IDB and the JBIC; therefore, this project would complement the MOPC's scheme by responding to the needs of the national and departmental unpaved road networks that would connect to the newly rehabilitated tertiary rural roads (caminos vecinales).

17. The following tables (A4-4, A4-5 and A4-6) present the segments to be upgraded and maintained under this Project component. The project will carry out, at the outset of the project, basic works of improvement (including improvement of drainage systems, increase of altitude in low areas, etc.) and change of surface in most cases from earth to gravel. These basic works will be followed by maintenance of the improved segments. It should be noted that eleven of the segments that have been selected to be targeted by this component are being improved by other MOPC's road projects. In the case of those segments this project will cover only the maintenance work.

Table A4-4
Segments in the Caazapá Department

Works Contracts	Road	Segment Description	Length (Km)
1	PY15	SAN JUAN NEPOMUCENO - RíÍO TAYAY	25.12
2	F05	CAAZAPA - BOQUERON	13.70
3	PY20	CRUCE A LIMA (F-01) - RIO TEBICUARY (LÍMITE DPTAL.)	30.70
4	F05	SAN FRANCISCO - SAN JUAN NEPOMUCENO	35.24
5	F01	SAN MIGUEL YARATI-I - YATAITY	32.20
4	F01	YATAITY (YUTY) - SAN FRANCISCO	9.70
6	F05	BOQUERÓN - SAN FRANCISCO	26.30
7	PY15	RÍO TAYAY - TAVAI	23.81
8	F01	ISLA YOBAI - BUENA VISTA	13.04
			209.82

Table A4-5
Segments in the Caaguazú Department

Works Contracts	Road	Segment Description	Length (Km)
1	PY18	SAN JOAQUIN - SAN ANTONIO - YHU	16.90
2		CAAGUAZÚ - AGUA - MOJÓN 7 - RUTA YHU CAAGUAZÚ (PY13)	27.50
3		PY13 - 3 DE FEBRERO - TORO CANGUE - YHOVY CENTRAL - RAÚL OVIEDO	60.60
4		YHOVY CENTRAL - J.E. ESTIGARRIBIA (CALLE 2)	19.20
5	E13	PY02 (KM 155) - CRUCE ALTO - R I 3 CORRALES	16.70
6	E01	CARAYAO - CLETO ROMERO (A° MOLINA)	28.20
5	E08	PY08 (KM 140) CRUCE CAPILLITA - DESVIO CALLE 4 SANTA LIBRADA	16.50
7	E11	JUAN MANUEL FRUTOS - 3 DE FEBRERO	16.60
6	E05	PY08 - CRUCE MAINUMBY	2.00
8	E03	NUEVA LONDRES - LA PASTORA	23.90
5	E08	CALLE 4 DESVIO SANTA LIBRADA - ESC. N° 810 CRUCE ALTO	5.10
8		PY08 - LA PASTORA	27.00
1		EMPALME YHU -TARUMAÍ-CARPA CUÉ-SAN JOAQUÍN	21.20
			281.40

Table A4-6
Segments in the San Pedro Department

Works Contracts	Road	Segment - Description	Length (Km)
1	B13	PY08 (RUTA OVIEDO SANTANI) - UNION	16.00
2	B09	CHORE - GRAL. AQUINO	54.60
3	B08	CHORE - COLONIA COCUERA	16.11
3	B08	COLONIA COCUERA - SAN PABLO	17.88
4	B07	VILLA DEL ROSARIO - ESTERO YETYTY	13.90
5	B03	SAN PEDRO - PIRI PUCU	32.20
6	PY11	SAN VICENTE - ARROYO ITANARA	19.92
7	PY11	LIMA - QUIINDY	24.76
7	PY11	QUIINDY - NARANJITO	4.60
			199.97

Sub-component 3.2: Capacity building of MOPC Districts and Local Governments.

18. A very important aspect of the program has been the design and implementation of participatory processes for the selection of priorities, and the selection of interventions in relation to the sustainability of the future network. These participative processes have been and will continue to be organized and implemented in accordance with the following principles:

a. Participatory mechanisms include the dissemination of information and popular consultation (both binding and non-binding) for the selection of roads, financing, and execution of works.
b. Participatory instances are open to the public both at the national and local level, including public institutions, civil society organizations, and the private sector.
c. Participatory processes are based on objective data and technical knowledge, which are communicated through a comprehensible and accessible language to the different interest groups.
d. Participatory processes are aimed at achieving transparency and the development of anti-corruption mechanisms.

19. Along these lines, the following activities have been carried out during project preparation:

Workshops at Regional/Local Level

20. The selection of the eligible unpaved road network and the Road Master Plan were made in participatory workshops in the departments concerned (Caazapá, Caaguazú, and San Pedro). Governmental officials (MOPC, Departmental and Municipal Authorities), and representatives of NGOs, social organizations, involved communities, and private sector attended these workshops. They were held in two major stages i) selection of the potential eligible Unpaved Road Network in the three departments concerned, and ii) the definition of the Road Master Plans.

21. The workshops were coordinated by MOPC staff with the assistance of specialized consultants and aimed at establishing technical details of works to be proposed and to respond technical queries of participants. In order to enhance understanding of nature and scope of the workshops by participants, presentations were made through audio-visual means, printed maps and easy understanding summaries. Special attention was put in avoiding the usage of technical terminology.

a) Three Workshops to Select the eligible Unpaved Road Network[21]

22. The first series of three workshops to select the possible *eligible Unpaved Road Network* for the development in each department to be supported by the project in line with fiscal realities was nourished with information on three scenarios. These scenarios were built from social indicators (population that lives along the road, poverty indicators, characteristics of production, availability of services of education and health among others), total length of road network, costs, and evaluation of availability of resources for each scenario.

23. The platform of discussion focused on the following principles and concepts in the road maintenance approach of the project: i) importance of services of infrastructure to support the development of a community or region; ii) incidence of the improvement of the rural road network to access to markets and other public services such as education and health, among others; iii) importance of improvement and rehabilitation of the secondary road network of the country, that allows to connect the main network with the rural road network; iv) need of strengthening institutional capacity at local and central levels in the implementation of rural infrastructure projects; v) need of employment generation in rural areas in order to diminish community poverty indicators; vi) administration and operation of such road network once project's implementation is concluded to assure sustainability; vii) strategies to apply to guarantee sustainability of the improvements implanted in the road network and the systems of transport.

24. A set of preliminary modalities of co-participation in maintenance of roads as foreseen in the *Unpaved Road Network Maintenance Collaborative Strategy* was also presented. After presentations, participants grouped by Municipalities worked on the selection of *Unpaved Road Network* from three scenarios. Each network was presented in a map with the reference of the adjacent populations, necessary amount of investments for its improvement, the annual cost for its sustainable maintenance, and the benefited population. Groups also explored together co-financing mechanisms, and presented their conclusions in a plenary session.

25. A period of 15 days was given to participants to present proposals of co-financing expansion of *Eligible Unpaved Road Network.*

26. After the first workshop and the 15 days period to present proposals, authorities and communities agreed on these basic aspects:

i. Selection of the *Unpaved Road Network* consistent with the available resources for its sustainable maintenance;
ii. Long tentative list of road projects in the prioritized *Unpaved Road Network* that will be implemented in the five following years;
iii. Confirmation or modification of the functional classification of the road network;
iv. Definition of responsibilities for the maintenance of the diverse sections of the road network
v. Preliminary confirmation of possible contributions of local communities to the roads rehabilitation, improvement and conservation.

[21] Eligible Road Network is the minimal network that allows for basic service to the area. It resulted from technical, economic and social analyses, such as number of inhabitants to be benefited, volumes of traffic, agricultural production, commercial movement and access to main highways, among other aspects

b) Three Workshops to Define the Master Plan of Selected Unpaved Road Network

27. In a second series of workshops, a detailed inventory of the three eligible Unpaved Road Networks scenarios, the results of previous workshops, submitted proposals, and the Master Plan for each one of the three scenarios were presented in each of the concerned departments.

28. The Road Master Plans included works of conservation and rehabilitation of existing roads for a period of 5 years, elaborated in narrow collaboration with the MOPC, the Central Administration and Departments, and the representatives of communities.

29. They were elaborated on the basis of rational criteria, clearly documented to inform and sustain the election of the Road Master Plan of each Department. Special emphasis was placed on the economic feasibility analyses, such as Cost - Efficiency (EC) for the roads with less than 50 vehicles AADT, and Cost - Benefit (CB) for the roads with transits of more than 50 vehicles AADT.

Sub-Component 3.3: Implementation of the Indigenous People Plan and Environmental Management Plan

30. A detailed description of this sub-component is included in Annex 10.

Annex 5: Project Costs

PARAGUAY: Road Maintenance

Table A5-1
Project Costs

Project Cost By Component and/or Activity	Local US $million	Foreign US $million	Total US $million
1. Road Management and Strategic Planning Improvement	**2.67**	**3.84**	**6.51**
a) Strengthening of Road Management and Planning	1.91	1.56	3.47
b) Strengthening of Budget Processing and Preparation	0.02	0.06	0.08
c) Road Maintenance strengthening	0.44	1.32	1.76
d) Project Monitoring and Evaluation	0.30	0.90	1.20
2. Improvement and Maintenance of the Paved Road Network	**23.35**	**20.00**	**43.35**
a) Performance-based maintenance contracts	21.60	17.67	39.28
b) Capacity building in performance-based contracts	0.11	0.33	0.44
c) Toll optimization	1.63	2.00	3.63
3. Improvement and Maintenance of the Unpaved Road Network	**23.12**	**19.78**	**42.90**
a) Upgrading and maintenance works	22.67	18.55	41.21
b) Capacity building of MOPC Districts	0.20	0.59	0.79
c) MOPC District office improvement and equipment	0.07	0.09	0.17
d) Capacity building for Local Governments	0.06	0.17	0.23
e) Pre-investment and preparation studies for future year programs	0.12	0.37	0.50
TOTAL BASELINE COST	**49.14**	**43.62**	**92.75**
Physical Contingencies (2)	4.58	4.21	8.79
Price Contingencies (2)	1.92	1.77	3.69
TOTAL PROJECT COST	**55.63**	**49.60**	**105.23**
Front end fee	0.00	0.19	0.19
Interest during Construction			
FINANCING REQUIRED	**55.63**	**49.79**	**105.42**

[1]Identifiable taxes and duties are US$9.57 million, and the total project cost, net of taxes, is US$95.85M. Therefore, the share of project cost net of taxes is approximately 90%.
2The amount of physical contingencies and price contingencies was determined according to the best information available at negotiations and agreed upon with the GOP.

Annex 6: Implementation Arrangements

PARAGUAY: Road Maintenance

A. Sector Policy Responsibility

1. The Ministry of Public Works and Communications has the overall responsibility for the transport sector and the policy framework for project implementation. Through the Viceministry of Public Works it coordinates the Directorate of Highways and the Directorate of Rural Roads. The latter is responsible for planning and performing construction and maintenance of rural roads by direct management, or by contractors and regional road boards (Law No. 167/93). It is particularly engaged with local governments in a program of maintenance and improvement of these roads through resource sharing mechanisms. The former is responsible for planning, budgeting and determining the basis for contracting and/or bidding the construction, rehabilitation, conservation and maintenance of national and main roads (Law No. 167/93). Departments and Municipalities participate in civil works of roads with no legal responsibility. Under this scenario, the MOPC decided to assign one specific unit for the coordination of the present project.

B. Project Implementation and Coordination Unit

2. The Borrower created a Project Preparation and Coordination Unit (UPGP) in September 2004 through Ministerial order No. 523. The Unit has been staffed with four Ministry staff members who have shown capacity and interest in the project. During project identification, two inter-ministerial workshops have been carried out to define within the Ministry the shape and attributions of the UPGP. Furthermore, it was decided that the UPP would be the main promoter of project activities and sole interlocutor with the Bank, but would use the MOPC divisions to prepare the project. Thus, the linkages and the responsible staff in all the MOPC dependencies that might deal with the project have been worked-out through this participatory process. The Ministry of Finance, which is carrying out its "Ministry of Excellence" initiative and promoting the improvement of pre-investment and project management in the line ministries, participated in the workshops and was an integral actor of the final definition of the shape and functions that the UPGP took.

3. The Project Preparation and Coordination Unit will coordinate the program implementation. This includes the elaboration of bidding schedules and communication documents; supervision of consultants and institutions involved in the implementation process; and the generation of training programs for local staff.

4. The implementation of individual components and activities will vary, in general the benefiting unit, division or department of the MOPC will oversee, and/or implement the task. (i) ***Component 1:*** will be solely in charge of the Viceministry of Public Works, and will be mainly carried out by the Directorates of Highways, Technical Assistance, Traffic Control and Security and Legal. ***(ii) Component 2:*** Will be jointly carried out by the Viceministry of Public Works, and the Viceministry of Administration and Finance; the latter is responsible for revenue collection and thus in charge of the tolling system. The directorates involved will be the Highway Directorate, and the Finance Directorate. ***(iii) Component 3:*** Will be carried out within the Viceministry of Public Works, by the Directorates of Highway and Rural Roads. A detail of all the Departments and Units involved is given in Table A6-1.

Table A6-1 : Implementation Responsible Entities

	Vice-ministry	Directorate	Department
1. Strengthening Strategic Planning and Road Management			
Subcomponent 1: Strengthening of strategic planning			
1.1. Georeference the alignment of the new road classification	VMOPC	DV	DEP
1.2. Produce a definition of road works for the Environmental Law	VMOPC	DAJ	UA
1.3. Update the Law regarding the right-of-way	VMOPC	DV	
1.4. Update the Expropriation Law	VMOPC	DAT/DAJ	UBI
1.5. Update the National Traffic Law	VMOPC	DCST/PC	
1.6. Create the Strategic Road Planning Unit, or assign these function to an existing unit	VMOPC		
1.6.1. Create the Strategic Road Planning Unit	MINISTER /VMOPC		
1.6.2. Equip the Strategic Road Planning Unit with software and hardware	VMOPC		UPE
1.6.3. Train the Strategic Road Planning Unit staff	VMOPC		UPE
1.7 Develop a traffic counting system	VMOPC		UPE
1.7.1. Prepare a diagnostic and proposal for the development of the traffic counting system	VMOPC		UPE
1.7.2. Implement the traffic counting system	VMOPC		UPE
1.8. Strengthen the VMOPC (training, organization, procedures, functions and road management)	VMOPC	DAT	UPGP
1.9 Strengthening of the Procurement Unit (UOC), training, software, hardware.	VMOPC		UOC
Subcomponent 2: Strengthening budget preparation and execution			
2.1 Develop and implement a results based budget programming and execution system	VMOPC	DAT	

2.2. Develop and implement a management of works and budget execution monitoring system	VMOPC	DAT	
2.3 Develop and implement an automated works certification and payment system	VMOPC	DAT	
Subcomponent 3: Fortalecer la Gestión del Mantenimiento vial			
3.1. Strengthening management of road improvement and maintenance contracts (Training, organization, equipment, etc)	VMOPC	DV	DC
3.2. Updating the road pavement management system (SIAMV) and the road inventory	VMOPC	DV	DC
3.3. Develop and implement a communication strategy on road maintenance	VMOPC		UPGP
Subcomponent 4:Gestión y monitoreo del Proyecto			
4.1.Diagnostic, proposal and implementation of methodology to calculate and monitor vehicle operating costs (baseline and indicators)	VMOPC		UPGP
4.2. Socioeconomic impact survey and monitoring	VMOPC		UPGP
4.3 Baseline indicators surveys (baseline)	VMOPC		UPGP
4.4. Strengthening and training of UPGP (organization, operating manual, functions, training, etc.)	VMOPC		UPGP
4.5 Implementation of IPP	VMOPC		UPGP
4.6. Implementation of EMP	VMOPC		UA
2. Improvement and maintenance of the paved road network			
Subcomponent 1: Performance-based contracting			
1.1 Establish quality standards for the road network	VMOPC		UGPG
1.2. Capacity building in MOPC on performance-based contracts	VMOPC	DV	DC
1.3. Performance-based contracts (works and supervision)	VMOPC	DV	DC
Subcomponent 2: Toll system optimization			
2.1. Toll System Master Plan	VMAF	DF	UP

2.2. Implementation of the toll system optimization plan	VMAF	DF	UP
3. Improvement and maintenance of the unpaved road network			
Subcomponent 1 : Road improvement and maintenance works			
1.1. Establish quality standards for the road network	VMOPC		UPGP
1.2. Define the intervention modalities (type of contracts, type of executioners)	VMOPC		UPGP
1.3. Implementation of works (works and supervision)	VMOPC	DV/DCV	DC
1.4. Preparation studies for future programs	VMOPC		UPGP
Subcomponent 2: Capacity building and development program for MOPC Districts& Local Governments			
2.1 Capacity building in (3) MOPC District Offices on monitoring and controlling road maintenance contracts	VMOPC	DV	DC
2.2 Improvement of infrastructure and equipment of (3) MOPC District Offices for management of road maintenance and works contracts	VMOPC	DV	DC
2.3 Capacity building program for Local Governments (3 Departments)	VMOPC		UPGP

Legend:

Project Preparation Unit	**UPGP**	Unidad de Preparación y Gestión del Proyecto
Toll Unit	**UP**	Unidad de Peajes y Pesajes
Pavement Evaluation Department	**EP**	Departamento de Evaluación de Pavimento
Road Maintenance Department	**DC**	Departamento de Conservación
Environmental Unit	**UA**	Unidad Ambiental
Patrimony Unit	**UBI**	Unidad de Bienes Inmobiliarios
Traffic Police	**PC**	Policía Caminera
Highway Directorate	**DV**	Dirección de Vialidad
Rural Roads Directorate	**DCV**	Dirección de Caminos Vecinales
Technical Assistance Directorate	**DAT**	Dirección de Asesoría Técnica
Legal Directorate	**DAJ**	Dirección de Asuntos Jurídicos
Traffic Security and Control Directorate	**DCST**	Dirección de Control y Seguridad de Tránsito
Finance Directorate	**DF**	Dirección de Finanzas

Vice-ministry of Public Works and Communications	**VMOPC**	ViceMinisterio de Obras Públicas y Comunicaciones
Vice-ministry of Administration and Finance	**VMAF**	ViceMinisterio de Administración y Finanzas

Annex 7: Financial Management and Disbursement Arrangements

PARAGUAY: Road Maintenance

A. Introduction

1. The Financial Management Assessment is carried out to determine whether the implementing entity of the proposed project has acceptable financial management arrangements in place according to the guidelines issued by the Financial Management Board on November 3, 2005. The objective of the assessment is to analyze that the FM system ensures: (i) funds are used only for the intended purposes in an efficient and economical way; (ii) the preparation of accurate, reliable and timely periodic financial reports; (iii) safeguarding the entities assets; and (iv) are subject to acceptable auditing arrangements.The Borrower would be the Republic of Paraguay, represented by the Ministry of Finance and Project implementation and administration will fall under the Ministry of Public Works and Communications (MOPC).

2. The Paraguay Road Maintenance Project (RMP) would be implemented by the MOPC through the mainstreamed Ministry's FM procedures. MOPC has a dedicated group (Dirección de Crédito Público - DCP) which is part of MOPC structure and uses internal MOPC procedures, only in the case of processing of transactions related with multilateral financed projects. DCP will be the Bank's counterpart in FM matters and will be responsible for the reporting and accounting functions of the project. Budgeting, flow of funds, accounting, budget execution and internal control will be carried out by the relevant MOPC areas through mainstreamed processes. Previous experience of the Ministry of Public Works and Communications (MOPC) in implementing Bank and other multilateral financed projects has been taken into consideration for this assessment.

3. The proposed financial management arrangements, taking into account the additional proposed measures, are considered adequate and the Borrower has sufficient capacity and resources to implement the proposed project. Additional fiduciary measures are designed to mitigate the identified risks.

4. All uses of funds would be processed by the MOPC, and supported by external third party documentary evidence and the related goods and services procured in line with Bank guidelines. MOPC would be responsible for Project financial management, making payments to suppliers, obtaining loan disbursements, and accounting and financial reporting.

5. The assets to be procured during the Project would be incorporated into the Project and recorded in its asset registry.

6. MOPC has internal policies and procedures that will be complemented by project documents. In case of road maintenance agreements with Civil Society Organizations (CSO) and Municipalities (Convenios) specific policies and administrative procedures will be established and documented in a Project Complementary Operational Manual which will be reviewed by the assigned FMS.

7. Lessons learned from other projects recently prepared in Paraguay and in the region were taken into consideration. Given this, the project has been prepared with a high degree of participation and transparency as attested in the social assessment of the project and its social and transparency framework. Workshops have been conducted throughout the preparation stage to improve project design and measures to improve implementation transparency, foster social control of project funds to be incorporated into project accountability mechanisms and mitigate project FM risk. Additionally, a series

of workshops to delineate an Anti-corruption Plan are underway within the MOPC with the participation of transparency advocacy NGOs to determine additional mechanisms to mitigate fiduciary risks.

B. Country Issues

8. The Paraguay Country Financial Accountability Assessment (CFAA), March 2004, and the Country Assistance Strategy (CAS), November 2003, disclose the problems of governance and institutional weaknesses in Paraguay. The CAS states that Paraguay is renowned for corruption and contraband. Perceptions of corruption have increased during the past decade and Transparency International consistently ranks Paraguay among the ten most corrupt nations.

9. Corruption is not the only institutional weakness in Paraguay. The basic management systems of Government are also weak or non-existent. Within the central administration, the personnel management system is in an embryonic stage and is only now pulling together a master roster of all civil servants. A much larger task is to embark on the professionalization of the civil service, including competitive entrance exams, career streaming and training."

10. The CFAA states the following as the most important weaknesses and shortcomings that threaten financial accountability in Paraguay:

- Weak-control environment.
- Numerous and serious exceptions in the application of the financial administration law and regulations.
- Weak internal audit and control, especially in revenue collection agencies.
- The disproportionate share of time and resources that the Auditor General devotes to ad hoc review requests.
- Congressional introduction of budget increases without provision for corresponding financing
- Trade liabilities incurred by execution agencies, but unrecorded if they exceed the authorized cash program.
- Many exceptions to generally accepted accounting standards and financial statements not prepared in compliance with the Government's own regulations.

11. The CFAA concludes that both inherent and control risks are high and states that the advances made in the modernization of Paraguay's public financial management system could not compensate for the effects of a deficient control environment.

12. The Bank's country portfolio review of October 2003 also identified the main issues affecting portfolio implementation, including: (i) high staff turnover in the project implementation units; (ii) lack of timely and adequate availability of counterpart funds; (iii) cumbersome procedures for the flow of funds to the projects; (iv) delays in effectiveness and project start-up; and (v) limited disbursement capacity and limited knowledge of Bank procedures.

13. Actions to address these issues have been agreed at the portfolio level and are under implementation, and these issues have been taken into account in the design of this proposed Project. To address the limited knowledge of Bank procedures, as part of Project execution, specific training on FM operational policies and procedures will be provided to MOPC relevant staff.

C. Risk Analysis

14. The FM risk assessment process aims through its process to identify and incorporate appropriate risk mitigating measures into project design, and to enable the Bank to make decisions on the appropriate level of assessment and supervision intensity and to allocate FM resources in a manner consistent with assessed risks.

15. In the context of its lending operations, one of the risks that the Bank is concerned with is fiduciary risk – the risk that loan proceeds will not be used for the purposes intended. FM risk, a component of fiduciary risk, is a combination of country, sector and project specific factors, and is assessed at entry for each project, and monitored and mitigated in a structured manner throughout the life of the project.

16. The mitigation measures are designed to address the identified risks. FM Inherent risk at Entity and Project level is considered substantial because of the Improved Governance Framework and Action Plan to be implemented. Control risk after the implementation of the proposed mitigation measures is considered modest. The detailed risk matrix is presented below:

Table A7-1: Financial Management Risks

Risk	Risk Rating after mitigation measures	Risk Mitigating Measures Incorporated into Project Design	Condition
Inherent Risk			
▪ Country Level Lack of law enforcement. Absence of transparency. Weak attitude towards control.	High		
▪ Entity/ Project Level Cumbersome payment procedures. Weak institutional capacity. Procedures embedded in MOPC procedures (as result project risk associated with entity's risk).	Substantial	- Improved Governance Framework for project implementation	
Control Risk			
▪ Budgeting Risk of state capture and political abuse.	Modest	- Separate Program budget line in Ministry's budget system to monitor budget allocations and cash plans allotments. - FM supervision will include specific review of the budget allocations and execution on this specific budget line. - Multi year budget for project implementation.	Legal clause
▪ Accounting	Modest	- Periodic reconciliation between project accounting and Public FM system (SICO)	

		will be implemented and described in operational manual. - Capacity Building activities for DCP part of project implementation (training and equipment).	
▪ Internal Control	Substantial	- Capacity building of Internal Audit Unit part of project implementation (training and equipment). - Capacity building activities would be monitored through FM supervision. - Complaint system to report/ prevent fraud and abuses part of the project implementation.	
▪ Funds Flow Cumbersome payment procedures and lack of clarity on payment prioritization is an opportunity for corruption.	Modest	- Automation of payments at MOPC to avoid discretion on payments prioritization part of project implementation. - Lack of knowledge on Bank procedures will be addressed by specific training to be provided by the Bank through its periodic training program that is offered in LCC7 countries periodically by LCOS.	
▪ Financial Reporting	Modest	- Unaudited interim financial reports would be produced also by the system.	
▪ Auditing Timely audited financial statements track record in Paraguay.	Modest.	External audit will be carried out following International Auditing Standards.	
Overall Residual Risk rating	**High**		

17. The FM risk model embodies a qualitative focus and is based on principles embodied in internationally recognized good practices, in particular "Enterprise Risk Management – Integrated Framework", issued by COSO (1); Under the model:

18. Inherent Risk: It is defined as that which arises from the environment in which the project is situated. It is the risk that the project financial management system does not operate as intended due to factors such as country governance environment, rules and regulations. Inherent risk comprises three elements: Country, Entity and Project level risk.

19. Control Risk: It is the risk that the project's financial management system is inadequate to ensure project funds are used economically and efficiently and for the purpose intended. Control risk is measured by reference to the six elements of Financial Management as defined in OP10.02 Financial Management, Paragraph 1. These are budgeting, accounting, internal control, funds flow, financial reporting and auditing.

20. Detection Risk: is the risk that a material misuse of loan proceeds fails to be detected. FM practices indicates that detection risk is lowered by proposing capacity strengthening measures for the weaknesses identified as posing unacceptable levels of risk; and/or o increasing supervision intensity.

21. Residual Risk: It is the combination of the project's inherent and control risks as mitigated by the combination of client control frameworks and Bank supervision effort. The overall project residual risk is the combination of inherent and control risk as result from a weight based on professional judgment.

22. In Paraguay, given the existence of corrupt practices and administrative lapses reported by the control entities and independent surveys, it can solely be concluded that the attitude toward control is weak and the country inherent risk is high. The mitigation measures are designed to address the identified risks. FM Inherent risk at Entity and Project level is considered substantial because of the Improved Governance Framework to be implemented. Control risk after the implementation of the proposed mitigation measures is considered modest.

23. The identified risks were taken into account in proposing the FM arrangements and controls, and if implemented as proposed, the impact on the project would be limited. But given the country environment and the uncertainty on how these measures will be affected by the environment and how they will be implemented and until there is evidence they will actually done and effectively working, we maintain the overall high risk rating.

24. As the overall risk assigned is high according to FM procedures, the package will be subject to the FM Regional Manager review.

D. Strengths and Weaknesses

25. Strengths: MOPC in line staff has administrative experience and internal procedures are well documented and mainstreamed into Ministry's operations.

26. Weaknesses: Residual risks are mostly associated with inherent country risk. Lack of expertise in bank financed operations will be addressed by specific operational training and Bank supervision.

E. Internal Control

27. As established in Paraguay's financial management Law # 1535-99, MOPC is subject to internal audit of the General Auditor and the Internal Audit Department (Departamento de Auditoría Interna) which conducts its annual audit plans, conducts independent audits on MOPC operations, verify obligations and payments and the satisfactory compliance by suppliers with contract terms.

F. External Audit Arrangements

28. MOPC would contract an annual project financial statement audit under Terms of Reference prepared in accordance with the Bank Guidelines to be performed according to international auditing standards by an independent auditor acceptable to the Bank. The audit would cover all funding and expenditures reported in Project financial statements.

G. Disbursement Arrangements

29. Loan proceeds will be withdrawn by MOPC using the advance method with supporting documentation based on Statements of Expenditures (SOEs). The option to request direct payment to suppliers from the Loan Account is also proposed. All consolidated SOEs documentation would be maintained by MOPC for post-review and audit purposes for up to one year after the final withdrawal from the loan. MOPC would request access to the Bank's Client Connection webpage to access the 1903 Form and to perform the reconciliation process periodically between their bank account and the resources received from the different sources.

30. The following Disbursement Methods may be used under the Loan:

- Reimbursement
- Advance
- Direct Payment

31. Supporting documentation for documenting project expenditures under advances and reimbursement method will be: Statements of Expenditures (SOEs) for all expenditures below the following thresholds: Payments for Civil Works against contract valued at US$3,000,000 or more; payments for Goods against contracts valued at US$250,000 or more; and payments for Consultant Services against contracts valued at US$200,000 for firms and 50,000 for individuals respectively.

32. The Minimum Value of Applications for direct payments and reimbursements is US$1, 000,000.

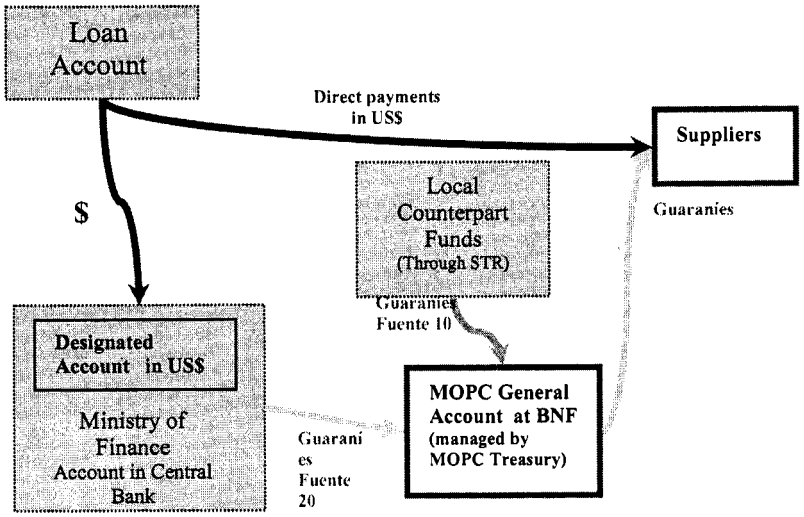
H. Designated accounts

33. The Government would open a segregated Designated Account (DA) in the Central bank of Paraguay (BCP) in U.S. dollars to be administrated by the Ministry of Finance (MOF). Funds would flow from the Loan account to BCP account as is mandatory for projects in Paraguay. Funds from BCP will be transferred to the MOPC general account as the expenses are incurred through mainstreaming budget execution through request of transfers (STR) that would be made for every payment.

34. The Designated Account will be held at the BCP and the recommended ceiling for the DA is US$7.0 million, estimated as sufficient for up to six moths of project execution.

35. The proposed flow of funds is detailed in the following graphic:

Chart A7-1
Flow of Funds



36. The Government has issued a decree to implement in the Central Government a direct payment mechanism to Government suppliers' bank accounts from the MoF that will simplify, once implemented, the flow of funds.

37. Loan proceeds would be disbursed against the following expenditure categories:

Table A7-2
Disbursements per Expenditure Category

Expenditure Category	Amount financed in US$ million	Bank's Financing Percentage
1. Works	52.40	70%
2. Goods	2.54	70%
3. Consulting Services, Supervision, Audits and Training	9.58	70%
4. Operating Costs	0.40	70%
5. Unallocated	8.38	70%
6. Front End Fee	0.18	100%
Total Project Costs	**73.48**	

I. Accounting and Reporting

38. MOPC would be responsible for keeping accounting records for Project activities at the Project level. Accrual accounting would be used following international accounting standards. Transactions would be recorded in Project Management Information System (MIS) and Government budget system (SIAF) as well, this would allow adequate reporting and SA and SOEs control.

39. MOPC would issue annual financial statements and semiannual Interim Unaudited Interim Financial Reports (UIFRs) for reporting purposes that will be part of the progress reports. The proposed format for the UIFRs would be part of the Complementary Operational Manual and would include:

- Disbursement performance: source and uses of funds, for each semester and cumulative (uses by category), and uses of funds by component;,
- Physical progress: Allocated budget and financial execution compared with physical progress and results achieved.

J. Staffing

40. The proposed organizational arrangement for this project comprising MOPC structure is considered satisfactory and DPC FM key is adequately skilled. FM training to the DCP and Internal Audit department staff is part of the FM action plan.

K. Action Plan

During appraisal and prior to negotiations, MOPC as part of the agreed FM action plan has presented to the Bank the following:

(i) A set of detailed administrative procedures for project execution.
(ii) The Project's Chart of Accounts;
(iii) Format and contents of the annual financial statements and unaudited Interim Financial Reports (UIFRs) format for monitoring and evaluation purposes;

(iv) A detailed plan satisfactory to the Bank on the capacity building actions for the Internal Audit Unit and DCP to be carried out as part of project implementation.

L. Supervision Plan

Table A7-4
Supervision Plan

Type	Timing	Mechanism	Objective
Visit	General Supervision: each fourth month.	Integrating supervision missions at least one time each semester. Others separately.	♦ Review FM system. ♦ SOE review ♦ Supervise Special Account Reconciliation. Use of funds. ♦ Follow up on External Auditors recommendations/ raised issues. ♦ Review staffing.
FMR Review	Quarterly	Over the Unaudited Interim Financial Reports submitted to the Bank.	♦ Review UIFRs information consistency. ♦ Raise issues disclosed in Progress reports/ UIFRs.
Audit Review	Once a Year	Over the Audit Reports submitted to the Bank	♦ Review Audit Report. ♦ Raise issues disclosed in Audit Report

Annex 8: Procurement Arrangements

PARAGUAY: Road Maintenance

A. General

1. Procurement for the proposed project would be carried out in accordance with the World Bank's "Guidelines: Procurement under IBRD Loans and IDA Credits" dated May 2004; and "Guidelines: Selection and Employment of Consultants by World Bank Borrowers" dated May 2004, and the provisions stipulated in the Legal Agreement. The various items under different expenditure categories are described in general below. For each contract to be financed by the Loan, the different procurement methods or consultant selection methods, the need for pre-qualification, estimated costs, prior review requirements, and time frame are agreed between the Borrower and the Bank in the Procurement Plan. The Procurement Plan will be updated at least annually or as required to reflect the actual project implementation needs and improvements in institutional capacity.

2. **Procurement of Works**: Works procured under this project would include three (3) performance-based contracts for the rehabilitation and maintenance of paved roads (GMANS) and contracts for the rehabilitation and conservation of unpaved national roads and secondary roads. The interventions in the main unpaved road network in each Department would include road rehabilitation and maintenance or contracting arrangements for specific works (bridges, drainage, etc.). ICB procedures would be followed for contracts estimated to cost US $3 million or more. Pre-qualification of civil works contracts is not expected for this project. NCB procedures may be followed for contracts estimated to cost less than US $3 million. Shopping procedures may be followed for contracts estimated to cost less than US $250,000. The procurement would be carried out using the Bank's Standard Bidding Documents (SBDs) for all ICB and National SBDs agreed with or satisfactory to the Bank. A performance-based contract bidding document will be prepared and approved by the Bank in order to be included in the Project Operational Manual (OM).

3. **Procurement of Goods:** Goods procured under this project would include IT, Communications and Safety Equipment. ICB procedures would be followed for contracts estimated to cost US $250,000 or more. NCB procedures may be followed for contracts estimated to cost less than US $250,000. Shopping procedures may be followed for contracts estimated to cost less than US $50,000. The procurement will be carried out using the Bank's SBDs for all ICB and National SBDs agreed with or satisfactory to the Bank.

4. All procurement notices shall be advertised in the website of *"Dirección General de Contratación Pública"* (DGCP) or in at least one local newspaper of national circulation. ICB notices and contract award information shall be advertised in the *UN Development Business online (UNDB online)* and in the *Development Gateways's dgMarket,* DGCP shall also be used to publish information on awarded contracts in accordance with provisions of paragraphs 2.60 of the Procurement Guidelines and as mandated by local legislation.

5. **Selection of Consultants:** Consulting Services under this project would include Technical assistance for: (i) strengthening strategic planning, (ii) road conservation, (iii) budget management, (iv) procurement information system, and (v) capacity building and development program for MOPC district offices and local Governments and contracts for the fiscal function of the civil works contracts. *Consulting Firms* would be selected following QCBS. Consulting services for financial auditing services and services to organize workshops would be procured using Least-Cost Selection (LCS) and / or Consultant's Qualifications (CQ) method and for contracts estimated to cost US $100,000 or less. Short

lists of consultants for services estimated to cost less than $200,000 equivalent per contract may be composed entirely of national consultants in accordance with the provisions of paragraph 2.7 of the Consultant Guidelines. The procurement will be carried out using Bank's standard RFP. The MOPC shall publicly advertise requests of expressions of interest as the basis for developing short list of firms and individual consultants. *Individual consultants* would be selected by comparison of qualifications of three candidates and hired in accordance with the provisions of paragraphs 5.1 through 5.3 of the Consultant Guidelines. The consultant's fee for the PPU staff may be in accordance with the range established in the Government law and their contracts signed for the expected duration of the assignment, with escape clauses in case of non-performance.

6. The DGCP and / or national newspaper shall also be used to advertise expressions of interest for consulting firms or individuals, and to publish information on awarded contracts in accordance with provisions of paragraphs 2.28 of the Consultants Guidelines and as mandated by local legislation. Contracts expected to cost more than US $200,000 shall be advertised in the *UNDB online* and in *dgMarket.*

7. **Training:** Workshops and study tours, do not involve procurement processes. This includes expenditures for cost of tuition, venue, food, per diem and traveling for the MOPC staff, and would be procured according to normal business principles. A detailed Annual Training Plan of activities, including cost estimate, would be submitted each year to the Bank for prior review. The procedures should be described in the OM.

8. **Prior Review for Works, Goods and Services (other than Consultant Services) Contracts:** ICB contracts for Works, contracts for Goods estimated to cost US$250,000 or more and all Direct Contracting, if any, would be subject to prior review by the Bank. The first two processes under each method of procurement and for each calendar year would be also subject to prior review regardless of amount.

9. **Prior review for Consultant Services Contracts:** Consultancy services to be assigned to firms estimated to cost above US $200,000 per contract, consultancy services to be assigned to individuals estimated to cost above US $50,000 per contract and Single Source selection of consultants firms would be subject to prior review by the Bank. The first two processes under each method of selection and for each calendar year would be also subject to prior review regardless of amount.

10. **Others**: None.

B. Assessment of the agency's capacity to implement procurement

11. The Procurement law 2051/2002 enacted in December 2002 established a Central Policy and Technical Unit (UCNT) in the Ministry of Finance (MH), and Operational Procurement Units (UOCs) in each agency. The UOC of the Ministry of Public Works and Communications (MOPC), which is the link with the MH, with the support of the Project Preparation Unit (PPU), which was created by the Borrower to be the main promoter of project activities and the interlocutor with the Bank, and the sub-operational procurement units (SUOCs) of the Directorate of Highways (DV), the Directorate of Rural Roads (DCR), the Directorate of Technical Assistance (DTA) and the Directorate of Finance (DF), will be responsible for carrying out procurement activities for the project. The structure of the PPU consists of five Ministry staff members and one of them will be responsible for supporting the SUOCs in the preparation of procurement plans and bidding documents. The procurement assessment was carried out as part of the pre-appraisal mission. The assessment reviewed the project's components and the organizational structure for implementing the project and the interaction among the PPU, the UOC and the SUOCs. The technical organization appears to have clearly identifiable responsibilities for the decisions, project ownership and

project management which results with a minimum of political interferences. The technical organization's priority is to ensure that the project delivery team is composed of high quality individuals, and that MOPC keeps these individuals in key roles. Past Bank project experience in the MOPC showed various risks related to project implementation, i.e. variations orders due to inadequate engineering designs, unrealistic completion periods, delayed access to the construction site, absence of procurement information system, weaknesses in contract supervision and contract management, absence of an operational manual, and lack of experience in World Bank guidelines and procedures. The assessment found that procurement staff in the different units has limited or no experience in Bank documents and procedures and that their salaries are low compared to those in the private sector and among other MOPC staff levels. The MOPC has a wide range of salaries among jobs of similar responsibilities. For instance, high-level employees receive "supplements", overtime, representation and other payments that can double their nominal salaries. On the other hand, salaries of mid level professionals are low when compared to those in the private sector; therefore, it is not uncommon that some of those individuals hold two or more jobs with the government or elsewhere. Those differences and the high staff turn over represents a high risk in the implementation of the project.

12. A meeting with members of the Paraguayan Chambers of Construction and Roads expressed the same concerns mentioned above, related to incomplete designs, lack of a procurement planning and information system, delays in payments, and weaknesses in the supervision, and added two concerns, one related to the need of having a price adjustment mechanism for civil work contracts, and the other was to the establishment of legal responsibility for: (i) a consulting firm who prepares a design, (ii) a consulting firm who carries out the supervision of a contract, and (iii) a civil work contractor, in order to correct the current situation that assigns all the responsibilities to the civil work contractor.

13. The following are the risks identified: (i) lack of planning specially in the preparation of a procurement plan and its implementation; (ii) weaknesses in the preparation of bidding documents, which include the selection of qualifications and eligibility requirements for contractors and suppliers; (iii) unfamiliarity with contract packaging, (iv) incomplete or vague technical specifications and terms of reference; (v) weaknesses in the bid evaluation process, contract management and supervision; (vi) incomplete procurement information and disorganized filing systems; (vii) delays in all the steps of a bidding process and contract's execution, including payments; (viii) lack of a complaints data base, and (ix) weaknesses in the enforcement of the code of ethics.

14. An Action Plan has been prepared to include the corrective measures. The plan has been agreed with the Borrower and it includes:

- Reinforcement of the PPU with a specialized procurement staff;
- Preparation of a comprehensive Project Operational Manual (OM), acceptable to the Bank and adopted prior to Project implementation. The manual would include both organizational and operational functions, with a specific chapter on procurement that details all the procedures, channels of responsibilities and flow of documentation with a maximum timeframe for the different internal office approvals. The recently approved Ministerial Resolution No. 146 of March 28, 2006, addressed the timeframe issues by reducing administrative steps; this detail should be included in the OM. The manual should also include a description of the functions for the PPU's procurement staff and of the designated staff within the UOC and SUOCs who will be responsible for preparing and processing project procurement documents.
- Hiring of a qualified procurement consultant, with World Bank experience, to: (i) organize the procurement functions within the PPU; (ii) provide training to all staff involved in the preparation, review and process of bidding documents, including staff from the Audit office and the Office of the Comptroller General (CGR); (iii) verify that the contract management / information system is well established and operational; (iv) improve or organize the filing

system; and (v) ensure that the use of the local procedures and approvals does not cause delays and if needed identify additional measures to address internal delays. The new measures have also been included in the OM.

- Review by the Bank of the Annual Procurement Plan and quarterly updates, including the Annual plan for training;
- Drafting of standard bidding documents for NCB (works, and goods) and request for proposals for consultant's services. These documents may be included in the OM;
- Implementation of a procurement information system acceptable to the Bank before disbursements;
- Preparation of a plan to improve the procurement filing system or establish a new secure filing system for the Project.
- Establishment of a complaint mechanism which should specify who will be the responsible office and the time limit for resolving the complaint.

15. The overall project risk for procurement is High, due to the limited Bank procurement experience of the PPU staff and the administrative staff of the UOC and SUOCs. If the above-mentioned measures should take place, the overall risk of procurement would be lowered to Average.

C. Procurement Plan

16. The Borrower, at appraisal, developed a procurement plan for project implementation which provides the basis for the procurement methods. This plan has been agreed between the Borrower and the Project Team on May 6, 2006 and is available at the MOPC webpage. It will also be available in the project's database and in the Bank's external website. The Procurement Plan will be updated in agreement with the Project Team annually or as required to reflect the actual project implementation needs and improvements in institutional capacity.

D. Frequency of Procurement Supervision

17. In addition to the prior review supervision to be carried out from the Bank's office, the capacity assessment of the Implementing Agency has recommended two supervision missions to visit the field to carry out post review of procurement actions, complemented with IPRs.

E. Details of the Procurement Arrangements Involving International Competition:

18. Special procurement conditions:

(i) Open competition will be allowed to contractors and suppliers to participate in the bidding processes of works and goods (Art 18, b, 3);
(ii) Selection of consultants shall be carried out on the basis of a request for proposals (RFP) to a limited number of qualified firms, and not on the basis of open competition;
(iii) The Conciliator in the resolution of contract disputes should be and independent third parties;

(1) Goods, Works, and Non Consulting Services

(a) List of contract packages to be procured following ICB and direct contracting (excluding contingencies):

1	2	3	4	5	6	7	8	9
Ref. No.	**Contract (Description)**	**Estimated Cost**	**Procure ment Method**	**P-Q**	**Domestic Preference (yes/no)**	**Review by Bank (Prior / Post)**	**Expected Bid-Opening Date**	**Comments**
1	Performance-based maintenance contract – 1 (GMANS 1)	10,000,000	ICB		No	Prior	04/20/2007	
2	Performance-based maintenance contract – 5 (GMANS 5)	10,000,000	ICB		No	Prior	05/22/2007	
3	Performance-based maintenance contract - 0 (GMANS 0)	6,000,000	ICB		No	Prior	07/12/2007	
4	Performance-based maintenance contract – 2 (GMANS 2)	6,000,000	ICB		No	Prior	09/12/2007	
5	Traffic counting system	2,200,000	ICB		No	Prior	03/20/2007	

(b) ICB contracts estimated to cost above $3,000,000 for works and above $250,000 for goods per contract and all direct contracting will be subject to prior review by the Bank.

(2) Consulting Services

(a) List of consulting assignments with short-list of international firms.

1	2	3	4	5	6	7
Ref. No.	**Description of Assignment**	**Estimated Cost**	**Selection Method**	**Review by Bank (Prior / Post)**	**Expected Proposals Submission Date**	**Comments**
1	Supervision of performance-based maintenance contract 1 (GMANS I)	700,000	QCBS	Prior	01/30/2007	Short list through Expressions of Interest
2	Supervision of performance-based maintenance contract 5 (GMANS 5)	700,000	QCBS	Prior	02/27/07	Short list through Expressions of Interest
3	Supervision of performance-based	420,000	QCBS	Prior	05/23/2007	Short list through

	maintenance contract 0 (GMANS 0)					Expressions of Interest
4	Supervision of performance-based maintenance contract 2 (GMANS 2)	420,000	QCBS	Prior	08/30/2007	Short list through Expressions of Interest
5	Supervision of road improvement works in Caaguazú Depart. (three contracts)	239,085 (1,159,520)	QCBS	Prior	01/24/2007	Short lists through Expressions of Interest
6	Supervision of road improvement works in San Pedro Depart. (four contracts)	300,685	QCBS	Prior	01/16/2007	Short lists through Expressions of Interest
7	Supervision of road improvement works in Caazapá Depart. (three contracts)	270,695	QCBS	Prior	01/30/2007	Short lists through Expressions of Interest
8	Preparation of 2nd and 3rd year works in unpaved road network	360,000	QCBS	Prior	07/17/2007	Short list through Expressions of Interest
9	Indigenous People's Plan implementation	300,000	QCBS	Prior	02/27/2007	Short list through Expressions of Interest
10	Update of the SIAMV and the road inventory	1,450,000	QCBS	Prior	02/21/2007	Short list through Expressions of Interest
11	Socioeconomic Impact survey - baseline	150,000	QCBS	Prior	11/30/2006	Short list through Expressions of Interest
12	Strengthening and training of UPGP	100,000	QCBS	Prior	11/30/2006	Short list through Expressions of Interest
13	Strengthening and training of Government and Municipalities staff	118,140	QCBS	Prior	04/27/2007	Short list through Expressions of Interest
14	Strengthening and improvement of the toll collection system	300,000	QCBS	Prior	03/08/2007	Short list through Expressions of Interest
15	Design of a Communication strategy of road maintenance	100,000	QCBS	Prior	02/19/2007	Short list through Expressions of Interest

(b) Consultancy services estimated to cost above $100,000 per contract and single source selection of consultants (firms) for assignments estimated to cost above $50,000 will be subject to prior review by the Bank.

(c) Short lists composed entirely of national consultants: Short lists of consultants for services estimated to cost less than $200,000 equivalent per contract may be composed entirely of national consultants in accordance with the provisions of paragraph 2.7 of the Consultant Guidelines.

Annex 9: Economic and Financial Analysis

PARAGUAY: Road Maintenance

A. Summary of Benefits and Costs

1. The Ministry of Public Works and Communication (MOPC) carried out an economic analysis of the civil works project components: (i) performance-based maintenance contracting program (GMANS contracts) and (ii) improvement and maintenance of unpaved road network and bridges program. The economic evaluation covered altogether the rehabilitation of bridges and road works on 1,164 km that consist of 968 km paved roads, comprising the four GMANS contracts financed under the project, and 338 km unpaved roads, comprising the first year improvement program of unpaved roads located at the Caaguazú, San Pedro, and Caazapá Departments, representing a total investment of US$30.1 million.

2. The main project economic benefits are savings in vehicle operating costs and travel time costs (road user costs) and maintenance costs as a result of the improved network condition in terms of ride quality and travel times. Net economic benefits of the GMANS contracts were evaluated using the Highway Development and Management Model (HDM-4) Version 1.3, which simulates life cycle conditions and costs and provides economic decision criteria for multiple road design and maintenance alternatives. Net economic benefits of the improvement and maintenance of unpaved roads program were evaluated using the Roads Economic Decision Model Version 3.2, which adopts the same economic analysis principles and road user costs as HDM-4 and is customized for the economic evaluation of unpaved roads. Net economic benefits of the bridge rehabilitation program were evaluated using unit road user costs computed using the HDM-4 model.

3. The GMANS contracts will insure the preservation of the current asset value of the core paved roads network of the country that is currently in good to fair condition. The road works on the unpaved roads and bridges located on the Caaguazú, San Pedro, and Caazapá Departments will substantially improve the quality of service provided by the basic unpaved road networks that are essential for the economic development of the Departments, the access of the rural population to social services, and an effective economic and social national integration. The economic analysis of the project indicates that the project economic benefits are substantial. The overall Net Present Value (NPV), at 12% discount rate, is US$47.9 million and the Economic Rate of Return (ERR) is 33%. Compared with the without project scenario, the MOPC will increase road works expenditures by US$22.7 million, in present value terms over the evaluation period, while road users will save US$70.5 million over the same period, which means that every dollar spent by the MOPC results in around 3.1 dollars saved by road users. Table A9-1 presents the economic analysis summary.

Table A9-1
Economic Analysis Summary

Benefits	Performance-based Contracts	37.9
(US$ million)	Unpaved Road Network and Bridges	32.6
	Total	70.5
Costs	Performance-based Contracts	7.3
(US$ million)	Unpaved Road Network and Bridges	15.4
	Total	22.7
Net Benefits	Performance-based Contracts	30.6
(US$ million)	Unpaved Road Network and Bridges	17.2
	Total	47.9
Economic Rate of Return	Performance-based Contracts	41%
(%)	Unpaved Road Network and Bridges	29%
	Total	33%

B. Main Assumptions

4. Vehicle fleet characteristics and economic unit costs were defined for seven vehicle types, economic road user costs being on average 85% of financial costs, based on the results of the study titled "Estructura de los Costos de Funcionamiento de Vehículos Automotores", by the Dirección Nacional de Transporte (DINATRAN), released in 2005. Table A9-2 presents the vehicle fleet characteristics and economic unit costs.

Table A9-2
Vehicle Fleet Characteristics and Economic Unit Costs

	Car	Pickup	Bus	Light Truck	Medium Truck	Heavy Truck	Articulated Truck
Economic Unit Costs							
New Vehicle Cost (US$/vehicle)	10,909	18,587	86,930	31,292	51,428	60,437	80,496
New Tire Cost (US$/tire)	32.20	80.49	282.44	133.17	282.44	282.44	339.51
Fuel Cost (US$/liter)	0.43	0.54	0.54	0.54	0.54	0.54	0.54
Lubricant Cost (US$/liter)	1.56	1.56	1.76	1.76	1.76	1.76	1.76
Maintenance Labor Cost (US$/hour)	2.87	2.87	2.87	2.87	2.87	2.87	2.87
Crew Cost (US$/hour)	0.00	0.00	3.31	1.35	2.79	2.79	5.03
Overhead (US$)	358	786	6488	1908	2678	3808	4622
Interest Rate (%)	12	12	12	12	12	12	12
Working Passenger Time (US$/hour)	1.05	1.40	0.21	0.00	0.00	0.00	0.00
Non-working Passenger Time (US$/hour)	0.52	0.70	0.11	0.00	0.00	0.00	0.00
Cargo Delay (US$/hour)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Utilization and Loading							
Kilometers Driven per Year (km)	20,000	30,000	103,479	51,000	54,000	66,600	70,200
Hours Driven per Year (hr)	384	720	2,484	1,224	1,440	1,598	1,685
Service Life (years)	7	7	10	7	10	10	10
Percent of Time for Private Use (%)	100	100	0	0	0	0	0
Number of Passengers	2	1	42	0	0	0	0
Work Related Passenger Trips (%)	85	85	85	0	0	0	0
Gross Vehicle Weight (tons)	1.09	2.88	14.29	4.60	10.56	18.45	33.30
ESA Loading Factor	0.00	0.02	1.61	0.12	1.46	3.12	4.91
Traffic Composition							
Paved Roads Average Composition (%)	28%	37%	12%	1%	12%	5%	6%
Unpaved Roads Average Composition (%)	22%	34%	10%	12%	11%	6%	5%

5. Table A9-3 presents the typical economic road user composition for a road in good condition, with roughness equal to 2.0 IRI, m/km, on a flat terrain.

Table A9-3
Unit Road User Costs Composition, Flat Terrain and Roughness 2.0 IRI (US$ per vehicle-km)

	Car	Pickup	Bus	Light Truck	Medium Truck	Heavy Truck	Articulated Truck
Fuel and Oil	0.05	0.07	0.16	0.09	0.13	0.24	0.35
Tires	0.00	0.00	0.02	0.01	0.02	0.03	0.06
Parts and Labor	0.03	0.04	0.13	0.07	0.14	0.18	0.25
Depreciation and Interest	0.08	0.04	0.09	0.09	0.10	0.10	0.12
Crew Time	0.00	0.00	0.04	0.01	0.03	0.03	0.05
Overhead	0.00	0.00	0.03	0.02	0.02	0.03	0.03
Passenger and Cargo Time	0.02	0.01	0.09	0.00	0.00	0.00	0.00
Total	0.17	0.17	0.56	0.28	0.43	0.60	0.86

6. Table A9-4 presents the sensitivity of road user costs to roughness.

Table A9-4
Unit Road User Costs per Roughness Level for Flat Terrain (US$ per vehicle-km)

Roughness (IRI)	Car	Pickup	Bus	Light Truck	Medium Truck	Heavy Truck	Articulated Truck
2	0.17	0.17	0.56	0.28	0.43	0.60	0.86
4	0.18	0.18	0.59	0.30	0.45	0.63	0.90
6	0.18	0.19	0.65	0.32	0.49	0.67	0.94
8	0.19	0.20	0.72	0.35	0.53	0.71	1.00
10	0.21	0.23	0.82	0.39	0.58	0.77	1.09
12	0.23	0.26	0.92	0.42	0.63	0.84	1.19
14	0.25	0.29	1.02	0.46	0.69	0.92	1.30
16	0.27	0.33	1.13	0.50	0.75	0.99	1.41

7. The analysis period is 15 years and the discount rate is 12%. The annual traffic growth rate was defined based on the observed annual traffic growth from 1994 to 2004 on seven toll stations and was set to 2.5% per year for all vehicle types for the duration of the analysis period. Maintenance and rehabilitation costs were estimated in financial and economic terms; economic costs being on average 80% of financial costs. Table A9-5 presents typical average road works unit costs.

Table A9-5
Average Road Works Unit Costs

Work Class	Work Type	Predominant Work Activity	Financial Cost (US$/km)	Financial Cost (US$/m2)
Routine Maintenance	Routine Maintenance	Routine Maintenance Unpaved	1,680	N.A.
		Routine Maintenance Paved	3,866	N.A.
	Grading	Grading	91	N.A.
Periodic Maintenance	Regravelling	Regravelling 75 mm	8,775	1.46
	Asphalt Mix Resurfacing	Micro Surfacing 12 mm	20,232	2.81
		Asphalt Overlay 50 mm	36,576	5.08
	Strengthening	Asphalt Overlay 80 mm	57,888	8.04
		Asphalt Overlay 120 mm	65,880	9.15
		Asphalt Overlay 160 mm	87,840	12.20
		Asphalt Overlay 200 mm	109,800	15.25
		Mill and Replace 20/50 mm	63,072	8.76
Improvement	Upgrading	Upgrading to Earth Surface	23,855	3.67
		Upgrading to Gravel Surface	33,300	5.12
		Upgrading to Stone Surface	50,000	7.69

Cost per km considers 7.2 m wide paved roads and 6.5 m wide unpaved roads

C. Performance-based Maintenance Contracting Program

8. The paved roads network of Paraguay totals around 4,000 km and carries 1,674 million vehicle-km per year. The MOPC defined a core paved roads network to be rehabilitated and maintained under a program of seven GMANS contracts, which totals 2,012 km and carries 1,403 million vehicle-km per year that represents 50% and 84% respectively of the total paved roads network. Table A9-6 presents the performance-based maintenance program road sections.

Table A9-6
GMANS Contracts Road Sections

GMANS Contract	Segment	Length (Km)
I	PY 02: Caacupé – Inicio concesión Tapé Porá PY 03: Limpio – Empalme PY 08 PY 08: Empalme PY 03 – Coronel Oviedo	381
II	South Access: Desvío a Guarambaré - Itá PY 01: Itá - Encarnación	337
III	PY 06: Encarnación – Empalme PY 07	250
IV	PY 08: Empalme PY 03 – Yby Yau PY 05: Pedro Juan Caballero - Concepción	393
V	PY04: San Ignacio- Pilar	142
VI	PY 07: Hernandarias – Empalme PY 03 PY 03: Empalme PY 08 – Saltos del Guaira	400
URBAN	PY 02: San Lorenzo – Caacupé Acceso Sur: 4 Mojones – Desvío a Guarambaré	108
Total		2,011

9. Table A9-7 presents the performance-based maintenance program network characteristics.

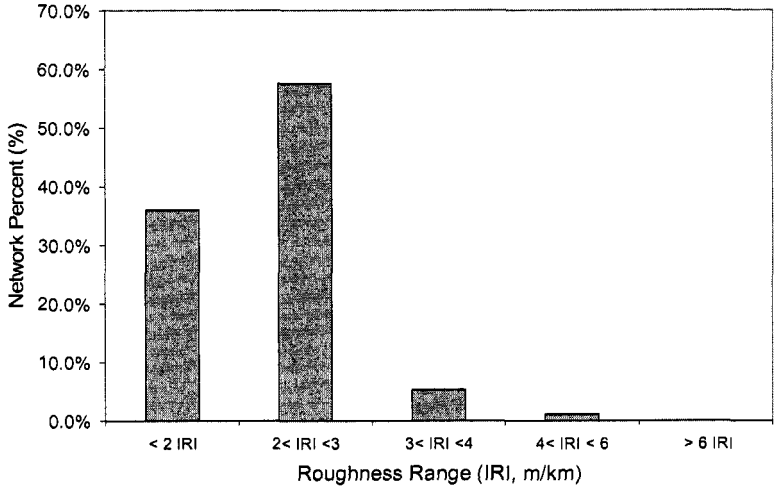
Table A9-7
GMANS Contracts Program Network Characteristics

Contract ID	Financing	Length (km)	Roughness (IRI)	Damage (%)	Traffic (AADT)	Utilization (M vehicle-km)	Utilization (%)
URBAN	IBRD	108	2.2	1	8,483	351	25%
GMANS I	IBRD	381	2.1	11	2,702	342	24%
GMANS II	IBRD	337	1.9	10	2,474	265	19%
GMANS III	Other	250	2.0	2	1,652	133	9%
GMANS IV	Other	393	2.2	19	827	122	9%
GMANS V	IBRD	142	2.2	10	267	14	1%
GMANS VI	Other	400	1.9	2	1,216	178	13%
Total		2,011	2.1	8	2,517	1,403	100%

Damage = Area of cracking, raveling and potholes

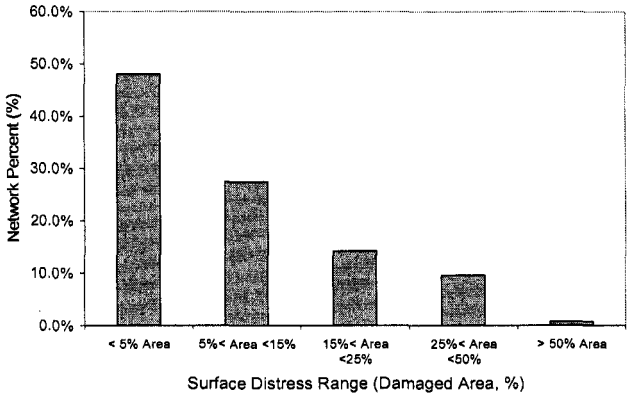
10. Figure A9-1 presents the current network length distribution by ride quality (roughness, IRI, m/km) which shows that 94% of the network is in good condition in terms of ride quality (roughness less than 3 IRI, m/km).

Figure A9-1
Network Roughness Distribution



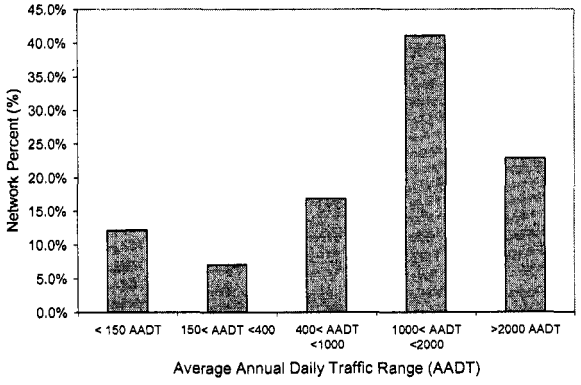
11. Figure 2 presents the current network length distribution by surface distress (total damage area equal to area of cracking, raveling and potholes) which shows that 48% of the network is in good condition in terms of surface distress (damaged area less than 5%), with the remainder of the network requiring periodic maintenance works.

Figure A9-2
Network Surface Distress Distribution



12. Figure 3 presents the current network length distribution by average annual daily traffic (AADT), which shows that the percentage of the network carrying more than 1,000 AADT is 64%.

Figure A9-3
Network Traffic Distribution



13. The project will finance the first three high priority GMANS contracts that carry the higher level of traffic; totaling 826 km. The MOPC collected detailed road characteristics, condition and traffic for each homogeneous road section that is part of the GMANS program and performed a Project level economic analysis of the three GMANS contracts to be financed under the project with the objective of: (i) identify sections that need only routine maintenance over the 2007-2010 period and sections in which periodic maintenance is needed; (ii) compare the net benefits of possible project-alternatives; (iii) ensure that all investments yield a rate of return higher than 12%; and (iv) estimate the network condition and road user costs over the evaluation period considering the Project investments. Table A9-8 presents the network characteristics in terms of road length, roughness, and average daily traffic of the three GMANS contracts to be financed under the project.

Table A9-8
Project GMANS Contracts Road Section Characteristics

Contract ID	Section Name	Length (km)	Roughness (IRI)	Damage (%)	Traffic (AADT)
GMANS URBAN	San Lorenzo A Pasaje Peatonal (Sur)	0.9	2.2	0	9,387
	San Lorenzo A Pasaje Peatonal (Norte)	0.9	2.2	0	9,387
	Pasaje Peatonal A Capiata (Sur)	4.9	1.9	5	9,387
	Pasaje Peatonal A Capiata (Norte)	4.9	2.1	0	9,387
	Capiata A Itaugua (Sur)	10.2	2.2	4	5,215
	Capiata A Itaugua (Norte)	10.2	2.2	2	5,215
	Itaugua A Ypacarai (Sur)	5.9	2.1	5	18,020
	Itaugua A Ypacarai (Norte)	5.9	2.0	0	18,020
	Travesía Ypacarai (Sur)	2.6	2.2	1	18,020
	Travesía Ypacarai (Norte)	2.6	2.2	1	18,020
	Ypacarai A Dv. San Bernardino (Sur)	1.7	2.2	1	14,840
	Ypacarai A Dv. San Bernardino (Norte)	1.7	1.9	0	14,840
	Dv. San Bernardino - Cerro	5.5	2.2	12	14,840
	Circunvalación Cerro (Sur)	2.5	2.4	0	14,840
	Circunvalación Cerro (Norte)	2.5	2.3	0	14,840
	Circunvalación Cerro - Km 35,10	1.7	2.2	0	29,680
	Km 35,9 A Caacupé	3.0	2.7	1	14,840
	Travesía Caacupé (Sur)	2.7	3.1	0	14,840
	Travesía Caacupé (Norte)	2.2	2.7	0	14,840
	Caacupé - Fin Crema	0.5	2.2	0	29,680
	Cuatro Mojones - Dv San Lorenzo (Sur)	7.4	2.2	0	22,991
	Cuatro Mojones - Dv San Lorenzo (Norte)	7.4	2.2	0	22,991
	Dv San Lorenzo (Sur) - Arroyo Ytororó	5.6	1.9	0	15,370
	Dv San Lorenzo (Norte) - Arroyo Ytororó	5.6	2.0	0	15,370
	Arroyo Ytororó - Empalme A Ypane	3.7	2.3	0	15,370
	Empalme A Ypane - Dv. Guarambaré	5.5	2.0	0	6,648
Subtotal		108.2	2.2	1	15,265
GMANS I	Caacupé A Eusebio Ayala	16.1	2.2	5	5,173
	Eusebio Ayala A Itacurubi De La Cordillera	15.9	2.3	2	4,665
	Itacurubi De La Cordillera A Límite Deptal.	7.5	2.5	0	4,665
	Limite Departamental A San Jose De Los Arroyos	7.6	2.1	0	4,665
	San José De Los Arroyos A Coronel Oviedo	29.4	2.1	19	4,665
	Cnel. Oviedo A Caaguazú	46.3	2.2	10	3,191

	Caaguazú A Inicio Concesión Tapé Porá	4.8	2.0	3	3,191
	Limpio A Emboscada	16.9	2.3	30	2,873
	Emboscada A Arroyos Y Esteros	27.2	2.0	1	2,873
	Arroyos Y Esteros A 25 De Diciembre	52.2	2.1	9	1,731
	25 De Diciembre A San Estanislao	31.6	1.9	19	1,731
	San Estanislao - Empalme Py09	19.8	1.9	15	1,731
	Empalme Py03 - Cruce Tacuara	12.3	2.0	9	969
	Tacuara (Cruce) A Limite Departamental	29.7	1.8	9	969
	Límite Departamental A Mbutuy	6.1	1.8	15	969
	Mbutuy A Carayao	26.4	1.9	29	1,048
	Carayao A Cnel. Oviedo	31.1	2.1	7	1,048
Subtotal		380.9	2.1	11	2,715
GMANS II	Guarambare A Ita	12.2	2.4	17	9,622
	Ita A Yaguaron	8.6	2.4	16	3,869
	Yaguaron A Paraguari	14.8	2.2	52	3,869
	Paraguari A Carapegua	20.3	2.0	9	3,869
	Carapegua A San Roque Gonzalez	13.6	1.9	9	13,462
	San Roque Gonzalez A Quiindy	12.4	1.9	5	125
	Quiindy A Caapucu	31.9	1.8	3	220
	Caapucu A Villa Florida	20.6	2.0	9	220
	Villa Florida A San Miguel Potrero	25.4	2.0	25	135
	San Miguel Potrero - San Juan Bautista	9.8	1.9	24	135
	San Juan Bautista - San Ignacio	34.1	1.8	0	135
	San Ignacio A Santa Rosa	15.8	1.8	0	135
	Santa Rosa A San Patricio	9.4	1.8	0	135
	San Patricio A Gral. Delgado	30.8	1.8	0	135
	Gral. Delgado A Cnel. Bogado	25.7	1.8	0	28
	Cnel. Bogado A Carmen Del Paraná	3.4	1.8	0	54
	Cnel. Bogado A Carmen Del Paraná-Encarnación	48.3	1.8	0	396
Subtotal		337.2	1.9	10	2,150
Total		826.3	2.1	7	6,710

14. Table A9-9 presents the investments on the road sections requiring periodic maintenance during 2007-2010 period. The roads sections requiring periodic maintenance totals 454 km that represents 55% of the overall network of 826 km. The predominant road works activities are the following: (i) 45% of the network requires only routine maintenance, (ii) 24% requires micro surfacing, (iii) 26% requires micro surfacing with shoulder repair, (iv) 2% requires micro surfacing with spot rehabilitation; and (v) 3% requires 50 mm overlays. The investments costs include the predominant road works activities, shoulder repair costs, spot road rehabilitation works, such as mill and replace activities, and. The estimated periodic maintenance investments during the 2007-2010 period totals US$9.09 million that represents an average expenditure of US$29,157 per km. The estimated recurrent expenditures for routine maintenance and vertical and horizontal signs for the period 2007-2010 period totals US$12.78 million that represents average annual expenditures of US$3,866 per km per year.

Table A9-9
Performance-Based Road Sections Requiring Periodic Maintenance

Contract ID	Section Name	Length (km)	Works (km)	Predominant Work Activity	Investment (M US$)	Investment (US$/km)
GMANS URBAN,	San Lorenzo A Pasaje Peatonal (Sur)	0.9	0.9	Overlay 50 mm	0.064	71,085
	Pasaje Peatonal A Capiata (Sur)	4.9	4.9	Overlay 50 mm	0.231	47,100
	Capiata A Itaugua (Sur)	10.2	10.2	M Surf. & Overlay	0.314	30,755
	Capiata A Itaugua (Norte)	10.2	1.0	Micro Surfacing 12 mm	0.020	19,516
	Itaugua A Ypacarai (Sur)	5.9	5.9	M Surf. & Overlay	0.247	41,842
	Itaugua A Ypacarai (Norte)	5.9	1.2	Micro Surfacing 12 mm	0.023	19,824
	Travesía Ypacarai (Sur)	2.6	2.6	Micro Surfacing 12 mm	0.062	23,688
	Travesía Ypacarai (Norte)	2.6	2.6	Micro Surfacing 12 mm	0.057	21,834
	Ypacarai A Dv. San Bernardino (Sur)	1.7	1.7	Micro Surfacing 12 mm	0.040	23,688
	Ypacarai A Dv. San Bernardino (Norte)	1.7	1.7	Micro Surfacing 12 mm	0.034	20,035
	Dv. San Bernardino - Cerro	5.5	5.5	Micro Surfacing 12 mm	0.114	20,794
	Circunvalación Cerro (Sur)	2.5	2.5	Micro Surfacing 12 mm	0.055	21,862
	Circunvalación Cerro (Norte)	2.5	2.5	Micro Surfacing 12 mm	0.051	20,288
	Circunvalación Cerro - Km 35,10	1.7	1.7	Micro Surfacing 12 mm	0.034	20,288
	Travesía Caacupé (Sur)	2.7	2.7	Micro Surfacing 12 mm	0.060	22,143
	Travesía Caacupé (Norte)	2.2	2.2	Micro Surfacing 12 mm	0.053	24,278
	Caacupé - Fin Crema	0.5	0.5	Micro Surfacing 12 mm	0.009	18,546
	Cuatro Mojones - Dv San Lorenzo (Sur)	7.4	7.4	Micro Surfacing 12 mm	0.171	23,154
	Cuatro Mojones - Dv San Lorenzo (Norte)	7.4	7.4	Micro Surfacing 12 mm	0.168	22,705
	Dv San Lorenzo (Sur) - Arroyo Ytororó	5.6	5.6	Micro Surfacing 12 mm	0.117	20,963
	Dv San Lorenzo (Norte) - Arroyo Ytororó	5.6	5.6	Micro Surfacing 12 mm	0.114	20,373
	Arroyo Ytororó - Empalme A Ypane	3.7	3.7	Micro Surfacing 12 mm	0.075	20,204
	Empalme A Ypane - Dv. Guarambaré	5.5	5.5	Micro Surfacing 12 mm	0.111	20,176
Subtotal		99.4	85.5		2.225	26,018
GMANS I	Caacupé A Eusebio Ayala	16.1	16.1	Micro Surfacing 12 mm	0.311	19,316
	Eusebio Ayala A Itacurubi De La Cordillera	15.9	15.9	M Surf. & Shoulder	0.965	60,680

	Itacurubi De La Cordillera A Líimite Deptal.	7.5	7.5	M Surf. & Shoulder	0.455	60,680
	Limite Departamental A San Jose De Los Arroyos	7.6	7.6	M Surf. & Shoulder	0.223	29,368
	San Jose De Los Arroyos A Coronel Oviedo	29.4	29.4	M Surf. & Shoulder	1.358	46,205
	Cnel. Oviedo A Caaguazú	46.3	9.3	M Surf. & Shoulder	0.264	28,495
	Caaguazú A Inicio Concesión Tape Pora	4.8	1.0	M Surf. & Shoulder	0.027	28,551
	Mbutuy A Carayao	26.4	7.1	Micro Surfacing 12 mm	0.126	17,733
	Carayao A Cnel. Oviedo	31.1	26.1	Micro Surfacing 12 mm	0.472	18,065
Subtotal		185.1	120.0		4.202	35,026
GMANS II	Guarambare A Ita	12.2	0.2	Micro Surfacing 12 mm	0.004	17,925
	Ita A Yaguaron	8.6	8.6	Micro Surfacing 12 mm	0.154	17,872
	Yaguaron A Paraguari	14.8	14.8	Micro Surfacing 12 mm	0.292	19,740
	Paraguari A Carapegua	20.3	2.6	Overlay 50 mm	0.187	70,800
	Carapegua A San Roque Gonzalez	13.6	7.1	M Surf. & Shoulder	0.132	18,688
	San Roque Gonzalez A Quiindy	12.4	9.9	M Surf. & Shoulder	0.188	18,934
	Quiindy A Caapucu	31.9	16.6	M Surf. & Shoulder	0.294	17,710
	Caapucu A Villa Florida	20.6	20.6	M Surf. & Shoulder	0.413	20,047
	Villa Florida A San Miguel Potrero	25.4	19.8	M Surf. & Shoulder	0.893	45,094
	San Miguel Potrero – San Juan Bautista	9.8	6.0	Micro Surfacing 12 mm	0.105	17,605
Subtotal		169.6	106.3		2.662	25,057
Total		454.1	311.7		9.089	29,157

15. The economic analysis was done considering a without project-alternative that includes routine maintenance and rehabilitation with a 200mm overlay when a road reaches 5.5 IRI, m/km. The project-alternatives evaluated micro surfacing, overlay and mill and replace road works. The economic analysis yields substantial economic returns. For the three GMANS contracts: (i) the net present value (NPV) is US$30.6 million; (ii) the economic rate of return (ERR) is 41%; and (iii) the discounted benefit cost ratio compared to the without project-alternative is 5.2. All road works yield and economic rate of return higher than 12%. Table A9-10 presents the economic analysis indicators.

Table A9-10
Performance-Based Maintenance Economic Analysis Results

Contract ID	Section Name	NPV (M US$)	ERR (%)	ERR Sensitivity Analysis (%)		
				Costs+20%	Benefits-20%	Both
GMANS URBAN	San Lorenzo A Pasaje Peatonal (Sur)	0.2	48%	43%	42%	38%
	Pasaje Peatonal A Capiata (Sur)	0.8	50%	45%	44%	39%
	Capiata A Itaugua (Sur)	2.2	83%	77%	76%	70%
	Capiata A Itaugua (Norte)	1.7	>100%	>100%	>100%	>100%
	Itaugua A Ypacarai (Sur)	0.6	41%	38%	38%	35%
	Itaugua A Ypacarai (Norte)	0.5	68%	64%	63%	58%
	Travesía Ypacarai (Sur)	0.2	33%	30%	30%	27%
	Travesía Ypacarai (Norte)	0.2	37%	34%	33%	31%
	Ypacarai A Dv. San Bernardino (Sur)	0.1	36%	33%	32%	30%
	Ypacarai A Dv. San Bernardino (Norte)	0.1	31%	28%	28%	25%
	Dv. San Bernardino – Cerro	0.5	41%	37%	37%	33%
	Circunvalación Cerro (Sur)	0.2	37%	34%	34%	31%
	Circunvalación Cerro (Norte)	0.2	35%	32%	32%	29%
	Circunvalación Cerro - Km 35,10	0.2	56%	52%	51%	46%
	Travesía Caacupé (Sur)	0.3	42%	38%	38%	35%
	Travesía Caacupé (Norte)	0.2	36%	33%	32%	30%
	Caacupé - Fin Crema	0.1	56%	52%	51%	47%
	Cuatro Mojones - Dv San Lorenzo (Sur)	1.8	74%	68%	66%	61%
	Cuatro Mojones - Dv San Lorenzo (Norte)	1.5	68%	62%	61%	56%
	Dv San Lorenzo (Sur) - Arroyo Ytororó	0.8	48%	44%	43%	40%
	Dv San Lorenzo (Norte) - Arroyo Ytororó	0.7	47%	43%	42%	39%
	Arroyo Ytororó - Empalme A Ypane	1.0	96%	88%	86%	79%
	Empalme A Ypane - Dv. Guarambaré	0.4	39%	35%	35%	32%
Subtotal		14.7	60%	55%	54%	49%
GMANS I	Caacupe A Eusebio Ayala	1.1	57%	52%	50%	45%
	Eusebio Ayala A Itacurubi De La Cordillera	0.8	22%	19%	19%	16%
	Itacurubi De La Cordillera A Limite Deptal.	0.4	23%	21%	20%	17%
	Limite Departamental A San Jose De Los Arroyos	0.5	31%	28%	27%	25%
	San Jose De Los Arroyos A Coronel Oviedo	2.8	32%	29%	28%	26%
	Cnel. Oviedo A Caaguazú	2.2	42%	40%	39%	36%
	Caaguazú A Inicio Concesión Tape Pora	0.3	45%	42%	41%	38%
	Mbutuy A Carayao	0.3	29%	27%	26%	24%
	Carayao A Cnel. Oviedo	0.2	17%	15%	15%	13%
Subtotal		8.7	31%	28%	27%	24%
GMANS II	Guarambare A Ita	1.1	>100%	>100%	>100%	>100%

	Ita A Yaguaron	0.8	43%	39%	39%	36%
	Yaguaron A Paraguari	1.8	55%	54%	54%	53%
	Paraguari A Carapegua	1.0	36%	34%	33%	32%
	Carapegua A San Roque Gonzalez	0.3	24%	23%	22%	21%
	San Roque Gonzalez A Quiindy	0.1	17%	16%	16%	15%
	Quiindy A Caapucu	0.5	24%	22%	22%	21%
	Caapucu A Villa Florida	1.0	27%	25%	25%	22%
	Villa Florida A San Miguel Potrero	0.5	21%	19%	18%	16%
	San Miguel Potrero - San Juan Bautista	0.2	24%	21%	21%	18%
Subtotal		7.3	32%	30%	29%	27%
Total		30.6	41%	38%	37%	34%

16. The proposed road works will keep the network in good condition during the 2007-2010 period and will defer the need of rehabilitation works. With the project, the estimated average network roughness is 2.3 IRI in 2010 and 2.8 IRI in 2016. If the project is not undertaken, the estimated average network roughness is 2.3 IRI in 2010 and 3.2 IRI in 2016, indicating than with the project in 10 years the network roughness will be 15% lower than if the project is not undertaken. Currently road users spend about US$302 million annually to operate their vehicles on this network. With the project, the road user costs savings during the 2007-2010 period totals US$4.9 million and during the 2007-2016 period totals US$63.8 million.

D. Optimization of the Toll System

17. The economic evaluation of this sub-component of the Project analyzed the economic return of the investment (US$3.3 million) for a period of five years and considered as benefits of the project the increase in revenues derived from the optimization of the toll fees collection as a result of the improvements and refurbishment of the toll post linked to the road segments to be maintained under the GMANS contracts for the paved road network, plus the additional benefits derived from the revenues resulting from the duplication of the toll fee collection system (changing from one-way to a two-way system). This assessment showed a positive economic return (NPV = US$10.3 million), which fully justifies the project (Table A9-11).

18. For this evaluation the following characteristics of the toll system were taken into consideration:

- Current collection revenues (information provided by the Department of Revenues) with an initial increment of 5% for improvement in the control systems, and increase of 80% due to the duplication of the fees collected through a two-way collection system, and an additional annual increase of 3% for increases in traffic levels;
- Current operating costs (information provided by the Department of Revenues) with an initial increment of 50% for the increase in costs derived from more sophisticated control systems, plus an increase of 70% due to the duplication of toll fees.
- An initial investment of US$0.3 million for consultancy services (for the design of a Toll Master Plan), US$3.2 million for improvement of facilities and equipments (electric and IT systems), and an annual investment of US$0.36 for maintenance costs.

Table A9-11
Toll Optimization Economic Analysis Results
(in US$)

		2006	2007	2008	2009	2010	2011	2012
Without project	Toll collection			6,166,936	6,351,944	6,542,503	6,738,778	6,940,941
Without project	Operation			562,500	562,500	562,500	562,500	562,500
With project	Toll collection			11,655,510	12,005,175	12,365,330	12,736,290	13,118,379
With project	Operation			1,434,375	1,434,375	1,434,375	1,434,375	1,434,375
PROJECT BENEFITS				**4,616,698**	**4,781,356**	**4,950,952**	**5,125,637**	**5,305,563**
Consultancy Services		300,000						
Improvement	Facilities		1,400,000					
Improvement	Equipment		1,800,000					
Maintenance				360,000	360,000	360,000	360,000	360,000
PROJECT COSTS		**300,000**	**3,200,000**	**360,000**	**360,000**	**360,000**	**360,000**	**360,000**
PROJECT NET COSTS		**-300,000**	**-3,200,000**	**4,256,698**	**4,421,356**	**4,590,952**	**4,765,637**	**4,945,563**

Discount Rate	12%
NPV	**10,277,379**

E. Improvement and Maintenance of Unpaved Road Network and Bridges Program

19. The road networks of the Departments of Caaguazú, San Pedro, and Caazapá were selected to be part of the program of improvement and maintenance of unpaved roads and bridges to be financed under the project. The departments were selected based on a multi-criteria analysis that considered the following factors: (i) unpaved roads density; (ii) rural population density; (iii) poverty index; and (iv) agricultural area density. Table A9-12 presents the multi-criteria analysis results, considering a weight of 0.5 to poverty index taking into account the focus of the program of improving access of low income population to economic and social centers.

Table A9-12
Secondary Roads Program Multi-Criteria Analysis

	Weight = 0.2		Weight = 0.2		Weight = 0.5		Weight = 0.1		
	Unpaved Roads Density (km/km2)		Rural Population Density (#/km2)		Poverty Index (%)		Agricultural Area Density (he/km2)		Priority
Department	Value	Factor	Value	Factor	Value	Factor	Value	Factor	Index
Caaguazú	61	1.00	27	0.35	61	0.97	35	0.52	0.81
San Pedro	56	0.92	13	0.17	62	1.00	14	0.20	0.74
Caazapá	45	0.74	12	0.16	48	0.77	27	0.39	0.60
Itapúa	43	0.70	19	0.25	37	0.59	43	0.63	0.55
Canindeyú	50	0.83	7	0.09	39	0.62	31	0.46	0.54
Concepción	36	0.60	6	0.08	48	0.77	5	0.07	0.53
Cordillera	45	0.74	31	0.41	35	0.57	6	0.09	0.52
Guairá	20	0.32	30	0.39	41	0.65	24	0.35	0.50
Central	39	0.65	77	1.00	21	0.33	5	0.08	0.50
Paraguari	42	0.69	20	0.26	35	0.55	8	0.12	0.48
Ñeembucú	47	0.77	3	0.04	37	0.59	2	0.03	0.46
Amambay	32	0.53	3	0.04	39	0.63	9	0.13	0.44
Misiones	26	0.43	5	0.07	38	0.62	6	0.08	0.42
Alto Paraná	19	0.31	13	0.17	26	0.42	68	1.00	0.41

20. The program will: (i) improve the ride quality and ensure all-weather access on a basic network of unpaved roads; (ii) ensure the sustainable maintenance of the basic network; and (iii) rehabilitate a series of bridges that are currently in critical condition. The MOPC performed a detailed inventory of all roads in the three Departments in terms of road condition, traffic, population served and relative importance of the roads determined with multi-criteria analysis. Based on this information, the MOPC defined a provisional basic unpaved roads network in each Department considering: (i) the financial resources available for the sustainable maintenance of the basic road network; (ii) technical aspects; (iii) beneficiary population; (iii) daily traffic; and (iv) access to main roads. The provisional basic unpaved roads network was then presented to the local population and decision makers in order to validate the selection of the basic unpaved roads network and make the necessary adjustments considering the inputs of the local consultation process. Table A9-13 presents the basic unpaved roads networks resulting from the local consultation workshops. The total length of the basic unpaved roads networks of the three Departments is 960 km, the average traffic is 87 vehicles per day and the direct beneficiaries are about 159,000 persons. The project will finance the improvement of 691 km, with the improvement of the remainder roads being considered for other sources of financing.

Table A9-13
Unpaved Road Networks

Department	Road Name	Financing	Length (km)	Traffic (AADT)	Beneficiary Population (#)		
					Direct	Indirect	Total
Caaguazú	PY08 - Cecilio Baez	Other	11.2	105	2,460	694	3,154
	Cecilio Baez - San Joaquín	Other	20.0	105	3,713	1,048	4,761
	San Joaquín - San Antonio – Yhu	IBRD	16.9	105	4,394	1,240	5,634
	Caaguazú - Agua - Mojón 7 - Yhu Caaguazú (PY13)	IBRD	27.5	94	10,755	4,102	14,857
	PY13-3 Feb-Toro Cangue-Yhovy Central-Raúl Oviedo	IBRD	60.6	95	8,715	4,482	13,197
	Yhovy Central - J.E. Estigarribia (Calle 2)	IBRD	19.2	45	6,599	1,147	7,746
	Punta de Empedrado (Repatriación) - Arroyito	Other	15.0	191	3,608	2,373	5,981
	PY02 (km 155) - Cruce Alto - R I 3 Corrales	IBRD	16.7	57	2,366	1,487	3,853
	Carayao - Cleto Romero (A° Molina)	IBRD	28.2	32	3,412	962	4,374
	PY08 (km 140) Cruce Capillita-Desvío Calle 4-Santa Librada	IBRD	16.5	69	2,400	1,680	4,080
	J.M. Frutos - 3 de Febrero	IBRD	16.6	48	5,024	1,283	6,307
	PY08 - Cruce Mainumby	IBRD	2.0	75	268	188	456
	Nueva Londres - La Pastora	IBRD	23.9	69	2,363	923	3,286
	Calle 4 - Desvío Santa Librada - Esc N° 810 Cruce Alto	IBRD	5.1	69	474	332	806
	Ruta 7 - 3 de Noviembre	Other	30.4	135	3,657	2,560	6,217
	PY08 - La Pastora	IBRD	27.0	35	3,531	2,243	5,774
	Empalme Ruta YHU - Tarumaí - Carpa Cué - San Joaquín	IBRD	21.2	30	2,827	1,622	4,449
Subtotal			358.0	80	66,566	28,366	94,932
San Pedro	PY08 (Ruta Oviedo - Santaní) – Unión	IBRD	16.0	158	1,776	652	2,428
	Unión - Guaica - PY03	Other	13.9	158	1,546	567	2,113
	Cruce Liberación (PY08) – Chore	Other	18.0	101	2,542	1,345	3,887
	Chore - Gral.Aquino	IBRD	54.6	101	7,709	4,080	11,789
	Chore - Colonia Cocuera	IBRD	16.1	131	3,773	1,875	5,648
	Colonia Cocuera - San Pablo	IBRD	17.9	131	3,558	960	4,518

	Villa del Rosario - Estero Yetyty	IBRD	13.9	77	3,053	346	3,399
	San Pedro - Piri Pucu	IBRD	32.2	109	4,822	601	5,423
	Naranjito - San Vicente	Other	18.7	46	3,446	2,412	5,858
	San Vicente - Arroyo Itanara	IBRD	19.9	46	3,446	2,412	5,858
	Lima – Quindy	IBRD	24.8	113	2,571	1,047	3,618
	Tacuatí – PY08	Other	37.0	60	2,592	423	3,015
	Quindy – Naranjito	IBRD	4.6	113	349	244	593
	Naranjito - Ruta PY08 (Gral. Resquín)	Other	21.4	109	2,588	1,134	3,722
Subtotal			308.9	104	43,771	18,098	61,869
Caazapá	San Juan Nepomuceno - Abai - Taruma – Tuna	Other	38.6	55	10,249	1,139	11,388
	San Juan Nepomuceno - Río Yatay	IBRD	25.1	78	8,677	1,218	9,895
	Caazapá – Boquerón	IBRD	13.7	76	7,054	745	7,799
	Cruce a Lima (F01) - Río Tebicuary	IBRD	30.7	124	1,617	622	2,239
	San Francisco - San Juan Nepomuceno	IBRD	35.2	76	5,216	1,708	6,924
	San Agustín - Tuparenda - Lim. Departamental	Other	45.0	60	2,099	1,029	3,128
	San Miguel Yarati-i-(PY08) – Yataity	IBRD	32.2	91	3,347	1,139	4,486
	Yataity (Yuty) - San Francisco	IBRD	9.7	91	2,426	495	2,921
	Boquerón (Caazapá) - San Francisco	IBRD	26.3	76	4,183	1,251	5,434
	Río Tayay – Tavaí	IBRD	23.8	78	1,576	644	2,221
	Isla Yobai - Buena Vista	IBRD	13	53	2,446	721	3,168
Subtotal			293.4	78	48,891	10,711	59,603
Total			960.4	87	159,228	57,176	216,404

21. The MOPC collected detailed road characteristics, condition and traffic for each homogeneous road section that will be part of the first year program of improvements and performed a Project level economic analysis with the objective of comparing the net benefits of possible project-alternatives and ensuring that the overall investments on a Department yield a rate of return higher than 12%. Table A9-14 presents the characteristics of the first year program of improvements of unpaved roads to be financed under the project. The first year program covers a network of 338 km and has a total budget of US$12.60 million, for the improvement of the unpaved roads to a gravel surface standard, excluding the bridge rehabilitation costs for bridges with a span larger than 6 meters.

Table A9-14
Improvement of Unpaved Roads First Year Program Characteristics

Department	Road Name	Length (km)	Traffic (AADT)	Surface	Financial Cost (M US$)	Financial Cost (US$/km)
Caaguazú	San Joaquín - San Antonio - Yhu	16.9	105	Gravel	0.57	33,600
	Caaguazú - Agua - Mojón 7 - Yhu Caaguazú (PY13)	27.5	94	Gravel	1.19	43,300
	PY13-3 Feb-Toro Cangue-Yhovy Central-Raúl Oviedo	60.6	95	Gravel	2.44	40,300
	J.M. Frutos – 3 de Febrero	16.6	48	Gravel	0.70	42,200
Subtotal		121.6	86		4.90	40,307
San Pedro	Chore - Gral.Aquino	54.6	101	Gravel	2.05	37,600
	Chore - Colonia Cocuera	16.1	131	Gravel	0.70	43,300
	Colonia Cocuera - San Pablo	17.9	131	Gravel	0.71	39,700
	Lima – Quindy	24.8	113	Gravel	0.74	29,700
Subtotal		113.4	119		4.20	37,016
Caazapá	Caazapá – Boquerón	13.7	76	Gravel	0.61	44,500
	Cruce a Lima (F01) - Río Tebicuary	30.7	124	Gravel	1.22	39,700
	San Miguel Yarati-i-(PY08) - Yataity	32.2	91	Gravel	0.93	29,000
	Boquerón (Caazapá) - San Francisco	26.3	76	Gravel	0.74	28,000
Subtotal		102.9	92		3.50	34,000
Total		337.9	99		12.60	37,282

Financial costs exclude the bridge rehabilitation costs

22. The economic analysis was done considering a without project-alternative that maintains the roads at their current poor condition and evaluating alternative improvement road works with different surface layers (earth, gravel or stones). For the three Departments: (i) the net present value is US$14.2 million; (ii) the economic rate of return is 32%; and (iii) the discounted benefit cost ratio compared to the without project-alternative is 2.4. The investments on all roads that carry more than 50 vehicles per day have an economic rate of return higher than 12%. The investment on the J.M. Frutos - 3 de Febrero road that has 48 vehicles per day, yields and economic rate of return of 11%, which was consider acceptable because the road serves a social function and is needed for regional integration. Table A9-15 presents the economic analysis indicators.

Table A9-15
Improvement of Unpaved Roads First Year Program Economic Analysis

Department	Road Name	NPV (M US$)	ERR (%)	ERR Sensitivity Analysis (%) Costs +20%	Benefits -20%	Both
Caaguazú	San Joaquín - San Antonio - Yhu	0.86	40%	34%	32%	27%
	Caaguazú - Agua - Mojón 7 - Yhu Caaguazú (PY13)	0.75	25%	20%	19%	15%
	PY13-3 Feb-Toro Cangue-Yhovy Central-Raúl Oviedo	1.54	24%	20%	20%	17%
	J.M. Frutos - 3 de Febrero	-0.03	11%	8%	7%	5%
Subtotal		3.12	24%	20%	20%	16%
San Pedro	Chore – Gral.Aquino	2.53	33%	29%	28%	24%
	Chore - Colonia Cocuera	0.97	38%	32%	30%	25%
	Colonia Cocuera - San Pablo	1.13	41%	35%	33%	28%
	Lima - Quindy	1.76	55%	46%	44%	37%
Subtotal		6.38	39%	33%	32%	27%
Caazapá	Caazapá - Boquerón	0.34	23%	19%	18%	14%
	Cruce a Lima (F01) - Río Tebicuary	2.20	43%	39%	38%	34%
	San Miguel Yarati-i-(PY08) - Yataity	1.14	34%	31%	30%	27%
	Boquerón (Caazapá) - San Francisco	1.01	38%	31%	30%	25%
Subtotal		4.70	36%	32%	31%	27%
Total		14.20	32%	28%	27%	23%

23. The economic analysis of the bridge rehabilitation program was done estimating the overall road user benefits of the bridge improvements n terms of savings in vehicle operating costs and time costs. The base case without project-alternative considers weight restrictions on the bridges implying that only light vehicles, with gross vehicle weight less than 10 tons, can circulate. In contrast, heavy vehicles, corresponding to an average of 30% of the total traffic, are forced to use alternative routes with an average increase in travel distance of 30% or reduce their loads by transferring some cargo to other vehicles. The project-alternative considers structural strengthening and widening or construction of new bridges that eliminates travel restrictions. The estimated cost of the bridge rehabilitation program is US$6.7 million for the three Departments. For the overall bridge rehabilitation program: (i) the net present value is US$3.0 million; (ii) the economic rate of return is 22%; and (iii) the discounted benefit cost ratio compared to the without project-alternative is 1.7.Table A9-16 presents the economic analysis indicators.

Table A9-16
Bridge Rehabilitation Program Economic Analysis

Department	Number Bridges (#)	Financial Cost (M US$)	Financial Cost (US$/bridge)	NPV (M US$)	ERR (%)	ERR Sensitivity Analysis (%) Costs+20% (%)	Benefits-20% (%)	Both (%)
Caaguazú	30	1.85	61,777	0.44	17%	13%	12%	9%
San Pedro	18	1.19	66,167	2.03	43%	36%	34%	29%
Caazapá	50	3.61	72,278	0.56	16%	12%	11%	8%
Total	98	6.66	67,941	3.03	22%	17%	16%	13%

24. The overall economic analysis of the first year unpaved roads improvement program and bridge rehabilitation program at the Caaguazú, San Pedro, and Caazapá Departments yields a net present value of US$ 17.2 million and an economic rate of return of 29%.

F. Sensitivity Analysis of Critical Items

25. The results are satisfactory relative to the main risks considered in the economic analysis, namely, higher investment costs and lower future benefits due mainly due to lower traffic. Under a worst-case scenario of benefits dropping to 80 of the current level and 20% increase in investment costs, the project yields a satisfactory rate of return of 24%. Table A9-17 presents the sensitivity analysis results.

Table A9-17
Sensitivity Analysis

Project Component	Base ERR (%)	ERR Sensitivity Analysis (%) A- Costs 20%	B- Benefit -20%	A + B
Performance-based Contracts	41%	38%	37%	34%
Unpaved Road Network and Bridges	29%	24%	23%	19%
Total	32%	28%	27%	24%

26. The analysis of switching values of critical items indicates that to yield a net present value equal to zero, for the performance-based contracts, investment costs need to be multiplied by 6.60 or benefits multiplied by 0.19. For the improvement and maintenance of the unpaved road networks and bridges, investment costs need to be multiplied by 2.22 or benefits multiplied by 0.45.

G. Fiscal Impact Analysis

27. The MOPC will benefit from the reduced future recurrent maintenance costs on the paved roads that will receive periodic maintenance under the GMANS contracts and from avoiding higher costs of future rehabilitation. With the project, the MOCP will expend for periodic maintenance and rehabilitation works, on the three GMANS contracts, around US$2,700 per km-year during the 2007-2010 period and US$2,800 per km-year during the 2011-2022 period. Without the project, the expenditures during the 2007-2010 period are zero and during the 2011-2022 period, they are estimated to be increase to around US$4,500 per km-year. For routine maintenance works, the estimated expenditures are US$3,900 per km-year.

Annex 10: Safeguard Policy Issues

PARAGUAY: Road Maintenance

Annex 10.A – Environmental Assessment

A. Environmental Assessment Conceptual Framework

1. The compilation and description of the different proceedings, practices and normative environmental instruments, among other documents, was required so that once they are part of one document they can serve as the Conceptual Framework for the environmental evaluation of road projects financed by the World Bank, BIRF, in Paraguay, specially for the project components on Improvement, Management and Maintenance of the Road Network in Paraguay.

2. This Conceptual Framework includes the proceedings that are followed for obtaining environmental licenses and to carry out environmental impact evaluations for road projects as the one being carried out as part of this Project.

Objectives

3. The general objective of this Conceptual Framework is to develop adequate proceedings that will allow the obtaining of Environmental Licenses and other normative requirements for the execution of the Project, in accordance with the environmental policies of Paraguay and the World Bank.

4. Other more specific objectives include: (i) to generate mechanism that can be replicated for obtaining Environmental Licenses for road segments or group of segments of the Project, as it may be required; (ii) to guarantee the compliance with normative and administrative environmental proceedings during project execution; and (iii) to comply with the administrative requirements of the World Bank.

Project Description

5. This section describes the Project in general and its components 2 and 3 in particular.

6. The MOPC, with the technical and financial assistance of the World Bank, is developing a series of studies for the preparation of the Project on Improvement, Management and Maintenance of the Road Network in Paraguay. The Project's objectives are as follows:

(1) The Institutional Strengthening component's objective is to train and provide the MOPC with the necessary tools for the planning and management of the National Road Network;

(2) The objective of the component on Improvement and conservation of the paved road network is to prevent the deterioration and improve the level of service to users of this network; this network is composed of approximately 2,000 kilometers of paved roads of international and regional integration (Routes 1, 2, 3, 4, 5, 6, 7 and 8);

(3) The objective of the component on Improvement and conservation of the unpaved road network is to improve the access of the poorest rural communities to the national paved road network in three Departments of Paraguay, for an effective economic and social integration of these communities. This will be done through the improvement and maintenance of unpaved national and departmental roads, benefiting those areas with the highest needs in terms of employment.

The overall project's development objective is to establish a sustainable road management strategy that ensures the upgrading and maintenance of the road network through a rational and balanced use of scarce resources. The guiding principle of this approach will be improved road conservation within a comprehensive vision that ensures the sustainability of road investments and the preservation of existing road assets.

Component 2: Paved Road Network

7. In the component of paved road network, the Project will finance the routine and periodic maintenance of the carriageway and shoulder pavement, the pavement marking and vertical signs, the outfall and the public dominion strip of the main national road network sections through performance-based management and maintenance contracts. These contracts will be able to include a variety of activities to increase and to maintain the level of service according to standards established by the MOPC in bidding documents.

8. This new contract modality is oriented to ensure a permanent quality service for users; in such contracts it will be able to include a variety of activities to increase the level of service according to standards established by the MOPC, and in which the contractor is obliged to do maintenance work, such as the pavements recoating and patching, cleaning of culverts, pavements painting, replacement of markings/ signs, removal of excesses of vegetation, removal of waste from the strips, etc. Also they will include works to improve the security conditions for neighbors and users (improvement of junctions, construction of docks for buses parking spots, etc.).

9. This proposal is expected to improve comfort and security conditions of the paved network in Paraguay, which will reduce the accidental rate; simultaneously, it will reduce the transport costs that will decrease the cost on internal consumer products and improve the competitiveness of export products.

10. The propose sections of the road network for performance-based maintenance contracts are:

Table A10.A-1
Paved Road Network included in the Project

SEGMENTS	LONG.
PY01: Asunción (Cuatro mojones) – Encarnación	381 km
PY02: Asunción (Distrito MOPC) – Caaguazú	173 km
PY03: Asunción (Limpio) – Salto del Guairá	411 km
PY04: Pilar – San Ignacio	142 km
PY05: Concepción – Pedro Juan Caballero	214 km
PY06: Encarnación – Empalme Ruta PY02	250 km
PY07: Ciudad del Este – Empalme Ruta Py03 (Carumbey)	153 km
PY08: Coronel Oviedo – Empalme Ruta 05 (Yby Yau)	287 km
TOTAL	**2011 km**

Source: Road Serviceability Direction

11. These sections were grouped in performance-based management and maintenance contracts (GMANS) described below:

Table A10.A-2.
Paved Road Network by GMANS Contracts

GMANS Contract	Segment	Length (Km)
I	PY 02: Caacupé – Inicio concesión Tapé Porá PY 03: Limpio – Empalme PY 08 PY 08: Empalme PY 03 – Coronel Oviedo	381
II	South Access: Desvío a Guarambaré - Itá PY 01: Itá - Encarnación	337
III	PY 06: Encarnación – Empalme PY 07	250
IV	PY 08: Empalme PY 03 – Yby Yau PY 05: Pedro Juan Caballero - Concepción	393
V	PY4: San Ignacio- Pilar	142
VI	PY 07: Hernandarias – Empalme PY 03 PY 03: Empalme PY 08 – Saltos del Guaira	400
URBAN	PY 02: San Lorenzo – Caacupé Acceso Sur: 4 Mojones – Desvío a Guarambaré	108
Total		2,011

12. All the selected segments of the paved network have many years of use. Regarding the environmental impacts it is possible to state at this stage that the project will not cause any significant indirect environmental liabilities on the natural resources or the concerned populations that could not be reverted or easily mitigated.

13. Additionally, the Environmental Unit of the MOPC has General Environmental Technical Specifications (ETAGs) that were approved by the Secretariat of the Environment (SEAM), and by the MOPC, the purpose of which is the protection of the environment throughout all road networks in the country. These ETAGs must be taken into consideration by the consultants, road works and maintenance contractors, and inspection consultants of all types of the road projects. The ETAGs are used in the bidding documents of the contractors, which contain the guidelines for the direct environmental management of the works to be executed.

14. Particular Technical Environmental Specifications, where they exist, must be added to the ETAGs.

Component 3: Unpaved Road Network

15. The component on unpaved national and departmental roads tries to take care of the primary and secondary unpaved road networks, which do not currently have any improvement or maintenance program in execution, with the elaboration of a Master Plan for each beneficiary department and the financing of execution of improvement and maintenance work in these road networks.

16. The program scope includes the following:

- Improvement of the unpaved road network to solve problems that prevent the transit throughout the year, ensuring access to markets and other public services (education, health, etc.)

- Maintenance works to ensure the circulation quality on unpaved roads during the duration of the project, strengthening the departments' and municipalities' maintenance capacities in the long term.
- Training Programs for local populations on road improvement and maintenance, as well as on use of equipment and road machinery for maintenance programs.
- Creation of the procurement mechanisms for the execution of civil works and maintenance works that are significant opportunities of employment for local population.

17. The Project is being prepared through participatory mechanisms which are guided by the following principles:

- Participatory mechanisms include dissemination of information and consultations (binding and non-binding) for the selection of roads, financing and the execution of works.

- Participation instances are open to the public at the national and local level, including public institutions, civil society organizations, and private sector.

- Participatory processes are based on objective data and technical knowledge that are available in a comprehensible and accessible language for all stakeholders.

- Promotion of transparency and development of anticorruption mechanisms.

18. The expected results are:

- An improvement in the quality of the roads leading to a significant circulation improvement throughout the year.

- An improvement of the rural communities' access to the production, trade, education, and health centers, etc.

- An improvement of the access to trade centers of the local population products, with a consequent reduction in the transport costs and a profit increase for workers and producers.

- An improvement in the education of the population and new stable sources of work.

19. The sections to be benefited through the program will be chosen and prioritized according to the following criteria:

(i) Eligibility: main unpaved roads and access roads to rural populations with the communitarian ratification.

(ii) Prioritization in relation to: (a) benefited population, (b) benefited agricultural production, (c) commercial and industrial movement, (d) vehicular transit, (e) health and education access, etc., (f) access to paved roads and (g) local contribution.

20. The MOPC's Preparation and Management Unit of the Project made a pre-selection of sections to be included in the improvement and maintenance of the unpaved road network program of the department of San Pedro, Caaguazú and Caazapá.

21. The criteria adopted by the MOPC to select the segments were:

- Technical aspects of the segments.

- Affected population.
- Access to main highways.
- Road traffic.

22. The road networks by department are as follows:

- SAN PEDRO: Total Road Network of the Department has a length of 7,317.65 km, where 333.50 km are paved, 231,06 are graveled, and 6,753, 09 km are earth roads.

- CAAGUAZU: Total Road Network of the Department has a length of 4,993.79 km, where 314.65 km are paved, 437.80 are graveled, and 4241.34 km are earth roads.

- CAAZAPA: Total Road Network of the Department has a length of 3,458.22 km, of which 26.01 km are paved, 240.53 km are graveled, and 3,186.34 km are earth roads.

Institutional and Organizational Framework

23. Within Paraguay's public administration, the environmental management of road projects is under the responsibility of the Environmental Unit (UA for its acronym in Spanish) of the MOPC, which is, more specifically, under the Vice-Ministry of Public Works and Communications, according to the governmental structure.

24. The UA's duties, which are specified in Ministerial Resolution No. 991 dated August 21 2000, include: (i) supervision of the process of execution, supervision and control of the environmental quality of road projects (through reports, and works fiscalization); (ii) coordination together with the MOPC of the execution and elaboration of bidding and procurement documents for road projects; (iii) environmental supervision of MOPC's public works as stated in Law No. 1533/2000 and Res. 991 mentioned above; (iv) participation in the elaboration of Terms of References for Studies including Environmental Impact, Design, Procurement, in addition to bidding documents, in road projects.

25. The legal framework includes:

1. Paraguay's National Constitution, Articles 6, 7, 8, Chapter V on Indigenous Peoples; Art. 6 on Ethnic Identity; and Art. 64 on Communitarian Property.
2. Law No. 294/93 on Evaluation of Environmental Impact, as modified by Law 345/94.
3. Decree No. 14,281/96, reglamenting Law No. 294/93.
4. Law No. 1,561/2000 on the Creation of the National System for the Environment and the National Council for the Environment, and the Secretariat for the Environment.
5. Law No. 716/96 on Penalization of Environmental Crimes.
6. Law No. 1,100/97 on Noise Pollution.
7. Law No. 426/94 establishing the Constitution of Departmental Government.
8. Municipal Organic Law No. 1,294/87.
9. Paraguay's "*Reglamento General Técnico de Seguridad, Higiene y Medicina en el Trabajo del Ministerio de Justicia y Trabajo*".
10. Municipal norms of the targeted municipalities.

B. Environmental Assessment

26. The Environmental Assessment of the Project was carried out pursuant to the guidelines of the environmental conceptual framework project and the guidelines of the World Bank.

This assessment qualified the project into category B, because its possible environmental impacts on human populations or on main ecological zones are minimal. These impacts are specific in relation to the space and places they affect; practically all of them are reversible; and in most cases mitigating measures can be implemented more easily than in projects of category A.

27. The assessment must examine possible negative and positive environmental impacts and must recommend necessary measures to prevent, reduce to the minimum, mitigate or compensate adverse repercussions, and improve the performance of the project from an environmental perspective.

Reasons for B Categorization

28. The proposed Project involves rehabilitation and maintenance works in existing roads, and as such no major environmental impacts that could endanger the natural surroundings of the area of influence, or the local population is expected. Due to the size and quantities involved in the works to be executed, no significant negative impacts are expected that might not be easily reverted or mitigated.

29. Additionally, due to the raising of environmental liabilities, it is expected that the project will help eliminate existing environmental impacts not attributable to the project, such as problems of erosion of accesses' sediments, canalization of entries and exits of draining works, among others.

30. The operation proceedings that the UA require for the development of construction activities, such as inspection and supervision of works are considered adequate for the modalities proposed in the routine improvement and maintenance of unpaved roads and the performance-based maintenance of paved roads.

31. However, these proceedings can be improved through the implementation of measures that can contribute to the institutional strengthening of the UA, as described below.

Important Environmental Aspects of the Project

32. The project is located in the eastern side of the country, both for the paved and unpaved road components. Even if this area presents the highest levels of rainfalls in the country (an average of 1500 mm/year), no major negative effects are expected to take place while improving and maintaining main and secondary roads.

33. The country presents a hilly topography, with some areas with high phreatic levels. The former is ideal for road construction, where few slope erosion and destabilization problems are seen. The project will undertake some of these activities.

34. The road sections located in the high phreatic level zones have been constructed over backfills which do not present erosion or sediment problems, mainly because no significant rivers or streams traverse them. This project will provide road maintenance in this area; therefore, no adverse effects are foreseen.

Targeted Areas

35. In order to provide a detailed description of the environmental effects and its socioeconomic repercussions, an account of the direct and indirect effects for both project components (National Paved Road Network and National and Departmental Unpaved Road Network) is provided.

Area of Direct Influence (AID)

36. The area of direct influence is determined by the following aspects: right of way, borrow pit zones, facilities spots (camping and workforce areas); works interference with surface or underground springs, interference with protected areas, expropriation zones, crossing of water beds, link to other road systems, hindrance with wildlife migration corridors.

37. A 500m strip to each side of the road axis has been adopted as the area of direct influence.

Area of Indirect Influence (AII)

38. The area of indirect influence is considered to be the one modified by the Road Project in its immediate geographic surroundings.

39. In order to determine the area of indirect influence not only social and economic efficiency criteria are being considered; topography, land and fertility distribution, climate change and its effects in the agricultural production, the presence or the projected existence of transport links near the road (competition), and other local factors are also taken into account.

40. Hence, the area of influence is mainly determined by the interrelation of the following set of factors:

- Topography and hydrography.
- Existing rural roads network nearby the analyzed road.
- Distance between the agricultural productive zones and the local markets.
- Modes of transport commonly used: animal's freight, animals of cargo, carts by animal traction, agricultural trucks, non-agricultural trucks, cars and buses.
- Origin and destiny axes for the road most frequent means of transport.

National Paved Road Network

41. The project is developed in the eastern region of the country, and includes communities in the following departments: Central, Cordillera, Caaguazú, San Pedro, Paraguari, Misiones, Ñeembucú, Itapúa, Alto Paraná, Canindeyú, Amambay and Concepción.

Area of Indirect Influence (AII)

42. It is expected that, due to the implementation of the National Road Network Component, land use, production costs, income, cost and distribution commercialization systems, and transport costs conditions, will not be altered.

43. In addition, it is expected that the above mentioned conditions will not be greatly affected, since these effects have already taken place, and because the road maintenance services will guarantee the provision of more efficient transport services.

National Unpaved Road Network

44. The Project is implemented in the eastern region of Paraguay, and includes the following Departments: San Pedro, Caaguazú and Caazapá.

Area of Indirect Influence (AII)

45. It is expected that, due to the implementation of the National and Departmental Unpaved Roads Component, land use, production costs, income, cost and distribution commercialization systems, and transport costs conditions, will be positively altered.

Alternatives

46. The design alternatives for the unpaved roads component are the ones proposed by the consultant company ITYAC and approved by the MOPC, which include environmental aspects, following the normative stated in the ETAGs for its final approval.

47. By the nature of the project in its paved road network maintenance component, it will be bidded in such way that the final design of the project will be done by the contractor, according to the level of service that is determined like optimal. For that reason, the proposal elected as winner must be the most efficient in terms of the activities to be carried out and the environmental costs.

Identification of Impacts and Measures of Environmental Mitigation

48. The World Bank's guidelines were followed for both components. Please see annexes on the Mitigation Plan for Environmental Liabilities kept in the Project's file.

49. The following Mitigation Measures were included in such plan:

- The first measures to be taken into account must be included in the ETAGs indicated by the SEAM, to which the contractor must become committed in the PASA and the Inspector by the Control and Follow-up Plan.
- The program to repair environmental liabilities is part of the mitigation measures, and includes measures of control of accesses' sediment erosion, cleaning of entry and exit canals of drain structures, re-vegetation, and complimenting drainage works.
- A strong Plan of Environmental Quality Control must be followed, which should take into account resources such as water, air, and soil, and which should include noise control, the prevention of proliferation of vectors transmitting diseases in stagnated waters, adequate management and disposal of pollutants.
- This Environmental Quality Control Plan should also take into account the management of oils, paintings, additives, or any other material kept in construction camps, for machinery, etc, and solids wastes and polluted waters from the contractors' and inspectors' camps.
- An Erosion Control plan defining measures of control of temporal and permanent erosion, structural and non-structural erosion, and vegetative erosion, throughout the project, and in borrowing sites, and waste disposal.
- Other plans to be included are those on re-vegetation for stabilization of soil movements, on fauna protection, on visual quality control, and on contingencies related to the use and transport of inflammable materials, explosives, etc with efficient channels of communication.
- Finally, there should be a plan for complete withdrawal from the project's corridor and camps, which should be returned clean and to the satisfaction of the inspector and supervisor.

C. Institutional Capacity

50. In accordance with the World Bank's guidelines, an evaluation of the institutional capacity of the UA of the MOPC was carried out as part of Project preparation. The following strengths and weaknesses were identified:

Table A10.A-3
Environmental Unit's Institutional Strengths and Weaknesses

STRENGTHS	WEAKNESSES
Participation in the cycle of the road projects with various modalities in recent years	It does not actively participate in the design of environmental road policies.
I was officially established by the MOPC and SEAM	Its influence in the MOPC is limited compared to other agencies.
It is responsible for the operative aspect of the inter-institutional agreement between MOPC and SEAM	It does not have enough staff or equipment.
It has an institutional environmental policy, although with scarce resources to implement it.	Its internal proceedings are not optimal in relation to the importance of this issue.
It has the ETAGs for road projects, elaborated in 2004.	It does not have enough budget, especially for trips.
Part of its staff has experience with WB's environmental policies.	Staff does not have enough incentives, especially due to lack of resources and equipment.
It staff has experience in evaluation of environmental impact.	It does not have a manual for road projects.
Its staff has participated in training programs on environmental protection.	It does not have experience in economic and financial analysis of road projects.
Its staff has participated in training programs on environmental evaluation and management, use of ETAGs, etc.	It does not have experience in projects such as the elaboration of a national road project.
Since 2005, it has incorporated mechanisms for communities' participation.	There is no well defined sector policy, nor staff in charge of relations with the public, or to deal with resettlement issues.
It has consultants for the policy framework on indigenous communities.	There is no enough staff to deal with this issue.
It has the capacity to deal with Plans for Socio-environmental Action (PASA) and Plans on Inspection and Control (PAVA)	There is a need for training on complementing and mitigating works' design, monitoring programs of water, air, soil and noise quality.
It has mechanisms for carrying out dissemination workshops for the public.	There are no specific policies or proceedings on resettlement projects.
There is a national environmental policy.	Few resources to be implemented as far as the UA is concerned.
The civil liability derived from environmental risks in under the responsibility of contractors and inspectors.	There is yet no major experience on policy or plans for contingency or national emergency for the prevention of damage deriving from natural disasters.
Each road project includes at least a PASA and a PAVA. Those projects requiring EIA include their respective Environmental Management Plan (PGA).	The process of environmental bidding is not yet optimal.

51. Following the identification of the strengths and weaknesses of the UA, the Task team defined the objectives, strategies, policies and measures to be implemented in the short, medium and long term in order to strengthen the institutional capacity of the UA at the MOPC.

Table A10.A-3
Institutional Strengthening of the Environmental Unit's Institutional Strengths and Weaknesses

OBJECTIVE	STRATEGIES	POLICIES	MEASURES
To improve the institutional capacity of the UA, for the purposes of this project, and other projects that may derive from this one.	• Institutional strengthening of the UA. • To reinforce the communication channels between the UA and the other agencies involved in the Project. • To optimize the environmental licensing process. • To improve the documents, processes, procedures, and mechanisms currently used to optimize the UA's performance.	• To establish permanent training programs to improve UA's management and operative capacity. • To strengthen the UA's managerial and administrative capacity. • To select those topics that generate significant impacts on the environmental management and administration of road projects. • To promote the ISO 14,000 certification of the different companies that will participate in the public biddings. • To implement programs on transfer of technology with other UAs and related agencies at the international level. • To promote the participation of the UA in the National Contingency Plans. • To develop work methodologies for the evaluation, supervision and auditing of road projects.	• To design an Environmental Manual for Road Projects, compiling al the information on the subject, improving the ETAGs and other current environmental procedures, mechanisms, and regulation, among other aspects. • To generate training programs, with this Manual as basis, on topics such as: (i) environmental impact evaluation; (ii) roads' environmental and landscape design; (iii) design of complementary works; (iv) mapping; (v) environmental audits; (vi) environmental quality control; (vii) geographic information systems, especially in relation to territorial management and protected areas; (viii) design of road projects that are environmentally apt; (ix) environmental economic and financial analysis of road projects; (x) socio-communitarian evaluation and management, including indigenous groups and historical heritage. • To optimize the UA's internal managerial and administrative procedures. • To establish a network of staff trained on environmental aspects located in the headquarters of the district's departmental maintenance. • To obtain electronic mail, and design a website in order to disseminate information on the UA and the project the UA is involved in. • To strengthen the mechanisms for collaboration between the UA and the Real State Unit to deal with the issues related to the individual and businesses located in the dominion's line. • To establish mechanisms for collaboration and information exchanges between the UA and the Technical Planning Secretariat (STP), the Social Action Secretariat (SAS), the National Tourism Secretariat, and Paraguay's National Institute for Indigenous People (INDI). • To promote modifications to the Environmental Impact Law No. 294/93 to facilitate the environmental licensing process of road projects. • To promote the presence of the UA at international events on environmental topics related to road projects. • To obtain the necessary equipment for the UA to adequately perform its duties.

Annex 10.B – Indigenous Peoples Plan

A. Background

1. The Indigenous Peoples Plan (IPP) was designed with the specific objective of effectively promoting the participation of indigenous peoples in the Program for the Improvement, Management and Maintenance of Paraguay's Road Network which will be carried out by the MOPC. This IPP identifies the way in which the Project can respond to some of the concerns raised by the indigenous peoples located in the intervention area. The IPP was designed in collaboration with representatives of the indigenous communities, indigenous associations, and non-governmental organizations working with indigenous peoples, in the Departments where the Project will be implemented, namely the Departments of Caazapá, Caaguazú and San Pedro.

2. Based on consultations with said indigenous associations and non-governmental organizations, the IPP identifies several priority topic areas in relation to the indigenous communities located in the Project's area of intervention, with issues related to training and organizational strengthening of the communities highlighted as prior steps to be taken in order to assure an active participation of the indigenous communities in the programs to be implemented by the MOPC in the region.

3. One of the basic principles of the IPP is to work with the beneficiary indigenous communities through participatory mechanisms, respecting the indigenous development process based on the principle of cultural pluralism and respect for the environment.

4. There are 98 communities that have been identified as beneficiary communities of the Project in the Social Evaluation (*see separate annex*) because of their proximity to the Project's targeted road network. However, the access of each of these communities to the different types of support that will be provided by the IPP will be differentiated and established at the Evaluation and Diagnosis Workshops that will be held annually in each of the three Departments involved in the Project, at the beginning of each year.

5. At these workshops the indigenous leaders will carry out an analysis and evaluation of their communities' and department's situation and will establish their own priorities in an Annual Action Plan. These Annual Action Plans will be consulted, socialized and designed through collaborative and participatory mechanisms by the communities and indigenous associations involved, and the technical team in charge of carrying out the IPP, in order to assure that all those involved in the Plan have clear and informed criteria for the selection of the participants of the training programs. These workshops will provide the necessary inputs for a participatory evaluation of the IPP and will allow for necessary adjustments of the IPP in accordance with the needs of the stakeholders.

B. Targeted Indigenous Communities

6. According to Paraguay's 2002 National Census of Indigenous Population, there are approximately 12,230 indigenous people in the three Departments involved in the Project, out of a country total of 85,674 indigenous people (1.5% of the country's total population). This departmental indigenous population occupies 2,512 households, and is distributed in 98 communities in the three Departments (See Table A10.B-1). These communities belong to the following ethnic indigenous groups, all of which are part of the Guarani Linguistic Family: the Mbya, the Avá Guaraní, and the Ache.

Table A10.B-1
Indigenous Communities in the Departments of
San Pedro, Caaguazú y Caazapá

Departments	No. of Communities	Households	Population		
			Men	Women	Total
San Pedro	26	562	1,438	1,324	2,762
Caaguazú	47	1,468	3,674	3,251	6,925
Caazapá	25	482	1,311	1,233	2,544

Source: *Dirección General de Estadística, Encuestas y Censos, II Censo Nacional Indígena de Población y Viviendas 2002.*

7. These indigenous communities live from a combination of economic activities that include subsistence agriculture, hunting, fishing and fruit collection, handcrafts, and occasional wage labor (see Table A10.B-2). It should be noted that the land occupied by these communities does no longer provide enough natural resources (due to deforestation and environmental degradation) for these communities to be able to subsist, which means that they need to resort to wage labor to survive.

Table A10.B-2
Indigenous Communities' Economic Activities per Department

SAN PEDRO	CAAZAPÁ	CAAGUAZÚ
26 communities Hunting: 92.3% Fruit Collection: 92.3% Handcrafts 73.1% Agriculture: 90.8% Agricultural wage labor: 10.3%	25 communities Hunting 96% Fruit Collection: 100% Handcrafts 92% Agriculture: 96.3% Agricultural wage labor: 1.8 %	47 communities Hunting: 91% Fruit Collection: 89.4% Handcrafts: 74.5% Agriculture: 90.3%. Agricultural wage labor: 7.6%

Author's Note: The data on the first three activities (hunting, fruit collection, and handcrafts) were obtained from the Community Questionnaire of the 2002 Indigenous Population Census. The others were obtained from the Individual Questionnaire.
Source: Author's calculation, based on the 2002 Indigenous Population Census, DGEEC: 2003.

8. Indigenous communities are amongst the poorest social groups of Paraguayan society, enduring persistent problems and deprivation such as a lack of adequate health assistance (only 26% of the indigenous communities in the country have access to a health center); high infant mortality (93.9 per thousand children); high analphabetism (94.1%); and 93% of these communities' households have no access to adequate sanitary installations or potable water.

9. The majority of the 98 communities that will participate in the IPP are located in the proximity of the targeted road segments (1 km to 5 km). The right of these communities to the land they occupy is not always protected by appropriate titles and registration (see Table A10.B-3), which worsens their precarious situation.

Table A10.B-3
Situation of Land Tenancy in the Targeted Departments

Department	Total of Communities	Communities without land property titles	Communities with land property titles
Caaguazú	47	40	7
San Pedro	26	9	17
Caazapá	25	6	19
Total	98	55	43

Source: 2002 National Census of Indigenous Population

10. These communities, like the poor farmers that live nearby, live in a situation of extreme poverty and environmental degradation. Considering the objectives of the Project, and the fact that there will be no construction of new roads, but rather the improvement of existing roads, no negative impact on these indigenous communities is expected. On the contrary, the Project, through its training programs, will benefit these communities, which are amongst the most vulnerable social groups in the country, responding to the specific demands identified by these communities themselves.

C. IPP's Objectives

11. The general objective of the IPP is the inclusion of the indigenous communities in the programs of execution of maintenance works in the rural road network, within a framework of equality and respect. As a result of this inclusion, the Plan aims at achieving a sustainable improvement of the quality of life of the targeted indigenous communities, through the strengthening of the organizational and managerial capacity of these communities, training their members and generating employment.

12. In order to increase the access of the beneficiary indigenous communities to information about the Project, the IPP will work with the technical human resources, the Plan's direct beneficiaries, as well as with the institutional human resources, in order to facilitate a change in attitude needed for the beneficiaries to gain full awareness and appropriation of their own situation, increasing their capacity to generate changes as part of a self-management process.

13. The IPP is expected to achieve a full integration of capacities from both the public and private sector, beyond a simple complementation of resources, based on the premise that the accumulation of their strengths can constitute the road towards development of a relegated segment of the population. In this line, the IPP proposes an inclusive and participatory process, that makes use of the MOPC's technical capacities, complemented by the capacities and experience of the NGOs, churches, and foundations working in this field, generating a positive transformation of the stakeholders involved in the process, and establishing the beneficiaries' demand as one of the central axis of the IPP, as a way of promoting, strengthening and consolidating the structures of the local and institutional organizations involved.

14. The IPP will try to comply with the following schedule and aims:

Table A10.B-4
IPP Activities

Activities/Year	1	2	3	4	5	6	Total
Communities' Analysis and Diagnosis, and elaboration of baseline data	98						98
Implementation of Annual Evaluation and Diagnosis Workshops	3	3	3	3	3	3	18
Construction and equipping activities of the Training Centers	3						3
Training Activities*		15	15	15	15	15	75
Organizational Strengthening Activities**		15	15	15	15	15	75

* The estimation is of a total of five training workshops per year with a total number of 45 trainees per workshops, totalizing 1,500 trainees from the indigenous communities at the end of the five-year period.
** The estimation is of a total of five training workshops per year, with a total of 150 indigenous leaders trained at the end of the five-year period.

Activities to be carried out as part of the IPP

15. The IPP has identified, based on consultations with indigenous communities and their representatives, different types of activities to be carried out as part of the Plan:

(1) Training programs in topics related to the rehabilitation and maintenance of roads, including maintenance of road equipment, which can serve as sources of employment and income, for at least 1,500 young indigenous people.

Some of topics to be included in these training programs that were identified during the Communities' Analysis and Diagnosis, and the Annual Evaluation and Diagnosis Workshops, are: (i) training in road maintenance, including general mechanics, carpentry, protection walls construction, use of machinery and equipment, etc.; (ii) training on environmental protection; (iii) training on information technology and PC skills, and secretarial skills, aimed at young people that could then work for the indigenous associations and communities; etc.

(2) Activities aimed at the organizational strengthening of indigenous associations, including training of at leas 150 leaders (men and women) members of said associations.

Some of the topics to be included in these training activities that were identified during the Communities' Analysis and Diagnosis, and the Annual Evaluation and Diagnosis Workshops are: (i) support for the creation and legalization of indigenous associations, including training on basic accountability, budget preparation, projects' and associations' management and administration, etc; (ii) training on citizenship, including courses on the functioning and organizational structure of Paraguay's Government (at the national, departmental and local levels) and society, duties of each agency in the government and ways to have access to the services they render; (iii) training on how to establish alliances and cooperation with governmental and non-governmental organizations; (iv) training on conflict resolution; etc.

These training activities will be aimed at providing indigenous communities with sufficient information on Paraguayan society in order to favor a more fluid relationship between them.

(3) Construction, installation and maintenance of three Centers for Indigenous Peoples' Training, located within an indigenous community, one in each of the three Departments included in the Project.

These centers will offer the appropriate space for the technical training programs that will be carried out as part of this IPP, but will also serve as centers for training on other areas of interest such as health, education, agro-forestry, etc. The centers will be built in indigenous communities to be selected by the indigenous associations and beneficiary communities and the technical team working in the IPP.

The maintenance of the centers and their installations will be under the responsibility of the indigenous associations in collaboration with the NGOs working in these communities. To this purpose, the indigenous associations and NGOs will enter into an Administration and Maintenance Agreement, where the rights and duties of each party will be established.

The expected schedule for this component of the IPP is as follows:

Table A10.B-5
IPP Activities Timeframe and Responsible Agency

Activity	Timeframe	Responsibilty for Activity
1. Preparation of blueprints and designs, and detailed budget	Month 1	MOPC
2. Acquisition of budget for materials	Month 2	MOPC
3. Selection of contractor	Month 2	MOPC
4. Acquisition of Materials	Month 3	MOPC
5. Construction of works	Months 3-6	Contractor
6. Inspection of works	Months 3-6	MOPC/Indigenous Associations
7. Reception of Works	Month 7	MOPC
8. Purchase of furniture and equipment	Month 7	MOPC
9. Installation, reception and inventory of furniture and equipment	Month 7	MOPC
10. Drafting of Administration and Maintenance Agreement and Centers' Operation Statutes	Month 7	MOPC, Indigenous Associations and NGOs
11. Maintenance of Centers	6 years	Indigenous Associations amd NGOs

To contribute to the achievement of the IPP's objectives, the training activities will be aimed at establishing a transformative educational process that is appropriate to the cultural characteristics of the indigenous peoples involved, using adequate methodologies and contents to this effect. The program will be guided by the following principles: (i) training should be participatory, and should be carried out in an environment adequate to the indigenous culture; (ii) the need to respect the indigenous communitarian tradition according to which no one is educated on his or her own, but only through a shared experience, and as part of an interrelationship with others; (iii) the need to use technical materials and dissemination materials that are adequate to the indigenous culture; (iv) and the need to respect traditional indigenous leadership structures, their traditional religion, and their cultural characteristics.

These activities will be carried out establishing alliances and partnerships with other projects and institutions with similar objectives, in order to coordinate and unify technical criteria, complement activities, and avoid duplication of efforts. To this effect, the team in charge of the IPP has already established contact with NGOs, churches and foundations working in this field in the three Departments targeted by the Project.

D. Institutional Framework

16. The MOPC will be the governmental agency responsible for the execution of the IPP with respect to the (i) training activities for young people and leaders; (ii) construction and equipping activities of the Training Centers; and (iii) the inclusion activities for the indigenous communities in the programs for road maintenance.

17. The IPP's Coordination Unit will be part of the Environmental Unit at the MOPC, and will use its offices and equipment for its operation. It is expected that the MOPC will designate a Coordinator for the IPP, who will be specialized in social sciences, and at least a technician for the training activities, who will be in charge of the planning and execution of the communities' diagnosis and the coordination and planning of the training workshops.

18. The IPP will have enough flexibility to establish alliances and partnerships with local and regional governments, as well as NGOs working in the area that might have more experience working with indigenous communities. In particular, the Plan will seek to coordinate its activities with the *Instituto Nacional del Indigena* (INDI), which pursuant to Law No. 904/81 is in charge of all issues related to indigenous communities, keeping this agency informed of all the activities carried out as part of this Plan. The Plan will also cooperate with the indigenous associations themselves, and with other projects and programs being carried out with indigenous peoples in Paraguay.

E. Monitoring and Evaluation

19. All the personnel and organizations working in activities pursuant to the IPP will be trained in basic monitoring and evaluation techniques, and will be in charge if presenting reports on the implementation of the IPP on a regular basis to the IPPs Coordination Unit.

20. An independent evaluation of the IPP's activities will be carried out at mid-term and at the end of the Project.

21. The Plan will be periodically evaluated and monitored, including: (i) periodic evaluations of the IPP's implementation; (ii) financial evaluations, to ensure appropriate adherence to the Project's financial and flux of funds regulations; (iii) continuous monitoring of indigenous participation at all levels of the plan; (iv) evaluation of personnel working in the IPP; (v) evaluation of the didactic material used in the training activities to ensure its adequacy to the indigenous cultural characteristics; and (vi) baseline data, socio-economic indicators, socio-organizational indicators; etc.

F. Budget

Table A10.B-6
IPP Budget

IPP's Budget							
Date	31-March-2006					Component's Cost	
Proposing Agency	MOPC						
Activity		**Unit**	**Quantity**	**Cost by Unity**	**Total Cost**	**Cost MOPC**	**Project's Cost**
Construction and Equipping Activities for 3 Training Centers (Project's responsibility)							
(A) Construction							
3 Training Centers							
3 Multi-use Room/Dining Room/Kitchen		m2	180	100	18,000		18,000
6 Bedrooms		m2	430	100	43,000		43,000
6 Bathrooms		global	6	2,500	15,000		15,000
Equipment		global	3	5,000	15,000		15,000
3 artesian wells		global	3	5,000	15,000		15,000
Subtotal Activity (A)					**106,000**	**0**	**106,000**
(B) Training Activities							
Trainer's Fees		Days	125	75	9,375		9,375
Trainer's expenses	Food, transport, insurance	Days	125	25	3,125		3,125
Participants' expenses	Food, transport, materials	People	1500	30	45,000		45,000
Sub total Activity (B)					**57,500**	**0**	**57,500**
(C) IPP's Coordination (MOPC's and Project's responsibility)							
C 1. Services							
Personnel							
Plan's Coordinator		Months	72	1500	108,000	108,000	
Anthropologist Consultancy		Months	57	1000	57,000		57,000
Per Diem		Days	570	20	11,400		11,400
Sub Total C1					176,400	108,000	68,400
C 2 Goods							
Central Office Equipment							
IT equipment (PC, printer)		set	1	1500	1,500	1,500	
Software (Office, Adobe, ArcView)		set	1	800	800	800	
GPS		Unit	1	350	350	350	
Furniture							
Desk, chairs, bookcase		set	1	500	500	500	
Office Materials		set	1	500	500	500	
Subtotal C 2					3,650	3,650	0

C 3 Other operative costs							
Use of material		Months	72	100	7,200	7,200	
Transport, maintenance, insurance		Months	57	500	28,500		28,500
Communication/internet		Months	72	100	7,200	7,200	
Sub total C 3					42,900	14,400	28,500
Total costo Activity (C)					**222,950**	**126,050**	**96,900**
IPP TOTAL COST					**386,450**	**126,050**	**260,400**
Total Cost for MOPC						**126,050**	
Total Cost for Project							**260,400**

Annex 10.C – Involuntary Resettlement Framework and Resettlement Action Plan

A. Background

1. The Ministry of Public Works (MOPC) prepared a Resettlement Action Plan (RAP) for the highway segments for which maintenance contracts will be let during Year I and the project and a Resettlement Framework (RF) for subsequent highway segments. The RF carefully balances the need to provide adequate highway maintenance and safety with the principle of minimizing the amount of involuntary resettlement and disruption to peoples' lives. As described above, the proposed project would support the management, improvement and maintenance of critical segments of Paraguay's national highway system including both paved highways and unpaved, gravel roads. The RF and RAP concern only the paved highways because there are no resettlement issues on the unpaved highways that would be maintained during the project.

B. Resettlement Framework:

2. The Resettlement Framework was prepared in accordance with OP4.12 (Involuntary Resettlement) with assistance from the Bank team and consultants. Since the project involves only maintenance of existing highways, there would be no resettlement of homes or businesses, or taking of land for new highway construction. However, along the paved highways, there are significant numbers of structures – mainly small businesses and a few homes – built on the highway right-of-way, often directly on the highway shoulder to attract passing motorists.

3. These structures range from simple awnings erected for shade to more elaborate structures constructed of wood or, in a few cases of masonry. Most of the existing structures are businesses designed to attract customers for the sale of fruits, vegetables, ornamental plants, snack foods, drinks and other merchandise, or services such as tire repair shops. Some businesses appear to have considerable clienteles, such as a few *chiperias* where a popular snack food is sold. In most cases, there are no parking facilities; vehicles patronizing these businesses stop directly on the highway shoulder, creating a hazard for passing traffic. Most businesses appear to be marginal and many are operated by teenagers or elderly people who may be working to supplement family incomes. Many of these small businesses are temporary, such as seasonal fruit stands. Many of the structures are linked to homes just outside the right-of-way. There are also a few homes located on the rights-of-way (perhaps 5 – 10% of the total), mainly linked to small businesses. There are virtually no stand-alone residences on the highway rights-of-way.

4. MOPC has conducted a preliminary census of structures erected along highway rights of way, classified them according to their use and construction, and has photographed the location of each one. A total of 647 illegal structures was identified. However, the census has not included demographic and social information because MOPC wanted to avoid setting off a speculative surge of new occupants. There are a few locations where there are several businesses clustered in a short stretch of highway, but, in general, occupations are spread out at long intervals along the highway.

5. Paraguayan law (Law 75/69) provides that the Highway Department (*Dirección General de Vialidad* or DGV) of the Ministry of Public Works has exclusive rights and jurisdiction over highway rights-of-way for national, departmental and municipal routes in rural zones. It also authorizes the DGV to issue permits to occupy highway rights-of-way. In practice, such permits have been issued only to allow construction of gates and linkages between private roads and public highways. DGV has not issued permits for private structures or businesses. Therefore, all of the structures built along the highway rights-of-way in Paraguay are technically illegal. This has not stopped some municipalities from issuing

business licenses and collecting taxes on business located on highway rights-of-way or power companies from providing electrical service.

6. Despite the illegality of these structures, MOPC recognizes that many of them have been in place for several years and that they are an important source of income for the occupants. MOPC has therefore taken a policy decision, consistent with Bank Policy, to displace the absolute minimum necessary number of these structures. In most cases, the displacement will involve simply setting the structure back from the highway far enough to allow maintenance works to be conducted or to remove structures that pose a safety risk to their occupants or to highway users.

7. The eligibility criterion adopted by MOPC are either that a structure occupying the right-of-way is (a) an impediment to maintenance works or (b) presents a safety risk. Shortly after maintenance contracts are let, all occupations along the highway segment covered by the contract will be censused and assessed. Occupants of the highway rights-of-way will be formally notified that they are illegally occupying the highway right-of-way and that their presence does not constitute a commitment by MOPC to allow them to continue in place indefinitely. They will further be notified that any construction of new facilities or the consolidation or expansion of existing structures occupying the highway right-of-way will not be permitted and may be subject to immediate removal by the authorities. Pursuant to MOPC's Resolution No. 147/2006, the MOPC has established that March 28, 2006 will be the cut-off date for each highway segment after which no new occupations will be permitted. According to this resolution, the MOPC will conduct a publicity campaign in a timely fashion for each highway segment to advise people of their rights and obligations, and to prevent any speculation by rent-seekers. MOPC will also provide a non-judicial grievance mechanism for occupants who feel that they have not been fairly treated. MOPC will provide a dedicated telephone number as a complaints line.

8. Entitlements will consist mainly of the provision of building materials, vehicles, and construction assistance necessary to set the occupant back further from the highway shoulder. Special provisions will be made for persons judged to be exceptionally vulnerable such as elderly persons, single mothers, invalids, etc. They will be attended to by MOPC social workers who will see to it that their special needs are met. In cases where it is logistically impossible to simply set back the business location, the occupant may be offered another location along the highway, preferably at a point where the right-of-way can accommodate the relocated business. MOPC will consider opening some "rest plazas" where motorists can pull completely off the highway and purchase goods or services. MOPC has decided that, in most cases, cash compensation will not be awarded because every effort will be made to allow people to remain on the highway right-of-way so that they can continue to conduct their businesses. Experience in Paraguay has shown that payment of cash compensation could touch off a speculative rush of rent-seekers to occupy places where they think they can qualify for cash entitlements.

9. For each highway segment included in the project a separate RAP will be prepared. The RF includes a detailed outline for RAPs that includes a census of structures, characterization of construction types, demographic census, estimate of numbers of occupants that would be required to move, the description of mitigation measures to be taken, the budget required including a special reserve for unforeseen events, the responsibilities of each institution or party, a timetable, provisions for monitoring and evaluation, a grievance mechanism, and mechanisms for insuring good communication between the affected parties, MOPC and the contractors.

10. Actual implementation of the resettlement plan will be delegated to contractors selected by competitive bidding. However, the census, special consideration for vulnerable people, project monitoring, grievance procedures, and ex post review will be managed by MOPC itself. The estimated cost of relocating structures off the highway will be incorporated into the cost estimates for highway maintenance and provisions will be made for adjustments in contract values should additional costs be

incurred. The contractors will be supervised by MOPC social workers. The project will be monitored at regular intervals and an evaluation conducted after the conclusion of a regular maintenance cycle.

C. Resettlement Action Plan for Road Sections 2, 3 and 8.

11. These road segments include some 490 km of roads along which a total of 258 illegal occupations were observed. Of these, 21 are homes 227 are businesses and 10 are unoccupied. A total of 419 people are linked to these structures as occupants or operators. A detailed RAP was prepared for the road segments that will be covered by the project during the first year of operation. The RAP applies the framework described above, indicating that the primarily eligibility criteria are to facilitate the maintenance works and to preserve public safety. This RAP involves some 860 km of highway along which there are four structures involving some ten people that meet the eligibility requirements. The RAP identifies the responsibilities involved and provides for a grievance committee to be composed of representatives of the Legal Department, the DGV, and the Project Management Unit of MOPC. The action plan is summarized in the following tables. The RAP also includes a timetable and time limits for each action required as well as provisions for field monitoring and evaluation.

Table A10.C-1

ROUTE #	LOCATION SEGMENT	KM	SIDE R/L	Placement on Right of Way	Use of structure	Observations
PY02	Caacupé - E. Ayala	58.600	R	Drainage ditch	Entrance to a home	
PY02	Caacupé - E. Ayala	64.850	R	Drainage ditch	Business	
PY02	Caacupé - E. Ayala	66.900	L	Shoulder	Business (plant nursery)	
PY02	E. Ayala - Itacurubí	84.900	R	Drainage ditch and embankment	Business	

Table A10.C-2

ROUTE #	LOCATION SEGMENT	KM	SIDE (R/L)	Permanent or seasonal	Construction Materials	Photograph number
PY02	Caacupé - E. Ayala	58.600	R	Permanent	Brick and concrete	DSC00047
PY02	Caacupé - E. Ayala	64.850	R	Permanent	Wood and sheet metal	DSC00056
PY02	Caacupé - E. Ayala	66.900	R	Permanent	Wood slats	DSC00060
PY02	E. Ayala - Itacurubí	84.900	R	Permanent	Sheet Metal	DSC00062

Table A10.C- 3

ROUTE #	LOCATION SEGMENT	KM	SIDE (R/L)	Number of occupants	Comments (vulnerability, etc.)
PY02	Caacupé - E. Ayala	58,600	R	4	
PY02	Caacupé - E. Ayala	64,850	R	0	Abandoned
PY02	Caacupé - E. Ayala	66,900	L	2	
PY02	E. Ayala - Itacurubí	84,900	R	4	

Table A10.C-4

ROUTE #	LOCATION SEGMENT	KM	LADO	Materials and labor required	New location	Comments
PY02	Caacupé - E. Ayala	58.600	D	• 2 5-meter iron pipes 3 man-days		Enlarge drainage section
PY02	Caacupé - E. Ayala	64.850	D	• 2 man-days • Transport	edge of right-of-way	
PY02	Caacupé - E. Ayala	66.900	I	• 2 man-days • Transport	edge of right-of-way	
PY02	E. Ayala - Itacurubí	84.900	D	• Demolition y Reconstruction	edge of right-of-way	

ROUTE #	LOCATION SEGMENT	KM	SIDE	Payment schedule	Budget (Guaranies)
PY02	Caacupé - E. Ayala	58.600	R	within 6 months of contract signing	5,000,000
PY02	Caacupé - E. Ayala	64.850	R	within 6 months of contract signing	3,000,000
PY02	Caacupé - E. Ayala	66.900	L	within 6 months of contract signing	3,000,000
PY02	E. Ayala - Itacurubí	84.900	R	within 6 months of contract signing	10,000,000

Contingencies	within 6 months of contract signing	5.000.000
TOTAL		**G$26.000.000.-**

Annex 11: Infrastructure and Public Spending

PARAGUAY: Road Maintenance

1. The challenge facing Paraguay in the infrastructure sector is how to maintain and expand infrastructure assets to support growth given fiscal constraints. Paraguay's needs in terms of infrastructure are high, both in terms of quality and coverage. A survey of business people rated Paraguay at the bottom of a number of Latin American countries in the level of development and service efficiency of its infrastructure. Only about a quarter of the road network is paved, and over 80 percent of the paved network is in fair or poor condition. The state of the unpaved network is even more precarious, with only 14 percent in good condition.

2. Additional resources to meet the country's large infrastructure gap are required. Fiscal resources are, however, scarce. Debt sustainability analysis and fiscal projections point to the limited fiscal space available to Paraguay to increase spending, notwithstanding the current Administration's efforts at better fiscal management and wide-ranging reforms.[22] In order to maintain the current debt-to-GDP ratio, assuming 3.5 percent GDP growth and an average interest rate of 6 percent (i.e., assuming, as is likely, that interest rates will rise in the near future), Paraguay would need to achieve primary surpluses of about 1.1 percent of GDP per annum. The required surplus would decline to 0.9 percent of GDP if economic growth were to increase to 4 percent, and would increase to 1.8 percent of GDP if growth declined to 2 percent. The Central Government's primary surplus in 2003-04 averaged 1.8 percent of GDP (with GDP growth rates of 3.8 and 4.1 percent, respectively). Therefore, primary surpluses adequate to stabilize or lower the public debt burden are an achievable target given the continuation of policies stressing fiscal prudence and economic growth.[23] However, there is little scope for upward adjustment in government salaries or for increases in social or infrastructure spending without structural increases in revenues.[24]

3. A high level of current expenditures, amounting to 75 percent of total revenues and 137 percent of tax revenues,[25] crowds out needed investments. The wage bill constitutes the bulk of this spending (55 percent), followed by transfers to cover the deficit of the public sector pension fund, the *Caja Fiscal*. In regional terms, Paraguay's wage bill is particularly high. The deficit of the public pension fund, the *Caja Fiscal*, covered by treasury resources, is substantial (1.5 percent of GDP in 2004). Other factors that make it difficult for Paraguay to generate significant primary surpluses include a relatively low tax burden and

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[22] The prospects for fiscal sustainability in Paraguay in the medium term have significantly improved over the past three years as a result of the current Administration's efforts at better fiscal management: tax revenues increased by nearly 3 percentage points of GDP, from 9.2 percent of GDP in 2002 to 11.9 percent in 2005, largely through efforts to improve tax administration; current and total expenditures were cut by 1 and 2 percentage points of GDP, respectively, between 2002 and 2005, through the passage of austere budgets supported by stringent cash management practices. The result has been a shift in the fiscal stance from deficit to surplus. Between 2002 and 2005, the overall balances improved from -2.3 percent of GDP to 0.6 percent at the Central Government level and from -3.1 to 0.9 percent of GDP for the Consolidated Public Sector. Primary balances improved from -1.7 to 1.9 percent of GDP and from -1.3 to 2.7 percent of GDP for the Central Government and Consolidated Public Sector, respectively.

[23] Despite Paraguay's limited access to capital markets and the very favorable terms the country enjoys on its debt -- the implicit interest rate paid on external public debt in the past five years has averaged 2.8 percent per year, approximately 98 percent of public external debt is owed to multilateral and bilateral institutions, 60 percent of loans are contracted with a maturity of over 15 years, and 30 percent are contracted at a fixed annual rate of less than 3 percent -- the external public debt-to-GDP ratio doubled between 1996 and 2002 (from 22 percent to near 43 percent) as a result of the combination external and internal factors including growing fiscal deficits, a slowdown in growth and exchange rate devaluation.

[24] The World Bank. (2006). Paraguay Public Expenditure Review. Report No. 32797-PY.

[25] Figures are for 2004.

substantial evasion, limited access to capital markets, a large informal sector and modest growth prospects.

4. In the face of fiscal constraints the resources to meet Paraguay's infrastructure needs will have to come primarily from an increased effectiveness and efficiency of infrastructure spending, complemented with new private sector investment and additional revenue mobilization. This annex focuses on ways in which the quality of sector expenditures can be improved, with a specific emphasis on bottlenecks related to the Paraguayan budget process.

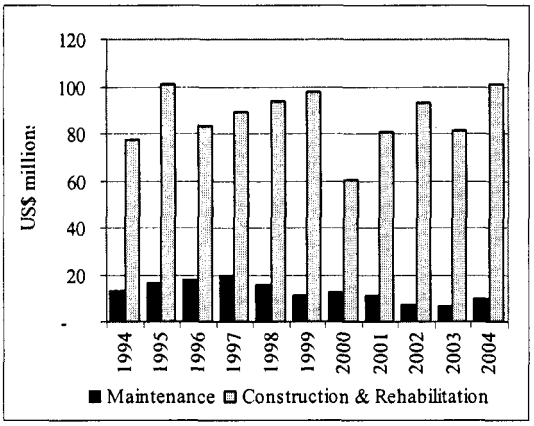
Key Issues in Public Infrastructure Spending

5. Poor planning and priority-setting. Although some advances have been made, investment planning in infrastructure is weak and is primarily undertaken as part of the due diligence process in project appraisal required to access bilateral and multilateral financing. The problem is compounded by the large number of donors financing the sector and the lack of coordination amongst them. International donors play a major role in the definition of investment priorities and thus, to some extent, deficiencies in infrastructure planning in Paraguay also reflect their lack of coordination. Despite important donor assistance in this area, the technical capacity required to formulate new investment plans could be strengthened significantly. Planning units within infrastructure agencies rarely engage in comprehensive planning activities. Their functions are largely limited to contracting out feasibility studies for donor-funded projects and preparing the administrative documentation required to get loans approved.

6. Insufficient infrastructure investment. Fiscal constraints and high personnel expenditures have reduced the envelope of public resources available for infrastructure investment. The lack of investment in road maintenance, for example, combined with poor planning and prioritization has led to the continued deterioration of the road network, particularly the unpaved network. At the same time, the road authority's (*Dirección de Vialidad*, DV) budget allocation dropped by about 35 percent between 2001 and 2004. It is estimated that Paraguay spends only 20 percent of the minimum needed to maintain its road network from further deterioration (Figure 11.1).

7. Human resource issues. The rigidity of public sector human resource policies hampers entities' ability to improve operational effectiveness and efficiency. In some agencies, there is a shortage of staff, and salaries are too low to attract and retain qualified personnel (e.g., in the Directorate of Roads). In others, there is excess staffing. The agencies are severely restricted in the options available to them for rationalizing their staffing levels by current personnel policies and practices.

Figure A11-1 - Budget Execution at the Roads Authority



Source: Roads Authority (*Dirección de Vialidad*)

The Budget Process and Infrastructure

8. Infrastructure investments have some unique characteristics, including their high capital intensiveness, their need for long-term planning horizons, unpredictable costs, and infrequent, lumpy payments. Infrastructure assets are long-lasting and have significant and continuous maintenance needs that can multiply exponentially, to the point of demanding full reconstruction, if they are not implemented. These characteristics accentuate the negative impact of inadequate budget policies, institutions and distorted incentives. In Paraguay, as in many other countries, infrastructure sectors are more vulnerable to the rigidities, arbitrary changes, cyclicality and negative incentives embedded in the budget process. This vulnerability is compounded by the overall context in which budgetary decisions are made, characterized by resource scarcity, weak institutions, and corruption. Infrastructure is particularly exposed to budget cuts during times of fiscal adjustment. It is usually less politically costly to cut infrastructure investment than to lay off public employees. When resources are available, as there is little political capital to be gained from investing in infrastructure maintenance, politicians tend to favor new investment over maintenance.

9. **Priority setting and planning**. Overall responsibility for the infrastructure sectors, including the formulation of policy, planning and coordination lies, *de facto*, with the Ministry of Public Works and Communications (*Ministerio de Obras Públicas y Comunicaciones*, MOPC). Although the STP (*Secretaría Técnica de Planificación*) is responsible for coordinating and prioritizing public investment, this role is not yet effective, and the vacuum has not been filled by the MOPC. This has led to a high degree of fragmentation in sector planning and a lack of policy coordination. This fragmentation and a lack of technical capacity have weakened the importance of planning as a tool for effective infrastructure investment.

10. The budget process does not promote the incorporation of investment planning into resource management. Budget policies do not require an ex-ante evaluation of investment projects and the nature of the interventions by the MoF and Congress – often resulting in ad-hoc cuts and resource allocation changes at the micro-level – as well as the limited capabilities of executing agencies, diminish the relevance of making conscious use of investment planning tools.

11. Despite these weakness, there are promising planning initiatives underway. For example, the MOPC is increasingly conscious of the need for a more integrated strategy for road network development due to competing needs and limited resources. As a result the MOPC has presented the skeleton of a strategy for the development and maintenance of the road network, defined around 5 major objectives: i) maintain the existing network; ii) complete the remaining missing links in the priority main road network; iii) develop the all-weather road network; iv) undertake minimal improvements on targeted earth roads; and v) improve the services to the population. The main novelty of this strategy is a management-of-assets approach whose most important aspect is to strengthen and preserve the existing network. Improvements such as these would need to be linked to the overall budget process in the sector.

12. **Budget preparation**. Despite laws to the contrary, current practice is to elaborate budgets incrementally, i.e., based on resources available for the previous year. One reason for this is the rigidity of expenditures in the infrastructure sector, with a high weight on personnel expenditures (e.g. DINATRAN). Moreover, decentralized entities must set aside part of their budgets for transfers to the Central Government. Beyond rigid expenditures and as noted above, the budget preparation process is not given adequate weight in some sector institutions, diminishing the credibility of the budget process as a tool for resource planning and management. The result is an annual budget that reflects neither the priorities of the agencies nor those of the sector as a whole.

13. Revenues are frequently overestimated, leading to the preparation of unrealistic budgets. This is the case in DINATRAN, which does not make transfers to the Central Government and thus tends to overestimate its revenues in order to introduce more flexibility in the re-allocation of spending during budget execution.

14. At the Ministry of Finance (MoF), entity budgets undergo important changes. The MoF introduces changes to the allocations included in the budget proposals at the line-item level, the lowest level of aggregation. The Ministry also reassigns the financing sources used for different budget items. Infrastructure sector entities often receive significant resources from both credits (*Fuente* 20, or FF20) and own revenues (FF30), in addition to treasury resources (FF10). For instance, during the 2005 exercise, although the MoF allowed a larger envelope for investment programs in the MOPC, it did this by shifting the source of funds from treasury resources to institutional funds dependent on the application of the SIVIPAR law and overall revenue collection.

15. The budget preparation process thus encourages entities to overestimate revenues in order to increase flexibility during execution although this in itself compels the MoF to subsequently cut inflated spending requests. Significant time and other resources are wasted in the process, and the final result does not adequately reflect either national or sector priorities.

16. **The budget approval process**. To some extent, Congressional changes to the budget during the approval process tend to rebalance the cuts imposed by the MoF on sector budgets. Line ministry representatives and agency heads lobby Congress during the approval stage in a series of informal discussions to increase their budgets. Congress paradoxically may act as the arbiter between entities within the Executive.

17. But Congress does not limit itself to overseeing overall budget envelopes and financing sources. It intervenes in budget line items, adding a layer of changes that frequently undermine effectiveness and efficiency considerations. In 2005, for instance, Congress increased personnel spending and reduced amounts requested for human resource management schemes such as early and voluntary retirement plans in some institutions. The legislature also modifies line-items within externally funded investment projects.

18. **Budget execution**. Under-execution of budgets is high in the infrastructure sector, particularly in the MOPC and entities under the Central Government, and somewhat less so in the decentralized agencies (Table A11-1). The difference in execution levels is explained by the relative size of the entities' capital investment programs. The MOPC and DV for instance, dedicate a much larger proportion of their budgets to investment (90 percent in the DV in 2004), the category that is substantially under-executed year after year. In addition to poor prioritization and planning at the entity level, operational capacity and the budget process both affect budget execution.

19. The cuts implemented by both MoF and Congress to entities' budget proposals, without the necessary discussion with the concerned line ministry, significantly affect program ownership and execution. For instance, resources for background investment and feasibility studies are frequently cut, while the funds for the corresponding construction are maintained. The modification of funding sources has a similar effect.

Table A11-1 Infrastructure Agencies: Budget Execution, 2002-04

	2002	2003	2004 *p*
MOPC		52%	56%
DV	86%	52%	61%
SENASA	65%	54%	59%
ANNP	87%	82%	63%[1]
DINAC	95%	75%	53%[1]
DINATRAN	94%	91%	n/a
ESSAP	n/a	73%	56%

[1] Budget execution to end-November

Source: Government of Paraguay

20. Execution delays have in the past resulted from the accumulation of arrears to suppliers because of the failure to record commitments in budget information systems. By law, arrears must then be paid during the first two months of the following year, occupying the bulk of the available budget in January and February. Activities can only begin in earnest in March. These delays are compounded by the frequent reprogramming of resources during the year. The deficiencies in budget planning are such that line ministries often prepare their reprogramming and budget supplement requests *before* the Budget Law is approved. The lengthy approval process for reprogramming requests generates further delays. Finally, cash management practices slow down execution even further, particularly in investment. Insufficient counterpart funds for externally funded projects are included in the monthly cash plans, hindering the execution of many projects for which external resources are available.

21. Limited operational capacity also affects resource execution. The lack of planning has meant that the introduction of a new procurement framework for the public sector is generating numerous delays as more stringent controls are imposed. As entities fail to adequately include provisions for procurement in their budgets, they must resort to resource reprogramming requests during the year, delaying program execution. Inadequate information systems hamper budget execution; most entities lack adequate financial information systems to control revenue collections, expenditures and disbursements and in none of the institutions analyzed are there linkages between financial and physical execution.

22. Weak operational capacity is also the result of shortcomings in human resource policies. The entities analyzed are not able to implement policies to create incentives or to improve the quality of human resources because of the legal rigidities that stipulate that only Congress can modify the size of the payroll and change salary ranges. Agencies have at times attempted to introduce programs to reduce staffing by voluntary retirement, but Congress has eliminated or reduced budgetary provisions for such activities. In many entities salaries are too low to attract and retain qualified personnel.

23. **Monitoring and evaluation**. Although some initiatives are underway, monitoring and evaluation is largely non-existent in infrastructure entities. Data collection and management, the basis of monitoring and evaluation, is weak, and there are few efforts to conduct monitoring activities that do not form part of externally financed projects.

Annex 12: Social Evaluation of the Component of Improvement and Maintenance of the Unpaved Road Network

PARAGUAY: Road Maintenance

1. Social Evaluation

1.1. Objectives and Scope of this Social Evaluation

The objective of this evaluation is to identify opportunities, barriers and risks that could complement previous social analysis carried out during Project preparation, or to serve as basis for reinforcing the social impacts of the unpaved road network and rural access component of the project. This is in consideration of the fact that the Project's expected social impacts will be greater in the beneficiary communities of this component. The evaluation, thus, focuses on the population of the three beneficiary Departments: Caaguazú, Caazapá, and San Pedro.

The previous social analysis that were carried out were: (i) the preparation of the Conceptual Framework of the Project and the definition of its components, which resulted in the incorporation of a special component for the most vulnerable communities in the country, the component on Improvement and Maintenance of the Unpaved Road Network (hereinafter Component 3); (ii) the identification of the Beneficiary Departments of Component 3, which resulted in the selection of the three poorest departments in the country: Caaguazú, Caazapá, and San Pedro; and (iii) the preparation and implementation of the terms of reference of the Component 3, specifically in the definition of the Master Plans of each of the three scenarios of eligible Unpaved Road Networks[26] of each selected Department. The principles of the Social Participation Framework (EPS), prepared at the outset of the project preparation, were applied to the different participatory spaces of the Unpaved Road Network definition.

This Social Evaluation will be complemented by: (i) the results of the studies and participatory processes, which include social criteria, in the definition of the Master Plans of the Eligible Unpaved Road Networks Rural Road Networks of the three Beneficiary Departments; (ii) the Indigenous Peoples Plan; (iii) the Conceptual Framework for the Social Impact Monitoring Plan; (iv) the Environmental Management Plan; and (v) the Resettlement Framework. The latter two will be designed for the entire Project.

1.3. Topics addressed in the Social Evaluation

The analysis performed pursuant to this Social Evaluation (i) revised the intervention strategy to reinforce the social impacts of Component 3, including recommendations to favor social inclusion; empower social actors; and identify social risks; (ii) verified the implications of the implementation framework and aspects that can improve social impact of the Component 3; and (iii) recommends indicators for the monitoring plan, and resources and requirements to develop it.

To achieve the evaluation's objective the analysis included topics related to the socio-economic and cultural, institutional, historical and political context, and some legal considerations, with emphasis on the three Beneficiary Departments targeted by Component 3. This evaluation analyzed data and information at two levels, in most cases, on averages corresponding to said three Departments (hereinafter **departmental total**); and on averages obtained from the 37 municipalities pre-selected in February 2006,

[26] Eligible Road Network is the minimal network that allows for basic service to the area. It resulted from technical, economic and social analyses, such as number of inhabitants to be benefited, volumes of traffic, agricultural production, commercial movement and access to main highways, among other aspects

which per Beneficiary Department are located as follows: 14 in Caaguazú, 14 in San Pedro, and 9 in Caazapá (hereinafter **municipal total**). (*See Diagnostic and detailed consulted sources in Project File*).

2. Findings and Proposed Measures

The findings and proposed measures for each of the five dimensions of the social evaluation (social diversity and gender; institutions, norms and behaviors; social actors, participation; and social risks) are as follows:

2.1. Diversity and Gender

2.1.1. Analysis

Representing almost 17% of the total population of the country, 893,572 persons are the most direct beneficiaries of this project component. Certain socio-economic indicators of this population, such as the preponderance of the primary sector in these three departments, agriculture as the majority's source of employment, unsatisfied basic needs, among others, result in the classification of the most direct beneficiary population into two very distinct intra and inter-groups categories: a reduced number of people with large economic and social assets, and larger group with unsatisfied basic needs, and even in extreme situations of need.

Poverty (measured through income) of the departmental total is approximately of 50% (San Pedro 51.1%, Caaguazú 48,7%, and Caazapá 46.5%). On average, almost 60% of the population is employed in agricultural activities (67.5% - 29,812 persons in Caazapá; 61.9% - 59,617 persons in San Pedro; and 50.1% - 70,971 persons in Caaguazú). This is framed by a participation in the primary sector of an average of around 60% of the Departmental GDP (78.7% in Caazapá, 70.4% in San Pedro, and 51.5% in Caaguazú), compared to a 30% at the country level. In the three selected departments, 48.4% of households (86,127) registered at least 1 unsatisfied basic need (NBI for its acronym in Spanish) compared to 39.5% at the national level. They have a departmental monthly income in average of US$97.50, and the Gini Coeficient in income distribution per capita reached a departmental total average of 0.54 in 2002.

The classification of the beneficiary population into two major categories was made possible by the fact that Paraguay is a country relatively homogenous in its ethnic and religious composition, but where facts such as living in an urban or rural area, or speaking only Guarani, can be sufficient criteria to ascribe to a specific social class, and which, despite not constituting a source of conflict, can be barriers to social mobility. A more detailed analysis of each of these groups, which will determine with more precision who are really in each category is being carried out as part of the preparation of Component 3, and will be completed before the Evaluation of the Project. However, it is anticipated that amongst these two major categories, the groups that will deserve special attention are children and youth, women, unemployed, and indigenous communities.

Within this framework, the main characteristics of the identified categories are:

***Beneficiaries with Large Economic and Social Assets*:** This category is comprised by people that could be amongst the 13% of the total departmental beneficiary population living in urban areas, amongst the 50% of the population without any unsatisfied basic need, amongst the 12% of the population whose main language is Spanish. This group is also comprised by large soy producers that cultivate around 2% of the 562 agricultural units of a surface starting from 100 hectares of the departmental total. In addition to large agricultural producers, this group also includes large cattle breeders, among others. These people are the ones with higher education level, broader access to public services, who are organized in associations

with strong intra-group capacities, and enough strength to have an impact on the decision making process of public policies.

***Beneficiaries with Scarce Economic and Social Assets*:** This category is comprised by almost 77% of the total departmental beneficiary population who lives in rural areas and is amongst the 49.6% of average level of poverty (measured by income). This group is also comprised by people with an average monthly income of US$94.76 (17.4% less than the national average), in a context where the GDP per capita measured by its average acquisitive power parity (or PPA for its acronym in Spanish) of the departmental total was of US$3,092 PPA in 2002 compared to the GDP per capita of US$4,610 PPA at the national level in that same year.

Also, this group includes the over 50% of households with at least one unsatisfied basic need (NIB), or the around 88,8% of the total of households that speaks only Guarani, a reason for which they can be marginalized. This group also includes in its majority small agricultural producers, and some few medium size producers that cultivate cotton and soy in approximately the 95% of the 52,497 agricultural units of up to 100 hectares of the departmental total, according to data of the year 2002. Many of these producers are those who cannot take out their products to markets due to lack of roads. This group is also comprised by those who do not own land; as well as around 2% of small farmers in the total beneficiary population that lives in *settlements*, according to data of the year 2001, who in their majority cultivate on pieces of land of between 1 to 2 hectares (whereas the agricultural statute considers 10 hectares as minimum surface to obtain the necessary levels of income for effective settlement and coverage of a family's basic needs). It should also be noted that these poor farmers have registered a total of almost 9% of migration between 1997 and 2002. This group also includes people who do not own any land, which can be considered with the following data: in 2001, over a country total of 31,962 lots, the departmental total is of 17,874 lots, of which 3% are under property titles, 71% are in the process of obtaining titles, and 26% are occupied, but without any on-going proceedings for obtaining titles.

Finally, this group also includes small farmers growing other grains, or raising bovine or equine cattle.

From the data indicated in this categorization, it can be inferred that this group is comprised by farmer families, including women and children that cultivate the land for daily consumption, and whose economic assets are limited to personal subsistence, or are unemployed or sub-employed. The agrarian statute considers 10 hectares as the minimum surface to obtain enough levels of income for effective settlement and coverage of a family's basic needs. It is also probable that this category includes children and youth who need to walk from 5 to 10 kilometers to go to school, or who in many cases not even go to school; as well as women who die during delivery for not having means of transportation to go to health units.

These people are, in many cases, organized –in work committees, mostly of production, that in turn are part of other intermediary associations of different sizes. Some of these associations have shown an increasing mobilization strength since the political transition initiated in the country in 1989, but have been unable to consolidate themselves. One of the reasons for this is that capture of their interests by their own leaders, who in many cases replicate the model of political leaders, disconnecting themselves form their local bases. In general, farmer organizations by themselves have limited capacity to influence relevant policies for their interests at the macro level.

Social Groups that Require Special Attention. A more detailed analysis of both categories mentioned above, which is being carried out during the preparation of the MCAR, will be able to determine with better precision those groups that have been historically considered as more vulnerable (women, children and youth, unemployed, and indigenous people) and that are described in this Evaluation due to the special attention that they might deserve in terms of social inclusion.

Women. Around 48% of the total departmental population are women (427,823) compared to 2,559,956 women at the country level, in an environment where economic, social, cultural and political discrimination have historically affected women, leading to their disadvantageous position with respect to men in almost all fields. Within this 48% there are 39,757 households with women as heads, mostly in the rural areas. Within that same 48% there are young women between the age of 15 and 24 that add to 80,505. Almost 4% (1,929 women) are unemployed in a framework of a departmental total of women Employed Population per Year (PEA for its acronym in Spanish) of 56,750.

Children and Youth. Almost 62% of the total departmental beneficiary population is under 24 years of age, out of which 174,808 are youngsters between the ages of 15 and 24 (19%), compared to a proportion at the national level of 20.3% (1.049.352) in this age group (figures per department are: 19.4% (61,763) in San Pedro; 19.7% (85,827) in Caaguazú; and 19.5% (27,219) in Caazapá. The matriculation rate in primary education (first to ninth grade) is of around 83%, and secondary education (first to third course) is of 10%. These rates are not far behind those at the national level, which are of 80.4% and 11.9%, respectively. However, the rate of school attendance in secondary education is below 50%.

Unemployed. Around 0.68% of the total departmental beneficiary population is unemployed (6,079 people), in a departmental total PEA of 289,455, and a country total PEA of 648,218 in 2002. Therefore, the total departmental unemployment rate is of 2.1% compared to 5.5% of unemployment at the national level, in 2002. However, available data for the year 2004 shows a total departmental unemployment rate of 7.4%. This group should also include the people in the 19.5% of sub-employment, since they show a similar level of vulnerability.

Indigenous Peoples. In the Project's intervention area there have been identified 98 indigenous communities, adding up to 12,230 indigenous persons out of a national total of 85,674, made up by five (Mbya, the Ava Guarani, and the Ache) out of 17 ethnic groups in the country, which belong to the Guarani Linguistic Family, one of the five in the country. These ethnic groups constitute political and social unites that cannot be divided by the same criterion of political division of the country without creating internal conflict that could affect negatively them. Therefore, all 98 indigenous communities will be considered beneficiaries of the Project. From these 98 communities, 43 communities have their land pursuant to a title in their name, while 55 communities occupy lands that belong to third parties, occupying 2,512 houses. These Guarani indigenous communities live from a combination of economic activities that include subsistence agriculture, hunting, fishing, collection, occasional waged employment, and handcrafts. The land they occupy no longer provides enough natural resources for these communities subsistence. (*See more under Indigenous Peoples Plan, in separate annex*).

2.1.2. Measures

Considering that this is an improvement and conservation road project, the primary objective of which is to facilitate the road access to several communities, and that the social impacts generated by it are indirect, the recommendations that are presented in terms of social inclusion are to take advantage of the project development in order to favor it. In this line, the Project could take as starting reference the economic and social capabilities of the different categories of beneficiaries to establish levels and types of participation in the different mechanisms included in the agreements of the *Collaboration Strategy of the RVRB* in three different areas:

(i) **Financing of the works in the Unpaved Road Network**. Include a clause in the agreements, establishing levels and types of differentiated contributions according to the two categories of economic and social assets. The contributions could vary from financial resources to labor.

(ii) **Participation through Positive Action or Positive Discrimination.** In the training activities, and other spaces generated by the implementation of this component, including decision making, participation quotas for women, youth and unemployed people should be established. In the case of children, priority in the timing of works could be given to those segments where there are more children.

(iii) **Indigenous Peoples Plan.** A Plan for indigenous communities is being prepared, the basic principle of which is to work with these communities using participatory mechanisms that respect the process of indigenous development pursuant to the principle of cultural pluralism and respect of the environment. For the preparation of this Plan, pertinent consultations with Indigenous Peoples in the Departments of Caazapá, Caaguazú, and San Pedro have been carried out. Certain types of support have been identified in the Plan for Indigenous Peoples for its beneficiaries, including training in road maintenance, and road machinery and equipment; assistance to participate in the programs of works execution y maintenance that could imply employment opportunities; creation and implementation of training centers for indigenous communities, located within an indigenous community, one for each department (*See Indigenous People Plan, in Annex 10*).

2.2. Institutional Framework, Norms and Behaviors

2.2.1. Analysis

With the objective of identifying alternatives that could improve the social impact of component 3 in terms of institutional framework, and the pertinence of foreseeing measures for the management of risks that have not yet been identified, the implementation mechanism under the form of inter-institutional agreements envisaged for this component was revised. The following items were analyzed: (i) previous experiences of the MOPC in the topic; (ii) lessons and perspectives of its broadening in terms of inter-institutional articulation; (iii) actors' duties and capabilities; and (iv) governance environment in which it will develop.

2.2.1.(i) Inter-institutional Agreements carried out by the MOPC

Eight agreements that were signed by the MOPC since 2003 have been revised. The modalities are varied, including agreements with the sub-national governments (departmental and municipal –on an individual basis or together—and in some cases, including civil society organizations), and also with civil society organizations only. The agreements are in the majority of cases with sub-national governments. All of them are on road maintenance, with lengths in average of 300 km, with a duration of one year, renewable by mutual consent of the parties. All agreements include provisions of budget rendition. The financing of half of the agreements has come out of the MOPC own resources and the sub-national governments, and the other half with a proportion of 60% of international lending resources, and 40% from the local counterpart. In both cases, the majority of the contributions of the sub-national governments are in the form of equipment and payment of overtime of field personnel. Only one of the agreements is under the modality of performance-based contract, and includes a mechanism for sustainability by establishing toll charges as a financing mechanism once the agreement is finished.

There is no evaluation of the performances of these agreements; however, the mere examination of the documents and some interviews reflect potentialities as well as limitations.

The potentialities are: the evidence of an interest in finding solutions to road problems in all three levels of government, crystallized in the inter-institutional agreements; and the existence of an accumulated experience in the topic which, however incipient, is growing.

Four aspects that stood out during the analysis, and that should be object of further and deeper evaluation in order to verify the levels of their limitations are: (a) the non-existence of a common criterion for the establishment of agreements; (b) the low levels of financial contributions of the sub-national governments; (c) the short duration of the agreements without any provision for mechanisms for sustainability ; and (d) only with reference to the agreements with civil society organizations: these agreements have shown administrative limitations regarding the impossibility of keeping documentation for an adequate budget rendition on the part of urban commissions since they lack legal standing, and linked to the absence of clarity regarding whether these agreements fall within the public or private arena in case of a judicial resolution of a conflict arising from it.

2.2.1.(ii) Lessons from the Inter-Institutional Agreements and Perspective for their Broadening during the Implementation of component 3

From a more strategic perspective, with the elements already identified and others that are developed in this Evaluation, the inter-institutional arrangement of the agreement for the implementation of this component faces three main challenges: (a) the complexity involved in articulating the intervening institutions; (b) the need to find measures o encourage the participation of sub-national governments due to the differences in capabilities and resources amongst them in relation to the MOPC; and (c) the requirement of identifying mechanisms to minimize the risks of its development in an environment of fragile governance.

2.2.1.ii.(a) Complexity of Inter-institutional Articulation.

The articulation of three levels of government (national, departmental and municipal) in relation to a topic in which the MOPC, as a national agency, is the only instance legally responsible for road infrastructure, requires the verification of certain aspects to simplify the relationship. All three levels of government, in addition to holding their own political and administrative autonomy, have different responsibilities, cultures in the rendition of services, defined public images, and above all different duties, capabilities and resources

The MOPC is an administrative organ, which is exercised by a Minister, named by the Executive branch, and is in charge of the specific field of national, departmental, and urban road infrastructure.

In turn, the sub-national governments are elective organs, and are exercised by governors and mayors with executive functions, that in most cases must be approved by the departmental and municipal assemblies, also elected by popular vote in each community. The duties of these sub-national government are multiple, from duties related to health, education, emergency situations, and among others, those related to civil works. These duties, in addition to other factors, are exercised within a context of strong political implications, in which clientelism is a very common way of having access to privileges. (*See more in Governance Environment item (c) in this section*).

2.2.1.ii.(b) Participation of Sub-national Governments. Capacities and Needs different to those of MOPC's.

Institutional Weaknesses. Two dimensions are observed in this realm. In the first dimension, it is the same for institutions in all three levels of government. None of them has a specialized organizational unit for developing the strategy of social participation that is a fundamental characteristic of this component of

the Project. The second dimension of weakness that becomes evident is that, under current conditions, the scarce management capacity of the departmental and local governments with respect to their functions and resources with which they currently count would deepen as a result of incorporating a new duty. The relationship between duties and resources of these governments has been historically uneven. For instance, in the period 1993-2001, of the total national general budget between 90% and 95% was administered by the central government, and only 2% has been transferred to the totality of the decentralized entities, which include departmental governments[27].

Financial, Legal, and Human Resources Capacities of Sub-national Governments. In general, sub-national governments have similar limitations. This is the case, for example, in the area of human resources. For instance, according to recent studies, a permanent governmental official must deal with, on average, 5,539 inhabitants, and in the case of municipalities the number is 1,524, without a common pattern according to which that relation is established. These governments do not have meritocracy procedures for the selection or promotion of public officials. Political clientelism is an important source of employment at all governmental levels. However, the comparison of both levels of government shows that the capacities of departmental governments are more limited than those of the municipal governments.

Departmental Governments. These governments do not have the competency neither to approve their own budgets nor to collect taxes, which means in practice that they do not count with their own resources, depending exclusively on resource transfers from the central government and municipalities (co-participation of taxes on gambling, real state, and value added), and royalties. The distribution of total income to departmental governments does not follow a common pattern for all departments. In the case of co-participation of taxes, it depends on the levels of collection performed by municipalities and the national government. According to a recent report, the structure of Departmental Governments' expenditures presents two stages. Since their creation in 1993 until 1998, there was equilibrium between the percentage participation of physical and personnel services investments. However, since 1999 physical investments have decreased, with an increase in personnel investments and transfers (in the majority of cases to non-for-profit entities without any normative to avoid a discretional use). In general terms, departmental governments allocate between 10% and 15% of their expenditures to physical investments.

Municipal Governments. In comparison, municipal governments have improved more quickly than departmental governments because they administer their own budget, and have the capacity to collect taxes (taxes, fees, and contributions). By law, municipal governments are classified in four groups, according to their financial capacities, which are defined by the Executive Branch before each municipal election. The first group is comprised by those municipal governments whose budgets are above the 50% of the annual average of the total budget of the municipal governments of the 17 departmental capitals in the country; while those below that 50% are divided into the other three groups with percentages that are lower than 3%. Following municipal elections held in 2001, of the total of municipal governments selected for the RVRB, only one belongs to Group 1 (2.7%), 9 belong to Group 2 (24.3%), 33 belong to Group 3 (64.8%), and 2 belong to Group 4 (5,4%). while the others belong, in most cases, to Group 3.

Behavior of the Municipalities' Total Income in 2001-2003. Though the total income of municipalities has increased in general, it should be noted that this is due, in most cases, to royalties, although in some cases, real state tax collection helped in that increase (and this has happened in many departmental capital

[27] This 2% does not include transfers to municipalities, since municipal budgets are not part of the National General Budget.

cities due to the fact that they have improved their real state registries with the support of international cooperation[28].

As a reference, and to have some sort of element for analysis of, on the one hand, the composition and behavior of the Municipalities' Total Income, and the participation of Current Income and Capital Income with respect to Total Income; and on the other hand, the Total Expenditures, and their composition by Personnel Expenditures and Capital Expenditures, four Municipalities were analyzed in the period 2001-2003, one for each Group and each Beneficiary Department. These Municipalities, per Department, are: (i) Caaguazú: Caaguazú in Group 1, the municipality with the largest size of the departmental total; (ii) San Pedro: San Pedro de Ycuamandiyú, the largest municipalities of the Department in Group 2; (iii) Caazapá: Abaí, one of the smallest municipalities, also in Group 2; and (iv) Caaguazú: Carayaó, the smallest municipality of the municipal total in Group 4[29].

In the preceding period, the total income of the Municipality of Caaguazú was of approximately US$0.34 million and US$0.48 million; San Pedro del Ycuamandiyú's, of US$0.17 million and US$0.2 million; Abaí's, of US$0.05 million and US$0,09; and Carayaó's, of US$0.04 million and US$0.09 million.

In the preceding period, the Muncipalities' Current Income is, in most cases, of more than 70% of Total Income, and Capital Income is of approximately 20%. A pattern that has been noted during the evaluation, and which should be subject of a deeper analysis, has been noted that the larger the Municipality, the lesser is the level of Capital Income. So, for instance, in the Municipality of Caaguazú (Caaguazú), the largest in the municipal total, Current Income has been of an average of 81% with respect to the Total Income, while Capital Income has been of almost 16%. At the same time, Carayaó (Caaguazú), the smallest in the municipal total, had a Current Income lower than 50%, while Capital Income was of approximately the same proportion.

Behavior of the Municipalities' Total Expenditures in 2001-2003. In the same period 2001-2003, the behavior of Total Expenditures in the same four Municipalities of the Beneficiary Departments has been as follows: the Total Expenditures of the Municipality of Caaguazú were of approximately US$0.36 million and of US$0.44 million, with an average of 52% of Personnel Services, and 13% of Physical Investments.

It is surprising to see that two Municipalities in Group 2, San Pedro de Ycuamandiyú (the largest municipality in the Department of San Pedro), and Abaí (Department of Caazapá) showed totally different behaviors. San Pedro de Ycuamandiyú showed a behavior similar to the Municipality of Caaguazú, except that it showed slightly lower proportions. However, Abaí showed a proportion of Personnel Services in average of 42%, and of 1.5% in Physical Investment.

Whereas, Carayaó, Caaguazú, showed Total Expenditures of approximately US$0.04 million and US$0.08 million, with a participation of Personnel Services of 31% and 22% respectively, and of 51% and 35% in Physical Investment respectively.

Indicator of Municipal Management Capacity. This indicator which was created in a recent study takes into consideration the budget, human resources, and services rendered by municipal governments, with a minimum value of 0.18 and a maximum of 0.66. San Pedro ranked last with a value of around 0.4; Caazapá was second with 0.5, and the best ranked was Caaguazú with 0.6. It is important to point out that this indicator does not include road management.

[28] Mainly with support from USAID.
[29] Source: COPLANEA/JICA: Local Governments in Paraguay, Asuncion 2002.

Royalties' transfers for Departmental and Municipal Governments. For both levels of government these transfers are important resources for capital expenditures since, by law, they can only be used for Expenditures of that nature, fiscal investments, and equipment. Total royalties transferred to all departmental governments in the country in 2001 amounted to US$1.78 million. Of this amount, San Pedro, Caaguazú and Caazapá received approximately 5% each (US$91,307.7). Up until 2005 the total of royalties transfers to departmental governments has duplicated, reaching almost US$2.96 millions. Out of this amount, San Pedro, Caaguazú, and Caazapá received around 5% each (US$174,302.8).

In turn, during the period 2001-2005, total income from royalties for the selected municipal governments increased over a 100%. Caazapá is first with the largest quantity, with its municipal governments increasing a 192%, from around US$0.24 million in 2001 to approximately US$0.71 million in 2005. Caaguazú is second, with increases of 183.87%, from around US$0.43 million in 2001 to US$1.2 million in 2005. In San Pedro, there were increases of 148%, from US$0.4 million in 2001 to US$1 million in 2005.

2.2.1.ii.(c) Governance Environment for the Implementation of Inter-institutional Agreements.

Inter-institutional agreements will develop in an environment of fragile governance, with very low levels of institutional capacity due to low levels of transparency, the lack of participatory mechanisms and budget rendition, the existence of corrupt practices, and the political or economic capture of the interests of the most vulnerable actors. Several national and international reports on these topics show similar diagnosis.

Transparency International usually positions Paraguay amongst the most corrupt countries, showing a ranking of 2.1 in a range that goes from 10 (highly transparent) to 0 (highly corrupt), in the year 2005. The World Bank's governance indicators have systematically ranked Paraguay in the lowest percentiles in the world, and in almost all cases below the 50th position in a scale of 0 to 100 where the higher values correspond to better levels of governance.

In the same line, national reports have reached similar conclusions. According to a recent report, corrupt practices in the country transcend the public arena. The majority of public officials believe that corruption is promoted by businessmen; and that the most negative and inefficient aspects of public administration are the absence of meritocracy in the selection and promotion of public official, and the inexistence of salary incentives, among others. Another important weakness that has been highlighted is the opacity of institutions both in the inside and the outside.

However, other findings of the same report show aspects that could be brought to full potential, such as the existence of a favorable environment for public sector reform amongst public officials, and the perception that communication media and neighborhood commissions are the civil society organizations most efficient in the fight against corruption.

The political composition of the different levels of power and their intra and inter-institutional relationships is a key aspect for the development of the Project. Since the last departmental elections in 2003, two of the departments are led by the ANR, and one by the PLRA. In turn, since the last municipal elections in 2001, 9 out of 17 districts in San Pedro are run by the ANR and 8 by the PLRA. In Caazapá, out of the 14 districts that were selected 8 are led by the ANR and 6 by the PLRA. And in Caaguazú, 13 districts are led by the ANR, 6 by the PLRA, and one district is run by an independent mayor.

2.2.2. Measures

According to the results of the analysis, the MOPC should:

(i) **Social Unit**. The MOPC should create or assign a Social Unit within its organizational structure to allow the development of the social processes of this component of the Project.

(ii) **Agreements as the basis for Categorization by Levels of Government.** The MOPC, as the agency responsible for all road infrastructure in the country, should lead a process aimed at constituting the agreements as basis for the institutionalization for the classification of roads and highways, and the assignment of responsibility of each level of government. These agreements should be seen as preparation for a situation that will necessarily occur at some point. To this effect, the MOPC should evaluate the agreements already signed. Special attention should be given to the agreements with Neighborhood/Urban Commissions due to the administrative and legal limitations that these agreements entail in relation to a public instance, in addition to the limitations that these agreements already have with regards to institutional strengthening.

(iii) **Program for the Strengthening of Local Capacity, and Result-based Incentives.** The MOPC should design the contents to be developed, and establish result-based incentives that include indicators such as: a) capacity to agglutinate municipalities, and other social actors; b) formalization of the unpaved road networks in public policy instruments through inter-institutional management; c) number, levels, and impacts of the offered training.

(iv) **Modalities of the Collaboration Strategy for the Unpaved Road Networks**. The MOPC should design these modalities, clearly defining objectives, activities, aims, products, deadlines, criteria, and incentives differentiated by levels of results.

(v) **Regulation for Inter-Institutional Cooperation of Public Character.** The MOPC should establish a statute on this issue with clear provisions on the role of each of the institutions participating in the Agreement. In the same way, it should provide for clear strategies of internal and external communication that facilitate the coordination amongst all the institutions parties to the Agreement, and which should be publicly presented in one instance with a clear common message related to the Project. Other mechanisms should also be envisaged such as intra and inter-institutional budget rendition. Mechanisms of conflict resolution, and of external monitoring for actors not directly intervening in the agreement, should also be envisioned.

(vi) **Implementation of the Project's Social Participation Framework (EPS)**. The MOPC should define mechanisms and activities in the agreements. (*See Section 2.4. Participation*).

(vii) **Training Program**. The MOPC should design training programs with differentiated strategies according to the needs and characteristics (including cultural characteristics) of the different actors involved.

(viii) **Actions' Dissemination**. The MOPC should publicize all actions taken using the most adequate means in relation to the characteristics of the actors involved, following the principles of the Project's Social Participation Framework (EPS). (*See Section 2.4. Participation*).

2.3. Social Actors

2.3.1. Analysis

The social actors involved are positively benefited by this Project, since by merely improving the road access to these communities a very important social need will be addressed, contributing an element that can improve many social aspects of the communities involved. Within this framework, the possibility of achieving the objective of this component and the project will be increased to the extent that the MOPC takes advantage of the potentials that each key actor involved has to offer.

2.3.2. Measures

The measures that will contribute to a larger social impact, in accordance to the role that each actor can play in the project, are as follows:

(i) **Principals of More Direct Beneficiaries.** The Project could have as starting reference the economic and social capacities of the beneficiaries' categories described in **paragraphs 12-20**, to establish differentiated levels and types of participation in the agreements of the Collaboration Strategy for the RVRB in three areas: Financing of civil works in the RVRB; Participation by Positive Action/Discrimination; and the Indigenous Peoples Plan.

(ii) **Associations of Large Agricultural Producers**. These associations have high levels of intra-group organization, strong capacity to influence decision-making processes of public policies, and experience in systems of road management, both private and in cooperation with MOPC. The Project should count on these associations at the outset of the project, starting with an exhaustive evaluation of their experiences in order to incorporate them to the operative design of the Collaboration Strategy for the Unpaved Road Network.

(iii) **Non-Governmental Organizations**. NGOs can be an important ally due to their specialized knowledge of rural organizations, and because they already have structures that have provided assistance to these communities. They could provide training services in topics such as community organization, and those relative to road maintenance, while at the same time they can become vehicles of communication of the more vulnerable groups, as well as channels of external monitoring. In order to increase NGO's collaboration capacities, the Project could envisage their training in road improvement and maintenance.

(iv) **Universities**. The Project should count on the experience of the Catholic University of Asunción, in the development of information materials and studies on road maintenance. The Project should use the possibility of engaging young university students in maintenance management through the internship program of the National University of Asunción.

(v) **Chambers of the Construction Sector.** A priori, these associations could be seen as the actors that will benefit the least from the Project, due to the change in the current type of contracts that they have for the construction of civil works, and the modality of contractual relationship with the MOPC. These actors influence capacity is very high; therefore, the Project continue the dialogue with the chambers initiated during the preparation of the Social Participation Framework. To this effect, the Project should take into consideration the fact that Construction Chambers had already expressed their concerns with respect to the MOPC management models in this component, wondering whether it would be outsourced to third parties, through concessions or another means. In addition, they showed skepticism with respect to the models of inter-institutional cooperation between the national and sub-national

governments, and even more if civil society organizations were included. Also, they reiterated that the maintenance of the paved and unpaved road networks should be addressed through an integral approach, and then outsourced to contractors. The continuation of a dialogue with these associations should focus on dissipating their main concerns, and offering them the possibility of training through exchanges with other chambers with more experience, among others.

2.4. Participation

2.4.1. Analysis

Context. Paraguay has historically been captured by political and economic elites that have affected the participation of those generally involved, and in particular of the more vulnerable social groups (either because of their lack of trust, or because they were captured by the elites). Paraguay started its political transition in 1989 after many years of dictatorship and the rule of economic and political elites. Only two political parties, the oldest and largest ones, the *Liberal Radical Autentico* (PLRA), and the *Asociacion Nacional Republicana "Partido Colorado"* (ANR), have exercised the Presidency of the Republic throughout the history of Paraguay. The ANR did so for 52 consecutive years.

It should be noted that the first direct elections of departmental and municipal governments were held in 1992. At the beginning of the political transition the level of electoral participation was considerable, but has systematically decreased since then. The persistence of low levels of governance, institutional performance, and corruption has generated some sort of fatigue with regards to political participation amongst Paraguayan society in general, and has led to an increase in the levels of mistrust towards public leadership. The national and departmental elections in 1998 showed an electoral participation of 81%, decreasing to 65% in 2003. At the same time, turnout in the elections for municipal authorities in 1996 was of 83%, decreasing to 55% in 2001.

Low but Increasing Levels of Citizens' Participation. Despite the decrease in political participation, several reports show that citizens' participation has increased, even influencing some public policies. Amongst the selected departments, there is an example of this increased participation of civil society, in the case of Caazapá. This Department has a Departmental Development Plan, which was designed through participatory mechanisms, and along which the departmental government and the associated municipal governments have shown progress towards the development of the activities included in such Plan, although still with a strong external support. It should be noted that the problem of poor road conditions in such Plan has arisen as a priority, but as of this date there are no concrete measures to face this problem. On the other hand, in San Pedro, the vehicle for the reactivation of the Association of Municipalities will be the RVRB.

2.4.2. Measures

(i) Participation in the Project Preparation.

During the project preparation phase, the World Bank team has made efforts to support the interest and commitment shown by the MOPC in incorporating a working culture different to the one reigning in the public administration throughout the country. In this line, several important steps were taken during project preparation in terms of participation, including around 11 participative meetings of various kinds, ranging from the dissemination of best practices on the project's topic to the definition of the eligible Unpaved Road Network, including: Workshops within the Ministry of Public Works and Communications (MOPC); International Workshop: "Dialogue on Road Management for the Development of Paraguay"; Three Preparatory Meetings for Project's Presentation Workshops at a

Regional/Local Level; Three Project's Presentation Workshops at a Regional/Local Level; Six Workshops at Regional/Local Level: Three to Select the Eligible Unpaved Road Network, and Three to Define the Master Plan of Selected Unpaved Road Network.

(ii) Project's Social Participation Framework (EPS).

The EPS was designed at the outset of the project preparation, and was the result of a series of consultations with different social sectors involved in the project. The EPS has been used during project preparation, and is expected to continue during the all its cycle. The EPS contemplates three components: (1) participation; (2) communication; and (3) transparency. The main objectives of the EPS are; (a) to create awareness of the importance of expenditures and mechanisms for road maintenance to preserve the road network's condition; (b) to disseminate the project's objectives and achievements, and its contribution to the strengthening of the MOPC; (c) to guarantee a broad participation of the different stakeholders aiming at, among other objectives, establishing their roles and responsibilities in realist and fair terms; and (d) to increase project's transparency and accountability throughout its cycle.

(iii) Participation Component of the EPS.

Guiding Principles. The participation component of the EPS includes the following principles: (a) to promote feasible mechanisms for participation, moving from unidirectional communication to common responsibility in order to assure continuous feedback; (b) to link participatory mechanisms to the basic objectives of transparency and accountability in the rendering of public service, designing them with an anti-corruption approach; (c) to promote spaces for dialogue based on realist and objective data and technical expertise, avoiding the creation of expectations that might not be satisfied; (d) to exercise positive action or "positive discrimination" with regards to the more vulnerable groups such as youth, women, unemployed, and indigenous communities.

More Direct Beneficiaries' Participation. Three modalities will be developed: (a) selection of the basic unpaved road network feasible for the development of each department in line with the fiscal situation of the country; (b) to broaden the existing basic unpaved road network according to one of the modalities of co-participation envisaged in the Collaboration Strategy for the Improvement and Maintenance of the Unpaved Road Network to increase the resources of the MOPC with other resources of co-financing; and (c) project's monitoring, evaluation and social responsibility. The first two activities have already been developed, and are currently under the responsibility of the UPGP of the MOPC, with the support of experts on social participation.

(iv) Communication Component of the EPS.

Objectives. The communication component of the EPS is aimed at: (a) disseminating information on the importance of Expenditures and mechanisms for road maintenance in order to preserve the road network's condition; (b) disseminating the project's objectives and achievements, and its contribution to the strengthening of the MOPC; (c) promoting the project's social interest and demand taking into consideration the fact that road maintenance is not a current practice in Paraguay, bringing it into the public agenda, and stimulating the ownership of this objective by decision-makers in order to facilitate the outset of the project. These activities will be under the responsibility of the UPGP, and the MOPC's team for institutional consolidation.

(v) Transparency Component of the EPS.

Objectives. The objectives of the transparency component of the EPS are: (a) to facilitate the access to information about the project's plans and achievements through the generation of flows of relevant and

timely information to all different stakeholders, using appropriate means in accordance with their cultural characteristics and their needs; (b) to open an interactive space amongst the groups involved, collecting their opinions, suggestions, etc., systematizing and delivering them to the decision-makers at the MOPC to assure a permanent feedback along the project's progress; and (c) to generate mechanisms for the project's budget rendition. These activities will be under the responsibility of the MOPC's directorate of communication, in coordination with the UPGP, and the MOPC's team for institutional consolidation. See more on this in section 3.4 of this Evaluation.

(vi) State of Implementation of the EPS.

Activities. During project preparation, many activities have been developed engaging the social sectors involved, in furtherance of the EPS. The objectives and results of these activities, and the decisions taken along the process have been carefully documented for reference in the next steps of the project. The achievements made in terms of participation during project preparation are encouraging; however, it should not be disregarded that this type of achievements might still be considered the result of the vision of some champions of a process of change, instead of an increase in their institutionalization.

(vi) Operative Measures. In order to guarantee the implementation of the EPS, the operative measures will be included in the Project's Operation Manual.

2.5. Social Risks

2.5.1. Analysis

World Bank's Safeguard Policies. The social risks that could appear in terms of the environment, resettlement, and indigenous populations have been considered. None of these issues raises any negative impact; however, plans for each of these issues have been prepared. (*See separate annex*)

Summary of Social Diversity and Gender Risks, Institutional Framework, Norms and Behavior, and Participation. In each of these sections, according to the findings of this evaluation, each of the challenges in social terms that could impair the social impact of this component have been pointed out, as well as the measures that could be taken to increase such impact, which can be summarized as follows:

(i) **Two beneficiary groups with different economic and social assets living in areas of high levels of poverty.** A small group with the highest assets coexists with a large number of poor people, the majority of which are small farmers, including vulnerable groups such as women, children, youth, unemployed, and indigenous communities.

(ii) **Institutional Arrangement for the Implementation of the component**. This issue raises important challenges due to the diversity of duties, capacities, and resources of the institutions that are part of this component.

(iii) **Social Actors**, including private contractors, with insufficient knowledge on road conservation and road maintenance through participatory processes.

(iv) **Low but increasing levels of social participation** in an environment of easy capture by economic and political elites.

2.5.2. Measures

The project will approach these issues through:

(i) **Differentiated Intervention Strategies** according to the needs and capacities of the different beneficiary groups;

(ii) **Institutional Strengthening Component** aimed at facilitating the institutional arrangement for implementation;

(iii) **Strengthening of the Communication Activities of the Social Participation Framework (EPS)**, which has been implemented since the start of project preparation with campaigns for public awareness, and training of the different sectors according to their needs and cultural characteristics;

(iv) **Consolidation of Participation Activities of the Social Participation Framework (EPS)** to guarantee the participation of the more vulnerable social groups, and increase transparency and accountability.

3. Indicators for the Monitoring of the Implementation of the Measures Recommended in this Evaluation

3.1 Diversity and Gender

3.1.(i) Levels of contribution by category of more direct beneficiaries in the Unpaved Road Network.

3.1.(ii) Levels of participation of the more vulnerable groups through positive action or "positive discrimination": women, children and youth, and unemployed.

3.2. Institutional Framework, Norms and Behavior

(i) Creation of the Social Unit in the MOPC.

(ii) Performance of the Social Unit in the MOPC.

(iii) Number of agreements achieved with the application of the lessons learned from current agreements, and adhesion on the need of road categorization and the assignment of responsibilities for each of the levels of government involved.

(iv) Levels of local capacity generation by type of result and incentive.

(v) Formalization of the Unpaved Road Network in public policy instruments.

(vi) Number of modalities ruled in the Collaboration Strategy for the Unpaved Road Network

(vii) Statute of Inter-Institutional Cooperation with public character.

(viii) Percentage of inclusion of the more vulnerable groups in all instances of decision making.

(ix) Training Program for the different social actors.

(x) Percentage of training offered to the different social actors.

(xi) Percentage of inclusion of the more vulnerable groups in the training programs.

3.3. Social Actors

(i) Number and levels of participation of the Associations, Universities, NGOs, or other organizations involved in the agreements.

3.4. Participation (Communication, Participation and Transparency)

(i) **Baseline for Social Impacts**. The baseline will be developed at the outset of the Project, and will serve as the basis of the measurement that will be performed at mid-term and the end of the Project.

(ii) **Transparency and Accountability System.** The system will be designed pointing to the generation of flows of timely and reliable information on the Project, making it accessible to all relevant social actors according to their needs and cultural characteristics. The means that could be used include websites, written press, radio, informative bulletins or newsletters (on paper or electronic versions), among others. In the same line, this system could include mechanisms for external observatories specialized by topic, allowing for different levels of knowledge of the stakeholders on topics such as:

a. Incorporation of measures envisioned in the following dimensions: social and gender diversity; institutions, norms and behaviors; and social actors.

b. Behavior of expenditures and mechanisms of road maintenance to preserve the road network's condition.

c. Project's progress, its objectives, activities, and achievements in its contribution to the strengthening of the MOPC.

d. Performance of the RVRB and information on its budget, including estimated and actual civil works' costs, and comparisons with other civil works.

e. Percentage of beneficiaries in road access, for access to social services and markets, using the baseline of social impacts, the conceptual framework of which is expected to be completed before the evaluation of the Project.

f. Percentage of satisfaction of the groups involved about the Unpaved Road Network, using participatory mechanisms for collection of data such as individual or community marking or ranking cards.

g. Contracts for the Unpaved Road Network, its outset, process and finalization.

Annex 13: Improved Governance Framework and Project Action Plan

PARAGUAY: Road Maintenance

1. Weak Governance, Efficiency and Corruption

The Ministry of Public Works of Paraguay (MOPC) receives around 10% of the total annual public expenditure. Within the Ministry more than 90% is dedicated to the road sector. As a result, improved efficiency in the use of these resources is essential for improving the overall public expenditure of the Paraguayan Government. The efficiency of public expenditure depends highly on the governance environment and practices, therefore any effort to improve resource allocation must be accompanied by an enabling governance framework[30] that facilitates accrued efficiency, and inhibits corrupt practices[31].

With this in mind and following the government's initiatives of improving transparency and governance overall, the MOPC wished to ensure that this project would be designed in a way as to guarantee its success and to contribute to improving the MOPCs governance practices. As a result, and with active MOPC participation it was decided to design and append to the project an Improved Governance Action Plan for the project.

This annex describes the major elements of weak governance in the Paraguay road sector, presents the Bank country wide risk assessments, describes the analyses carried out by the Task Team during project preparation regarding the governance context of MOPC, presents the series of workshops and discussions conducted by MOPC at the initiative of the project's UPGP to delineate the major components of the framework and Action Plan, and finally presents the resulting Improved Governance framework (IGF) and Improved Governance Action plan for the project (IGAP)[32].

2. Paraguay Country Context

2.1 Paraguay Governance Diagnostic

Paraguay was ranked in the 147th place among 159 nations surveyed for the Transparency International's 2005 Corruption Perceptions Index with a score of 2.1, improving the one obtained in 2004 of 1,9 (within a range that goes from 1 to 10 - 1 being very corrupt and 10 being very clean). The World Bank Institute's

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[30] The World Bank Institute (WBI) defines governance as the traditions and institutions by which authority in a country is exercised for the common good. This includes (i) the process by which those in authority are selected, monitored and replaced, (ii) the capacity of the government to effectively manage its resources and implement sound policies, and (iii) the respect of citizens and the state for the institutions that govern economic and social interactions among them.

[31] Corruption on the other hand, is generally defined as "the abuse of public office for private gain." In an environment of weak governance, it is believed that corruption may thrive; however, weak governance itself is not necessarily synonymous with corruption, but rather with different manners of inefficiencies. For instance, due to budgetary structure, bureaucracy, scarce resources or inadequate management capacity, agencies may distort priorities, or otherwise not adequately prepare programs – all of which lead to inefficiency, but not necessarily as a result of or to corrupt practices. Thus it is important to distinguish between corruption – intent and action for private gain with public goods – and a range of governance failures that likely provide opportunities if not inducements for rent seeking behavior.

[32] The framework and the action plan are a result of a two year preparation process that has accompanied this project and which has also seen the preparation of a Social Assessment (Annex 12) which led to establishing a Social Participation Framework for the whole project from early on in the preparation stage. The focus on improved efficiency and performance the project has adopted, has naturally led to analyzing the governance environment in which the project will evolve. This analysis has resulted in the framework and action plan presented below.

six indicators of governance, which include control of corruption, have placed systematically Paraguay in the lowest percentiles of the countries of the world. In 2004, the control of corruption indicator, which had a slight improvement from 1998, obtained the last place of several Latin American countries, with 12,8 (in a scale from 0 to 100 in which greater values imply better degrees of governance).

National reports are no less severe in the assessment. Political Culture and Democratic Governance Practices published in 2004 by CIRD2, a renowned civil society organization of the country that works on democracy and civil society issues, indicated that almost 80% of the surveyed population considers that politicians are not doing enough to improve the national situation. Transparency Paraguay published an Index of Transparency, Integrity and Efficiency, and a National Survey on Corruption in 2005. The main conclusions of both reports indicate that there is ample room for improvement regarding these issues in Paraguay.

Several governance and institutional weaknesses may be at the core of the low confidence and legitimacy in public institutions present today in Paraguay; particularly salient are the absence of transparency, voice and accountability, and weak basic management systems of government.

2.2 Current Governance Improvement Initiatives

The current government's development strategy led by President Duarte Frutos and his team when they took office in August 2003, foresaw as one of its main objectives the recovery of confidence in state institutions through among others measures a sustained fight against corruption and a greater participation of civil society in the formulation of public policy and the control of public expenditure.

In this respect, Paraguay has been undertaking some formal actions to tackle corruption, such as the ratification of the Interamerican Convention against Corruption in 1996, and the United Nations Convention against Corruption in 2004[33]. Furthermore, in 1999, an anti-corruption commission, which is now known as the National Integrity Council (CISNI), was formed with the technical support of the World Bank Institute. The commission comprised an equal number of government and civil society representatives. Based on the results of the WBI survey of public opinion, the Commission decided to concentrate on three areas - customs, the judiciary and public procurement. The Commission developed a program of institutional reforms for each area during 2000 and, following a period of public consultation, has been actively promoting the adoption of these reforms since then. The main achievement of the Commission to date has been shepherding the approval of a new procurement law through Congress and raising awareness within Government and the public at large that concrete steps can be taken to reduce the institutional opportunities for corruption.

The most outstanding progress of this strategy is the implementation of the Law of Public Procurement. The public presentation of declaration of assets of the highest level of the Executive Branch officials, including the President and Vice President, can be considered as a good basis for the enactment of the legislation, currently stagnated in the legislative branch. The implementation of the new customs code has been crucial for substantive improvement of an area repeatedly assessed as the most corrupt in Paraguay.

Nevertheless, increase of governmental transparency is still limited to initiatives such as that of the Ministry of Finance and its Unit of Transparency and Citizen Participation, the electronic system of

[33] These Conventions foresee several measures that impose the countries an agenda to improve their fight against corruption such as to enable greater public access to information, specially about public expenditures; civil service reform including public declaration of assets and code of ethics for governmental officials, improvement of institutions of control, judiciary system, procurement, customs, enhancement of accountability of political parties, increase civil society participation in public sphere, among other measures.

public procurement, other units of transparency in the Contraloría General de la República, some agreements with Transparency Paraguay to monitor some bidding processes, supported by IDB. Also, some institutions, among them the Parliament and the Supreme Court have recently adopted a code of ethics.

Some basic legislation to improve public accountability such as the access to the public information, regulation of declaration of public officials assets, civil society participation, modification of civil service have not passed despite the close lobby and follow up of several civil society organizations. Civil court procedures for the enforcement of private property rights and contracts are slow at best and corrupt at worst. Public prosecutors are inexperienced, understaffed and function without basic equipment.

In 2005, CISNI and the institution in charge of the statistical information in Paraguay - Dirección General de Estadística, Encuestas y Censos (DGEEC), with the technical assistance of the World Bank Institute, embarked on a follow-up of the Governance and Diagnostic work carried in 1999. This effort will provide information to the Government of Paraguay to monitor progress made on the areas of governance and anti-corruption since 1999, and it will provide information to update the governance strategy - Plan Nacional de Integridad. The World Bank has been supporting the process with technical assistance. The new system of public procurement being implemented from 2003, is perceived as an effective element tackling corruption. On the other hand, civil society's participation may increase, since some indicators such as their knowledge about mechanisms to denounce corruption improved 100 % in the last five years (from 15% that answered knew about them in 1999 to 30% in 2005).

3. Assessment of weak Governance risk areas in the roads sector

In the context of the Paraguay road sector, improving governance means mainly focusing on improving the capacity of the government to effectively manage its resources and implement sound policies, which is what the design of the proposed project aims at. Additionally, as has been described above, to ensure improved efficiency, and so the success of the project, it is also necessary to tackle the governance environment in which the project will evolve. Consequently from the outset of the preparation of this project an assessment of the Governance context of Paraguay in relation to the proposed project and the specific issues relating to the road sector and the Ministry of Public Works (MOPC) was carried out.

The assessment of the governance contest is based on the assessments carried out in preparation of the Country Assistance Strategy (CAS), and in particular the Country Financial Accountability Assessment (CFAA), the Country Procurement Assessment Review (CPAR) of 2002, documents produced by CISNI[34] and WBI, amongst others. Furthermore, specific documentation of the road sector in particular the ICR of the Eighth Highway project (last Bank project in the sector), and all the Aide Memoires on file were analyzed. On this basis, the sector lessons regarding the Governance environment were apprehended and integrated into project preparation and design.

On the basis of this accumulated knowledge the Task Team and MOPC decided to further explore the status of the governance environment in the road sector and MOPC in particular, and how it may relate to the project itself, through a series of discussions and three internal workshops.

This section presents the main findings of country wide assessments, sector specific assessments, and the results of the MOPC internal workshops on governance as it may relate to the project cycle of the proposed Road Maintenance project.

[34] Herramientas Básicas para una Gestión Pública Integra, CISNI, 2004. Desempeño Institucional, Gobernabilidad y Corrupción, CISNI, 1999; Desempeño Institucional, Gobernabilidad y Corrupción, CISNI, 2004.

3.1 Country wide governance assessment

The Task Team analyzed the background country assessments, such as, the CAS risk assessment, the CPAR and CFAA carried out in 2002, which contributed to the risks identified in the Country Assistance Strategy. The country risks as defined in the CAS may be summarized as follows: (i) existence of corruption; (ii) weak basic governmental management systems; (iii) special groups pressure the government to accommodate their interests; (iv) decline of the rule of law[35]. The CAS base case included as triggers the satisfactory progress on the Procurement Action Plan and the Financial Action Plan[36]. The proposed Road Maintenance project is in the CAS base case Scenario.

According to the last CPAR of 2002, Paraguay was considered to have a high procurement risk at that time, and a specific procurement improvement action plan was developed. The CPAR findings regarding specific Bank project implementations included: (i) weak implementation capacity, weak information systems; (ii) restrictive practices tending to favor participation of certain construction companies, including indications of collusion in some agencies; (iii) difficulties for land acquisition, due to expropriation system, which includes a one on one price negotiation with the land owner and (iv) PIU staff paid in dollars, establishing preferential treatment.

The CFAA carried out at the same time found that there was a high inherent risk of funds being used for unauthorized purposes despite advances in the central administration systems, and there was a high control risk given the weak control environment that prevails. As a result, the CFAA proposed reforms in the areas of internal control, external audit, and control of decentralized entities needed to achieve acceptable level of risk.

At the recent Country risk assessment it was concluded that significant progress had been made on procurement and that in all GOP institutions the situation today had significantly changed, and progress had been satisfactory on financial management issues. Third party analyses indicate the same opinion. As a result, at present Paraguay remains in the base case scenario.

Additionally, the recommendations regarding budgeting issues of the recent Public Expenditure Review (PER) were internalized. The link and importance of the budget preparation cycle and its impact on the ability of the Ministry of Works to adequately plan activities and resource allocations was clearly identified; there exists a vicious circle whereby the external mechanisms that affect the budget negatively affect the incentives of the line Ministry to adequately prepare budgets and link the to plans and results. At the same time, the lack of adequately planned and prepared budgets increases the probability of budgets that will be inapplicable and thus will affect negatively the incentives for future planning.

3.2 Sector and project specific governance assessment

In addition to the background analysis, and several discussions with a range of stakeholders and experiences in other countries, a series of workshops were held in the MOPC in order to determine the governance weaknesses that may affect the project cycle. The workshops were attended by the Viceminister, Directors of the main Directorates, Department heads and staff[37] The workshop discussions were framed in such a way as to identify what, in the opinion of the participants, were the areas of weak governance vulnerable to corruption within the MOPC affecting project cycles and what may be the suggested way of strengthening those weaknesses. The discussion was channeled by asking the attendants to provide inputs for improvements in three critical and inter-related elements: (i) mitigating tools and incentives; (ii) sanctions and remedies; (iii) transparency and empowerment of stakeholders.

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[35] Paraguay Country Risk Review, World Bank, April, 2006.

[36] To pass to the high case, the triggers included: (i) the enactment of revised legislation on asset declaration, including enforcement mechanism; (ii) revision of the civil service law and establishment of merit-based criteria.

[37] In all this amounted to around 30 people in two workshops.

On this basis, the discussions were carried out, producing a significant number of suggestions per project area and included in the appendix. These results, arrived at by Ministerial staff in open discussion are the basis of the proposed Improved Governance Action Plan.

Furthermore, the project fiduciary assessments (procurement and financial management, in annexes 7 and 8) carried out during project preparation and their conclusions were integrated as inputs to the project specific governance assessment and the recommendations are an integral part of the proposed Improved Governance Action Plan.

3.2.1 Project fiduciary assessments

An in-depth assessment of procurement capacity of the MOPC's UPGP capacity was carried out by the team and a detailed action plan was prepared to address all risks identified and to provide some opportunities for improvement in procurement matters, as discussed in Annex 8. The overall procurement risk has been rated as high. The assessment mainly indicates the lack of experience with Bank's procurement guidelines, procedures, documents and in contract information to ensure implementation of the project according to Bank requirements. The key features of the procurement risk mitigation strategy for this project include: (i) actions designed to strengthen the MOPC's procurement capacity, procedures and controls, and (ii) enhanced technical and fiduciary supervision during implementation. As a result of the mitigating measures the procurement risk assessment for the project is average.

Similarly, the in-depth financial management capacity assessment for the project carried out by the team and presented in annex 7, concludes that given the country level assessments and the existence of corrupt practices, administrative lapses reported by the control entities, and independent surveys, the inherent risk at country level is high. The inherent risk at entity and project level is considered to be substantial because of weak institutional capacity, cumbersome payment procedures and weak controls. The control risk after the implementation of mitigation measures is considered to be modest. The key features of the mitigation measures for the project include: (i) review of budget allocation to project; (ii) new dedicated FM information system; (iii) periodic reconciliation between project accounting and FM system (SICO); (iv) capacity building programs for MOPC staff; (v) automation of payments to avoid discretion of payment priorization; (vi) unaudited interim financial reports, as well as audits following international standards; (vii) implementation of the Improved Governance Action Plan. . As a result, of the mitigation measures the inherent entity and project risk is reduced to substantial but the overall assigned risk is maintained high due to country considerations.

Both procurement and financial management mitigation measures and recommendations have been incorporated into the Improved Governance Action Plan.

3.2.2 Sector level finding

Sector level findings stem essentially from the lessons learnt from previous projects and the discussions and workshops carried out within the MOPC on the specific issue of possible governance weaknesses believed to exist in the MOPC that may affect the project throughout its cycle. As a result of the workshops a detailed matrix was prepared that provided an identification of the areas considered to be vulnerable to governance weaknesses in the project cycle, possible mitigation measures, possible transparency measures, possible sanctions, and possible controls. This matrix as well as the discussions are the basis for the Improved Governance Action plan proposed. A summary of the findings are presented in an appendix to this annex.

4. A Proposed Improved Governance Framework

The MOPC has successfully implemented the modernization of its procurement system, and by all counts it is a significant improvement form the past. Additionally, the acting Minister has embraced the issues of transparency[38] and social participation that are the basis of this project and have been implemented during preparation. Both Ministers and Viceministers have participated in a significant number of workshops with participants from civil society, local governments and other stakeholders, demonstrating their commitment with the transparency and accountability pillars of this project. In essence, this project has induced accrued involvement and development of accountability and transparency activities by the MOPC staff, and all have shown interest and commitment to this approach which is an important base on which to develop further governance improvement initiatives.

As a result, the Paraguay Road Maintenance project has been designed from the outset, on building an integrated approach that seeks to enhance the efficacy and impact of the project on the one hand, and on the other to introduce practices and processes to the MOPC that would help it to gradually reform to improved governance practices.

To this end, its preparation has closely followed the three pillars that underpin this project's approach which, already at identification stage, were declared to be: ***(i) results-focus***, through the identification of targets for each intervention, to be monitored and whose attainment will be the indication of success; ***(ii) accountability***, through planning and programming all interventions based on objective and quantifiable criteria that can be presented for public scrutiny; and ***(iii) participation and transparency*** through mechanisms designed to facilitate stakeholders' having the opportunity to influence and share control over the decision making about the initiatives that affect them and with the flow of timely and reliable information on the progress of activities and results obtained made accessible to all relevant stakeholders. This approach was developed to strengthen some of MOPC's most important weaknesses and to create a stronger link to the Ministry of Finance, given the importance of the MOPC in the total public budget, and the GOP interest in improving public expenditure.

Consequently, it can be said that from the outset the project has been designed with a governance lens. The project elements incorporated into the design constitute the overall Improved Governance Framework of the project, and the first leg of the approach. The second leg is the Improved Governance Plan which complements the project designs with more specific measures. Below are presented the main features of the project design that are the main Improved Governance Framework elements:

(i) Increasing Accountability by setting budgetary linked policy commitments:

- ***Road Sector Strategy;*** The project is actively pursuing the issuance of a Road Sector Strategy that requires that the MOPC's future road management plans are linked to resource availability and budgets and discussed with all stakeholders.
- ***Linking priorities to budget;*** The project seeks to develop a five-year plan that will link activities to the budget and guide future MOPC decision on this basis, limiting discretionary decisions in the future.
- ***Road Social Contract;*** The project seeks to develop a Road Social Contract which consists of bilateral agreements between the MOPC and the major stakeholders (users, contractors, local governments, and national government) whereby the MOPC commits itself to improving the road sector through its new Road Sector Strategy in general, but commits to specific elements to each stakeholder according to the stakeholders interests after several consultations. In turn, the

[38] Recently, there were excessive unit costs of works financed by IDB and CAF detected and the MOPC carried out a one day conference to address the issues and to present technical explanations and proposed measures to an audience of around 100 people

stakeholder also commits itself to contributing to enhance the Road Strategy. The document is signed as a Contract amongst parties, and reflects the MOPC's commitment to a new strategy and renders it accountable to the main stakeholders concerned.

(ii) Sharpening the focus on results and performance:

- ***Performance-based contracts;*** A major element of the project is to introduce performance-based contract as a new tool of carrying out maintenance works. This approach provides the advantage of using indicators that may be easily monitored that will determine the level of the results obtained by MOPC projects and individual contractors. The advantage here is that the indicators are easily understood by the general public and can to a great extent be "monitored" by users empowering them to seek accountability.
- ***Results Framework and Socio-economic Impact;*** The project not only includes a robust results framework that will monitor the progress of the project outcomes, and outputs, but it will also measure socio-economic impacts through a specially designed survey that will create a poverty, and socio-economic baseline that will be measured twice during the life of the project, to ascertain what the real impact is of the project on income and poverty.

(iii) Ensuring transparency of the decision making process, objectives and commitments:

- ***Social Participation and Transparency Framework:*** The unpaved road network component has been designed and prepared through a participatory process with local communities, local governments, and civil society, both for the definition of the basic road network to include in the project and also to define the contributions, in-kind or monetary that the community could make to the project. This approach has had the advantage of involving potential beneficiaries from the beginning, empowering them, making them participate in decisions that will influence final outcomes and also promoting transparency and commitment of MOPC with the project.
- ***Dissemination and consultation:*** Throughout project design consultations have been carried out with stakeholder and institutions in order to arrive at a consensus regarding the project. Equally, a campaign of dissemination of the project objectives and design has been carried out throughout preparation. The campaign will continue during implementation to disseminate progress, results and indicators throughout the projects life-span.

Nonetheless, and given the risk assessment carried out it was felt that, in addition to the specific design of the project and the guarantees its design may provide as an Improved Governance Framework, a specific Action Plan could accompany the project to deal with details in processes that were identified as weak, and to provide a structure to monitor and evaluate progress.

5. Project's Improved Governance Action Plan

The objective of this Plan is to strengthen governance around the project and as a result prevent corruptive practices, so that the full impact potential of the project is attained.

The proposed plan has been designed specifically for the Paraguay Road Maintenance project, nonetheless the structure of the proposed plan would be applicable to any donor financed project, or to the MOPC as a whole. Preliminary discussions with IDB, who is a major donor of the road sector, show that IDB would be willing to adopt the same plan for its future operation. The implementation of this plan by the MOPC on two major loans would contribute greatly to the overall governance environment and would permeate to activities financed by other sources.

5.1 Action Plan Structure

The Plan is essentially a tool to improve the impact of the project and to transfer a number of methods and practices improving governance that may be adopted by the MOPC at large to improve efficiency of

operations throughout. The plan will be thus composed of preventive actions, deterrents, and detection mechanisms.

The prevention measures are specific to the project cycle and processes and will address these processes within larger areas of weakness that were identified. All measures are presented as mitigation measures and incentives. Additionally, particular transparency measures are proposed that enhance mitigation and increase accountability.

The deterrence measures will include control mechanisms and sanctions. Sanctions will be applied in the case of proven corruptive practices. The sanctions will need to be applicable for them to be a real deterrent, and consequently will require actionable proof. The difficulty, however, lies in finding actionable proof that can without a shadow of a doubt prove the intent to acquire personal gain, which will clearly demonstrate corruption. To this end, specific investigative tools, skills and teams are needed to analyze actions that have already taken place. It is clear then, that a Task Team has neither the vocation nor skills to carry-out any sort of post-act investigation to seek proof of corruptive practices. Thus, INT will be in charge of this investigation[39].

The detection measures will include two phases and corresponding indicators: (i) indicators of potential inefficiencies – "alerts"; (ii) indicators of potential fraud – "red flags". In the last few years significant research has gone into analytical work regarding indicators of suspicion of corruptive practices. The anti-corruption literature has produced a certain number of indicators that have been denominated commonly as "red flags" and that are supposed to alert of possible corruption or fraud.[40] These indicators are very vast in their scope as shown in Box 1, and in a sense, although they may only be a potential indication of fraud, they are already proof of inefficiency at best or anti-competitive and unfair practices at worst. In other words, there is significant incentive to avoid reaching this stage in the first place.

5.2 Implementing Arrangements

This plan is to be implemented by the MOPC counterparts and the Bank Task Team. The supervisory strategy will be the following: (i) the project preparation unit (UPGP), will coordinate so that all MOPC divisions involved in the project are providing the identified indicators of the Plan, through follow-up of task completion; (ii) the MOPC units involved in the project will maintain relevant IGAP indicators; (iii) the internal audit department of MOPC will prepare a periodic report on the basis of indicators kept and monitored and send it to the Minister, the Bank Task Team and a third party (NGO); (iv) the Task Team will monitor the plan on the basis of this periodic report containing indicators.

Regular project supervision will be strengthened by undertaking: i) regular technical supervision to determine quality and progress of implementation in each scheduled supervision; and ii) integrated fiduciary spot checks.

The Bank Task Team will monitor the progress of the Improved Governance Action Plan, with its MOPC counterpart, UPGP and Internal Audit, following the Framework in Table A13-1.

Supervision and Monitoring process

The Task Team will monitor the IGAP on the basis of the Monitoring Framework proposed. The mechanism will be a six-monthly report sent to Task Team by UPGP and Internal Audit Unit of MOPC and the review will be conducted during supervision missions.

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[39] Sanctions will be applied as the final outcome of the Action Plan, and as such are not presented in the Deterrence column of Table A13-2.

[40] See Box A13-1 for examples of fraud indicators, as described in: "Fraud Indicators", Office of the Inspector General Investigations, USAID.

Additionally, an in-depth review of the Improved Governance Action Plan will be carried out at the mid-term review; the UPGP and Internal Audit will make a formal presentation of the progress made during the project life up to that point. Both reporting mechanisms will be reporting covenants.

When any indicator reaches an Alert level, the Task Team will seek explanations from the UPGP and MOPC Internal Audit Unit, who will present a written response. If the answer is satisfactory to the Task Team – the alert will be lifted. If not, the Task Team will conduct an enhanced supervision.

If more than half of the indicators reach an Alert level the Task Team will conduct a specific enhanced supervision mission.

Enhanced supervision

The scope of the enhanced supervision will be defined by the Task Team according to the Alert indicators and levels detected. The enhanced supervision team will include the fiduciary team (Financial Management and Procurement Specialist) and whatever additional skill mix is warranted according to the Alerts detected.

During the Task Team's enhanced supervision, it will endeavor to analyze the issues and look out for the "Red Flags" indicated in Box A13-1.

If no red flags are detected as a result of the enhanced supervision, the Task Team will update the risk assessment of the project and propose project implementation improvement measures if necessary.

If Red Flags are found, the team will seek a written explanation from the Internal Audit of the MOPC and report to INT.

The explanations given by MOPC and the gravity and extent of Red Flags detected will be discussed with INT in an analysis meeting. If it is deemed that the explanations are satisfactory, the alert will be lifted. If the explanations are not satisfactory or the gravity of the Red Flags is deemed high then INT will proceed to an investigation.

Red Flag Investigation

INT will investigate the reported Red Flags through a mission to the country, and discussions with MOPC Internal Audit and Transparencia Paraguay, who will be monitoring as third party.

If no proof of corruption is found the alert will be lifted, and INT will provide recommendations for improvements to be made to the monitoring framework, IGAP and project implementation process.

If proof of corruption is found the Bank will officially ask the MOPC to take the measures that the existing legal framework allows:

- legal proceedings against individuals;
- black listing of firms.

Table A13 -1 Improved Governance Implementation Arrangements

ENTITY	STAGES AND TRIGGERS					
	Supervision	Trigger	Enhanced Supervision	Trigger	Investigation	Outcomes
MOPC	• Maintenance of databases • Biannual Reporting of core indicators • Fiduciary requirements	Alert Indicator[1]	• Written explanations of Alert Indicator causality • Provision of information	Red Flag[2]	• Written explanations of Red Flags • Provision of information	• Application of sanctions
MH						• Application of sanctions
WB Task Team	• At least two annual integrated technical/ fiduciary team missions; • Monitoring IGAP for Alert Indicators		• Additional fiduciary/governance supervision(team skill mix, scope and timing of mission): - IGAP update - Project Implementation Improvement Plan • Monitoring for Red Flags		• Call INT for joint analysis of detected Red Flags and MOPC explanations	
INT					• Joint analysis with Task Team of Red Flags • Investigation of Red Flags	• Debarrement • Referral

1) Alert Indicators are presented in Table A13-2
2) Red Flags are presented in Box A13-1

Reporting

The reporting requirements will include a semi-annual report produced by MOPC, Internal Audit Unit, on the Alert Indicators established in the IGAP. This procedure will be streamlined into the normal supervision missions.

MOPC will have to provide written explanations to Task Team requests when Alert Indicators warrant explanations.

Once a year, streamlined into internal Bank reporting the Task Team will send a report to INT indicating how many the number of enhanced supervision missions, if any, and if there has been the detection of any Red Flags.

Figure A13-1: Structure of Improved Governance and Anti-corruption Action Plan

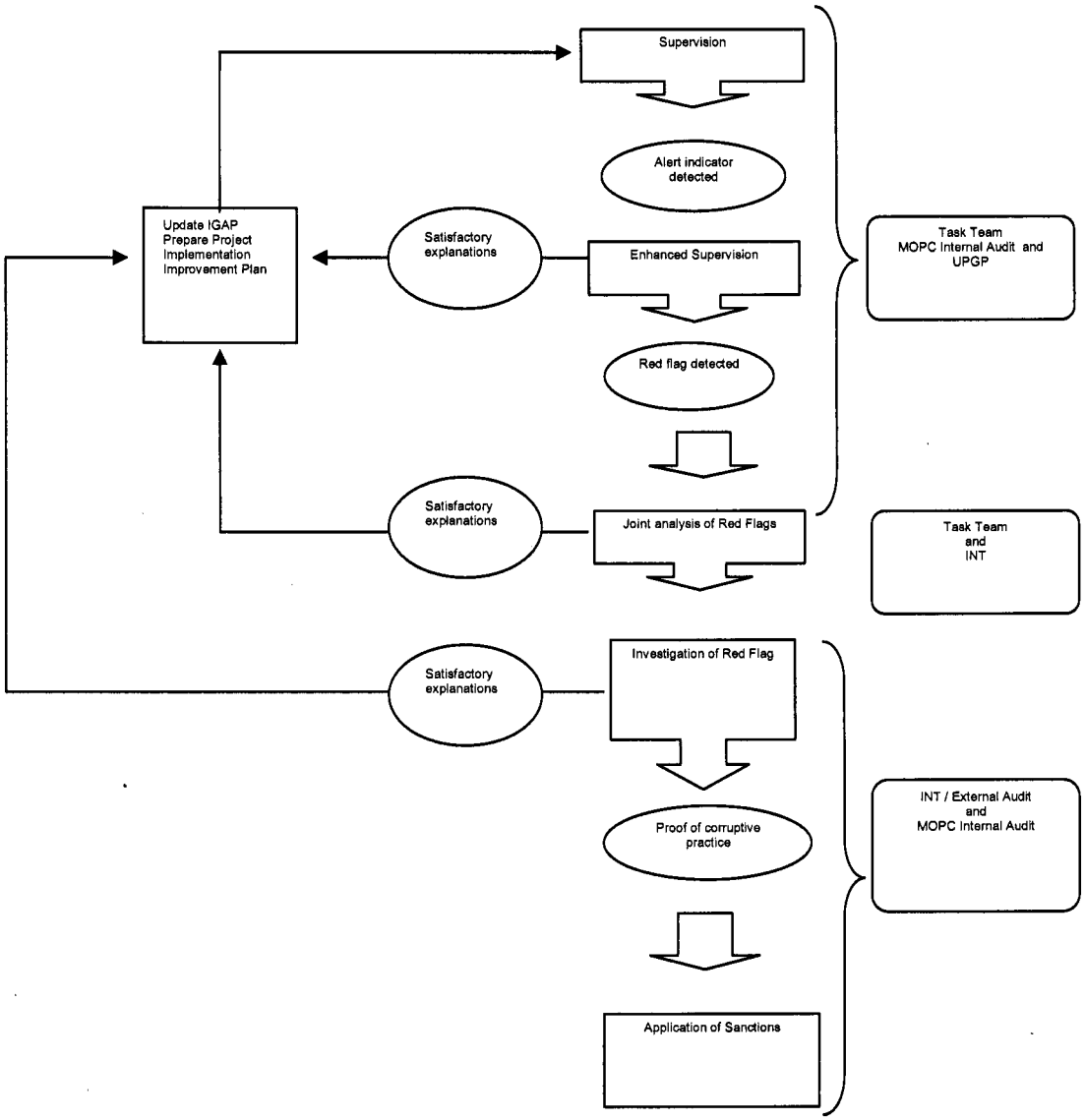


Table A13 -2 Improved Governance Control Framework

Governance Risks	Mitigation Actions and Controls			Detection
	Prevention	Transparency	Deterrence	Alert Indicator
1) POLICY FORMULATION, AND PLANNING				
Unclear road strategy and mid-term and long-term choices allowing for inefficient short term choices	Establishment of a Road Sector Strategy and signature of a Road Social Contract with stakeholders Satisfactory Implementation of Road Sector Strategy and Road Social Contract	Broad dissemination of Road Strategy and Road Social Contract.		Road Social Contract developed, negotiated and signed with stakeholders Road Social Contract implemented
Lack of planning which allows discretionality and arbitrary selection of projects	Development of planning tools such as Five year-term road plan, introduction of HDM4 for road section selection	Broad dissemination of Five-year road plan		Approval of Five-year road plan.
Lack of institutional setup to gather , maintain and update information for informing the planning effort	Creation of a Strategic Planning Unit	Dissemination among relevant stake-holders of the creation of the SPU and its activities		Effective creation of Strategic Planning Unit.
Unclear and inefficient project implementation procedures and cycle	Preparation and effective use of a project Operational Manual	Dissemination of relevant aspects of Operational Manual among relevant stakeholders		Deviation from procedures in Operational Manual
2) BUDGETING				
Inadequate budget preparation, lack organization, lack of adequate databases	Execution of a capacity building program in budget preparation, restructuring of data relative to budget and concentration of preparation activities in one unit.			Adequate database and budget monitoring system created.
Lack of linkage between the budget and the policy priorities of projects of MOPC	Commitment to Road Sector Strategy investment and physical targets stated by the MOPC through the adoption of the strategy.	Dissemination of proposed budget in terms of Road Strategy Targets	Report to Bank Team on proposed annual budget in relation to Road Strategy target.	Deviation of budget from Road Strategy annual budget targets
Inadequate budget itemization	Development of a new budget itemization proposal to include road maintenance activities			
3) PROCUREMENT				
Unclear, diverse, unofficial technical specifications	Preparation and use standard bidding documents for NCB contracting	Disseminate new standard bidding documents for NCB contracting		Standard bidding documents prepared and approved for NCB projects financed under this loan
Information leaks of bidding documents		Make available all technical and background information and preparation studies to all as soon as produced		Communication of extent, location and type of background information available to all potential participating firms

Governance Risks	Mitigation Actions and Controls			Detection
	Prevention	**Transparency**	**Deterrence**	**Alert Indicator**
Insufficient technical staff or capacity of staff for adequate preparation of technical specifications	Increase in the number of agents assigned to the control, and supervision of design and other consulting services	Publish the tender documents in international media such as Digimarket, Development Business and newspaper of other countries		
Collusion	Increase the number of participating firms to the maximum possible and reduce the prequalification instances.			Number of firms participating in bids
	Elimination of prequalification phase for performance-based maintenance contracts			
	Use of international tenders for performance-based contracts with a longer period to present bids given (90 days)			
	Declaration in detail of subcontractors in all bidding documents.			Subcontractors do not have adequate qualifications.
	Dimensioning the size and number of contracts to generate more competition			
		Publish grey list on MOPC webpage and other media		Price variance of submitted proposals
Inadequate bid evaluation process	Applying Ministerial resolution adapting procurement procedures to those compatible with those of the Bank			Length of procurement process, by steps until designation of winner
Weakness in norms for contract negotiation	Use of Bank procedures, training in Bank procedures			Length of contract negotiation until signature
Excessive delays for contract awards	MOPC will set-up a procurement database, so that all steps are monitored, and may serve as post-procurement analysis.	Make public contract process and expected dates by steps, using means such as the webpage		
Lack of experience in World Bank procedures	Execution of a capacity building program and training for staff on World Bank procedures		Assessment of procurement capacity in first year of implementation	
Inefficient planning of procurement needs and cash plan or budgetary allocations	Preparation and adherence to a procurement plan for the project	Publish the procurement plan in the MOPC webpage		

Governance Risks	Mitigation Actions and Controls			Detection
	Prevention	Transparency	Deterrence	Alert Indicator
4) EXECUTION AND SUPERVISION				
Inefficiency of supervision during the execution of works	Create and implement mechanisms for public information and complaints, including free call system	Publish in webpage and on road side signs and other media all complaint mechanisms, and channels		Number of public complaints
Inefficiency of MOPC control of firms carrying-out the supervision	Preparation and implementation of agreements with University group/ locals to monitor works and levels of service	Publish in webpage and other media all third party monitoring agreements, and their progress		Number of unsatisfactory reports from voluntary third party monitors
Collusion between construction firm and supervision firm to create delays	Design supervision contracts to mitigate incentives for delays	Publish grey list on MOPC webpage and other media		Timely conclusion of works
Lack of objective criteria for the final acceptance of works resulting in opaque acceptance procedure	Design works contracts with objective criteria for the final reception of works			Objective criteria for works acceptance in contracts
5) FINANCIAL MANAGEMENT				
Risk of budgeting abuse and state capture	Separate program budget line in Ministry's budget system to monitor budget allocations and cash plans allotments		FM supervision will include specific review of the budget allocations and execution on this specific budget line	
	Prepare a multiyear budget for project implementation	Dissemination of multiyear budge		
Accounting weakness may lead to improper allocation of funds	Implement a new dedicated FM information and accounting system part of project implementation		Periodic reconciliation between project accounting and Public FM system (SICO) will be implemented and described in operational manual	
	Execute capacity Building activities for DCP part of project implementation (training and equipment			
Weakness in internal controls	Execute capacity building of Internal Audit Unit part of project implementation (training and equipment).		External audit will be carried out using international standard every year	
Excessive internal bureaucracy for payment causing delays		Monitor payment delays and make them public via the webpage		Payment compliance with standard payment processing times
Weak financial reporting practices	Implement a new FM information system part of project implementation		Unaudited interim financial reports would be produced also by the system	

Governance Risks	Mitigation Actions and Controls			Detection
	Prevention	**Transparency**	**Deterrence**	**Alert Indicator**
6) REVENUE COLLECTION				
The toll system lacks adequate control mechanisms , facilitating evasion and reducing potential revenues	Modernization of the tolling control system (automating traffic counting, modernizing toll booths, and creating a centralized control center	Publishing toll revenues in a regular publication and webpage	Unaudited annual interim financial reports	Toll revenues
Obsolete and inadequate equipment and procedures in the toll system. Reducing efficiency and productivity	Modernization of operating system and procedures (operating manuals, surveillance system, etc)			
7) POLITICAL INTERFERENCE				
Political interference and influence to define resource allocation, road sections and activities. Resulting in allocation according to political sympathy	Road Strategy and Road social contract. Preparation of Departmental Road Master Plans through a participatory planning process with all local stakeholders	Dissemination of Road Strategy, Road Social Contract, and dissemination of Departmental Road Master Plans that were prepared in a participatory planning manner with all local stakeholders		Execution according to Road Master Plan.
Equipment and human resources in decentralized MOPC respond to local pressures by local governments and representatives and attend the lower level road networks	Execute capacity building and modernization program in MOPC Districts.			
8) LIMITED CAPACITY				
Lack of experience in MOPC in performance-based contracts may limit supervision of contracts	Capacity building program, and consultant advisors during preparation and supervision of first contracts			Compliance by the contractors with level of service requirements
Lack of experience in participatory planning exercises and social monitoring, may reduce ability to replicate preparation of future Department's Road Master Plans	One staff member with relevant capacity designated to follow all participatory planning and social monitoring activities from preparation on, with appropriate technical background			Designation of staff member to social participatory activities.
Lack of experience and capacity in safeguards may reduce monitoring and control ability	Execute capacity building program for Environmental Unit, and/or any other instance in charge of the safeguards	Dissemination of Environmental Assessments, Mitigation Measures, Social Assessment, Indigenous People's Plan, and Resettlement Framework and Action Plan.		Satisfactory execution of proposed safeguard mitigation plans
Lack of experience of local contractors of performance-based maintenance contracts may	Road Social Contract signed with contractors. . Training sessions to be	Dissemination of Road Social Contract with Contractor,		

produced unsatisfactory results	held with contractor on the performance-based maintenance contract bids and bidding documents.	including workshops and training sessions.on performance-based maintenance contract bids and bidding documents		
Reduced training and knowledge of road engineering and maintenance by local governments may hamper planned supervision and monitoring of activities	Road Social Contract signed with local governments. Cooperation Strategy with Local Governments implemented to train them so that they can perform road works monitoring tasks	Dissemination of Road Social Contract and Cooperation Strategy with Local Governments, using media according to the needs of relevant stakeholders, including workshops		
9) PROJECT IMPACT ENHANCEMENT				
Lack of impact of project on local populations because of road maintenance modalities chosen	Road maintenance modalities in rural networks chosen through a participatory approach. Local microenterprises, associations etc. privileged	Dissemination of Participatory Approach of road maintenance modalities using media according to the needs of relevant stakeholders, including workshops		
Discrimination of indigenous population, women and children in participating and benefiting from project.	Positive discrimination measures in favor of indigenous peoples, women and children, in Indigenous People's Plan (IPP) and Social Assessment	Dissemination of Positive Discrimination Approach, using media according to the needs of relevant stakeholders		Deviation from proposed actions in IPP and social assessment recommendations.

Table A13 -3 Indicators and Triggers

Alert Indicator	Data origin	Time Frame	Alert level
Road Social Contract developed, negotiated and signed with stakeholders	- UPGP	During first year of effectiveness.	- failure
Road Social Contract implemented	- UPGP		- not satisfactory
Approval of Five-year road plan.	- VMOPC	Between first and second year of effectiveness.	- failure
Effective creation of Strategic Planning Unit.	- VMOPC	During first year of effectiveness.	- failure
Adequate database and budget monitoring system created.	- DAT	During first year of effectiveness.	- failure
Deviation of budget from Road Strategy annual budget targets	- DAT		- failure
Standard bidding documents prepared and approved for NCB projects financed under this loan	- UOC/UPGP	During first year of effectiveness.	- failure
Communication of extent, location and type of background information available to all potential participating firms	- UPGP		- failure
Number of firms participating in bids	- UOC		- less than 3 and repetition
Subcontractors have adequate qualifications.	-UOC		-failure
Price variance of submitted proposals	-UOC	-	-significant variance between lowest and highest bidders
Length of procurement process, by steps until designation of winner	-UOC	-	-significant deviation from the maximum amount as per Ministerial resolution
Length of contract negotiation until signature.	- UPGP		- significant deviation from maximum amount as per Ministerial resolution
Number of public complaints per project	- UPGP	-	- significant deviation from maximum amount as per Ministerial resolution
Number of unsatisfactory reports from voluntary third party monitor	- UPGP		- significant deviation from maximum amount as per Ministerial resolution
- Timely conclusion of works	-UPGP		- significant deviation from stipulated calendar in contract
- Objective criteria for works acceptance in contracts	- UPGP		- failure
- Payment compliance with standard payment processing times	- UPGP		- - significant deviation from standard payment processing

			time.
- Toll revenues	- DF		- decreasing revenues
Indicator- "Alert"	**Type of Indicator**		**Alert level**
Execution according to Road Master Plan.	- VMOP		- not satisfactory.
Compliance by the contractor with level of service requirements	- UPGP		- not satisfactory.
Designation of staff member to social participatory activities.	- UPGP	During first year of effectiveness.	- failure
Execution of proposed safeguard mitigation plans	- UA		- not satisfactory.
Deviation from proposed actions in IPP and social assessment recommendations.	- UPGP		- not satisfactory

BOX A13-1
Fraud Indicators or Red Flags

Fraud indicators are best described as clues or hints that a closer look should be made at an area or activity. Although poor management decisions or negligence may give rise to possible indications of fraud, the difference between fraud and negligence is a fine line called intent. All that indicators can do is to point the way for further detailed inquiry.

PRE-SOLICITATION PHASE

- Release of information by firms participating in project design to contractors competing for the contract
- Designing "pre-qualification" standards or specifications to exclude otherwise qualified contractors
- Splitting up requirements to get under small purchase requirements or to avoid prescribed levels of review or approval
- Information leaks to contractors or their representatives by technical or contracting personnel
- Justifications for sole source or negotiated procurement signed by officials without authority or bypassing required levels of review

SOLITATION PHASE

1) Preparation of Bidding Documents:

- Rigged specifications to meet the qualifications of one particular contractor
- Placing any restrictions in the solicitation documents to restrict competition
- Restricting procurements to exclude or hamper any qualified contractor
- Limiting the time for submission of bids so that only those with advance information have adequate time to prepare bids
- Revealing any information about procurement to one contractor, which is not revealed to all (from either technical or contracting personnel)
- Conducting bidders' conference in a way which invites bid rigging or price fixing, or permits improper communications between contractors
- Failure to assure that a sufficient number of potential competitors is aware of the solicitation
- Improper communication with contractors, or improper social contact with contractor representatives
- Government personnel or their families acquiring a financial interest or employment in a contractor or subcontractor
- Special assistance to a contractor in preparing bid
- Referring a contractor to a specific subcontractor

2) Proposal Submission

- Improper acceptance of late bid
- Falsification of documents or receipts to get a late bid accepted
- Withdrawal of the low bidder (who may become a subcontractor to the higher bidder who gets the contract)
- Collusion or bid rigging between bidders (Indicators of bid rigging: (i) identical bids are received; (ii) a number of bids are received that are much higher than published costs of previous contracts of the same type, or of previous bids by the same firms for similar contracts; (iii) fewer firms bid than would normally be expected from that industry; (iv) there is an inexplicably large gap between the winning bid and all other bids; (v) apparent recurring patterns of low bids, such as corporations always winning a bid in a certain geographical area, or other patterns indicating collusive division of territory, or in a particular rotational sequence vis-à-vis other bidders; (vi) the successful bidder subcontracts work to companies that submitted higher bids on the same project; (vii) bids are very close on non-standard items with no suggested retail price; (viii) correlation between contractors that win the bids and the size of the contracts; (ix) certain contractors always bid against each other or conversely certain contractors do not bid against one another; (x) competing contractors regularly socialize, or contractors and government procurement personnel socialize.
- False certifications/information of contractor (size of business certification; certification of independent price determination; financial capabilities; performance; companies conducting business under several names; etc)
- Change in bid after other bidders' prices are known

3) Bid Evaluation

- Improperly disqualifying or discarding the bid or proposal of a contractor
- Accepting non-responsive bids from preferred contractors
- Unnecessary contact with contractor personnel by persons other than the contracting officer during solicitation, evaluation and negotiation processes
- Any unauthorized release of information to a contractor or other person
- Any exercise of favoritism toward a particular contractor during the evaluation process

- Use of biased evaluation criteria or biased individuals on the evaluation panel
- Documents from competing firms contain similar or identical: (i) company names; (ii) handwriting/signatures; (iii) company stationary; (iv) invoice numbers (in sequence); (v) telephone numbers.

POST-SOLICITATION PHASE

4) Contract Attribution and Signature

- Award of a contract to a contractor who is not the lowest responsible, responsive bidder
- Disqualification of any qualified bidder
- Allowing a bidder to withdraw without justification
- Failure to forfeit bid bonds when a contractor withdraws improperly
- Material changes in the contract shortly after award
- Awards made to contractors with an apparent history of poor performance
- Awards made to the lowest of a very few bidders without re-advertising considerations or without adequate publicity
- Awards made that include items other than those contained in bid specifications
- Awards made without adequate documentation of all pre-award and post-award actions including all understandings or oral agreements
- "Back-dated" or after-the-fact justifications may appear in the contract file or may be signed by persons without the authority to approve noncompetitive procurement
- Contractor misrepresentation as to costs during negotiations
- Failure of government personnel to obtain and rely upon pricing data

5) Execution, Supervision and Control

- Receipt of works and services is certified even though physical inspections have not been performed
- Contractors fail to meet the contract terms but nothing is done to force compliance
- Unsuccessful bidders become subcontractors after the contract is awarded
- The labor of government employees is used to perform parts of contracted work
- Contract files are either incomplete or missing required documents
- Contract documents are altered, backdated, or modified to cover deficiencies
- Fictitious or inordinate time frames and dates are entered on contractor records (e.g. maintenance; inspection; receipt of reports)
- Contract deviations by means of changes requested and granted immediately after contract award
- Used or inferior products are substituted for the product actually ordered
- Defective pricing, which might include: (i) persistent defective pricing; (ii) repeated defective pricing involving similar patterns or conditions; (iii) failure to correct known system deficiencies; (iii) indications of falsification or alteration of supporting data; (iv) protracted delay in release of data to government to preclude possible price reductions; (v) identical or nearly identical high salary history data on employees or consultants.
- Employment of people known to have previously perpetrated fraud against the government

6) Payments

- Contractors are overpaid or paid twice for the same items/services and there is no attempt to recoup the overpayments
- Accounting reconciliation is not performed regularly relative to (i) contract payments; (ii) daily transactions; (iii) inventory.
- Cost proposal data that is incorrect or less than current or complete
- Billings (including progress payments) not adequately supported by project status or reliable cost data (including duplicate or altered invoices; double billing; etc)
- Significant increase in price without corresponding increase in work
- Substantial subcontracting without the knowledge and approval of contracting officer
- Failure to meet specifications

Source: Fraud Indicators: Office of the Inspector General Investigations; USAID

Appendix 13A

MOPC Governance Workshop Findings

On 25th April and 12th, May 2006, two MOPC Governance Workshop took place in order to identify areas of governance weakness risks throughout the project cycle. The identification of risk areas were grouped into sector level, project specific level, and project component specific level.

Sector level findings

The Budgeting Process: Each year, congressmen and senators modify the budget presented to them by the executive to the point of entering into the most detailed budget items and even the salaries of individuals in line ministries. The process goes through several stages from commissions to the plenary in both houses. The practice then includes lobbying by the line ministry for budgets not to be cut and negotiations ensue between individual congressmen and ministerial staff that involve transactions that will affect the final outcome of the budget structure. This activity renders itself to the possibility of Ministry staff making more "efforts" in favor of some parts of the budget then others, particularly for projects that will benefit a more "generous" contractor.

Additionally, the role of Congress renders budget planning and resource allocation in MOPC irrelevant, and introduces the practice of discretional decision on what is to be done and with what resources at that level. This practice is equally relayed by line Ministers as common practice within their institutions.

The influence of the existing budget preparation process on discretional allocation of budget and activities within ministries is thus significant and renders accountability difficult. The MOPC is, within the framework of this project, making efforts to link road strategy and budget preparation in a way as to limit discretionality and increase accountability. In parallel, and as a result of the Bank Public Expenditure Review the Government is seeking measures to deal with the impacts of the existing budgeting process.

Subjective criteria for project selection In addition to the above, and perhaps both cause and consequence of the budgeting process, selection of projects and major investments are decided by the Minister based on essential the electoral platform and political motivation. From then on project financing is sought and preparation and implementation take place. The projects, if financed by IFIs, are submitted to the scrutiny and evaluation methods of IFIs, but an overall strategy and arbitration mechanism between competing demands does not exist. There is no room for optimum resource allocation in this framework and planning initiatives and sensible budget programming become obsolete. The discretionary project selection against no clear criteria rather than the will of the political authority creates an adverse climate for seeking optimal resource allocation.

Planning in the transport sector remains weak. The traditional lack of correlation between priorities (determined by some consistently applied criteria, whether economic or not) and budgets and investments still exists. In fact, a mixture of tradition, institutional set-up, lack of capacity, lack of resources, has rendered the planning process irrelevant. This is evident as planning has been handicapped by the following situations: (i) instinctive pre-eminence on the superior authority as a matter of tradition for project selection, as explained above, (ii) difficulty in attracting young professionals to MOPC staff so that the existing technical staff is of an advanced average age and has had a long permanence in the Ministry. Besides of the possible benefits this type of staff may have, it is less inclined to adopt new planning tools, sophisticated software and in general tends to work in a more traditional way, and (iii) remunerations of MOPC staff are very low, obliging staff to keep a second job in the afternoons and significantly reducing productivity in MOPC. This situation causes further tension with temporary staff

contracted by multilateral financing entities for the implementation of externally financed projects, whose remunerations are higher[41].

Transport expenditures show various types of imbalances that warrant a rethinking of transport sector planning. The bulk of the investment has concentrated on paving the main roads, too little has been invested on improving the secondary and rural roads, resulting in vast areas of the country still lacking adequate access to the main transportation corridors. More critically, road maintenance has never been considered an expenditure priority.

Project level specific findings

As a result of all assessments, the main areas identified to date that may present governance weakness and may affect projects implemented by the MOPC are the following:

a) Preparation of Bidding Documents:

Unclear, diverse, unofficial technical specifications The MOPC suffers from a lack of official or standard technical specifications for works, which lead to situations of difficulties for evaluators, negotiators from MOPC when dealing with contractors, especially in conflictual contexts.

Lack of sufficient expertise of staff preparing technical specifications The above point is compounded with insufficient staff with all the technical knowledge to deal with many technical specifications, and thus the preparation of specifications from the outset may be faulty, obliging post bidding negotiations with the winning contractor on unfavorable terms to the MOPC.

Technical specifications directed to a specific audience The lack of clarity, competency and expertise may additionally lead to making it easier to tailor technical specifications for a specific audience, making the process vulnerable to corruption.

Deficient project designs Deficient project designs may lead to the same situation as the technical specifications may be correct but corresponding to erroneous concepts and leading to the same above mentioned results. When the designs, which is usually the case, are carried out by consulting firms, this denotes a lack of adequate supervision by the MOPC of the consulting firm. Additionally it was noted that the lack of overall efficiency and bureaucracy may also lead to delays in bidding that make the original designs obsolete when it comes to tendering the corresponding works contract.

Information leaks There may be information leaks regarding the contents of the bidding documents in favor of a particular bidder, putting competitors at a disadvantage.

Lack of knowledge and expertise with Bank procurement guidelines. The procurement assessment for the projects determined that there exists little knowledge among MOPC staff regarding Bank procedures, as the previous Bank project was quite a while ago. This may lead to delays and errors.

b) Proposal submission:

Collusion by contractors During bid submission collusion has been suspected to occur, also during contract attribution, where the winning bidder who has bid a lower price does not sign the contract and leaves it to the second best bidder; MOPC staff suspects collusion between first and second placed bidders.

c) Bid evaluation

[41] MOPC remunerations range approximately from $150 to $800 (Director of Roads). The IDB contracted staff for the loan supporting Rural Roads has contracted staff at $1500 to $2000. All figures are indicative.

Lack of coherence of the evaluation team The MOPC official evaluation team is made up of the Viceministers and the corresponding Directors and Legal Director. The actual evaluators are technical staff that do not sign the evaluation, and take a certain amount of time as they are not solely carrying out that task. Additionally, the evaluation goes from one person to another as after the technical evaluation takes place, the legal and economic evaluations follow; it is not all carried out at the same time. The process is particularly prone to delays and leaks.

Leakage of information regarding the evaluation process This was clearly identified as an issue and the above point explains one of the possible causes of leakage.
Lack of set evaluation criteria It was felt that the lack of official norms regarding the evaluation criteria could lead to manipulation of bid documents to favor a given candidate during the bidding process.

Absence of responsibility of the evaluator The technical evaluator during bid evaluation does not sign the evaluation he or she carries out, it is the corresponding Viceminister and Director who make up the committee who are ultimately responsible who sign. This approach, which involves signatures at three stages by the evaluating committee is prone to delays, and is opaque regarding the capacity and quality of the evaluator.

d) Contract attribution and signature
Weaknesses in norms and procedures for negotiations Lack of preparation and a normative framework that defines the procedures during negotiations, lead to frailty of the negotiating team on the MOPC. Together with the fact that the preceding step is prone to errors, the likelihood of negotiations is high at contract attribution – making this particular step very vulnerable.

Delays in contract attribution Delays in contract attribution stem from the bureaucratic process through which the MOPC goes through to "validate" the evaluation of proposals. It involves the verification of all of the evaluation work by the legal department and special advisors to the Minister before he signs the contract attribution; this follows already a lengthy process of Viceministerial verification of bid evaluation and signature.

e) Execution, supervision and control
Inefficiency of supervision firms It is felt that supervision firms may be inefficient leading to insufficient control of works, acceptance of undue modifications and overall deficient results.

Inefficiency of MOPC control of supervision firms It is felt that MOPC is being inefficient in the way it monitors and controls the work of supervision firms, leading to the above negative results and making it difficult to apply necessary correction measures.

Collusion between contractors and supervision firms Usually supervision firms are paid on a time-based contract as it is difficult to predict accurately the time a specific construction work will take place. As a result, there is an advantage for the supervision firm for the construction to drag on. This framework may be prone to collusion between contractor and supervision firm to extend the duration of the job.

Lack of objective criteria for final acceptance of works Objective criteria for final acceptance of works are usually absent from contracts thus leaving significant discretion at the final reception of the contracted work, and making it difficult to make objective decisions.

Excessive internal bureaucracy for payments The payment circuit within the Ministry is manual and undergoes several controls and steps by different Departments that do not add value to the process of verification and approval of payments. This produces unnecessary delays, and gives discretionary power to different individuals along the process. As a result, firms are paid late and they have to spend time and

effort following their invoices around the MOPC. The process lends itself to unnecessary contact between MOPC staff and firm members which may be vulnerable to corruption.

Lack of criteria to prioritize payments All line ministries are subject to a cash plan (plan de caja) which limits the available funds for disbursements in any given month, irrespective of the adopted budget. As a result, contracted on-going activities may not be paid due to this limitation. An arbitration of who gets paid and who waits is imminently discretionary lending itself for corruption.

Weak internal controls. Financial assessment for the project indicates that the existing internal controls in MOPC are weak and may lead to substantial risk. Other financial management risks are modest.

Project component specific findings

a) Component 2: Improvement and Maintenance of Paved roads (performance-based maintenance contracts)

Limited expertise in performance based maintenance contracts by MOPC staff. There is limited operational experience with this type of contract and approach within the MOPC. Therefore, delays and or errors in preparation, execution and supervision are probable.

Limited expertise in performance based maintenance contracts by local contractors. There is limited expertise by local contractors with this type of contract and approach, risking limited participation and/or lobby against this modality before execution. During execution there is risk of low performance and conflicts with the supervisory team.

Limited expertise in performance based maintenance contracts by local consultants There is limited expertise by local consultants with type of contract and approach, risking limited participation and/or lobby against this modality before execution. During execution there is risk of low performance and conflicts with the contractor's team, and even collusion.

The term of the contract of four to five years may be an incentive for collusion The length and stability of the contracts may incite contractors to seek collusive practices to ensure participation in the program.

b) Component 3: Improvement and Maintenance of Unpaved Road Network.

Political pressures and elite capture may adversely effect the selection of the order in which Departments are chosen and the choice of road sections which to improve. Traditionally the road sector in many countries is seen as a vehicle with which to distill favors in exchange for political support. This includes road construction and simple road improvements to communities that in turn may support the governing party or individual. Additionally, the lack of objective criteria to choose road sections that need improvement and the level of improvement leaves too much discretionary power on MOPC side that may be swayed by non technical arguments.

Inefficient use of MOPC decentralized District equipment and staff. The inefficient programming of the use of MOPC resources will result in bad maintenance practices. Furthermore, without clear plans District staff easily respond to local pressures and emergencies that dictate the use of equipment rather than an objective plan.

Limited human resources capacity There is limited training and capacity in road maintenance and construction anywhere other than MOPC staff. The increased participation of local governments, local communities and civil society will inevitably increase difficulties due to their lack of specialized training.

c) Toll system subcomponent

Lack of adequate control mechanisms, procedures and equipment in the toll operating system facilitate evasion. The toll system is paper based without any centralized control or monitoring system, and rough estimates indicate that there is a minimum of 30% revenue loss.

Annex 14: Project Preparation and Supervision

PARAGUAY: Road Maintenance

	Planned	Actual
PCN review	04/2005	06/2005
Initial PID to PIC		11/02/2005
Initial ISDS to PIC		11/02/2005
Appraisal	04/2006	06/12/2006
Negotiations	06/2006	07/11/2006
Board/RVP approval	07/2006	08/22/2006
Planned date of effectiveness	01/01/2007	
Planned date of mid-term review	06/01/2009	
Planned closing date	06/30/2006	

Key institutions responsible for preparation of the project:

Bank staff and consultants who worked on the project included:

Name	Title	Unit
Andres Pizarro	Senior Transport Specialist	LCSFT
Jose Maria Alonso Biarge	Highway Engineer, Consultant	LCSFT
Graciela Sanchez Martinez	Civil Society and Social Development Specialist	LCCPY
Maria Lucy Giraldo	Procurement Specialist	LCOPR
Rodrigo Archondo-Callao	Highway Engineer	TUDTR
Antonio Blasco	Financial Management Specialist	LCSFM
Alejandro Alcala Gerez	Counsel	LEGLA
Verónica Raffo	Consultant	LCSFT
Roberto Muñoz	Environmental Specialist, Consultant	
Daniel Gross	Resettlement Consultant	IEGSG
Lucio Caceres	Institutional Development Consultant	LCSFT
Rodrigo Garcia Verdu	Economist	LCSPP
Xiomara Morel	Sr. Finance Officer	LOAG1
Jose Luis Irigoyen	Sector Manager	LCSFT
Juan Gaviria	Sector Leader	LCSFP
Emmanuel James	Lead Highway Specialist (Peer Reviewer)	LCSFT
William Paterson	Lead Highway Specialist (Peer Reviewer)	EAST
Jerry Lebo	Sr. Transportation Specialist (Peer Reviewer)	EAST
Zeinab Partow	Country Economist (QER Reviewer)	LCSPE
Stephen Brushet	Lead Highway Specialist (QER Reviewer)	LCSFT
Susana Sanchez	Senior Financial Economist (QER Reviewer)	LCSFP

Bank funds expended to date on project preparation:

1. Bank resources: 317,664.59
2. Trust funds: US$930,240
3. Total: US$1,247,904.5

Annex 15: Documents in the Project File

PARAGUAY: Road Maintenance

1. Project Implementation Plan

Draft Operational Manual, July 2006

Evaluación Social del Componente Mejoramiento y Mantenimiento de la Red Vial No Pavimentada (Project's Social Evaluation), March 2006

Evaluación Ambiental del Proyecto (Project's Environmental Evaluation), March 2006

Plan para los Pueblos Indígenas (Indigenous Peoples' Plan), March 29, 2006

Marco para Asistencia para el Traslado de Ocupaciones Ilegales de la Franja de Dominio (Resettlement Plan), March 2006

2. Bank Staff Assessments

PROJECT DOCUMENTS

Project Appraisal Document (Appraisal Stage) – Road Maintenance – P082026, June 6, 2006

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MOPC, Vice-Ministerio de Obras Publicas y Comunicaciones, Unidad de Preparación de Proyectos. *Informe de Avance No.* 1. Diciembre 2004

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Annex 16: Statement of Loans and Credits

PARAGUAY: Road Maintenance

Project ID	FY	Project Name	Original Amount in US$ Millions IBRD	IDA	GRANT	Cancel.	Undisb.	Orig.	Difference between Expected and Actual Disbursements Frm Rev'd
P039983	1998	PY 4th Rural Water Supply & Santitation	40.00	0.00	0.00	0.00	7.32	7.32	1.99
P080961	2005	PY Financial Sector T.A. Project	5.70	0.00	0.00	0.00	5.70	0.47	
P087036	2005	PY Modernization Ministry of Finance	7.50	0.00	0.00	0.00	7.50	2.00	
P039994	2005	PY Prog. Financial Sector Adj. Loan	15.00	0.00	0.00	0.00	15.00	15.00	
P069269	2002	PY Pilot Community Development Project	9.00	0.00	0.00	0.00	5.28	5.28	5.28
P073526	2004	PY- Education Reform Project	24.00	0.00	0.00	0.00	19.56	17.21	
P082056	2006	PY-Mother & Child Basic Health Insurance	22.00	0.00	0.00	0.00	22.00		
		Total	123.20	0.00	0.00	0.00	82.37	47.29	7.27

PARAGUAY
STATEMENT OF IFC's
Held and Outstanding Portfolio
As of 05/31/2006
In Millions of US Dollars

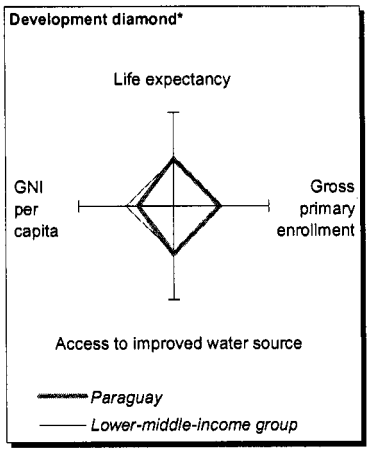
		Committed				Outstanding			
		IFC				IFC			
FY Approval	Company	Loan	Equity	Quasi	Partic.	Loan	Equity	Quasi	Partic.
2005	Telecel Paraguay	15.00	0.00	0.00	0.00	15.00	0.00	0.00	0.00
	Total portfolio:	15.00	0.00	0.00	0.00	15.00	0.00	0.00	0.00

Paraguay

Road Maintenance

Annex 17: Country at a Glance

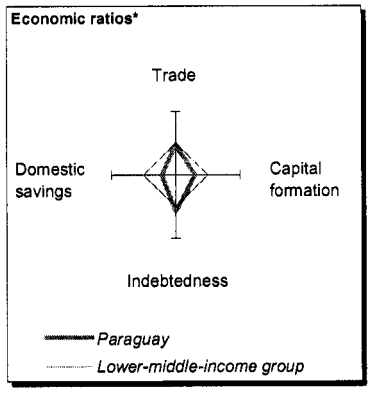
POVERTY and SOCIAL	Paraguay	Latin America & Carib.	Lower-middle-income
2004			
Population, mid-year *(millions)*	5.8	541	2,430
GNI per capita *(Atlas method, US$)*	1,170	3,600	1,580
GNI *(Atlas method, US$ billions)*	6.8	1,948	3,847
Average annual growth, 1998-04			
Population *(%)*	2.3	1.4	1.0
Labor force *(%)*	3.0	0.9	0.7
Most recent estimate (latest year available, 1998-04)			
Poverty *(% of population below national poverty line)*	..	..	..
Urban population *(% of total population)*	58	77	49
Life expectancy at birth *(years)*	71	71	70
Infant mortality *(per 1,000 live births)*	25	28	33
Child malnutrition *(% of children under 5)*	..	..	11
Access to an improved water source *(% of population)*	83	89	81
Literacy *(% of population age 15+)*	92	89	90
Gross primary enrollment *(% of school-age population)*	110	123	114
Male	112	126	115
Female	108	122	113



KEY ECONOMIC RATIOS and LONG-TERM TRENDS

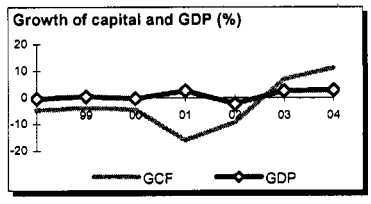
	1984	1994	2003	2004
GDP *(US$ billions)*	4.4	7.9	6.0	7.1
Gross capital formation/GDP	22.9	23.4	19.8	18.4
Exports of goods and services/GDP	15.9	34.2	32.3	29.4
Gross domestic savings/GDP	16.7	4.8	5.4	14.2
Gross national savings/GDP	16.7	6.5	7.8	16.8
Current account balance/GDP	-3.0	-3.5	2.6	1.3
Interest payments/GDP	1.4	0.9	1.7	2.1
Total debt/GDP	33.5	26.8	53.1	46.3
Total debt service/exports	15.7	6.3	11.0	15.8
Present value of debt/GDP	..	..	51.5	..
Present value of debt/exports	..	..	105.6	..

	1984-94	1994-04	2003	2004	2004-08
(average annual growth)					
GDP	3.6	1.0	2.6	2.9	3.3
GDP per capita	0.9	-1.3	0.1	0.4	1.1
Exports of goods and services	15.1	-6.2	13.2	4.9	4.0

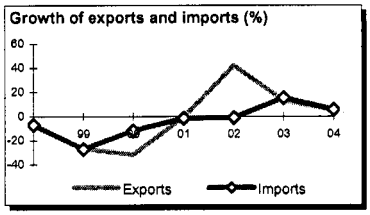


STRUCTURE of the ECONOMY

	1984	1994	2003	2004
(% of GDP)				
Agriculture	28.7	23.7	27.2	27.2
Industry	22.7	25.6	24.2	24.2
Manufacturing	14.3	15.2	13.6	13.6
Services	48.6	50.6	48.5	48.5
Household final consumption expenditure	76.8	88.4	87.7	78.8
General gov't final consumption expenditure	6.5	6.8	6.9	7.0
Imports of goods and services	22.1	52.8	46.7	33.6

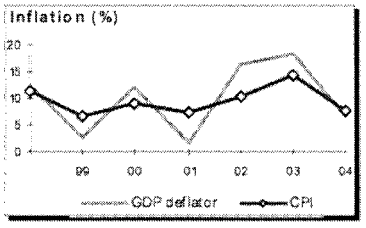


	1984-94	1994-04	2003	2004
(average annual growth)				
Agriculture	3.6	2.8	9.5	2.1
Industry	5.0	0.6	-0.3	2.1
Manufacturing	-1.5	0.0	-1.0	2.1
Services	3.0	0.1	-0.4	3.9
Household final consumption expenditure	4.2	0.9	4.8	1.8
General gov't final consumption expenditure	11.2	0.4	-7.2	3.5
Gross capital formation	4.9	-3.8	7.0	11.2
Imports of goods and services	16.4	-6.1	15.4	5.5



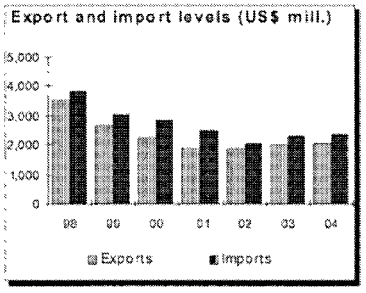
PRICES and GOVERNMENT FINANCE

	1984	1994	2003	2004
Domestic prices				
(% change)				
Consumer prices	..	20.6	14.3	7.5
Implicit GDP deflator	26.9	21.0	18.3	6.7
Government finance				
(% of GDP, includes current grants)				
Current revenue	..	15.1	15.4	16.0
Current budget balance	..	14.0	2.6	3.5
Overall surplus/deficit	..	1.16	-0.4	0.2



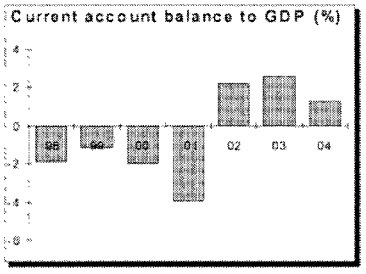
TRADE

	1984	1994	2003	2004
(US$ millions)				
Total exports (fob)	560	3,423	2,014	2,096
Soy products	102	222	625	625
Cotton	131	171	65	68
Manufactures	..	2,606	518	539
Total imports (cif)	740	3,604	2,323	2,395
Food	..	1,463	663	622
Fuel and energy	..	159	244	314
Capital goods	..	755	508	524
Export price index *(2000=100)*	..	114	77	78
Import price index *(2000=100)*	..	101	72	71
Terms of trade *(2000=100)*	..	113	107	107



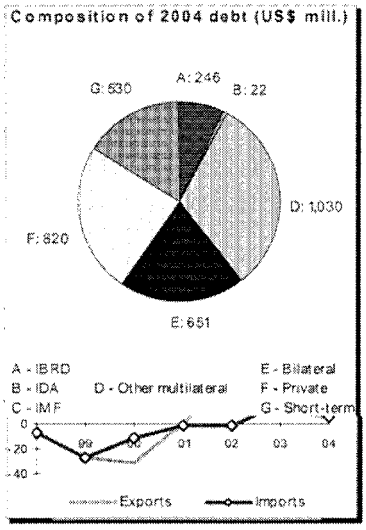
BALANCE of PAYMENTS

	1984	1994	2003	2004
(US$ millions)				
Exports of goods and services	756	3,786	2,636	2,643
Imports of goods and services	948	4,197	2,631	2,733
Resource balance	-191	-411	5	-90
Net income	49	114	8	6
Net current transfers	9	23	139	176
Current account balance	-133	-274	155	93
Financing items (net)	38	613	-481	-192
Changes in net reserves	96	-339	326	99
Memo:				
Reserves including gold *(US$ millions)*	..	1,041	983	1,080
Conversion rate *(DEC, local/US$)*	243.7	1,905.0	6,435.6	5,976.0



EXTERNAL DEBT and RESOURCE FLOWS

	1984	1994	2003	2004
(US$ millions)				
Total debt outstanding and disbursed	1,470	2,105	3,202	3,299
IBRD	179	162	256	246
IDA	47	37	24	22
Total debt service	136	256	323	478
IBRD	27	59	31	47
IDA	1	1	2	2
Composition of net resource flows				
Official grants	11	19	18	..
Official creditors	137	31	7	-69
Private creditors	27	23	30	-124
Foreign direct investment (net inflows)	5	137	91	..
Portfolio equity (net inflows)	0	0	0	..
World Bank program				
Commitments	30	165	54	0
Disbursements	59	7	42	17
Principal repayments	14	44	21	35
Services	3.0	6.1	10.4	3.9
Household final consumption expenditure	4.2	0.9	4.8	1.8
General gov't final consumption expenditure	11.2	0.4	-7.2	3.5
Gross capital formation	4.9	-3.8	7.0	11.2
Imports of goods and services	16.4	-6.1	15.4	5.5



Annex 18: Letter of Sector Policy

PARAGUAY: Road Maintenance

Paraguay Road Sector Strategy

1. Road Policy Framework and Guidelines

Transport infrastructure in Paraguay is still in a development phase, especially in the western part of the country known as "El Chaco".

Rural access and mobility are highly dependent on road infrastructure exerting particular pressure on its maintenance, especially of unpaved roads, that represent the majority of the country's roads network. At the same time, agricultural and cattle production, which constitute the principal component of Paraguay's economy, also generate significant pressure on the road network infrastructure, particularly due to the recent expansion of the oleaginous production.

All of the above announces a backlog in supply with respect to demand, arriving at a time of high fiscal restrictions and considerable social and political demands in other sectors such as, education, health, social integration, etc.

Finally, although the above mentioned comments are related to road infrastructure, due to the country's landlocked conditions, coordination and harmonization with other infrastructure modes (especially fluvial) is necessary in order to better serve the export and import flows.

In the face of this assessment where a growing country is handicapped by insufficient infrastructure and transport services, it would be ideal that the sector evolve following a National Transport Plan consistent with a solid Regional Land Use and Development Plan.

While the National Transport Plan is formulated, Paraguay's road policy seeks to align itself with objectives of economic growth and enhancement of social equality within a fiscally and environmentally sustainable framework, and in the context of institutional transparency promoted by the National Government.

In this context, Paraguay's Road Strategy seeks to support:

1) The improvement of production and export competitiveness, in order to overcome the handicaps of a landlocked country, through the consolidation of the international road network and the main paved road network
2) The expansion of employment opportunities and market access through a slow but sustainable development of the improved and unpaved network in order to improve the accessibility in rural areas to the paved network, and
3) The improvement in the quality of life and access to basic services in the country's least developed areas through more and better roads for urban and rural populations.

2. Existing Situation of the Road Network

A general assessment of the national road network reveals the country's main needs in this respect:

1) A road network with insufficient coverage with respect to the country's size. The indicators corresponding to area coverage of the paved or improved road networks are low compared to those from other countries in the region. The recently built paved networks have improved mobility but are still insufficient. The improved unpaved national and local networks are also insufficient in terms of coverage and oblige users to travel long distances on bad conditioned roads before reaching the paved network.

2) An incomplete regional integration road network that needs to be completed not only due to reasons of sovereignty and integration with regional markets, but also to overcome the country's landlocked condition which affects trade and the price of commercialized products in the external markets.

3) A road network without maintenance, that risks rapid deterioration if no adequate policies are implemented, that needs to be modernized through initial interventions and subsequent regular, planned and continuous maintenance in the paved as well as the unpaved networks.

Taking into account the existing situation, the proposed strategy seeks to give priority to the above mentioned priorities in accordance with the road policy framework and guidelines previously presented.

3. MOPC's Road Strategy

Given the situation of Paraguay's road network, the National Government and MOPC's Road Strategy propose the following General Objectives:

1) Complete the country's principal network in its international and bi-oceanic corridors.
2) Improve general conditions of roads serving production and populations through the extension of the paved road network and improvement of the quality of the unpaved network.
3) Introduce new road maintenance practices for all road networks that can assure the quality and safety of road networks, minimize transport costs and facilitate accessibility of the population to basic services.
4) Ensure fiscal sustainability of the programs to be developed, in other words, that they are in accordance with the available budgetary resources.
5) Improve the efficiency in the use of resources, in order to achieve better results with the available resources.

4. Lines of Action

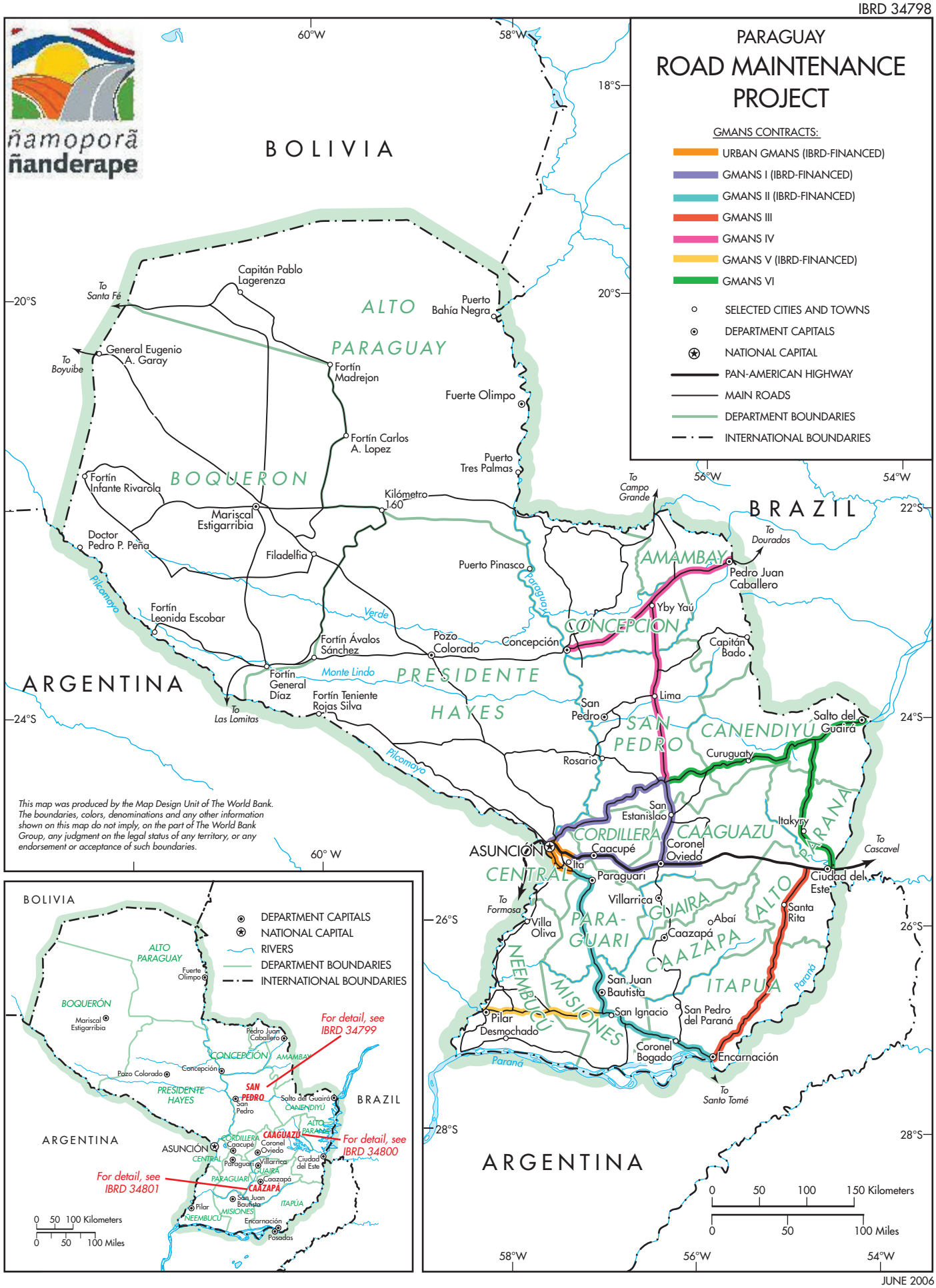
The proposed strategy, although it seems self evident conceptually, will require a significant institutional effort for its implementation. As a consequence, a major reform in MOPC procedures and planning methodologies will be necessary, in order to adapt to the Paraguayan reality, successful managerial practices that have proved successful in other countries of the region.

As a result, the specific objectives and targets of this strategy are being formulated, particularly due to the National Government's intention to formulate them after consultation with principal sectors and society in general.

Further to the above mentioned reasons, the Government and MOPC's projected actions destined to implement the planned strategy are described below:

1) Develop a policy to complete the missing links of the regional integration road network, connecting production and distribution centers with principal borders.

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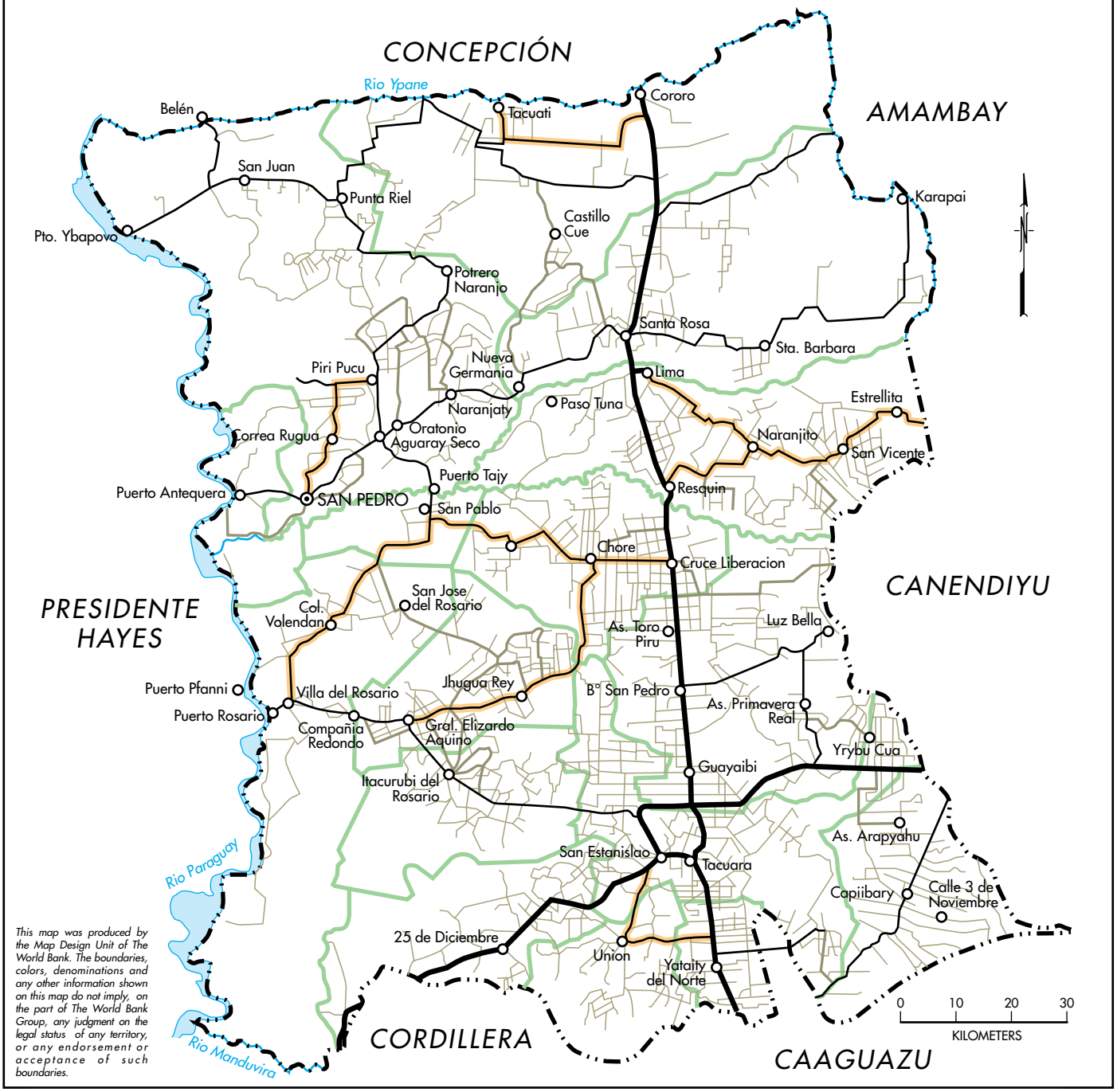
PARAGUAY

ROAD MAINTENANCE PROJECT

DEPARTMENT OF SAN PEDRO

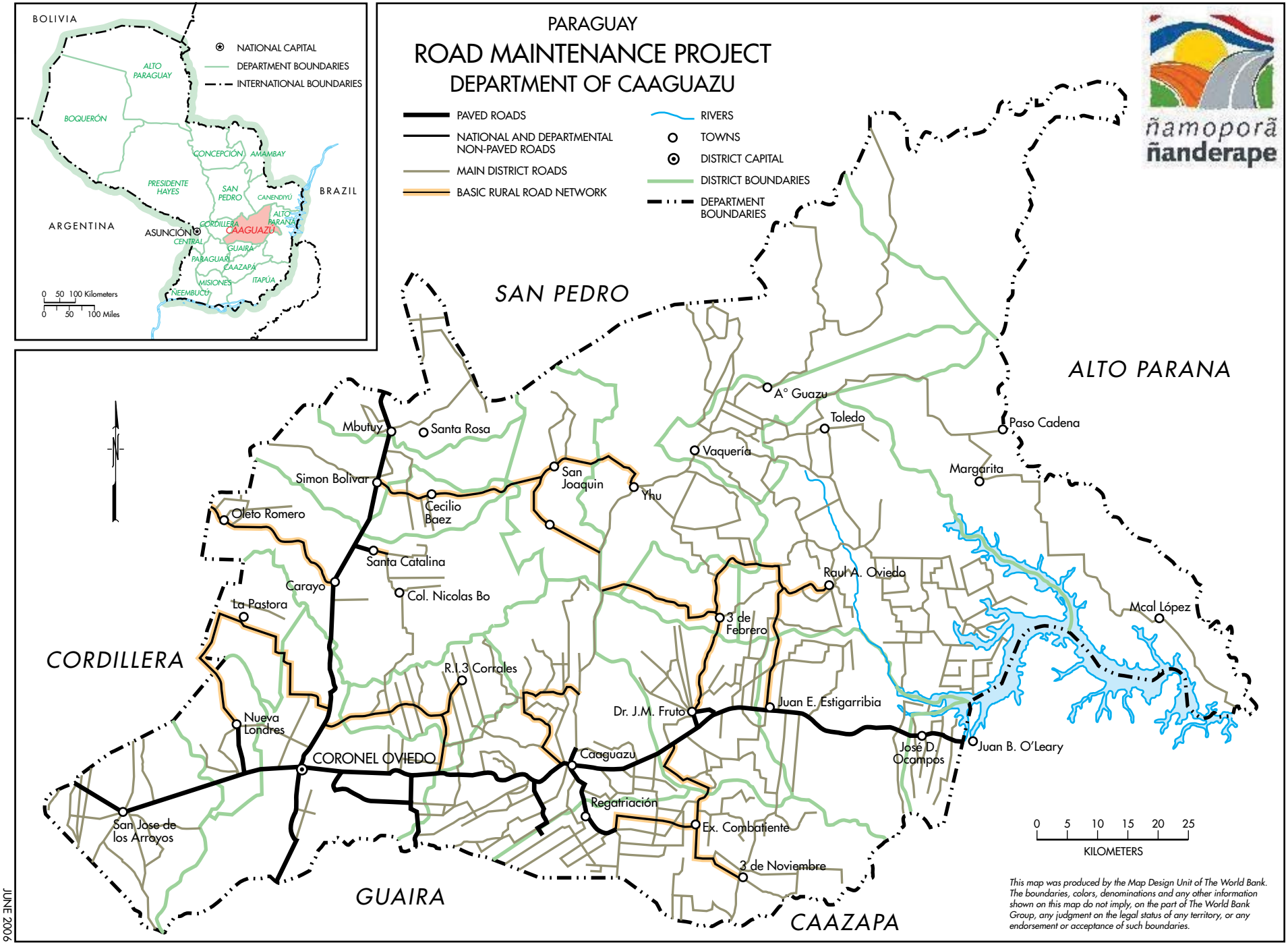
- PAVED ROADS
- NATIONAL AND DEPARTMENTAL NON-PAVED ROADS
- MAIN DISTRICT ROADS
- SECONDARY DISTRICT ROADS
- BASIC RURAL ROAD NETWORK
- RIVERS
- TOWNS
- DISTRICT CAPITAL
- DISTRICT BOUNDARIES
- DEPARTMENT BOUNDARIES

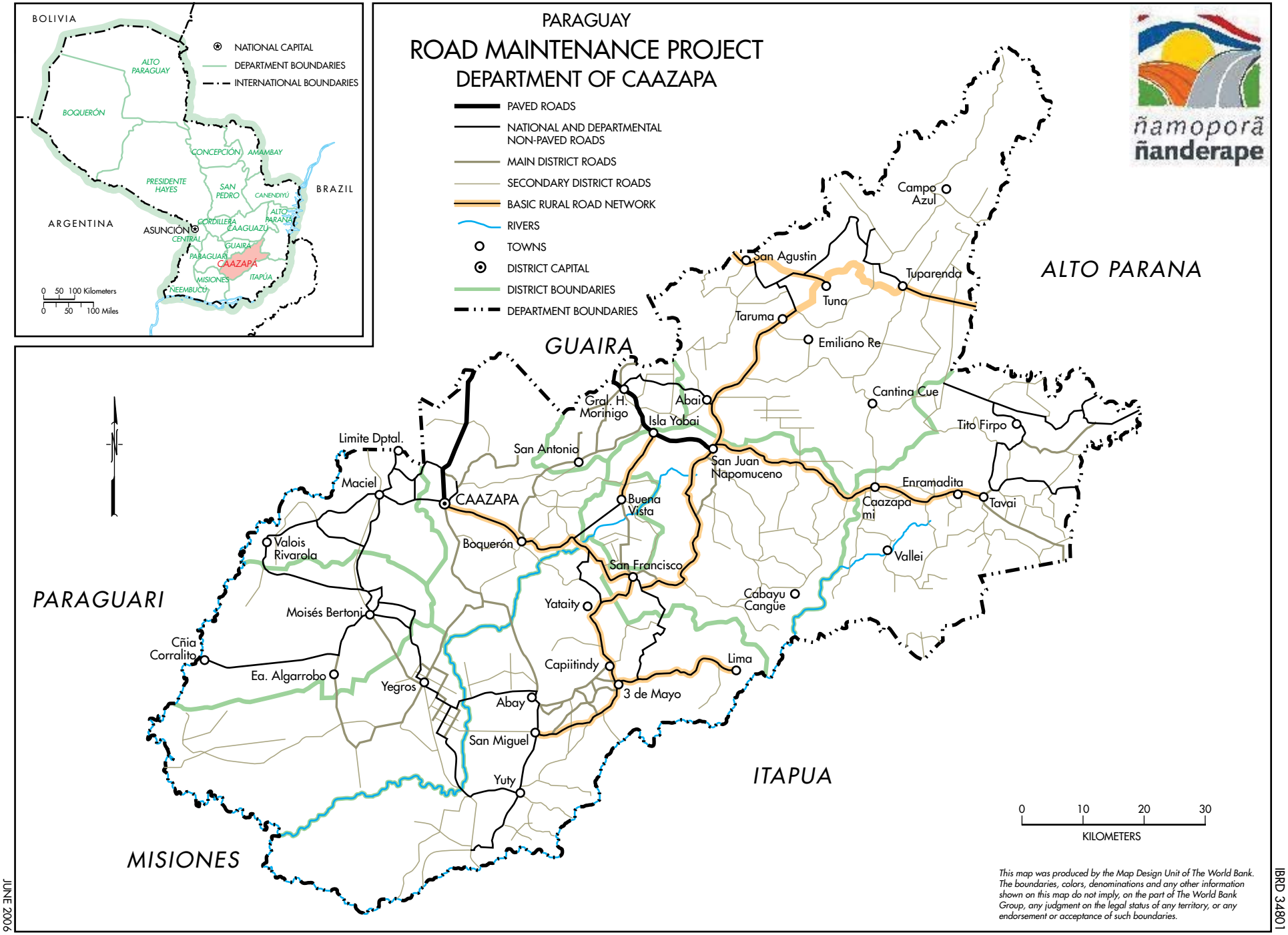




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PARAGUAY
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NATIONAL AND DEPARTMENTAL NON-PAVED ROADS
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BASIC RURAL ROAD NETWORK
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BRAZIL
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AMAMBAY
PRESIDENTE HAYES
SAN PEDRO
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Mbutuy
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Cecilio Baez
Oleto Romero
Santa Catalina
Carayo
Col. Nicolas Bo
La Pastora
R.I.3 Corrales
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3 de Febrero
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Dr. J.M. Fruto
Juan E. Estigarribia
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Juan B. O'Leary
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ROAD MAINTENANCE PROJECT
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NATIONAL AND DEPARTMENTAL NON-PAVED ROADS
MAIN DISTRICT ROADS
SECONDARY DISTRICT ROADS
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MISIONES
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Tuparenda
Tuna
Taruma
Emiliano Re
Cantina Cue
Tito Firpo
Gral. H. Morinigo
Abai
Isla Yobai
Limite Dptal.
San Antonio
San Juan Napomuceno
Maciel
CAAZAPA
Buena Vista
Enramadita
Caazapami
Tavai
Valois Rivarola
Boquerón
San Francisco
Vallei
Cabayu Cangüe
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